Hartshorne, Chapter IV: Curves

Solutions to the exercises

Conventions. As in the book, k is an algebraically closed field and "curve" means a complete nonsingular curve over k unless an exercise says otherwise. Results of the text are cited by their numbers in Hartshorne ("Thm. 1.3", "Prop. 4.6", ...), and results of other chapters as "II, ..." or "III, ...". The numbering and statements of the exercises follow the edition scanned in this repository; each exercise is restated before its solution, and starred exercises are marked "Starred in the book" at the end of the statement.

Honest notes. Several printed statements need a hypothesis or correction, and the solutions say what is assumed or proved instead: characteristic assumptions (char $\neq 2$ in much of Section 4; characteristic 0 in Ex. 5.2, 5.3, 5.5, 5.7 and part of 6.7); the parenthetical remark in Ex. 2.3(h) about a tacnode is wrong (the dual branch has orders $(3, 4)$); Ex. 4.16(d) as printed fails for supersingular curves; Ex. 5.5(b) is false as printed in the cuspidal case. **Ex. 6.7 is only partly proved:** the first half is proved in full, and the second half is proved for plane curves and curves on a quadric, while for the remaining nondegenerate curves it is proved conditionally on one lemma (the tangent line at a general point meets the curve nowhere else, in characteristic 0) which I could not prove; the solution says so precisely. Some solutions use standard facts beyond the text (EGA smoothness criteria, miracle flatness, Weierstrass preparation), which are named where used.

1. The Riemann-Roch theorem

Conventions. Throughout, k is algebraically closed, a *curve* is a complete nonsingular curve over k (so projective), K denotes a canonical divisor (and also, when no confusion is possible, a function field), and for a divisor D on a curve we write $\mathcal{L}(D)$ for the associated invertible sheaf, $H^0(X, \mathcal{L}(D)) = \{f \in K(X)^* : (f) + D \geq 0\} \cup \{0\}$ and $l(D) = \dim_k H^0(X, \mathcal{L}(D))$. Results of the text are cited as "Thm. 1.3", "Lemma 1.2", ..., those of other chapters as "II, Ex. 6.11" etc. Recall the following facts, used repeatedly: if $f : X \to \mathbf{P}^1$ is the morphism defined by a nonconstant $f \in K(X)$, then f is finite, $f^*(\infty) = (f)_\infty$ is the divisor of poles of f, and $\deg f = [K(X) : k(f)] = \deg(f)_\infty$ (II, 6.8, 6.9, and $e_P = -\operatorname{ord}_P(f)$ at a pole, because $1/f$ is a local parameter at ∞). For a point P the sheaf sequence $0 \to \mathcal{L}(D - P) \to \mathcal{L}(D) \to k(P) \to 0$ gives $l(D - P) \leq l(D) \leq l(D - P) + 1$.

Exercise 1.1

> **Problem.** Let X be a curve, and let $P \in X$ be a point. Then there exists a nonconstant rational function $f \in K(X)$, which is regular everywhere except at P.

Solution. Let g be the genus and put $n = 2g + 1$. For $m \geq 2g - 1$ we have $\deg(K - mP) = 2g - 2 - m < 0$, so $l(K - mP) = 0$ by Lemma 1.2, and the Riemann–Roch theorem (Thm. 1.3) gives $l(mP) = m + 1 - g$. Hence $l(nP) - l((n-1)P) = 1$ (both n and $n - 1 = 2g$ are $\geq 2g - 1$), so $H^0(X, \mathcal{L}((n-1)P))$ is a proper subspace of $H^0(X, \mathcal{L}(nP))$. Pick f in the difference. Then $(f) + nP \geq 0$ but $(f) + (n-1)P \not\geq 0$, which means $\mathrm{ord}_P(f) = -n$ and $\mathrm{ord}_Q(f) \geq 0$ for all $Q \neq P$. Thus f is regular on $X - \{P\}$ and has a pole of order $n = 2g + 1 \geq 1$ at P; in particular f is nonconstant. □

Exercise 1.2

> **Problem.** Again let X be a curve, and let $P_1, \ldots, P_r \in X$ be points. Then there is a rational function $f \in K(X)$ having poles (of some order) at each of the P_i, and regular elsewhere.

Solution. For each i choose, by Ex. 1.1, a rational function f_i regular on $X - \{P_i\}$ with a pole at P_i, and put $f = f_1 + \cdots + f_r$ (if $r = 0$ take $f = 0$). At P_i, the function f_i has a pole while $\sum_{j \neq i} f_j$ is regular, so f has a pole at P_i (a sum of a function with a pole and a regular function has a pole). At any $Q \notin \{P_1, \ldots, P_r\}$ every f_j is regular, hence so is f. □

Exercise 1.3

> **Problem.** Let X be an integral, separated, regular, one-dimensional scheme of finite type over k, which is *not* proper over k. Then X is affine. [*Hint*: Embed X in a (proper) curve $\overline{X}$ over k, and use (Ex. 1.2) to construct a morphism $f : \overline{X} \to \mathbf{P}^1$ such that $f^{-1}(\mathbf{A}^1) = X$.]

Solution. *Step 1: X is an open subscheme of a curve $\overline{X}$.* Let $L = K(X)$; it has transcendence degree 1 over k. Let $\overline{X}$ be the (unique) nonsingular projective curve with function field L (I, 6.9 and its scheme-theoretic form II, 6.7), so the closed points of $\overline{X}$ are the discrete valuation rings of L/k. For a closed point $P \in X$, the local ring $\mathcal{O}_P$ is a regular local ring of dimension 1, hence a DVR of L/k. Since $\overline{X} \to \operatorname{Spec} k$ is proper, the valuative criterion (II, 4.7) shows that the generic point $\operatorname{Spec} L \to \overline{X}$ extends to a morphism $\operatorname{Spec} \mathcal{O}_P \to \overline{X}$. This extension is defined on an open neighborhood U_P of P in X: if $\operatorname{Spec} B \subseteq \overline{X}$ is an affine open containing the image of the closed point and $\operatorname{Spec} A \subseteq X$ is an affine neighborhood of $P = \mathfrak{p}$, the morphism is a k-algebra map $B \to A_{\mathfrak{p}}$, the finitely many generators of B go to elements of $A_{\mathfrak{p}} \subseteq L$ with a common denominator $s \in A - \mathfrak{p}$, so it comes from $B \to A_s$, which is compatible with the inclusions in L. Two such morphisms agree on the overlaps (they agree at the generic point, X is reduced and $\overline{X}$ is separated, II, 4.1 and its proof), so they glue to a morphism $g : X \to \overline{X}$ which is the identity on function fields. For each closed point P the map $\mathcal{O}_{\overline{X}, g(P)} \to \mathcal{O}_P$ is an inclusion of DVRs of L with the second dominating the first; since a valuation ring is maximal for domination (I, Thm. 6.1A) it is an isomorphism. If $g(P) = g(Q)$, then $\mathcal{O}_P = \mathcal{O}_Q$ as subrings of L, and the two morphisms $\operatorname{Spec} \mathcal{O}_P \to X$ centered at P and Q agree on $\operatorname{Spec} L$; as X is separated, the uniqueness part of the valuative criterion (II, 4.3) gives $P = Q$. So g is injective on closed points (and sends the generic point to the generic point, so g is injective). Next, g is a local isomorphism near each closed point P: take affine opens $\operatorname{Spec} A \ni P$ and $\operatorname{Spec} B \ni g(P)$ with $g(\operatorname{Spec} A) \subseteq \operatorname{Spec} B$, and let $P, g(P)$ correspond to $\mathfrak{p} \subset A$, $\mathfrak{q} = \mathfrak{p} \cap B$. Then $B \subseteq A \subseteq L$ and A is a finitely generated B-algebra, $A = B[a_1, \ldots, a_m]$; as $A_{\mathfrak{p}} = B_{\mathfrak{q}}$ we can write $a_i = b_i / s_i$

with $b_i, s_i \in B$, $s_i \notin \mathfrak{q}$. With $s = \prod s_i \notin \mathfrak{p}$ we get $A_s = B_s$, i.e. g maps the open set $D(s) \ni P$ of $\operatorname{Spec} A$ isomorphically onto the open set $D(s)$ of $\operatorname{Spec} B$. The set of points near which g is an open immersion is open and contains all closed points, hence is all of X. An injective local open immersion is an open immersion. We regard $X \subseteq \overline{X}$ as an open subscheme. (This is the standard embedding of a one-dimensional separated scheme into its complete model.)

Step 2. Since $\overline{X}$ is proper over k and X is not, $X \neq \overline{X}$. The complement $\overline{X} - X$ is a proper closed subset of an irreducible curve, hence a nonempty finite set $\{P_1, \ldots, P_r\}$ of closed points. By Ex. 1.2 there is $f \in K(\overline{X})$ with a pole at every P_i and regular elsewhere. The corresponding morphism $f : \overline{X} \to \mathbf{P}^1$ is finite (nonconstant morphism of curves, II, 6.8) and $f^{-1}(\infty) = \{P_1, \ldots, P_r\}$ (the points where f has a pole). Hence $f^{-1}(\mathbf{A}^1) = \overline{X} - \{P_i\} = X$, and the restriction $X = f^{-1}(\mathbf{A}^1) \to \mathbf{A}^1$ of a finite morphism is finite, in particular affine. As $\mathbf{A}^1$ is affine, X is affine. □

Exercise 1.4

> **Problem.** Show that a separated, one-dimensional scheme of finite type over k, none of whose irreducible components is proper over k, is affine. [*Hint*: Combine (Ex. 1.3) with (III, Ex. 3.1, Ex. 3.2, Ex. 4.2).]

Solution. Let X be such a scheme; it is noetherian. Recall (III, Ex. 3.1, 3.2) that a noetherian scheme is affine if and only if each of its irreducible components, with the reduced induced structure, is affine (first pass to X_{red}, then to the components). So let X_i be an irreducible component with its reduced structure. It is integral, separated, of finite type over k, and $\dim X_i \leq 1$; it cannot have dimension 0, because a zero-dimensional integral scheme of finite type over k is $\operatorname{Spec} k$, which is proper. So $\dim X_i = 1$.

Let $\pi : \widetilde{X}_i \to X_i$ be the normalization (II, Ex. 3.8). It is a finite surjective morphism, $\widetilde{X}_i$ is integral, normal of dimension one, hence regular, and separated and of finite type over k (being finite over X_i). It is not proper over k: otherwise X_i, the image under the surjective map π of a proper scheme, would be proper (X_i is separated and universally closed, since for every Z the map $\widetilde{X}_i \times Z \to X_i \times Z$ is surjective and $\widetilde{X}_i \times Z \to Z$ is closed), contrary to the hypothesis. By Ex. 1.3, $\widetilde{X}_i$ is affine. By Chevalley's theorem (III, Ex. 4.2: if $f : X' \to Y'$ is finite surjective of noetherian separated schemes and X' is affine, so is Y'), X_i is affine. Hence all irreducible components of X are affine, so X is affine. □

Exercise 1.5

> **Problem.** For an effective divisor D on a curve X of genus g, show that $\dim |D| \leq \deg D$. Furthermore, equality holds if and only if $D = 0$ or $g = 0$.

Solution. Write $D = P_1 + \cdots + P_n$ with $n = \deg D$ (points may repeat) and put $D_j = P_1 + \cdots + P_j$, $D_0 = 0$. Since $l(D_j) \leq l(D_{j-1}) + 1$ (see the conventions above, with $D_{j-1} = D_j - P_j$) and $l(0) = \dim H^0(X, \mathcal{O}_X) = 1$ (X is a complete variety), we get $l(D) \leq n + 1$, i.e. $\dim |D| = l(D) - 1 \leq \deg D$.

If $D = 0$ then $\dim |D| = 0 = \deg D$. If $g = 0$, then $\deg(K - D) = -2 - \deg D < 0$, so $l(K - D) = 0$ (Lemma 1.2) and Riemann–Roch gives $l(D) = \deg D + 1$, i.e. equality.

Conversely suppose equality holds, $D \neq 0$ and $n \geq 1$. Then $l(D) = n + 1$, which forces $l(D_j) = l(D_{j-1}) + 1$ for all j; in particular $l(P_1) = 2$. Take $f \in H^0(X, \mathcal{L}(P_1))$ nonconstant. Its divisor of poles is $\leq P_1$ and nonzero, so $(f)_\infty = P_1$, and the finite morphism $f : X \to \mathbf{P}^1$ has degree $\deg(f)_\infty = 1$. Thus $K(X) = k(f) \cong K(\mathbf{P}^1)$ and $X \cong \mathbf{P}^1$ (I, 6.12), so $g = 0$. Hence equality holds only for $D = 0$ or $g = 0$. □

Exercise 1.6

> **Problem.** Let X be a curve of genus g. Show that there is a finite morphism $f : X \to \mathbf{P}^1$ of degree $\leq g + 1$. (Recall that the degree of a finite morphism of curves $f : X \to Y$ is defined as the degree of the field extension $[K(X) : K(Y)]$ (II, Section 6).)

Solution. Fix a point P and let $D = (g + 1)P$. By Riemann–Roch, $l(D) \geq \deg D + 1 - g = 2$ (as $l(K - D) \geq 0$). So $H^0(X, \mathcal{L}(D))$ contains a nonconstant function f. Then $(f)_\infty \leq D$, and $(f)_\infty \neq 0$ because f is nonconstant. The associated morphism $f : X \to \mathbf{P}^1$ is finite of degree $\deg f = \deg(f)_\infty \leq \deg D = g + 1$. □

Exercise 1.7

> **Problem.** A curve X is called *hyperelliptic* if $g \geq 2$ and there exists a finite morphism $f : X \to \mathbf{P}^1$ of degree 2.
>
> (a) If X is a curve of genus $g = 2$, show that the canonical divisor defines a complete linear system $|K|$ of degree 2 and dimension 1, without base points. Use (II, 7.8.1) to conclude that X is hyperelliptic.
>
> (b) Show that the curves constructed in (1.1.1) all admit a morphism of degree 2 to $\mathbf{P}^1$. Thus there exist hyperelliptic curves of any genus $g \geq 2$.
>
> *Note.* We will see later (Ex. 3.2) that there exist nonhyperelliptic curves. See also (V, Ex. 2.10).

Solution. (a) By Example 1.3.3, $\deg K = 2g - 2 = 2$ and $l(K) = g = 2$, so $|K|$ is a complete linear system of degree 2 and dimension $l(K) - 1 = 1$. Suppose P were a base point of $|K|$: then every section of $\mathcal{L}(K)$ vanishes at P, i.e. $l(K - P) = l(K) = 2$. By Riemann–Roch applied to the divisor P, $l(P) - l(K - P) = 1 + 1 - 2 = 0$, so $l(P) = 2$. But $l(P) = 2$ forces $X \cong \mathbf{P}^1$ (see the proof of Ex. 1.5), contradicting $g = 2$. So $|K|$ has no base points. Hence $\mathcal{L}(K)$ is generated by its two-dimensional space of global sections, and by (II, 7.8.1) it defines a morphism $\varphi : X \to \mathbf{P}^1$ with $\varphi^*\mathcal{O}(1) \cong \mathcal{L}(K)$. As $\deg \mathcal{L}(K) = 2 > 0$, φ is not constant, hence finite, and $2 = \deg \varphi^*\mathcal{O}(1) = \deg \varphi \cdot \deg \mathcal{O}(1) = \deg \varphi$ (II, 6.9). So X is hyperelliptic.

(b) Let $Q \cong \mathbf{P}^1 \times \mathbf{P}^1 \subseteq \mathbf{P}^3$ be a nonsingular quadric surface and let $X \subseteq Q$ be an irreducible nonsingular divisor of type $(g + 1, 2)$, $g \geq 2$, which has $p_a = g$ (Remark 1.1.1 and III, Ex. 5.6); so $X = Z(F)$ for a bihomogeneous form F of bidegree $(g + 1, 2)$ in $((x_0, x_1), (y_0, y_1))$, and F is irreducible. Consider the first projection $\mathrm{pr}_1 : X \to \mathbf{P}^1$ (to the factor $(x_0 : x_1)$, in which F has degree $g + 1$, while F has degree 2 in (y_0, y_1)). It is a morphism and is not constant on the integral curve X unless X lies in a fibre $\{a\} \times \mathbf{P}^1$, which would mean that the linear form defining a divides F, impossible since F is irreducible of bidegree $(g+1, 2)$ with $g+1 \geq 3$. So it is finite.

Its degree is $[K(X) : k(x)]$ where $x = x_1/x_0$. In the chart $x_0 y_0 \neq 0$ we have $K(X) = k(x)(y)$ with $y = y_1/y_0$ and y satisfies the dehomogenization $f(x, y) = a(x)y^2 + b(x)y + c(x) = 0$ of F, where a, c are not identically zero (otherwise y_1 or y_0 would divide F). This polynomial is irreducible in $k(x)[y]$, since by Gauss' lemma a factorization would give a factorization of F. Hence $[K(X) : k(x)] = 2$, and $\mathrm{pr}_1 : X \to \mathbf{P}^1$ has degree 2. Since $g \geq 2$, X is hyperelliptic of genus g for every $g \geq 2$. □

Exercise 1.8

Problem. *p_a of a Singular Curve.* Let X be an integral projective scheme of dimension 1 over k, and let $\widetilde{X}$ be its normalization (II, Ex. 3.8). Then there is an exact sequence of sheaves on X,

$$0 \to \mathcal{O}_X \to f_*\mathcal{O}_{\widetilde{X}} \to \sum_{P \in X} \widetilde{\mathcal{O}}_P/\mathcal{O}_P \to 0,$$

where $\widetilde{\mathcal{O}}_P$ is the integral closure of $\mathcal{O}_P$. For each $P \in X$, let $\delta_P = \mathrm{length}(\widetilde{\mathcal{O}}_P/\mathcal{O}_P)$.

(a) Show that $p_a(X) = p_a(\widetilde{X}) + \sum_{P \in X} \delta_P$. [*Hint*: Use (III, Ex. 4.1) and (III, Ex. 5.3).]

(b) If $p_a(X) = 0$, show that X is already nonsingular and in fact isomorphic to $\mathbf{P}^1$. This strengthens (1.3.5).

(c) If P is a node or an ordinary cusp (I, Ex. 5.6, Ex. 5.14), show that $\delta_P = 1$. [*Hint*: Show first that δ_P depends only on the analytic isomorphism class of the singularity at P. Then compute δ_P for the node and cusp of suitable plane cubic curves. See (V, 3.9.3) for another method.]

(Part (c) is starred in the book.)

Solution. Here $f : \widetilde{X} \to X$ is the normalization, a finite birational morphism; $\widetilde{X}$ is an integral normal, i.e. nonsingular, projective curve. The quotient $\mathcal{T} = \sum_P \widetilde{\mathcal{O}}_P/\mathcal{O}_P$ is a coherent sheaf supported on the finite set of singular points, and $\dim_k H^0(X, \mathcal{T}) = \sum_P \mathrm{length}_{\mathcal{O}_P}(\widetilde{\mathcal{O}}_P/\mathcal{O}_P) = \sum_P \delta_P$, because every composition factor of an $\mathcal{O}_P$-module of finite length is $k(P) = k$. Also $H^i(X, \mathcal{T}) = 0$ for $i > 0$ (the sheaf is a finite sum of skyscraper sheaves, hence flasque).

(a) For an integral projective scheme of dimension 1 one has $p_a = 1 - \chi(\mathcal{O})$ (III, Ex. 5.3). The Euler characteristic is additive in exact sequences (III, Ex. 5.1), so $\chi(f_*\mathcal{O}_{\widetilde{X}}) = \chi(\mathcal{O}_X) + \chi(\mathcal{T}) = \chi(\mathcal{O}_X) + \sum_P \delta_P$. Since f is finite, hence affine, $H^i(X, f_*\mathcal{O}_{\widetilde{X}}) \cong H^i(\widetilde{X}, \mathcal{O}_{\widetilde{X}})$ (III, Ex. 4.1), so $\chi(f_*\mathcal{O}_{\widetilde{X}}) = \chi(\mathcal{O}_{\widetilde{X}})$. Therefore $\chi(\mathcal{O}_{\widetilde{X}}) = \chi(\mathcal{O}_X) + \sum \delta_P$, and applying $1 - (\cdot)$ gives $p_a(X) = p_a(\widetilde{X}) + \sum_P \delta_P$.

(b) By Prop. 1.1, $p_a(\widetilde{X}) = g(\widetilde{X}) \geq 0$, and each $\delta_P \geq 0$. If $p_a(X) = 0$, part (a) forces $\delta_P = 0$ for all P and $g(\widetilde{X}) = 0$. Now $\delta_P = 0$ means $\mathcal{O}_P = \widetilde{\mathcal{O}}_P$, so $\mathcal{O}_X \to f_*\mathcal{O}_{\widetilde{X}}$ is an isomorphism. As f is finite, $\widetilde{X} = \mathrm{Spec}_X\, f_*\mathcal{O}_{\widetilde{X}} = \mathrm{Spec}_X\, \mathcal{O}_X = X$, i.e. f is an isomorphism. So $X \cong \widetilde{X}$ is nonsingular of genus 0, hence $X \cong \mathbf{P}^1$ by Example 1.3.5.

(c) *Reading of the terms.* In I, Ex. 5.6 and 5.14 a node is a double point of a plane curve with two distinct tangent directions, and an ordinary cusp is a double point whose tangent cone is a double line and whose cubic term is nonzero on that line, i.e. the singularity of $y^2 = x^3$. Concretely: in suitable coordinates x, y at P (so that $\widehat{\mathcal{O}}_P = k[[x, y]]/(h)$ for the local equation h of the curve), $h = xy +$ (terms of order ≥ 3) for a node, and $h = y^2 +$

(cubic form with nonzero coefficient a of x^3) + (terms of order ≥ 4) for an ordinary cusp. We use only these descriptions, in any characteristic. (In characteristic $\neq 2, 3$ they say exactly that $\widehat{\mathcal{O}}_P \cong k[[x,y]]/(xy)$, resp. $k[[x,y]]/(y^2 - x^3)$.)

Step 1: δ_P depends only on $R = \widehat{\mathcal{O}}_P$. Put $\mathcal{O} = \mathcal{O}_P$, $\widetilde{\mathcal{O}} = \widetilde{\mathcal{O}}_P$ (a semilocal Dedekind domain, finite over $\mathcal{O}$ with the same fraction field). The quotient $\widetilde{\mathcal{O}}/\mathcal{O}$ has finite length, so it is unchanged by the faithfully flat completion: $\widetilde{\mathcal{O}}/\mathcal{O} \cong \widehat{\widetilde{\mathcal{O}}}/R$, where $\widehat{\widetilde{\mathcal{O}}} = \widetilde{\mathcal{O}} \otimes_{\mathcal{O}} R = \prod_{Q \mapsto P} \widehat{\mathcal{O}}_{\widetilde{X},Q}$ is a finite product of complete DVRs (the $\mathfrak{m}_P$-adic topology on $\widetilde{\mathcal{O}}$ coincides with the topology of its Jacobson radical). Choose $0 \neq c \in \mathcal{O}$ with $c\widetilde{\mathcal{O}} \subseteq \mathcal{O}$; c is a nonzerodivisor on R by flatness, and $R \subseteq \widehat{\widetilde{\mathcal{O}}} \subseteq c^{-1}R \subseteq Q(R)$ (total ring of fractions), with $Q(\widehat{\widetilde{\mathcal{O}}}) = Q(R)$. A finite product of DVRs is integrally closed in its total ring of fractions, and it is integral over R; therefore $\widehat{\widetilde{\mathcal{O}}}$ is exactly the integral closure $\overline{R}$ of R in $Q(R)$. Hence

$$\delta_P = \dim_k \overline{R}/R,$$

which depends only on the isomorphism class of the complete local k-algebra R, i.e. on the analytic type of the singularity.

Step 2: node. Let $h = xy + h_3 + h_4 + \cdots \in k[[x,y]]$ with h_i homogeneous of degree i. Write $h = xy + g$ with $g \in (x,y)^3 = x(x,y)^2 + y(x,y)^2$, say $g = xa + yb$ with $a, b \in (x,y)^2$. Then $h = (x+b)(y+a) - ab$ with $ab \in (x,y)^4$. Putting $x_1 = x + b$, $y_1 = y + a$ (new coordinates, congruent to x, y modulo $(x,y)^2$) we get $h = x_1 y_1 + g_1$ with $g_1 \in (x,y)^4$. Repeating, we obtain coordinates x_n, y_n with $x_{n+1} - x_n,\ y_{n+1} - y_n \in (x,y)^{n+1}$ and $h = x_n y_n + g_n$, $g_n \in (x,y)^{n+2}$. The sequences converge in $k[[x,y]]$ (this works in every characteristic), and $h = x_\infty y_\infty$ with (x_∞, y_∞) a coordinate system. So $R \cong k[[x,y]]/(xy)$. Since $(x) \cap (y) = (xy)$ the map $R \to k[[x]] \times k[[y]]$ is injective; the target $k[[x]] \times k[[y]]$ is a product of DVRs, finite over R (generated as an R-module by $(1,0), (0,1)$), with total ring of fractions $k((x)) \times k((y)) = Q(R)$; hence it is $\overline{R}$. The image of R is $\{(a(x), b(y)) : a(0) = b(0)\}$ (it contains $x \mapsto (x,0)$, $y \mapsto (0,y)$, $1 \mapsto (1,1)$). So $\dim_k \overline{R}/R = 1$, i.e. $\delta_P = 1$.

Step 3: ordinary cusp. Now $h = y^2 +$ (cubic with x^3-coefficient $a \neq 0$) + (order ≥ 4). Then $h(0,y) = y^2 \cdot$(unit), so by the Weierstrass preparation theorem $h = U \cdot (y^2 + \alpha(x)y + \beta(x))$ with U a unit and $\alpha, \beta \in xk[[x]]$. Comparing the terms of degree 2 and 3 of h shows $\operatorname{ord}\alpha \geq 2$, $\operatorname{ord}\beta \geq 3$, and the coefficient of x^3 in β equals $a/U(0) \neq 0$; so $\operatorname{ord}\beta = 3$. Thus $R \cong k[[x]][y]/(y^2 + \alpha y + \beta) = k[[x]] \oplus k[[x]]y$. Write $\alpha = x\alpha'$, $\beta = x^2\beta'$ with $\alpha' \in xk[[x]]$, $\operatorname{ord}\beta' = 1$, and set $w = y/x$. Then $w^2 + \alpha' w + \beta' = 0$ is an Eisenstein equation over $k[[x]]$, so $A = k[[x]][w] = k[[x]] \oplus k[[x]]w$ is a complete DVR (with uniformizer w), finite over R, with $Q(A) = A[1/x] = Q(R)$; hence $A = \overline{R}$. Since $y = xw$, $R = k[[x]] \oplus k[[x]]xw$ and $A/R \cong k[[x]]w/xk[[x]]w \cong k$. Thus $\delta_P = 1$.

(As a check, the plane cubics $y^2 z = x^3 + x^2 z$ and $y^2 z = x^3$ have $p_a = 1$, normalization $\mathbf{P}^1$, and exactly one singular point, a node resp. cusp, so (a) gives $\delta = 1$ for them, in agreement.) $\square$

Exercise 1.9

Problem. *Riemann-Roch for Singular Curves.* Let X be an integral projective scheme of dimension 1 over k. Let X_{reg} be the set of regular points of X.

(a) Let $D = \sum n_i P_i$ be a divisor with support in X_{reg}, i.e., all $P_i \in X_{\text{reg}}$. Then define

$\deg D = \sum n_i$. Let $\mathcal{L}(D)$ be the associated invertible sheaf on X, and show that

$$\chi(\mathcal{L}(D)) = \deg D + 1 - p_a.$$

(b) Show that any Cartier divisor on X is the difference of two very ample Cartier divisors. (Use (II, Ex. 7.5).)

(c) Conclude that every invertible sheaf $\mathcal{L}$ on X is isomorphic to $\mathcal{L}(D)$ for some divisor D with support in X_{reg}.

(d) Assume furthermore that X is a locally complete intersection in some projective space. Then by (III, 7.11) the dualizing sheaf ω_X is an invertible sheaf on X, so we can define the *canonical divisor* K to be a divisor with support in X_{reg} corresponding to ω_X. Then the formula of (a) becomes

$$l(D) - l(K - D) = \deg D + 1 - p_a.$$

(Starred in the book.)

Solution. Let $\mathcal{K}$ be the sheaf of rational functions on X. The set $X - X_{\text{reg}}$ is a proper closed subset of the integral one-dimensional scheme X, hence a finite set of closed points. For $P \in X_{\text{reg}}$ the local ring $\mathcal{O}_P$ is a DVR with uniformizer t_P; a divisor $D = \sum n_P P$ supported in X_{reg} is a Cartier divisor, and $\mathcal{L}(D) \subseteq \mathcal{K}$ is the subsheaf with $\mathcal{L}(D)_P = t_P^{-n_P}\mathcal{O}_P$ for $P \in X_{\text{reg}}$ and $= \mathcal{O}_P$ at all other points (on a neighborhood of a regular point it is the usual $\mathcal{L}(D)$).

(a) Let $P \in X_{\text{reg}}$. Then $\mathcal{L}(D) \subseteq \mathcal{L}(D+P)$ and the quotient is a skyscraper sheaf at P with stalk $t_P^{-n-1}\mathcal{O}_P/t_P^{-n}\mathcal{O}_P \cong k(P)$ (where $n = n_P$); so there is an exact sequence $0 \to \mathcal{L}(D) \to \mathcal{L}(D+P) \to k(P) \to 0$. By additivity of χ (III, Ex. 5.1) and $\chi(k(P)) = 1$, we get $\chi(\mathcal{L}(D+P)) = \chi(\mathcal{L}(D)) + 1$, while $\deg(D+P) = \deg D + 1$. So the formula holds for D iff it holds for $D+P$ (and likewise for $D-P$). Any divisor supported in X_{reg} is reached from 0 by finitely many steps $\pm P$ with $P \in X_{\text{reg}}$, so it suffices to treat $D = 0$, where the formula says $\chi(\mathcal{O}_X) = 1 - p_a$; this is (III, Ex. 5.3) (for an integral projective scheme of dimension r, $p_a = (-1)^r(\chi(\mathcal{O}_X) - 1)$).

(b) We first record: (i) if $\mathcal{M}$ is very ample and $\mathcal{N}$ is generated by global sections, then $\mathcal{M} \otimes \mathcal{N}$ is very ample (II, Ex. 7.5). Indeed, $\mathcal{M}$ gives a closed immersion $i : X \to \mathbf{P}^N$, $\mathcal{N}$ gives a morphism $j : X \to \mathbf{P}^M$; then $(i, j) : X \to \mathbf{P}^N \times \mathbf{P}^M$ is a closed immersion (it is the graph of j followed by $i \times \mathrm{id}$), and composing with the Segre embedding gives an embedding by $\mathcal{M} \otimes \mathcal{N}$. In particular $\mathcal{A}^m$ is very ample for $m \geq 1$ if $\mathcal{A} = \mathcal{O}_X(1)$ is.

Let D be a Cartier divisor and $\mathcal{A} = \mathcal{O}_X(1)$ a very ample sheaf. By Serre (II, 5.17) $\mathcal{L}(D) \otimes \mathcal{A}^n$ is generated by global sections for $n \gg 0$. Take such an n. Then $\mathcal{M}_1 = \mathcal{L}(D) \otimes \mathcal{A}^{n+1}$ and $\mathcal{M}_2 = \mathcal{A}^{n+1}$ are very ample by (i), and $\mathcal{L}(D) \cong \mathcal{M}_1 \otimes \mathcal{M}_2^{-1}$. As X is integral, every invertible sheaf is $\mathcal{L}(E)$ for a Cartier divisor E (II, 6.15), so $\mathcal{M}_i \cong \mathcal{L}(D_i)$ with Cartier divisors D_i. Then $\mathcal{L}(D) \cong \mathcal{L}(D_1 - D_2)$, so $D = D_1 - D_2 + (\varphi)$ for a rational function φ; put $D_1' = D_1 + (\varphi)$, which is linearly equivalent to D_1, so $\mathcal{L}(D_1') \cong \mathcal{M}_1$ is very ample. Then $D = D_1' - D_2$ with both very ample.

(c) Let $\mathcal{L}$ be invertible. As above $\mathcal{L} \cong \mathcal{L}(D')$ for a Cartier divisor D', and by (b) $D' = E_1 - E_2$ with $\mathcal{L}(E_i)$ very ample, giving closed immersions $X \hookrightarrow \mathbf{P}^{N_i}$ with $\mathcal{O}(1)|_X \cong \mathcal{L}(E_i)$. Choose hyperplanes $H_i \subseteq \mathbf{P}^{N_i}$ containing none of the finitely many points of $X - X_{\text{reg}}$ (possible: the hyperplanes through one given point form a proper linear subspace of the dual projective space, and k is infinite). Since X is integral and $X \not\subseteq H_i$, $E_i' = H_i \cap X$ is an effective Cartier divisor,

with $\mathcal{L}(E_i') \cong \mathcal{O}(1)|_X \cong \mathcal{L}(E_i)$ and support in X_{reg}. A Cartier divisor with support in X_{reg} is a divisor $\sum n_P P$ as in (a). Hence $\mathcal{L} \cong \mathcal{L}(E_1' - E_2')$ with $E_1' - E_2'$ supported in X_{reg}.

(d) A locally complete intersection is Cohen–Macaulay (II, 8.21A, 8.23), so Cor. 7.7 of Ch. III applies to the projective Cohen–Macaulay scheme X of dimension 1 and to the locally free sheaf $\mathcal{F} = \mathcal{L}(D)$: $H^1(X, \mathcal{L}(D)) \cong H^0(X, \mathcal{L}(D)^\vee \otimes \omega_X)'$ (dual vector space). Here $\omega_X \cong \mathcal{L}(K)$ and $\mathcal{L}(D)^\vee \otimes \omega_X \cong \mathcal{L}(K - D)$. Therefore $\dim H^1(X, \mathcal{L}(D)) = l(K - D)$, and $\chi(\mathcal{L}(D)) = l(D) - l(K - D)$; combined with (a) this is the asserted formula. □

Exercise 1.10

> **Problem.** Let X be an integral projective scheme of dimension 1 over k, which is locally complete intersection, and has $p_a = 1$. Fix a point $P_0 \in X_{\text{reg}}$. Imitate (1.3.7) to show that the map $P \to \mathcal{L}(P - P_0)$ gives a one-to-one correspondence between the points of X_{reg} and the elements of the group $\operatorname{Pic}^0 X$. This generalizes (II, 6.11.4) and (II, Ex. 6.7).

Solution. *Reading.* The superscript 0 is lost in the printed statement ("Pic X"); as in Example 1.3.7 the group is $\operatorname{Pic}^0 X$, the subgroup of invertible sheaves of degree 0 (the full group $\operatorname{Pic} X$ is much bigger, e.g. $\mathbf{Z} \oplus \operatorname{Pic}^0 X$ for a plane cubic). Here the degree of an invertible sheaf $\mathcal{M}$ is defined by $\deg \mathcal{M} = \chi(\mathcal{M}) - \chi(\mathcal{O}_X)$; by Ex. 1.9(a),(c) every $\mathcal{M}$ is some $\mathcal{L}(D)$ with D supported in X_{reg} and then $\deg \mathcal{M} = \deg D$, so deg is additive on $\mathcal{L}(D) \otimes \mathcal{L}(D') = \mathcal{L}(D + D')$ and $\operatorname{Pic}^0 X = \{\mathcal{M} : \deg \mathcal{M} = 0\}$ is a subgroup. Clearly $\mathcal{L}(P - P_0) \in \operatorname{Pic}^0 X$ for $P \in X_{\text{reg}}$.

Lemma. Let $\mathcal{M}$ be an invertible sheaf on X with $H^0(X, \mathcal{M}) \neq 0$. Then $\deg \mathcal{M} \geq 0$, and if $\deg \mathcal{M} = 0$ then $\mathcal{M} \cong \mathcal{O}_X$. Indeed, a nonzero section s gives an injection $s : \mathcal{O}_X \to \mathcal{M}$ (X is integral and $\mathcal{M}$ is locally free of rank 1), whose cokernel $\mathcal{T}$ vanishes at the generic point, hence has finite support, so $\chi(\mathcal{T}) = h^0(\mathcal{T}) = \sum \operatorname{length} \mathcal{T}_x \geq 0$. Then $\deg \mathcal{M} = \chi(\mathcal{T}) \geq 0$, with equality iff $\mathcal{T} = 0$ iff s is an isomorphism.

The canonical divisor has degree 0. By Ex. 1.9(d) and $p_a = 1$: for $D = 0$, $l(0) - l(K) = 0$, and $l(0) = \dim H^0(\mathcal{O}_X) = 1$, so $l(K) = 1$; for $D = K$, $l(K) - l(0) = \deg K$, so $\deg K = 0$. (By the Lemma, even $\omega_X \cong \mathcal{O}_X$.)

Surjectivity. Let $\mathcal{M} \in \operatorname{Pic}^0 X$; by Ex. 1.9(c), $\mathcal{M} = \mathcal{L}(D)$ with D supported in X_{reg} and $\deg D = 0$. Put $D' = D + P_0$, of degree 1. By Ex. 1.9(d), $l(D') - l(K - D') = 1 + 1 - 1 = 1$. The invertible sheaf $\mathcal{L}(K - D')$ has degree $-1 < 0$, so it has no nonzero sections by the Lemma, i.e. $l(K - D') = 0$ and $l(D') = 1$. Let s be a nonzero section of $\mathcal{L}(D')$ (unique up to scalar) and $\mathcal{T}$ the cokernel of $s : \mathcal{O}_X \to \mathcal{L}(D')$. Then $\chi(\mathcal{T}) = \deg D' = 1$, so $\mathcal{T}$ is a skyscraper sheaf $k(Q)$ at a closed point Q. Tensoring with $\mathcal{L}(D')^{-1}$ gives $0 \to \mathcal{L}(D')^{-1} \to \mathcal{O}_X \to k(Q) \to 0$, i.e. the ideal sheaf $\mathfrak{m}_Q$ of Q is isomorphic to the invertible sheaf $\mathcal{L}(D')^{-1}$. A local ring of dimension 1 whose maximal ideal is principal is a DVR, so $Q \in X_{\text{reg}}$ and $\mathfrak{m}_Q = \mathcal{L}(-Q)$. Hence $\mathcal{L}(D') \cong \mathcal{L}(Q)$, i.e. $\mathcal{L}(D) \cong \mathcal{L}(Q - P_0)$.

Injectivity. Suppose $\mathcal{L}(Q - P_0) \cong \mathcal{L}(Q' - P_0)$ with $Q, Q' \in X_{\text{reg}}$. Then $\mathcal{M} = \mathcal{L}(Q) \cong \mathcal{L}(Q')$, and $h^0(\mathcal{M}) = 1$ by the computation above (degree 1). The canonical inclusion $1 : \mathcal{O}_X \to \mathcal{L}(Q)$ has cokernel supported exactly at Q, and similarly for Q'; under the isomorphism these are two nonzero sections of $\mathcal{M}$, hence proportional, with cokernels that are isomorphic, so $Q = Q'$.

Hence $P \mapsto \mathcal{L}(P - P_0)$ is a bijection $X_{\text{reg}} \to \operatorname{Pic}^0 X$. □

2. Hurwitz's theorem

Conventions. k is algebraically closed, "curve" means complete nonsingular curve over k, and $\mathcal{L}(D)$, $l(D)$, K are as in Section 1. For a finite morphism $f : X \to Y$ of curves and $Q \in Y$ we use constantly $\sum_{P \mapsto Q} e_P = \deg f$ (II, 6.9). A finite morphism of nonsingular curves is flat, so $f_*\mathcal{M}$ is locally free of rank $\deg f$ for every invertible $\mathcal{M}$ on X. Hurwitz's theorem (Cor. 2.4) is used in the form $2g(X) - 2 = n(2g(Y) - 2) + \deg R$ for f separable of degree n, and $\deg R = \sum(e_P - 1)$ if all ramification is tame, in particular always in characteristic 0.

Exercise 2.1

Problem. Use (2.5.3) to show that $\mathbf{P}^n$ is simply connected.

Solution. "Simply connected" means that every etale covering $f : X \to \mathbf{P}^n$ (a finite etale morphism) is trivial, i.e. X is a finite disjoint union of copies of $\mathbf{P}^n$ (Example 2.5.3). We use induction on n; $n = 0$ is trivial and $n = 1$ is Example 2.5.3.

Let $n \geq 2$ and let $f : X \to \mathbf{P}^n$ be etale and finite. Since f is open and closed and $\mathbf{P}^n$ is connected, each connected component X' of X maps onto $\mathbf{P}^n$ with some degree d', and $X' \to \mathbf{P}^n$ is again finite etale. It suffices to show $d' = 1$ for each X', because a finite etale morphism of degree 1 is an isomorphism. So assume X' is connected. It is smooth over k (etale over $\mathbf{P}^n$), hence regular, connected, so an integral normal projective variety of dimension $n \geq 2$ (finite over $\mathbf{P}^n$).

The sheaf $f^*\mathcal{O}(1)$ is ample on X': for coherent $\mathcal{F}$ on X', since f is finite (hence affine), (III, Ex. 4.1) and the projection formula give $H^i(X', \mathcal{F} \otimes f^*\mathcal{O}(m)) \cong H^i(\mathbf{P}^n, f_*\mathcal{F} \otimes \mathcal{O}(m)) = 0$ for $i > 0$, $m \gg 0$ (III, 5.2), and then Serre's criterion (III, 5.3) applies. Let $H \subseteq \mathbf{P}^n$ be a hyperplane. Then $f^*H = f^{-1}(H)$ is an effective Cartier divisor on X' with $\mathcal{L}(f^*H) = f^*\mathcal{O}(1)$ ample, so by (III, 7.9) its support $f^{-1}(H)$ is connected. On the other hand $f^{-1}(H) \to H \cong \mathbf{P}^{n-1}$ is finite etale of degree d' (base change), so by induction it is a trivial covering: $f^{-1}(H) \cong \bigsqcup_{d'} H$. Being connected, $d' = 1$. Hence $X' \cong \mathbf{P}^n$, and $\mathbf{P}^n$ has no nontrivial etale covering. □

Exercise 2.2

Problem. *Classification of Curves of Genus 2.* Fix an algebraically closed field k of characteristic $\neq 2$.

(a) If X is a curve of genus 2 over k, the canonical linear system $|K|$ determines a finite morphism $f : X \to \mathbf{P}^1$ of degree 2 (Ex. 1.7). Show that it is ramified at exactly 6 points, with ramification index 2 at each one. Note that f is uniquely determined, up to an automorphism of $\mathbf{P}^1$, so X determines an (unordered) set of 6 points of $\mathbf{P}^1$, up to an automorphism of $\mathbf{P}^1$.

(b) Conversely, given six distinct elements $\alpha_1, \ldots, \alpha_6 \in k$, let K be the extension of $k(x)$ determined by the equation $z^2 = (x - \alpha_1) \cdots (x - \alpha_6)$. Let $f : X \to \mathbf{P}^1$ be the corresponding morphism of curves. Show that $g(X) = 2$, the map f is the same as the one determined by the canonical linear system, and f is ramified over the six points $x = \alpha_i$ of $\mathbf{P}^1$, and nowhere else. (Cf. (II, Ex. 6.4).)

(c) Using (I, Ex. 6.6), show that if P_1, P_2, P_3 are three distinct points of $\mathbf{P}^1$, then there exists a unique $\varphi \in \operatorname{Aut}\mathbf{P}^1$ such that $\varphi(P_1) = 0$, $\varphi(P_2) = 1$, $\varphi(P_3) = \infty$. Thus in (a), if we order the six points of $\mathbf{P}^1$, and then normalize by sending the first three to $0, 1, \infty$, respectively, we may assume that X is ramified over $0, 1, \infty, \beta_1, \beta_2, \beta_3$, where $\beta_1, \beta_2, \beta_3$ are three distinct elements of k, $\neq 0, 1$.

(d) Let Σ_6 be the symmetric group on 6 letters. Define an action of Σ_6 on sets of three distinct elements $\beta_1, \beta_2, \beta_3$ of k, $\neq 0, 1$, as follows: reorder the set $0, 1, \infty, \beta_1, \beta_2, \beta_3$ according to a given element $\sigma \in \Sigma_6$, then renormalize as in (c) so that the first three become $0, 1, \infty$ again. Then the last three are the new $\beta_1', \beta_2', \beta_3'$.

(e) Summing up, conclude that there is a one-to-one correspondence between the set of isomorphism classes of curves of genus 2 over k, and triples of distinct elements $\beta_1, \beta_2, \beta_3$ of k, $\neq 0, 1$, modulo the action of Σ_6 described in (d). In particular, there are many non-isomorphic curves of genus 2. We say that curves of genus 2 depend on three parameters, since they correspond to the points of an open subset of $\mathbf{A}^3_k$ modulo a finite group.

Solution. (a) By Ex. 1.7(a), $|K|$ is base point free of degree 2 and dimension 1 and defines a finite morphism $f : X \to \mathbf{P}^1$ of degree 2. Since $\operatorname{char} k \neq 2$, the degree-2 extension $K(X)/k(f)$ is separable. For each $Q \in \mathbf{P}^1$, $\sum_{P \mapsto Q} e_P = 2$, so $e_P \in \{1, 2\}$; all ramification is tame since $\operatorname{char} k \neq 2$. Hurwitz's theorem (2.4) gives $2 \cdot 2 - 2 = 2 \cdot (0 - 2) + \deg R$, so $\deg R = 6 = \sum_P (e_P - 1) = \#\{P : e_P = 2\}$. Thus f is ramified at exactly 6 points, each with $e_P = 2$; each lies alone in its fibre ($e_P = 2 = \deg f$), so the six ramification points map to six distinct branch points $B \subseteq \mathbf{P}^1$.

A different choice of basis of $H^0(X, \mathcal{L}(K))$ changes f by an element of $\mathrm{PGL}_2 = \operatorname{Aut}\mathbf{P}^1$ (I, Ex. 6.6): f is the morphism defined by a basis of $H^0(X, \mathcal{L}(K))$, and changing the basis composes it with a projective linear transformation. Hence f is unique up to $\operatorname{Aut}\mathbf{P}^1$, and X determines the 6-element set $B \subseteq \mathbf{P}^1$ up to $\operatorname{Aut}\mathbf{P}^1$.

(b) Put $h = \prod_{i=1}^6 (x - \alpha_i)$. It is squarefree of degree 6, hence not a square in $k(x)$, so $K = k(x)(z)$, $z^2 = h$, is a field of degree 2 over $k(x)$. Let X be the nonsingular projective curve with function field K and $f : X \to \mathbf{P}^1$ the morphism given by $k(x) \subseteq K$; it is finite of degree 2 and separable.

The affine part. Let $A = k[x, z]/(z^2 - h)$. The Jacobian criterion: the partial derivatives are $2z$ and $-h'(x)$, which vanish simultaneously with $z^2 = h$ only if $z = 0$, $h(x) = 0 = h'(x)$, impossible since h has simple roots. So $\operatorname{Spec} A$ is a nonsingular affine curve, A is integrally closed, finite and free of rank 2 over $k[x]$ (basis $1, z$); thus A is the integral closure of $k[x]$ in K and $\operatorname{Spec} A = f^{-1}(\mathbf{A}^1)$. For $a \in k$ we have $\dim_k A/(x-a)A = 2 = \sum_{P \mapsto a} e_P$ (since $A/(x-a)$ has this dimension and $\mathfrak{m}_P^{e_P}$-primary parts add up). The fibre ring is $k[z]/(z^2 - h(a))$: if $h(a) \neq 0$ it has two maximal ideals, so two points with $e = 1$; if $h(a) = 0$ it is $k[z]/(z^2)$, one point $P_a = (a, 0)$ with $e = 2$. At P_a the maximal ideal is $(x - a, z)$ and $x - a = z^2 \big/ \big(h/(x - a)\big)$ with $h/(x - a)$ a unit at a, so z is a local parameter at P_a, with $v_{P_a}(z) = 1$.

The point at infinity. Put $x' = 1/x$, $w = z/x^3$; then $w^2 = \prod(1 - \alpha_i x') =: h^*(x')$, with $h^*(0) = 1$ and h^* squarefree (its roots are the $1/\alpha_i$, $\alpha_i \neq 0$). The same argument applied to $B = k[x', w]/(w^2 - h^*)$ (nonsingular, integral closure of $k[x']$) shows: over $x' = 0$ the fibre ring is $k[w]/(w^2 - 1)$, so there are two points $\infty_\pm$ ($w = \pm 1$), unramified.

Hence f is ramified exactly at the six points P_{α_i} (with $e = 2$), over the six points $x = \alpha_i$, and nowhere else. Hurwitz: $2g(X) - 2 = 2(-2) + 6 = 2$, so $g(X) = 2$.

Comparison with $|K|$. Let $F = f^*(\infty) = \infty_+ + \infty_-$; any fibre of f is linearly equivalent to F. The function z has $v_{P_{\alpha_i}}(z) = 1$, is a unit at the other finite points, and at $\infty_\pm$: $z = w/x'^3$ with w a unit and x' a local parameter, so $v_{\infty_\pm}(z) = -3$. Hence $(z) = R - 3F$, where $R = \sum P_{\alpha_i}$ is the ramification divisor, i.e. $R \sim 3F$. By Prop. 2.3, $K_X \sim f^*K_{\mathbf{P}^1} + R \sim -2F + 3F = F$. Then $\mathcal{L}(K_X) \cong \mathcal{L}(F) = f^*\mathcal{O}(1)$, and $f^* : H^0(\mathbf{P}^1, \mathcal{O}(1)) \to H^0(X, f^*\mathcal{O}(1))$ is injective with 2-dimensional source, while $\dim H^0(X, \mathcal{L}(K_X)) = g = 2$. So it is an isomorphism: $|K_X| = f^*|\mathcal{O}(1)|$, and the morphism defined by the complete linear system $|K_X|$ is exactly f (for the basis f^*s_0, f^*s_1).

(c) By (I, Ex. 6.6), $\operatorname{Aut}\mathbf{P}^1 = \mathrm{PGL}_2(k)$ acting by $x \mapsto (ax+b)/(cx+d)$. Existence: for $P_1, P_2, P_3 \in k$ take $\varphi(x) = \dfrac{(x-P_1)(P_2-P_3)}{(x-P_3)(P_2-P_1)}$; if one of the P_i is ∞ omit the two factors containing it (e.g. $P_3 = \infty$: $\varphi = (x-P_1)/(P_2-P_1)$). One checks directly $\varphi(P_1) = 0, \varphi(P_2) = 1, \varphi(P_3) = \infty$. Uniqueness: if $\psi \in \mathrm{PGL}_2$ fixes $0, 1, \infty$, then $\psi(\infty) = \infty$ gives $c = 0$, $\psi(0) = 0$ gives $b = 0$, $\psi(1) = 1$ gives $a = d$, so $\psi = \mathrm{id}$.

Now order the six points B of (a) as $(p_1, \ldots, p_6)$ and let φ send p_1, p_2, p_3 to $0, 1, \infty$. Replacing f by $\varphi \circ f$ (again a degree-2 finite morphism, ramified exactly over $\varphi(B)$, by the uniqueness in (a)) we get a curve ramified over $0, 1, \infty, \beta_1, \beta_2, \beta_3$ with $\beta_i = \varphi(p_{3+i})$ distinct elements of $\mathbf{P}^1 - \{0, 1, \infty\} = k - \{0, 1\}$.

(d) Let S be the set of ordered 6-tuples $(p_1, \ldots, p_6)$ of distinct points of $\mathbf{P}^1$, and T the set of ordered triples $(\beta_1, \beta_2, \beta_3)$ of distinct elements of $k - \{0, 1\}$. By (c), PGL_2 acts on S freely (a transformation fixing three points is the identity), and every orbit contains exactly one normalized tuple $(0, 1, \infty, \beta_1, \beta_2, \beta_3)$; so $T \cong S/\mathrm{PGL}_2$. The group Σ_6 acts on S by permuting the entries, and this commutes with the PGL_2-action; hence it induces an action on $S/\mathrm{PGL}_2 \cong T$. Concretely, $\sigma \in \Sigma_6$ sends the normalized tuple to the permuted tuple, which is then renormalized by the unique φ of (c): this is exactly the procedure of (d), so it is a genuine action. Its orbit set is $T/\Sigma_6 = \{\text{6-element subsets of } \mathbf{P}^1\}/\mathrm{PGL}_2$.

(e) Define Θ from the set of isomorphism classes of genus-2 curves to T/Σ_6 by $X \mapsto$ class of the branch set $B(X)$ (well defined by (a), (c), (d); isomorphic curves have the same canonical map up to $\operatorname{Aut}\mathbf{P}^1$).

Surjective. Given a 6-element set $B \subseteq \mathbf{P}^1$, pick $c \in k - B$ and apply the automorphism $x \mapsto 1/(x-c)$ of $\mathbf{P}^1$, which moves B into k (the only point sent to ∞ is c). So we may assume $B = \{\alpha_1, \ldots, \alpha_6\} \subseteq k$, and the curve of (b) has genus 2 and branch set B.

Injective. Let X, X' be genus-2 curves with branch sets $B, B' = \sigma(B)$, $\sigma \in \operatorname{Aut}\mathbf{P}^1$. Replace f' by $\sigma^{-1}f'$ so that both have branch set B, and (as above) move B into k. Then $K(X) \supseteq k(x)$ is a quadratic extension, and since $\operatorname{char} k \neq 2$ it is $k(x)(\sqrt{u})$ with $u \in k(x)^*$; as k is algebraically closed we may take $u = \prod_j (x - \gamma_j)$ with distinct $\gamma_j \in k$ (modify u by squares and constants). The extension is ramified over a finite $a \in k$ iff $v_a(u)$ is odd, i.e. iff $a \in \{\gamma_j\}$, and over ∞ iff $\deg u$ is odd. (If $v_a(u)$ is odd and $P \mapsto a$ had $e_P = 1$, then $v_P(u) = v_a(u)$ would be odd, but u is a square in $K(X)$, so $v_P(u)$ is even. If $v_a(u) = 2m$, then $y = \sqrt{u}/(x-a)^m$ satisfies $y^2 = w$ with $w = u/(x-a)^{2m}$ a unit of $\mathcal{O}_a$; the fibre of $\mathcal{O}_a[y] = \mathcal{O}_a[Y]/(Y^2 - w)$ over a is $k[Y]/(Y^2 - w(a)) \cong k \times k$ since $\operatorname{char} k \neq 2$, so the integral closure has two points over a, each with $e = 1$. The same applies at ∞ with $x' = 1/x$ and $u\,x'^{\deg u}$.) As $B \subseteq k$ is the branch set, $\{\gamma_j\} = B = \{\alpha_1, \ldots, \alpha_6\}$ and $\deg u = 6$ is even. Thus $K(X) \cong k(x)(z)$ with $z^2 = \prod(x - \alpha_i)$, the same for X'; since a curve is determined by its function field (I, 6.12), $X \cong X'$. So Θ is a

bijection.

Finally T is infinite (k is infinite) and Σ_6 is finite, so T/Σ_6 is infinite: there are infinitely many non-isomorphic curves of genus 2, depending on 3 parameters (T is an open subset of $\mathbf{A}^3_k$). $\square$

Exercise 2.3

Problem. *Plane Curves.* Let X be a curve of degree d in $\mathbf{P}^2$. For each point $P \in X$, let $T_P(X)$ be the tangent line to X at P (I, Ex. 7.3). Considering $T_P(X)$ as a point of the dual projective plane $(\mathbf{P}^2)^*$, the map $P \to T_P(X)$ gives a morphism of X to its dual curve X^* in $(\mathbf{P}^2)^*$ (I, Ex. 7.3). Note that even though X is nonsingular, X^* in general will have singularities. We assume $\operatorname{char} k = 0$ below.

(a) Fix a line $L \subseteq \mathbf{P}^2$ which is not tangent to X. Define a morphism $\varphi : X \to L$ by $\varphi(P) = T_P(X) \cap L$, for each point $P \in X$. Show that φ is ramified at P if and only if either (1) $P \in L$, or (2) P is an *inflection point* of X, which means that the intersection multiplicity (I, Ex. 5.4) of $T_P(X)$ with X at P is ≥ 3. Conclude that X has only finitely many inflection points.

(b) A line of $\mathbf{P}^2$ is a *multiple tangent* of X if it is tangent to X at more than one point. It is a *bitangent* if it is tangent to X at exactly two points. If L is a multiple tangent of X, tangent to X at the points $P_1, \ldots, P_r$, and if none of the P_i is an inflection point, show that the corresponding point of the dual curve X^* is an *ordinary r-fold point*, which means a point of multiplicity r with distinct tangent directions (I, Ex. 5.3). Conclude that X has only finitely many multiple tangents.

(c) Let $O \in \mathbf{P}^2$ be a point which is not on X, nor on any inflectional or multiple tangent of X. Let L be a line not containing O. Let $\psi : X \to L$ be the morphism defined by projection from O. Show that ψ is ramified at a point $P \in X$ if and only if the line OP is tangent to X at P, and in that case the ramification index is 2. Use Hurwitz's theorem and (I, Ex. 7.2) to conclude that there are exactly $d(d-1)$ tangents of X passing through O. Hence the degree of the dual curve (sometimes called the *class* of X) is $d(d-1)$.

(d) Show that for all but a finite number of points of X, a point O of X lies on exactly $(d+1)(d-2)$ tangents of X, not counting the tangent at O.

(e) Show that the degree of the morphism φ of (a) is $d(d-1)$. Conclude that if $d \geq 2$, then X has $3d(d-2)$ inflection points, properly counted. (If $T_P(X)$ has intersection multiplicity r with X at P, then P should be counted $r-2$ times as an inflection point.) If $r = 3$ we call it an *ordinary inflection point*. Show that an ordinary inflection point of X corresponds to an ordinary cusp of the dual curve X^*.

(f) Now let X be a plane curve of degree $d \geq 2$, and assume that the dual curve X^* has only nodes and ordinary cusps as singularities (which should be true for sufficiently general X). Then show that X has exactly $\frac{1}{2}d(d-2)(d-3)(d+3)$ bitangents. [*Hint*: Show that X is the normalization of X^*. Then calculate $p_a(X^*)$ two ways: once as a plane curve of degree $d(d-1)$, and once using (Ex. 1.8).]

(g) For example, a plane cubic curve has exactly 9 inflection points, all ordinary. The line joining any two of them intersects the curve in a third one.

(h) A plane quartic curve has exactly 28 bitangents. (This holds even if the curve has a tangent with four-fold contact, in which case the dual curve X^* has a tacnode.)

Solution. *Standing assumptions and notation.* $\operatorname{char} k = 0$, and $X = Z(F) \subseteq \mathbf{P}^2$ is a nonsingular (hence irreducible) curve of degree d. For $d = 1$ everything is degenerate (X is a line, equal to

all its tangents, the map φ of (a) is constant and X^* is a point), so we assume $d \geq 2$ throughout; then no tangent line is contained in X, so every tangent meets X at P with a finite multiplicity. Genus: $g = (d-1)(d-2)/2$ (I, Ex. 7.2). We identify points of $\mathbf{P}^2$ with vectors of k^3 (up to scalars), and lines with covectors, written as vectors via $\ell \cdot p = 0$; the dual plane is $\mathbf{P}(k^3)$ again, a point ℓ being the line $\{p : \ell \cdot p = 0\}$ and a point p of $\mathbf{P}^2$ being the line $p^* = \{\ell : \ell \cdot p = 0\}$ of $(\mathbf{P}^2)^*$. We write $[a,b,c] = \det(a,b,c) = a \cdot (b \times c)$. The tangent map is $\gamma : X \to X^* \subseteq (\mathbf{P}^2)^*$, $\gamma(P) = [\nabla F(P)]$ (I, Ex. 7.3). For a line $L = \{\lambda \cdot p = 0\}$ and a point ℓ of the dual plane (a line) the point $\ell \cap L$ is $\ell \times \lambda$, and the line through two points p, q is $p \times q$.

Local computations (L1)–(L3). Let $P \in X$, let t be a local parameter at P, identify $\widehat{\mathcal{O}}_{X,P} = k[[t]]$ and let $r(t) \in k[[t]]^3$ be a lift of the arc $\operatorname{Spec} k[[t]] \to X \subseteq \mathbf{P}^2$, with $r(0) = P$. Primes are d/dt.

(L1) $r(0), r'(0)$ are linearly independent. Indeed in an affine chart $r = (x(t), y(t), 1)$; $k[[x,y]] \to k[[t]]$ is onto, so x or y has order exactly 1, i.e. $(x'(0), y'(0)) \neq 0$, and $r'(0) = (x'(0), y'(0), 0)$.

Put $\ell(t) = r \times r'$ and $D(t) = [r, r', r'']$. Then $\ell(0) \neq 0$, and

(L2) $[\ell(t)] = \gamma(P(t))$ (the tangent line at $P(t)$) and $\ell \cdot r = \ell \cdot r' = \ell' \cdot r = 0$. Indeed $F(r(t)) = 0$; differentiating, $\nabla F(r) \cdot r' = 0$, and Euler's relation gives $\nabla F(r) \cdot r = dF(r) = 0$; so $\nabla F(r(t))$ is orthogonal to r, r' and over $k((t))$ it is a multiple $c(t)\ell(t)$ of $\ell = r \times r'$; since $\nabla F(r(0)) \neq 0$ (nonsingularity) and $\ell(0) \neq 0$, c is a unit of $k[[t]]$. The relations $\ell \cdot r = \ell \cdot r' = 0$ are clear, and differentiating $\ell \cdot r = 0$ gives $\ell' \cdot r = -\ell \cdot r' = 0$. In particular $\ell(0) = T_P(X)$.

(L3) $\ell \times \ell' = D\, r$ (use $\ell' = r \times r''$ and $(a \times b) \times (a \times c) = [a,b,c]a$), and $\operatorname{ord}_t D = m_P - 2$, where $m_P = \operatorname{ord}_t(\ell(0) \cdot r(t)) = (T_P(X) \cdot X)_P \geq 2$ is the intersection multiplicity of the tangent line with X at P (I, Ex. 5.4; for a line $\{\lambda \cdot p = 0\}$ the multiplicity at $P \in X$ is $\operatorname{ord}_t(\lambda \cdot r(t))$). To compute $\operatorname{ord} D$ note that D is changed by a unit factor if r is replaced by ur ($D \mapsto u^3 D$), by a change of parameter $t = \tau(s)$ ($D \mapsto \tau'^3 D$), or by a linear change of coordinates ($D \mapsto \det \cdot D$). So we may take affine coordinates with $P = (0,0)$ and $T_P(X) = \{y = 0\}$; then y has order $m_P \geq 2$ on X at P, so x has order 1 and we may take $t = x$, $r = (t, g(t), 1)$ with $\operatorname{ord} g = m_P$. Then $D = \det \begin{pmatrix} t & g & 1 \\ 1 & g' & 0 \\ 0 & g'' & 0 \end{pmatrix} = g''$, which has order exactly $m_P - 2$ because $\operatorname{char} k = 0$.

(a) The morphism: with $L = \{\lambda \cdot p = 0\}$, $\varphi(P) = \nabla F(P) \times \lambda$, given by forms of degree $d-1$ with no common zero on X (as $\nabla F(P) \times \lambda = 0$ iff $T_P = L$). Let $v(t) = \ell(t) \times \lambda \in k[[t]]^3$, so $[v(t)] = \varphi(P(t))$, with $v(0) \neq 0$ since $\ell(0) \neq \lambda$. As $v \cdot \lambda = 0$, v lies in the plane $\lambda^\perp$ representing L; choose a basis e_1, e_2 of $\lambda^\perp$ so that $v = v_0 e_1 + v_1 e_2$ with $v_0(0) \neq 0$. Then $u = v_1/v_0$ is the pull-back of an affine coordinate on L at $\varphi(P)$, so

$$e_P(\varphi) = \operatorname{ord}_t\big(u - u(0)\big) = 1 + \operatorname{ord}_t u' = 1 + \operatorname{ord}_t(v_0 v_1' - v_1 v_0') = 1 + \operatorname{ord}_t(v \times v'),$$

using $\operatorname{char} k = 0$ and $v \times v' = (v_0 v_1' - v_1 v_0')\, e_1 \times e_2$ with $e_1 \times e_2 \neq 0$ constant. By the identity $(a \times b) \times (c \times e) = [a,b,e]c - [a,b,c]e$,

$$v \times v' = (\ell \times \lambda) \times (\ell' \times \lambda) = -[\ell, \lambda, \ell']\lambda = \pm(\ell \times \ell') \cdot \lambda\, \lambda = \pm D\, (r \cdot \lambda)\, \lambda$$

by (L3). Hence, with $(L \cdot X)_P = \operatorname{ord}_t(\lambda \cdot r) \geq 0$ (> 0 iff $P \in L$):

$$e_P(\varphi) = 1 + (m_P - 2) + (L \cdot X)_P. \qquad (*)$$

(In particular $u' \neq 0$, so φ is nonconstant.) Both summands $m_P - 2$ and $(L \cdot X)_P$ are ≥ 0, so $e_P(\varphi) > 1$ iff $m_P \geq 3$ (i.e. P is an inflection point) or $P \in L$. Since φ is a finite separable morphism of curves, it has only finitely many ramification points (Prop. 2.2(a)), and every inflection point is one of them. So X has only finitely many inflection points.

Lemma B ($\gamma : X \to X^*$ is birational; tangent to the dual branch). Let $d \geq 2$. The map γ is not constant (otherwise all tangent lines coincide with one line $L_0 \ni P$ for all P, so $X \subseteq L_0$). So $\gamma : X \to X^*$ is a finite surjective morphism onto an integral plane curve. By (L2), $\ell(0) \cdot r(0) = 0 = \ell'(0) \cdot r(0)$, so $\operatorname{span}(\ell(0), \ell'(0)) \subseteq r(0)^\perp$, and by (L3) $\ell(0), \ell'(0)$ are linearly independent iff $\ell \times \ell'(0) = D(0)r(0) \neq 0$ iff $m_P = 2$. So at a non-inflection point P, the arc $\gamma(P(t)) = [\ell(t)]$ is smooth at $t = 0$ (an immersion), with tangent line equal to $\operatorname{span}(\ell(0), \ell'(0)) = r(0)^\perp$, i.e. the line $P^* \subseteq (\mathbf{P}^2)^*$ of lines through P. To see $\deg \gamma = 1$: choose $Q \in X^*$ with X^* smooth at Q and with $\gamma^{-1}(Q)$ consisting of $\deg \gamma$ points (char 0), none of which is an inflection point (finitely many exceptions). If $P \neq P'$ both map to Q, the tangent line of X^* at Q equals both P^* and P'^*, so $P = P'$. Hence $\deg \gamma = 1$, and γ is birational: since X is nonsingular, γ is the normalization of X^*.

(b) Let L be a multiple tangent, tangent at exactly $P_1, \ldots, P_r$ ($r \geq 2$) and none an inflection point; let $Q = [L] \in X^*$. The points of X over Q are exactly $P_1, \ldots, P_r$, and by Lemma B the branches of X^* at Q (the images of the formal neighborhoods of the P_i) are r smooth branches with distinct tangent lines P_i^*. For a line $\Lambda \subseteq (\mathbf{P}^2)^*$ through Q, with local equation μ, the intersection multiplicity is $(\Lambda \cdot X^*)_Q = \sum_i \operatorname{ord}_{P_i}(\gamma^* \mu)$: since $\mathcal{O}_{X^*,Q} \subseteq \widetilde{\mathcal{O}}$ is finite birational with finite-length quotient, $\dim \mathcal{O}/\mu\mathcal{O} = \dim \widetilde{\mathcal{O}}/\mu\widetilde{\mathcal{O}}$ (multiplicities $e(\mu; \cdot)$ are additive and vanish on finite-length modules), and the right side is the sum above. Each summand is ≥ 1, and $= 1$ iff Λ is not the tangent line P_i^* of the i-th branch. Thus the multiplicity of X^* at Q (the minimum of $(\Lambda \cdot X^*)_Q$ over $\Lambda \ni Q$, I, Ex. 5.4(b)) is r, and the tangent cone consists of the r distinct lines $P_1^*, \ldots, P_r^*$: Q is an ordinary r-fold point.

Consequence: $r \geq 2$ means X^* is singular at Q, and a plane curve has finitely many singular points; so there are finitely many multiple tangents without an inflection point of tangency. A multiple tangent tangent at an inflection point is the tangent line at one of the finitely many inflection points (part (a)). So there are finitely many multiple tangents.

(c) Choose coordinates with $O = (0 : 0 : 1)$ and $L = \{z = 0\}$, so $\psi(x : y : z) = (x : y)$; ψ is finite and separable (char 0). Let $Q \in L$, say $Q = (1 : b)$ (the other chart is symmetric); the fibre $\psi^{-1}(Q)$ is the set of points of X on the line $OQ = \{y = bx\}$, and $x \neq 0$ at those points (else the point is $O \notin X$). The local parameter $s = y/x - b$ at Q pulls back to $(y - bx)/x$, and x is a unit, so for $P \in \psi^{-1}(Q)$:

$$e_P(\psi) = \operatorname{ord}_P(y - bx) = (OP \cdot X)_P,$$

the intersection multiplicity of the line OP with X at P. In particular $\deg \psi = \sum_{P \mapsto Q} e_P = d$. A line through P has multiplicity ≥ 2 iff it is the tangent $T_P(X)$, so ψ is ramified at P iff OP is tangent to X at P. In that case $O \in T_P(X)$ and, as O is on no inflectional tangent, P is not an inflection point, so $e_P = m_P = 2$.

By Hurwitz (2.4, tame since char 0): $2g - 2 = d(-2) + \deg R$ with $\deg R = \#\{P : e_P = 2\}$, and $2g - 2 = d(d-3)$ by (I, Ex. 7.2). So $\deg R = d(d-3) + 2d = d(d-1)$. Distinct ramification points have distinct tangent lines through O (otherwise that line would be a multiple tangent through O), and every tangent line through O is tangent at some ramification point. So exactly

$d(d-1)$ tangent lines of X pass through O.

Class: the line $O^* \subseteq (\mathbf{P}^2)^*$ meets X^* in the points $\gamma(P)$ with $O \in T_P(X)$, i.e. $\ell(0) \cdot O = 0$. The pull-back to X of the equation of O^* is $\ell(t) \cdot O$; at such a P it vanishes to order exactly 1, since $\ell(0) \cdot O = \ell'(0) \cdot O = 0$ would force $O \perp \operatorname{span}(\ell(0), \ell'(0)) = r(0)^\perp$ (Lemma B), i.e. $O = P \in X$. So $\gamma^*\mathcal{O}(1)$ has a divisor of $d(d-1)$ distinct simple points; thus $\deg \gamma^*\mathcal{O}(1) = d(d-1)$. As γ is birational onto X^* (Lemma B), $\deg X^* = d(d-1)$.

(d) Let $O \in X$ be such that O is not an inflection point and does not lie on any inflectional tangent or multiple tangent (these are finitely many lines by (a),(b), each meeting X in finitely many points since X contains no line, and finitely many inflection points): this excludes only finitely many O. Let $L \not\ni O$ be a line and $\psi_O : X \to L$ the projection from O, extended over O by $\psi_O(O) = T_O(X) \cap L$ (a morphism since X is a nonsingular curve). A general line through O meets X in O (simply) and $d-1$ other points, so $\deg \psi_O = d-1$. For $P \neq O$ the computation of (c) gives $e_P(\psi_O) = (OP \cdot X)_P$. At $P = O$ take affine coordinates with $O = (0,0)$, X parametrized by $r = (t, g(t), 1)$, $\operatorname{ord} g = m_O = 2$: then $\psi_O = (x : y) = (1 : t^{m_O - 1}u(t))$ with u a unit, so $e_O = m_O - 1 = 1$: ψ_O is unramified at O. For $P \neq O$, ψ_O is ramified iff $OP = T_P(X)$, and then P is not an inflection point (else O would lie on an inflectional tangent) and $e_P = m_P = 2$; moreover distinct such P have distinct tangents through O (no multiple tangent through O), and $T_P(X) \neq T_O(X)$ for $P \neq O$ since $T_O(X)$ is not a multiple tangent. By Hurwitz,

$$d(d-3) = 2g - 2 = (d-1)(-2) + \deg R, \qquad \deg R = d^2 - d - 2 = (d+1)(d-2),$$

and $\deg R$ is the number of points $P \neq O$ with $O \in T_P(X)$, i.e. the number of tangents through O other than $T_O(X)$. This is $(d+1)(d-2)$, for all but finitely many $O \in X$.

(e) Take L as in (a). The fibre $\varphi^{-1}(Q)$ over $Q \in L$ is $\{P : Q \in T_P(X)\}$. For all but finitely many $Q \in L$ the point Q is not on X and not on any inflectional or multiple tangent (these are finitely many lines, different from L because L is not tangent, so each meets L in one point), so by (c) there are exactly $d(d-1)$ tangents through Q, tangent at $d(d-1)$ distinct points. Thus $\deg \varphi = d(d-1)$ (φ is separable).

Hurwitz for $\varphi : X \to L \cong \mathbf{P}^1$: $d(d-3) = 2g-2 = d(d-1)(-2) + \deg R_\varphi$, so $\deg R_\varphi = d(d-3) + 2d(d-1) = 3d^2 - 5d$. By $(*)$, $\deg R_\varphi = \sum_P (e_P - 1) = \sum_P (m_P - 2) + \sum_P (L \cdot X)_P = \sum_P (m_P - 2) + d$ (Bezout for a line: $L \cdot X$ has degree d; L is not contained in X). Hence

$$\sum_{P \in X} (m_P - 2) = 3d^2 - 5d - d = 3d(d-2),$$

i.e. X has $3d(d-2)$ inflection points counted with weight $m_P - 2$.

Ordinary inflection point ($m_P = 3$): take coordinates as in (L3), $r = (t, g, 1)$ with $g = at^3 + \dots$, $a \neq 0$. Then $\ell = r \times r' = (-g',\ 1,\ tg' - g)$, so in the chart of $(\mathbf{P}^2)^*$ where the middle coordinate is 1 the branch of X^* at $\gamma(P)$ is $(u, v) = (-g', tg' - g) = (-3at^2 + \cdots,\ 2at^3 + \cdots)$: orders $(2, 3)$. Reparametrize by s with $u = s^2$ (a square root of a unit exists in char 0); then $v = \alpha(s^2) + s^3\beta(s^2)$ with $\beta(0) \neq 0$ and $\operatorname{ord} \alpha \geq 2$, so with $v' = v - \alpha(u)$: $v'^2 = u^3\beta(u)^2$. So the branch has equation $v'^2 - u^3\beta(u)^2 = 0$: multiplicity 2, tangent cone the double line v'^2, nonzero x^3-coefficient: an ordinary cusp (in the sense of Ex. 1.8(c)). (If $T_P(X)$ happens to be tangent to X at other points too, the point of X^* has further branches; the assertion concerns the branch from P, and the point is an ordinary cusp when T_P is tangent only at P.)

(f) By Lemma B, X is the normalization of X^*, so $p_a(\widetilde{X^*}) = g = (d-1)(d-2)/2$. By (c) X^* has degree $d(d-1)$, so $p_a(X^*) = \frac{(D-1)(D-2)}{2}$ with $D = d(d-1)$. By Ex. 1.8(a),(c) (nodes and ordinary cusps have $\delta = 1$, smooth points $\delta = 0$):

$$p_a(X^*) = g + \nu + \kappa,$$

where ν is the number of nodes and κ the number of cusps of X^*.

Determination of ν, κ. Let $Q = [L] \in X^*$, with $P_1, \dots, P_r$ the points of tangency of L ($r \geq 1$). By Lemma B the branch at P_i is smooth iff P_i is not an inflection point; for an inflection point with contact m the branch has orders $(m-1, m)$ (computed as in (e)), multiplicity $m - 1 \geq 2$. So the multiplicity of X^* at Q is $\sum_i \mu_i$ with $\mu_i = 1$ or $m_{P_i} - 1$. Singular points have multiplicity 2 by hypothesis, so either $r = 2$ and both branches are smooth, meeting with distinct tangents $P_1^* \neq P_2^*$ (a node), or $r = 1$ and $m_P = 3$ (a cusp, as in (e)). Conversely every bitangent gives a point with $r = 2$, hence a singular point, hence (by the hypothesis) a node with both tangency points non-inflectional; every inflection point gives a singular point, hence a cusp with $m_P = 3$ and $r = 1$. Therefore: nodes of $X^* \leftrightarrow$ bitangents of X ($\nu =$ number of bitangents), and cusps $\leftrightarrow$ inflection points, all ordinary, so by (e) $\kappa = 3d(d-2)$. Then

$$\nu = \frac{(D-1)(D-2)}{2} - \frac{(d-1)(d-2)}{2} - 3d(d-2).$$

With $(D-1)(D-2) = d^2(d-1)^2 - 3d(d-1) + 2$ and $(d-1)(d-2) = d^2 - 3d + 2$ the difference of the first two terms is $\frac{1}{2}(d^4 - 2d^3 - 3d^2 + 6d)$; subtracting $\frac{1}{2}(6d^2 - 12d)$ gives $\nu = \frac{1}{2}(d^4 - 2d^3 - 9d^2 + 18d) = \frac{1}{2}d(d-2)(d^2-9) = \frac{1}{2}d(d-2)(d-3)(d+3)$.

(g) Let $d = 3$. A line meets X in a divisor of degree 3, so $m_P \leq 3$ for every P (and X contains no line). By (e), $\sum(m_P - 2) = 3 \cdot 3 \cdot 1 = 9$, with weights 1 at the inflection points ($m_P = 3$): so there are exactly 9 inflection points, all ordinary.

Let $P \neq Q$ be inflection points, and H the hyperplane class. The line PQ is not $T_P(X)$ (which meets X only at P, with multiplicity 3), nor $T_Q(X)$, so it meets P and Q with multiplicity 1 and $PQ \cdot X = P + Q + R$ with a third point $R \neq P, Q$. We show that R is an inflection point. Take an inflection point O as origin of the group law on X (genus 1, Example 1.3.7), so that $\mathcal{L}(P - O)$ etc. are the elements of $\mathrm{Pic}^0 X$. A point S is an inflection point iff $3S \sim H$ (if $3S = T_S \cdot X \sim H$; conversely, $l(H) = 3 = h^0(\mathbf{P}^2, \mathcal{O}(1))$ by Riemann–Roch, so every divisor $\sim H$ is a line section, and $3S \sim H$ gives a line meeting X only at S). Since $H \sim 3O$, S is an inflection point iff $3(S - O) \sim 0$. Now $P + Q + R \sim H \sim 3O$ gives $(R - O) \sim -(P - O) - (Q - O)$, so $3(R - O) \sim 0$, and R is an inflection point. (Here $d = 3$ and char 0 are used.)

(h) For a general quartic $d = 4$ and (f) gives $\frac{1}{2} \cdot 4 \cdot 2 \cdot 1 \cdot 7 = 28$ bitangents. For *every* nonsingular quartic one can argue directly as follows (characteristic 0). A tangent line $T_P(X)$ meets X in a divisor of degree 4 which is of one of the forms $2P+Q+R$ ($Q \neq R$), $2P+2Q$ (bitangent), $3P+Q$ (ordinary flex), $4P$ (hyperflex, $m_P = 4$), with P, Q, R distinct. By Lemma B and (b), (e), the singular points of X^* are: a node at each bitangent, a cusp of orders $(2, 3)$ at each ordinary flex (its tangent line meets X only at P and simply at Q, so there is a single branch), and a unibranch point with orders $(3, 4)$ at each hyperflex. For orders $(3, 4)$ the value semigroup of $k[[u, v]] \subseteq k[[t]]$ is $\{0, 3, 4, 6, 7, 8, \dots\}$ (no cancellation can create the values $1, 2, 5$), so $\delta = \#\text{gaps} = 3$ (Step 1 of Ex. 1.8(c): $\delta = \dim k[[t]]/\widehat{\mathcal{O}}$). Let ν be the number of (honest) bitangents, κ the number of ordinary flexes, h the number of hyperflexes. By (e), $\kappa + 2h = 3 \cdot 4 \cdot 2 = 24$. By Ex. 1.8(a):

$p_a(X^*) = \frac{11\cdot 10}{2} = 55 = g + \nu + \kappa + 3h = 3 + \nu + \kappa + 3h$. Hence $\nu = 52 - \kappa - 3h = 28 - h$. So the number of bitangents, counting a hyperflex line (contact $2P + 2Q$ with $P = Q$) as a degenerate bitangent, is always $\nu + h = 28$. With the literal definition ("tangent at exactly two points") the number is $28 - h$, which is 28 exactly when there is no hyperflex. (Remark: the singularity of X^* at a hyperflex line is not a tacnode as stated in the book, but a unibranch singularity with orders $(3, 4)$; the count 28 in the sense above is unaffected.) □

Exercise 2.4

> **Problem.** *A Funny Curve in Characteristic p.* Let X be the plane quartic curve $x^3y + y^3z + z^3x = 0$ over a field of characteristic 3. Show that X is nonsingular, every point of X is an inflection point, the dual curve X^* is isomorphic to X, but the natural map $X \to X^*$ is purely inseparable.

Solution. Let $F = x^3y + y^3z + z^3x$ over the algebraically closed field k of characteristic 3. Then $F_x = 3x^2y + z^3 = z^3$, $F_y = x^3 + 3y^2z = x^3$, $F_z = y^3 + 3z^2x = y^3$. These vanish simultaneously only at $x = y = z = 0$, so $X = Z(F)$ is nonsingular (and irreducible, as a nonsingular plane curve is irreducible: two components would meet by Bezout).

The tangent line. At $P = (a : b : c) \in X$ the tangent line is $T_P = \{c^3x + a^3y + b^3z = 0\}$, so the dual map is $\gamma(P) = (c^3 : a^3 : b^3)$.

Every point is an inflection point. Let $P = (a, b, c) \in X$, and let $R = (x', y', z') \in T_P$ be any point of the tangent line different from P; the points of the line are $P + sR$. Since the characteristic is 3, $(x + sx')^3 = x^3 + s^3x'^3$, hence

$$F(P + sR) = F(P) + s\,(a^3y' + b^3z' + c^3x') + s^3\,(bx'^3 + cy'^3 + az'^3) + s^4F(R).$$

Here $F(P) = 0$ and $a^3y' + b^3z' + c^3x' = 0$ because $R \in T_P$. So $F(P + sR) = s^3(\cdots)$ is divisible by s^3, i.e. the intersection multiplicity of T_P with X at P is ≥ 3. Thus every point of X is an inflection point (in the sense of Ex. 2.3: contact ≥ 3).

The dual curve. Let $(u : v : w) = (c^3 : a^3 : b^3) = \gamma(P)$ in $(\mathbf{P}^2)^*$ with dual coordinates. Since F has coefficients in $\mathbf{F}_3$, $F(a, b, c)^3 = F(a^3, b^3, c^3)$, so $0 = F(a^3, b^3, c^3) = F(v, w, u) = v^3w + w^3u + u^3v = F(u, v, w)$ (note $F(v, w, u) = F(u, v, w)$). So $\gamma(X) \subseteq Z(F) =: X' \subseteq (\mathbf{P}^2)^*$, an irreducible curve; γ is not constant, so $X^* = \gamma(X) = X'$, which is the same quartic $F = 0$ written in the dual coordinates: $X^* \cong X$ (via the identity of the coordinates $(u : v : w) \leftrightarrow (x : y : z)$).

Pure inseparability. Under this identification γ is $(a : b : c) \mapsto (c^3 : a^3 : b^3)$, that is, the Frobenius morphism $(a : b : c) \mapsto (a^3 : b^3 : c^3)$ of X composed with the cyclic permutation of coordinates, an automorphism of X (as $F(z, x, y) = F$). Equivalently, on function fields γ^* sends $u/w \mapsto (c/b)^3$, so $\gamma^*K(X^*) = K(X)^3$, and $K(X)/K(X)^3$ is purely inseparable of degree 3 (II, Example 2.4.3 / Prop. 2.5). So $X \to X^*$ is purely inseparable (of degree 3). □

Exercise 2.5

> **Problem.** *Automorphisms of a Curve of Genus ≥ 2.* Prove the theorem of Hurwitz [1] that a curve X of genus $g \geq 2$ over a field of characteristic 0 has at most $84(g-1)$ automorphisms.

We will see later (Ex. 5.2) or (V, Ex. 1.11) that the group $G = \operatorname{Aut} X$ is finite. So let G have order n. Then G acts on the function field $K(X)$. Let L be the fixed field. Then the field extension $L \subseteq K(X)$ corresponds to a finite morphism of curves $f : X \to Y$ of degree n.

(a) If $P \in X$ is a ramification point, and $e_P = r$, show that $f^{-1}f(P)$ consists of exactly n/r points, each having ramification index r. Let $P_1, \ldots, P_s$ be a maximal set of ramification points of X lying over distinct points of Y, and let $e_{P_i} = r_i$. Then show that Hurwitz's theorem implies that

$$(2g-2)/n = 2g(Y) - 2 + \sum_{i=1}^{s}(1 - 1/r_i).$$

(b) Since $g \geq 2$, the left hand side of the equation is > 0. Show that if $g(Y) \geq 0$, $s \geq 0$, $r_i \geq 2$, $i = 1, \ldots, s$ are integers such that

$$2g(Y) - 2 + \sum_{i=1}^{s}(1 - 1/r_i) > 0,$$

then the minimum value of this expression is $1/42$. Conclude that $n \leq 84(g-1)$. See (Ex. 5.7) for an example where this maximum is achieved.

Note. It is known that this maximum is achieved for infinitely many values of g (Macbeath [1]). Over a field of characteristic $p > 0$, the same bound holds, provided $p > g+1$, with one exception, namely the hyperelliptic curve $y^2 = x^p - x$, which has $p = 2g+1$ and $2p(p^2-1)$ automorphisms (Roquette [1]). For other bounds on the order of the group of automorphisms in characteristic p, see Singh [1] and Stichtenoth [1].

Solution. We work over k algebraically closed of characteristic 0 and take as given (as the statement does) that $G = \operatorname{Aut} X$ is finite, of order n. By Artin's theorem $K(X)/L$ is Galois with group G and $[K(X) : L] = n$; $f : X \to Y$ is the corresponding finite morphism of degree n (Y the nonsingular projective curve with function field L), separable since char 0. Moreover $f \circ \sigma = f$ for every $\sigma \in G$.

(a) Let $Q \in Y$. The points of $f^{-1}(Q)$ correspond to the extensions to $K(X)$ of the valuation ring $\mathcal{O}_Q \subseteq L$. In a Galois extension, the Galois group acts transitively on the set of such valuation rings (Atiyah–Macdonald, Ex. 5.13), i.e. G acts transitively on $f^{-1}(Q)$. For $\sigma \in G$ and $P \in X$, σ induces an isomorphism $\mathcal{O}_{X,\sigma P} \to \mathcal{O}_{X,P}$ compatible with $f^{\#}$ (as $f\sigma = f$), so $e_{\sigma P} = e_P$. Hence all points of a fibre have the same ramification index r, and $n = \sum_{P \mapsto Q} e_P = r \cdot \#f^{-1}(Q)$, so $\#f^{-1}(Q) = n/r$.

Let $Q_i = f(P_i)$, $i = 1, \ldots, s$ be the branch points (a maximal set of ramification points over distinct points of Y has one point P_i over each branch point Q_i). By Hurwitz (2.4; char 0 so the ramification is tame), $2g - 2 = n(2g(Y) - 2) + \sum_P (e_P - 1)$. The ramification points lie in the fibres over $Q_1, \ldots, Q_s$, and the fibre over Q_i has n/r_i points, each with $e - 1 = r_i - 1$, contributing $(n/r_i)(r_i - 1) = n(1 - 1/r_i)$. Dividing by n:

$$\frac{2g-2}{n} = 2g(Y) - 2 + \sum_{i=1}^{s}\Big(1 - \frac{1}{r_i}\Big).$$

(b) Let $\gamma = g(Y)$ and $E = 2\gamma - 2 + \sum_{i=1}^{s}(1 - 1/r_i)$, assumed > 0, with each $r_i \geq 2$ so each term $1 - 1/r_i \geq \frac{1}{2}$.

- $\gamma \geq 2$: $E \geq 2$.
- $\gamma = 1$: then $E = \sum(1 - 1/r_i) > 0$ forces $s \geq 1$, and $E \geq \frac{1}{2}$.
- $\gamma = 0$, $E = -2 + \sum(1 - 1/r_i)$. If $s \leq 2$ then $E < 0$. If $s \geq 5$ then $E \geq -2 + \frac{5}{2} = \frac{1}{2}$. If $s = 4$: $E = 2 - \sum 1/r_i > 0$ and $\sum 1/r_i \leq 2$ with equality only for $(2,2,2,2)$, so some $r_i \geq 3$ and $\sum 1/r_i \leq \frac{3}{2} + \frac{1}{3} = \frac{11}{6}$, whence $E \geq \frac{1}{6}$.
- $\gamma = 0$, $s = 3$: $E = 1 - (\frac{1}{r_1} + \frac{1}{r_2} + \frac{1}{r_3}) > 0$ with $r_1 \leq r_2 \leq r_3$. We maximize $S = \sum 1/r_i < 1$. If $r_1 \geq 3$: $S \leq 1$ with equality only for $(3,3,3)$, so otherwise $S \leq \frac{1}{3} + \frac{1}{3} + \frac{1}{4} = \frac{11}{12}$. If $r_1 = 2$: $\frac{1}{r_2} + \frac{1}{r_3} < \frac{1}{2}$; $r_2 = 2$ is impossible; $r_2 = 3$ forces $r_3 \geq 7$, so $S \leq \frac{1}{2} + \frac{1}{3} + \frac{1}{7} = \frac{41}{42}$ with equality for $(2,3,7)$; $r_2 = 4$ forces $r_3 \geq 5$, $S \leq \frac{1}{2} + \frac{1}{4} + \frac{1}{5} = \frac{19}{20} < \frac{41}{42}$; $r_2 \geq 5$ gives $S \leq \frac{1}{2} + \frac{2}{5} = \frac{9}{10} < \frac{41}{42}$.

So in every case $E \geq \frac{1}{42}$, and the minimum value $\frac{1}{42}$ is attained for $\gamma = 0$, $s = 3$, $(r_1, r_2, r_3) = (2,3,7)$ (and only there). By (a), $\frac{2g-2}{n} = E \geq \frac{1}{42}$, i.e. $n \leq 42(2g-2) = 84(g-1)$. (For $g = 3$ the Klein quartic of Ex. 5.7 has $168 = 84 \cdot 2$ automorphisms, so the bound is sharp.) □

Exercise 2.6

Problem. *f_* for Divisors.* Let $f : X \to Y$ be a finite morphism of curves of degree n. We define a homomorphism $f_* : \operatorname{Div} X \to \operatorname{Div} Y$ by $f_*(\sum n_i P_i) = \sum n_i f(P_i)$ for any divisor $D = \sum n_i P_i$ on X.

(a) For any locally free sheaf $\mathcal{E}$ on Y, of rank r, we define $\det \mathcal{E} = \bigwedge^r \mathcal{E} \in \operatorname{Pic} Y$ (II, Ex. 6.11). In particular, for any invertible sheaf $\mathcal{M}$ on X, $f_*\mathcal{M}$ is locally free of rank n on Y, so we can consider $\det f_*\mathcal{M} \in \operatorname{Pic} Y$. Show that for any divisor D on X,

$$\det(f_*\mathcal{L}(D)) \cong (\det f_*\mathcal{O}_X) \otimes \mathcal{L}(f_*D).$$

Note in particular that $\det(f_*\mathcal{L}(D)) \neq \mathcal{L}(f_*D)$ in general! [*Hint*: First consider an effective divisor D, apply f_* to the exact sequence $0 \to \mathcal{L}(-D) \to \mathcal{O}_X \to \mathcal{O}_D \to 0$, and use (II, Ex. 6.11).]

(b) Conclude that f_*D depends only on the linear equivalence class of D, so there is an induced homomorphism $f_* : \operatorname{Pic} X \to \operatorname{Pic} Y$. Show that $f_* \cdot f^* : \operatorname{Pic} Y \to \operatorname{Pic} Y$ is just multiplication by n.

(c) Use duality for a finite flat morphism (III, Ex. 6.10) and (III, Ex. 7.2) to show that

$$\det f_*\Omega_X \cong (\det f_*\mathcal{O}_X)^{-1} \otimes \Omega_Y^{\otimes n}.$$

(d) Now assume that f is separable, so we have the ramification divisor R. We define the *branch divisor* B to be the divisor f_*R on Y. Show that

$$(\det f_*\mathcal{O}_X)^2 \cong \mathcal{L}(-B).$$

Solution. All sheaves $f_*\mathcal{M}$ below are locally free of rank n (f is finite and flat).

A determinant lemma. Let $0 \to \mathcal{E}' \to \mathcal{E} \to \mathcal{T} \to 0$ be exact on the curve Y, with $\mathcal{E}', \mathcal{E}$ locally free of the same rank n and $\mathcal{T}$ of finite support. Then $\det \mathcal{E}' \cong \det \mathcal{E} \otimes \mathcal{L}(-B_\mathcal{T})$, where $B_\mathcal{T} = \sum_Q \operatorname{length}(\mathcal{T}_Q)\, Q$. (This is a special case of the determinant calculus of II, Ex. 6.11(b),(d):

$\det \mathcal{T} = \mathcal{L}(B_{\mathcal{T}})$ and det is multiplicative.) Direct proof: $\wedge^n \mathcal{E}' \to \wedge^n \mathcal{E}$ is an injection of invertible sheaves, so $\wedge^n \mathcal{E}' = \wedge^n \mathcal{E} \otimes \mathcal{L}(-B)$ for an effective divisor B. At a point Q with uniformizer t the elementary-divisor theorem over the DVR $\mathcal{O}_Q$ gives bases e_i of $\mathcal{E}_Q$ and $t^{a_i} e_i$ of $\mathcal{E}'_Q$; then $\mathcal{T}_Q \cong \bigoplus \mathcal{O}_Q / t^{a_i}$ has length $\sum a_i$, which is the order of vanishing of $\wedge^n \mathcal{E}'_Q \to \wedge^n \mathcal{E}_Q$. So the coefficient of Q in B is length $\mathcal{T}_Q$.

(a) Let $\mathcal{A}$ be an invertible sheaf on X and $P \in X$. Apply f_* (exact, f finite) to $0 \to \mathcal{A}(-P) \to \mathcal{A} \to \mathcal{A} \otimes k(P) \to 0$: we get $0 \to f_*\mathcal{A}(-P) \to f_*\mathcal{A} \to k(f(P)) \to 0$, where the cokernel is a skyscraper sheaf at $f(P)$ of length 1. The lemma gives

$$\det f_*(\mathcal{A}(-P)) \cong \det f_*\mathcal{A} \otimes \mathcal{L}(-f(P)). \tag{1}$$

(For an effective divisor E one gets the same way $\det f_*\mathcal{A}(-E) \cong \det f_*\mathcal{A} \otimes \mathcal{L}(-f_*E)$: the cokernel $f_*(\mathcal{A} \otimes \mathcal{O}_E)$ has length at Q equal to $\sum_{P \mapsto Q} \mathrm{mult}_P(E)$, as residue fields are k; this is the case suggested by the hint.)

Claim: $\det f_*\mathcal{L}(D) \cong \det f_*\mathcal{O}_X \otimes \mathcal{L}(f_*D)$ for every D. For $D = 0$ it is trivial. If the claim holds for $D + P$, apply (1) with $\mathcal{A} = \mathcal{L}(D + P)$: $\det f_*\mathcal{L}(D) \cong \det f_*\mathcal{L}(D + P) \otimes \mathcal{L}(-f(P)) \cong \det f_*\mathcal{O}_X \otimes \mathcal{L}(f_*(D + P) - f(P)) = \det f_*\mathcal{O}_X \otimes \mathcal{L}(f_*D)$; conversely if it holds for D, (1) shows it for $D + P$. Every divisor is reached from 0 by adding and subtracting points, which proves the claim.

In general $\det f_*\mathcal{O}_X \ncong \mathcal{O}_Y$ (see (d): e.g. $f : \mathbf{P}^1 \to \mathbf{P}^1$ of degree 2 has $\deg B = 2$, so $\deg \det f_*\mathcal{O}_X = -1$), so $\det f_*\mathcal{L}(D) \neq \mathcal{L}(f_*D)$ already for $D = 0$.

(b) If $D \sim D'$ then $\mathcal{L}(D) \cong \mathcal{L}(D')$, so $\det f_*\mathcal{L}(D) \cong \det f_*\mathcal{L}(D')$, and by (a) $\mathcal{L}(f_*D) \cong \det f_*\mathcal{L}(D) \otimes (\det f_*\mathcal{O}_X)^{-1} \cong \mathcal{L}(f_*D')$; since $\mathrm{Pic}\, Y \cong \mathrm{Cl}\, Y$ (II, 6.16), $f_*D \sim f_*D'$. So f_* descends to $\mathrm{Pic}\, X = \mathrm{Cl}\, X \to \mathrm{Cl}\, Y = \mathrm{Pic}\, Y$. For $Q \in Y$, $f_*f^*Q = f_* \sum_{P \mapsto Q} e_P P = (\sum_{P \mapsto Q} e_P) Q = nQ$ (II, 6.9). So $f_*f^* = n$ on $\mathrm{Div}\, Y$, hence on $\mathrm{Pic}\, Y$.

(c) For the finite flat morphism f, by (III, Ex. 7.2(a)) the sheaf $f^!\omega_Y$ is a dualizing sheaf for X, hence $\cong \omega_X$ ($\omega_X = \Omega_X$ is the dualizing sheaf, III, 7.12), where $f^!\mathcal{G}$ is the $\mathcal{O}_X$-module corresponding to $\mathcal{H}om_Y(f_*\mathcal{O}_X, \mathcal{G})$ (III, Ex. 6.10(a)). Taking $\mathcal{F} = \mathcal{O}_X$ in (III, Ex. 6.10(b)): $f_*\Omega_X \cong f_*f^!\Omega_Y \cong \mathcal{H}om_Y(f_*\mathcal{O}_X, \Omega_Y) = (f_*\mathcal{O}_X)^\vee \otimes \Omega_Y$, as $f_*\mathcal{O}_X$ is locally free. Since $\det(\mathcal{E} \otimes \mathcal{N}) = \det \mathcal{E} \otimes \mathcal{N}^{\otimes \mathrm{rk}\, \mathcal{E}}$ for $\mathcal{N}$ invertible, and $\det(\mathcal{E}^\vee) = (\det \mathcal{E})^{-1}$,

$$\det f_*\Omega_X \cong \det\big((f_*\mathcal{O}_X)^\vee\big) \otimes \Omega_Y^{\otimes n} \cong (\det f_*\mathcal{O}_X)^{-1} \otimes \Omega_Y^{\otimes n}.$$

(d) By Prop. 2.3, $\Omega_X \cong f^*\Omega_Y \otimes \mathcal{L}(R)$. By the projection formula $f_*\Omega_X \cong \Omega_Y \otimes f_*\mathcal{L}(R)$, so using (a),

$$\det f_*\Omega_X \cong \Omega_Y^{\otimes n} \otimes \det f_*\mathcal{L}(R) \cong \Omega_Y^{\otimes n} \otimes \det f_*\mathcal{O}_X \otimes \mathcal{L}(B).$$

Comparing with (c): $(\det f_*\mathcal{O}_X)^{-1} \otimes \Omega_Y^{\otimes n} \cong \Omega_Y^{\otimes n} \otimes \det f_*\mathcal{O}_X \otimes \mathcal{L}(B)$. Cancelling $\Omega_Y^{\otimes n}$: $(\det f_*\mathcal{O}_X)^{-2} \cong \mathcal{L}(B)$, i.e. $(\det f_*\mathcal{O}_X)^2 \cong \mathcal{L}(-B)$. □

Exercise 2.7

Problem. *Etale Covers of Degree* 2. Let Y be a curve over a field k of characteristic $\neq 2$. We show there is a one-to-one correspondence between finite etale morphisms $f : X \to Y$ of degree 2, and 2-torsion elements of $\mathrm{Pic}\, Y$, i.e., invertible sheaves $\mathcal{L}$ on Y with $\mathcal{L}^2 \cong \mathcal{O}_Y$.

(a) Given an etale morphism $f : X \to Y$ of degree 2, there is a natural map $\mathcal{O}_Y \to f_*\mathcal{O}_X$. Let $\mathcal{L}$ be the cokernel. Then $\mathcal{L}$ is an invertible sheaf on Y, $\mathcal{L} \cong \det f_*\mathcal{O}_X$, and so $\mathcal{L}^2 \cong \mathcal{O}_Y$ by (Ex. 2.6). Thus an etale cover of degree 2 determines a 2-torsion element in $\operatorname{Pic} Y$.

(b) Conversely, given a 2-torsion element $\mathcal{L}$ in $\operatorname{Pic} Y$, define an $\mathcal{O}_Y$-algebra structure on $\mathcal{O}_Y \oplus \mathcal{L}$ by $\langle a, b\rangle \cdot \langle a', b'\rangle = \langle aa' + \varphi(b \otimes b'), ab' + a'b\rangle$, where φ is an isomorphism of $\mathcal{L} \otimes \mathcal{L} \to \mathcal{O}_Y$. Then take $X = \operatorname{Spec}(\mathcal{O}_Y \oplus \mathcal{L})$ (II, Ex. 5.17). Show that X is an etale cover of Y.

(c) Show that these two processes are inverse to each other. [*Hint*: Let $\tau : X \to X$ be the involution which interchanges the points of each fibre of f. Use the trace map $a \mapsto a + \tau(a)$ from $f_*\mathcal{O}_X \to \mathcal{O}_Y$ to show that the sequence of $\mathcal{O}_Y$-modules in (a) is split exact.]

Note. This is a special case of the more general fact that for $(n, \operatorname{char} k) = 1$, the etale Galois covers of Y with group $\mathbf{Z}/n\mathbf{Z}$ are classified by the etale cohomology group $H^1_{\text{et}}(Y, \mathbf{Z}/n\mathbf{Z})$, which is equal to the group of n-torsion points of $\operatorname{Pic} Y$. See Serre [6].

Solution. Throughout, "2-torsion" correspondence means: isomorphism classes of finite etale $f : X \to Y$ of degree 2 (isomorphism over Y) $\leftrightarrow$ isomorphism classes of invertible $\mathcal{L}$ with $\mathcal{L}^2 \cong \mathcal{O}_Y$. (The trivial cover $Y \sqcup Y$ corresponds to $\mathcal{L} = \mathcal{O}_Y$.) We use residue fields $k(Q) = k$ at closed points, k algebraically closed (the argument works for any k with char $\neq 2$ using geometric fibres).

(a) Let $\mathcal{A} = f_*\mathcal{O}_X$, locally free of rank 2 with the structure map $\mathcal{O}_Y \to \mathcal{A}$, $1 \mapsto 1$. For a closed point Q the fibre $\mathcal{A} \otimes k(Q)$ is the ring of the scheme-theoretic fibre $f^{-1}(Q)$, which is nonzero, so $1 \neq 0$ in it. Therefore 1 is part of a local basis of $\mathcal{A}$, the map $\mathcal{O}_Y \to \mathcal{A}$ is a locally split injection, and the cokernel $\mathcal{L} = \mathcal{A}/\mathcal{O}_Y$ is invertible. By the multiplicativity of det in short exact sequences of locally free sheaves (II, Ex. 5.16(d) / Ex. 6.11), $\det \mathcal{A} \cong \det \mathcal{O}_Y \otimes \det \mathcal{L} = \mathcal{L}$. Since f is etale, it is unramified and separable, $R = 0$ and $B = 0$, and Ex. 2.6(d) gives $(\det \mathcal{A})^2 \cong \mathcal{L}(-B) = \mathcal{O}_Y$. So $\mathcal{L}^2 \cong \mathcal{O}_Y$. (If X is disconnected, then each component maps onto Y because f is flat and finite, hence open and closed, so each has degree 1 and $X = Y \sqcup Y$, giving $\mathcal{A} = \mathcal{O}_Y^2$ and $\mathcal{L} \cong \mathcal{O}_Y$ directly. If X is connected it is regular, hence integral, i.e. a curve, and Ex. 2.6 applies as above.) Isomorphic covers give isomorphic $\mathcal{L}$, so we get a well-defined map.

(b) Let $\mathcal{L}$ be 2-torsion and $\varphi : \mathcal{L} \otimes \mathcal{L} \to \mathcal{O}_Y$ an isomorphism. Note φ is symmetric, since $\mathcal{L}$ is locally free of rank 1 and the swap on $\mathcal{L} \otimes \mathcal{L}$ is the identity (locally $b = \beta e$, $b' = \beta' e$). Locally $\mathcal{L} = \mathcal{O}_Y e$ and $u := \varphi(e \otimes e)$ is a unit, and the algebra $\mathcal{A} = \mathcal{O}_Y \oplus \mathcal{L}$ is locally $\mathcal{O}_Y[e]/(e^2 - u)$, which is commutative and associative; so $\mathcal{A}$ is a commutative associative $\mathcal{O}_Y$-algebra (these are local identities), finite locally free of rank 2. Hence $f : X = \operatorname{Spec} \mathcal{A} \to Y$ is finite, flat, of degree 2. At a closed point Q the fibre is $k[e]/(e^2 - u(Q))$ with $u(Q) \neq 0$; as $\operatorname{char} k \neq 2$, $e^2 - u(Q)$ has two distinct roots, so the fibre is $k \times k$, reduced, i.e. f is unramified. A flat unramified morphism is etale. So $X \to Y$ is an etale cover of degree 2.

(c) *(b) then (a).* Starting from $(\mathcal{L}, \varphi)$ and $\mathcal{A} = \mathcal{O}_Y \oplus \mathcal{L}$, the cokernel of $\mathcal{O}_Y \to \mathcal{A}$ is $\mathcal{A}/\mathcal{O}_Y = \mathcal{L}$. Also the isomorphism class of X over Y does not depend on φ: another choice is $\varphi' = c\varphi$ with $c \in H^0(\mathcal{O}_Y^*) = k^*$ (isomorphisms $\mathcal{L} \otimes \mathcal{L} \to \mathcal{O}_Y$ differ by units of $\mathcal{O}_Y$, which are constants since Y is a complete variety); take $\lambda \in k$ with $\lambda^2 c = 1$; then $\langle a, b\rangle \mapsto \langle a, \lambda b\rangle$ is an isomorphism of algebras $(\mathcal{O} \oplus \mathcal{L}, \varphi) \to (\mathcal{O} \oplus \mathcal{L}, \varphi')$. So the composite $\mathcal{L} \mapsto X \mapsto \mathcal{L}$ is the identity.

(a) then (b). Let $f : X \to Y$ be etale of degree 2 and $\mathcal{A} = f_*\mathcal{O}_X$. Since char $\neq 2$, the trace $\operatorname{tr} : \mathcal{A} \to \mathcal{O}_Y$ (trace of multiplication, an $\mathcal{O}_Y$-linear map) satisfies $\operatorname{tr}(1) = 2$, a unit; so $\frac{1}{2}\operatorname{tr}$

splits $\mathcal{O}_Y \to \mathcal{A}$ (this is the trace $a \mapsto a + \tau(a)$ of the hint, divided by 2), and $\mathcal{A} = \mathcal{O}_Y \oplus \mathcal{A}_0$ with $\mathcal{A}_0 = \ker \mathrm{tr} \cong \mathcal{A}/\mathcal{O}_Y = \mathcal{L}$ (split exactness of the sequence of (a)). Locally $\mathcal{A}_0 = \mathcal{O}_Y e$ with $\mathrm{tr}\, e = 0$; by Cayley–Hamilton, $e^2 - \mathrm{tr}(e)e + N(e) = 0$, so $e^2 = -N(e) =: u \in \mathcal{O}_Y$. Hence $\mathcal{A}_0 \cdot \mathcal{A}_0 \subseteq \mathcal{O}_Y$, and the multiplication of $\mathcal{A} = \mathcal{O}_Y \oplus \mathcal{L}$ is $\langle a, b\rangle\langle a', b'\rangle = \langle aa' + \varphi(b \otimes b'), ab' + a'b\rangle$ with $\varphi : \mathcal{L} \otimes \mathcal{L} \to \mathcal{O}_Y$ the multiplication. The map φ is an isomorphism: locally it is $e \otimes e \mapsto u$, and u is a unit, because otherwise at a point Q with $u(Q) = 0$ the fibre $k[e]/(e^2)$ would be nonreduced, contradicting that f is etale. Thus $\mathcal{A} \cong \mathcal{O}_Y \oplus \mathcal{L}$ as in (b), and $X \cong \mathrm{Spec}(\mathcal{O}_Y \oplus \mathcal{L})$ over Y. So the composite $X \mapsto \mathcal{L} \mapsto X'$ gives $X' \cong X$.

Hence the two processes are mutually inverse bijections. □

3. Embeddings in projective space

Conventions. k is algebraically closed, "curve" means complete nonsingular curve over k (as throughout Chapter IV), $\ell(D) = \dim_k H^0(X, \mathcal{L}(D))$ and $\dim |D| = \ell(D) - 1$. The degree of a curve $X \subseteq \mathbf{P}^n$ is the degree of the divisor $X \cdot H$ cut out by a hyperplane $H \not\supseteq X$ (Example 3.3.2). Put $\delta(d) = \frac{1}{2}(d-1)(d-2)$; a nonsingular plane curve of degree d has genus $\delta(d)$ (I, Ex. 7.2). Besides the results of Section 3 I use the following standard facts.

(F1) (Ex. 1.5) If E is an effective divisor then $\dim |E| \le \deg E$, with equality only for $E = 0$ or $g = 0$. In particular, *if E is a divisor of degree 1 with $\ell(E) \ge 2$, then $g = 0$* (then $E \sim E'$ with E' effective, and $\dim |E'| \ge 1 = \deg E'$).

(F2) (Ex. 1.8) If $\nu : X \to \Gamma$ is the normalization of an integral projective curve Γ, then $p_a(\Gamma) = g(X) + \sum_P \delta_P$ with $\delta_P = \mathrm{length}(\tilde{\mathcal{O}}_P/\mathcal{O}_P)$, and $\delta_P > 0$ exactly at the singular points; a node has $\delta_P = 1$.

(F3) Let $\nu : X \to \Gamma \subseteq \mathbf{P}^n$ be a finite birational morphism from a curve X onto an integral curve Γ (so ν is the normalization). Then $\deg \Gamma = \deg \nu^* \mathcal{O}_\Gamma(1)$. Indeed, tensoring $0 \to \mathcal{O}_\Gamma \to \nu_* \mathcal{O}_X \to \mathcal{Q} \to 0$ ($\mathcal{Q}$ of finite length $\sum \delta_P$) with $\mathcal{O}(m)$ and using the projection formula, $\chi(\mathcal{O}_\Gamma(m)) = m \deg \nu^* \mathcal{O}_\Gamma(1) + 1 - g(X) - \sum \delta_P$, and the left side is $m \deg \Gamma + 1 - p_a(\Gamma)$.

Lemma A. Let $X \subseteq \mathbf{P}^n$ be an integral curve of degree d not contained in any hyperplane. Then $d \ge n$.

Proof. Choose n distinct points $P_1, \dots, P_n \in X$. There is a hyperplane H through them (they impose at most n linear conditions on the $n+1$ coefficients), and $H \not\supseteq X$. The effective divisor $X \cdot H$ has degree d and multiplicity ≥ 1 at each $P_i \in H$, so $d \ge n$. □

Exercise 3.1

> **Problem.** If X is a curve of genus 2, show that a divisor D is very ample $\Leftrightarrow \deg D \ge 5$. This strengthens (3.3.4).

Solution. ($\Leftarrow$) $2g + 1 = 5$, so this is (3.2)(b).

($\Rightarrow$) Let D be very ample, $d = \deg D$. Then $d > 0$ (3.3) and $D \sim X \cdot H$ for a hyperplane section of an embedding, so D may be taken effective. By the proof of (3.1) the complete system $|D|$ is base point free and defines a closed immersion $\varphi : X \to \mathbf{P}^n$, $n = \dim |D|$; its image is

not contained in a hyperplane (the coordinates form a basis of $H^0(X, \mathcal{L}(D))$, so are linearly independent), and by Example 3.3.2 it has degree d.

If $n = 0$, φ is not an immersion of a curve. If $n = 1$, $\varphi(X)$ is a closed irreducible 1-dimensional subset of $\mathbf{P}^1$, so $X \cong \mathbf{P}^1$, contradicting $g = 2$. Hence $n \geq 2$, and Lemma A gives $n \leq d$.

$d \leq 2$ *is impossible:* then $n = d = 2$, so $\ell(D) = 3$. For a point P, $\ell(D - P) \geq 2$ while $\deg(D - P) = 1$, so $g = 0$ by (F1), a contradiction.

$d = 3, 4$ *are impossible:* since $d > 2g - 2 = 2$, D is nonspecial and Riemann–Roch gives $\ell(D) = d + 1 - g = d - 1$, so $n = d - 2$. For $d = 3$ this is $n = 1$, excluded above. For $d = 4$ we get $n = 2$: X is isomorphic to a plane curve of degree 4, whose genus is $\delta(4) = 3 \neq 2$.

Hence $d \geq 5$. □

Exercise 3.2

Problem. Let X be a plane curve of degree 4.

(a) Show that the effective canonical divisors on X are exactly the divisors $X.L$, where L is a line in $\mathbf{P}^2$.

(b) If D is any effective divisor of degree 2 on X, show that $\dim |D| = 0$.

(c) Conclude that X is not hyperelliptic (Ex. 1.7).

Solution. X is a hypersurface, so a complete intersection, and by (II, Ex. 8.4(e)) $\omega_X \cong \mathcal{O}_X(4 - 2 - 1) = \mathcal{O}_X(1)$; also $g = \delta(4) = 3$. Restriction of linear forms gives a map $H^0(\mathbf{P}^2, \mathcal{O}(1)) \to H^0(X, \mathcal{O}_X(1))$, injective because X lies on no line. Both sides have dimension 3 ($\dim H^0(X, \omega_X) = g = 3$), so it is an isomorphism: every section of $\omega_X \cong \mathcal{O}_X(1)$ is the restriction of a linear form.

(a) An effective canonical divisor is the divisor of zeros of a nonzero section of $\omega_X \cong \mathcal{O}_X(1)$, hence of the restriction of a linear form, i.e. it is $X.L$ for a line L (and $L \not\supseteq X$). Conversely every $X.L$ is the divisor of a section of $\mathcal{O}_X(1) \cong \omega_X$, hence canonical, effective.

(b) Let $D \geq 0$ have degree 2. By Riemann–Roch, $\ell(D) = 2 + 1 - 3 + \ell(K - D) = \ell(K - D)$. Fix $K = X.L_0$ ($L_0 = \{\lambda_0 = 0\}$); then $\mathcal{L}(K - D) \cong \{\text{sections } s \text{ of } \mathcal{O}_X(1) \text{ with } \operatorname{div}(s) \geq D\}$ via $f \mapsto f\lambda_0$, and by the first paragraph such sections are restrictions of linear forms λ with $\operatorname{div}(\lambda|_X) \geq D$. If $D = P + Q$ with $P \neq Q$ this means $\lambda(P) = \lambda(Q) = 0$, two independent conditions on the 3-dimensional space of linear forms, so the space is 1-dimensional. If $D = 2P$ it means $\lambda(P) = 0$ and $\operatorname{ord}_P(\lambda|_X) \geq 2$, i.e. the line $\{\lambda = 0\}$ is the tangent line of X at P (X is nonsingular at P); again a 1-dimensional space. So $\ell(K - D) = 1 = \ell(D)$ and $\dim |D| = 0$.

(c) A hyperelliptic curve has a finite morphism $f : X \to \mathbf{P}^1$ of degree 2. For distinct $a, b \in \mathbf{P}^1$ the effective divisors $f^*(a) \neq f^*(b)$ of degree 2 are linearly equivalent (pull-back of $a \sim b$, II, Ex. 6.8), so $\dim |f^*(a)| \geq 1$, contradicting (b). □

Exercise 3.3

Problem. If X is a curve of genus ≥ 2 which is a complete intersection (II, Ex. 8.4) in some $\mathbf{P}^n$, show that the canonical divisor K is very ample. Conclude that a curve of genus 2 can never be a complete intersection in any $\mathbf{P}^n$. Cf. (Ex. 5.1).

Solution. Let X be the complete intersection of hypersurfaces of degrees $d_1, \ldots, d_{n-1}$. By (II, Ex. 8.4(e)), $\omega_X \cong \mathcal{O}_X(m)$ with $m = \sum d_i - n - 1$. Since $g \geq 2$, $\deg K = 2g - 2 > 0$, while $\deg \mathcal{O}_X(m) = m \deg X$ with $\deg X > 0$; so $m \geq 1$. Compose the inclusion $X \hookrightarrow \mathbf{P}^n$ with the m-uple embedding $\mathbf{P}^n \hookrightarrow \mathbf{P}^N$ (II, Ex. 5.13): this is a closed immersion of X whose pull-back of $\mathcal{O}(1)$ is $\mathcal{O}_X(m) \cong \omega_X$. Hence $\omega_X = \mathcal{L}(K)$ is very ample.

If $g = 2$, this would make K, a divisor of degree $2g - 2 = 2$, very ample, contradicting (3.1) (very ample needs degree ≥ 5). So a curve of genus 2 is not a complete intersection in any $\mathbf{P}^n$. □

Exercise 3.4

Problem. Let X be the d-uple embedding of $\mathbf{P}^1$ in $\mathbf{P}^d$, for any $d \geq 1$ (I, Ex. 2.12). We call X the *rational normal curve of degree d* in $\mathbf{P}^d$.

(a) Show that X is projectively normal, and that its homogeneous ideal can be generated by forms of degree 2.

(b) If X is any curve of degree d in $\mathbf{P}^n$, with $d \leq n$, which is not contained in any $\mathbf{P}^{n-1}$, show that in fact $d = n$, $g(X) = 0$, and X differs from the rational normal curve of degree d only by an automorphism of $\mathbf{P}^d$. Cf. (II, 7.8.5).

(c) In particular, any curve of degree 2 in any $\mathbf{P}^n$ is a conic in some $\mathbf{P}^2$.

(d) A curve of degree 3 in any $\mathbf{P}^n$ must be either a plane cubic curve, or the twisted cubic curve in $\mathbf{P}^3$.

Solution. Write $X = \{(s^d : s^{d-1}t : \cdots : t^d)\} \subseteq \mathbf{P}^d$, $x_i \mapsto s^{d-i}t^i$.

(a) *Projective normality.* $X \cong \mathbf{P}^1$ is normal, so by (II, Ex. 5.14(d)) it suffices to show that $H^0(\mathbf{P}^d, \mathcal{O}(l)) \to H^0(X, \mathcal{O}_X(l))$ is surjective for all $l \geq 0$. Now $\mathcal{O}_X(l) \cong \mathcal{O}_{\mathbf{P}^1}(dl)$, with H^0 spanned by the monomials $s^{dl-j}t^j$, $0 \leq j \leq dl$. Given j, write $j = j_1 + \cdots + j_l$ with $0 \leq j_i \leq d$ (possible as $j \leq dl$); then $s^{dl-j}t^j = \prod_i s^{d-j_i}t^{j_i}$ is the image of the monomial $x_{j_1} \cdots x_{j_l}$. So the map is onto.

Ideal. The ideal I of X is the kernel of $\theta : k[x_0, \ldots, x_d] \to k[s,t]$, $x_i \mapsto s^{d-i}t^i$ (a form vanishes on X iff its image vanishes, k being infinite). Every monomial $m = x_{i_1} \cdots x_{i_l}$ goes to a monomial $s^{dl-w}t^w$, $w = \sum i_j$ (the *weight*). Hence I is spanned by the binomials $m - m'$ with $\theta(m) = \theta(m')$, i.e. with the same degree and weight (group the monomials of an element of $\ker\theta$ by image; the coefficients in each group sum to 0). Let $J \subseteq I$ be the ideal generated by the quadrics $x_i x_j - x_{i+1}x_{j-1}$ for $0 \leq i$, $i + 2 \leq j \leq d$. Take a monomial $x_{i_1} \cdots x_{i_l}$, and suppose two of its indices satisfy $i \leq j - 2$. Replacing $x_i x_j$ by $x_{i+1}x_{j-1}$ changes it modulo J and decreases $\sum i_r^2$ by $2(j - i) - 2 \geq 2$; so after finitely many steps we reach a monomial congruent to it mod J whose indices all lie in $\{a, a+1\}$ for some a. Given (l, w), the multiset of indices in $\{a, a+1\}$ is unique: if $q \in \{0, \ldots, l-1\}$ of the indices equal $a+1$ (and the others a), then $w = la + q$, which determines a and q. Hence any two monomials of the same degree and weight are congruent mod J, so $I = J$ is generated by quadrics. (For $d = 1$, $I = 0$.)

(b) By Lemma A, $d \geq n$, so $d = n$. Let $D = X \cdot H$ (degree d). The restriction $H^0(\mathbf{P}^n, \mathcal{O}(1)) \to H^0(X, \mathcal{L}(D))$ is injective, as X lies in no hyperplane, so $\ell(D) \geq n + 1 = d + 1$. Choose distinct points $P_1, \ldots, P_{d-1} \in X$; since $\ell(E - P) \geq \ell(E) - 1$ always, $\ell(D - P_1 - \cdots - P_{d-1}) \geq 2$, while this divisor has degree 1. By (F1), $g(X) = 0$, so $X \cong \mathbf{P}^1$.

Then $\mathcal{L}(D) \cong \mathcal{O}_{\mathbf{P}^1}(d)$ and the embedding is given by linearly independent sections $\sigma_0, \ldots, \sigma_d \in H^0(\mathbf{P}^1, \mathcal{O}(d))$. This space has dimension $d+1$, so the σ_i form a basis, and $\sigma_i = \sum_j a_{ij} s^{d-j} t^j$ with $A = (a_{ij})$ invertible. Thus the embedding is the d-uple embedding followed by the automorphism A of $\mathbf{P}^d$.

(c) Let X have degree 2 and let $\mathbf{P}^m$ be the linear span of X (smallest linear subspace containing it). Then $X \subseteq \mathbf{P}^m$ is not in a hyperplane, so $m \leq 2$ by Lemma A. $m = 0$ is absurd and $m = 1$ would make $X = \mathbf{P}^1$ of degree 1. So $m = 2$: X is a plane curve of degree 2, a conic.

(d) Similarly for degree 3 with span $\mathbf{P}^m$: $m \leq 3$, $m = 1$ is impossible, $m = 2$ gives a plane cubic, and $m = 3 = d$ gives, by (b), the rational normal curve of degree 3 up to an automorphism of $\mathbf{P}^3$, i.e. the twisted cubic. □

Exercise 3.5

Problem. Let X be a curve in $\mathbf{P}^3$, which is not contained in any plane.

(a) If $O \notin X$ is a point, such that the projection from O induces a birational morphism φ from X to its image in $\mathbf{P}^2$, show that $\varphi(X)$ must be singular. [*Hint:* Calculate $\dim H^0(X, \mathcal{O}_X(1))$ two ways.]

(b) If X has degree d and genus g, conclude that $g < \frac{1}{2}(d-1)(d-2)$. (Use (Ex. 1.8).)

(c) Now let $\{X_t\}$ be the flat family of curves induced by the projection (III, 9.8.3) whose fibre over $t = 1$ is X, and whose fibre X_0 over $t = 0$ is a scheme with support $\varphi(X)$. Show that X_0 always has nilpotent elements. Thus the example (III, 9.8.4) is typical.

Solution. Throughout, φ is given by the three linear forms vanishing at O; these are sections of $\mathcal{O}_{\mathbf{P}^3}(1)$ with no common zero on X, so $\varphi^* \mathcal{O}_{\mathbf{P}^2}(1) \cong \mathcal{O}_X(1)$. Put $\Gamma = \varphi(X)$, an integral plane curve of degree e (it is not a point, since X lies on no plane through O).

(a) Suppose Γ is nonsingular. Then $\varphi : X \to \Gamma$ is a finite birational morphism onto a normal curve, hence an isomorphism ($\mathcal{O}_\Gamma \to \varphi_* \mathcal{O}_X$ is a finite birational extension of integrally closed rings). Computation 1: since X lies in no plane, restriction $H^0(\mathbf{P}^3, \mathcal{O}(1)) \to H^0(X, \mathcal{O}_X(1))$ is injective, so $\dim H^0(X, \mathcal{O}_X(1)) \geq 4$. Computation 2: $\mathcal{O}_X(1) \cong \varphi^* \mathcal{O}_{\mathbf{P}^2}(1)$ and φ is an isomorphism, so $H^0(X, \mathcal{O}_X(1)) \cong H^0(\Gamma, \mathcal{O}_\Gamma(1))$. From $0 \to \mathcal{O}_{\mathbf{P}^2}(1-e) \to \mathcal{O}_{\mathbf{P}^2}(1) \to \mathcal{O}_\Gamma(1) \to 0$ and $H^1(\mathbf{P}^2, \mathcal{O}(m)) = 0$ (III, 5.1), $H^0(\mathbf{P}^2, \mathcal{O}(1)) \to H^0(\Gamma, \mathcal{O}_\Gamma(1))$ is surjective, so $\dim H^0(\Gamma, \mathcal{O}_\Gamma(1)) \leq 3$. Contradiction. So Γ is singular.

(b) Here X is in no plane (for a plane curve $g = \delta(d)$, so the inequality needs this hypothesis). By (3.10) there is $O \notin X$ such that φ is birational onto its image Γ. Then X is the normalization of Γ, so by (F3) $e = \deg \varphi^* \mathcal{O}(1) = \deg \mathcal{O}_X(1) = d$. By (F2), $\delta(d) = p_a(\Gamma) = g + \sum_P \delta_P$, and $\sum \delta_P \geq 1$ because Γ is singular by (a). Hence $g < \delta(d) = \frac{1}{2}(d-1)(d-2)$.

(c) In coordinates with $O = (0:0:0:1)$ the family is $X_a = \sigma_a(X)$, $\sigma_a(x_0 : x_1 : x_2 : x_3) = (x_0 : x_1 : x_2 : a x_3)$ for $a \neq 0$, extended to a flat family over $\mathbf{A}^1$ (III, 9.8.3). All fibres have the same Hilbert polynomial (III, 9.9; $\mathbf{A}^1$ is connected). For $a = 1$ this is $P_X(m) = \chi(\mathcal{O}_X(m)) = dm + 1 - g$ (Riemann–Roch). (As in (a), (b), φ is taken birational onto Γ, which is possible by (3.10).) Suppose X_0 had no nilpotents. Then X_0 would be the reduced scheme on Γ (a plane curve in the plane $x_3 = 0$), whose Hilbert polynomial is that of a plane curve of degree e: from $0 \to \mathcal{O}_{\mathbf{P}^2}(m-e) \to \mathcal{O}_{\mathbf{P}^2}(m) \to \mathcal{O}_\Gamma(m) \to 0$ one gets $P_\Gamma(m) = em + 1 - \delta(e)$. Comparing with

$dm + 1 - g$: $e = d$ and $g = \delta(d)$, contradicting (b). So X_0 has nilpotent elements. (For the twisted cubic: $d = 3$, $g = 0$, and the nodal cubic has $p_a = 1$, so the missing $p_a - g = 1$ is carried by the embedded point, as computed in III, 9.8.4.) □

Exercise 3.6

Problem. *Curves of Degree 4.*

(a) If X is a curve of degree 4 in some $\mathbf{P}^n$, show that either (1) $g = 0$, in which case X is either the rational normal quartic in $\mathbf{P}^4$ (Ex. 3.4) or the rational quartic curve in $\mathbf{P}^3$ (II, 7.8.6), or (2) $X \subseteq \mathbf{P}^2$, in which case $g = 3$, or (3) $X \subseteq \mathbf{P}^3$ and $g = 1$.

(b) In the case $g = 1$, show that X is a complete intersection of two irreducible quadric surfaces in $\mathbf{P}^3$ (I, Ex. 5.11). [*Hint:* Use the exact sequence $0 \to \mathcal{I}_X \to \mathcal{O}_{\mathbf{P}^3} \to \mathcal{O}_X \to 0$ to compute $\dim H^0(\mathbf{P}^3, \mathcal{I}_X(2))$, and thus conclude that X is contained in at least two irreducible quadric surfaces.]

Solution. (a) Let $\mathbf{P}^m$ be the linear span of X; X lies in no hyperplane of $\mathbf{P}^m$, so $m \leq 4$ by Lemma A. $m = 1$ is impossible (X would be $\mathbf{P}^1$, of degree 1).

$m = 4$: by Ex. 3.4(b), $g = 0$ and X is the rational normal quartic.

$m = 2$: X is a nonsingular plane quartic, so $g = \delta(4) = 3$.

$m = 3$: $X \subseteq \mathbf{P}^3$ is not in a plane. By (3.10) there is a projection $O \notin X$ birational onto its image, so by Ex. 3.5(b), $g < \delta(4) = 3$. If $g = 2$: $D = X \cdot H$ has degree $4 > 2g - 2$, so $\ell(D) = 4 + 1 - 2 = 3$, but $\ell(D) \geq 4$ as the four coordinates are independent sections of $\mathcal{L}(D)$ (X in no plane). So $g \in \{0, 1\}$. If $g = 0$, X is a nonsingular curve $\cong \mathbf{P}^1$ of degree 4 in $\mathbf{P}^3$ not in a plane, so $\mathcal{L}(D) \cong \mathcal{O}_{\mathbf{P}^1}(4)$ and the embedding is given by a base-point-free 4-dimensional subspace of $H^0(\mathbf{P}^1, \mathcal{O}(4))$ (of dimension 5), i.e. X is the projection of the rational normal quartic from a point of $\mathbf{P}^4$: a rational quartic curve in $\mathbf{P}^3$ of the type of (II, 7.8.6) (the one there is the projection of the normal quartic from $(0 : 0 : 1 : 0 : 0)$, the omitted monomial s^2t^2). This is case (1). If $g = 1$, case (3).

(b) Let $X \subseteq \mathbf{P}^3$, $g = 1$, $d = 4$; X is in no plane (a plane quartic has $g = 3$). From $0 \to \mathcal{I}_X(2) \to \mathcal{O}_{\mathbf{P}^3}(2) \to \mathcal{O}_X(2) \to 0$ we get the exact sequence $0 \to H^0(\mathcal{I}_X(2)) \to H^0(\mathcal{O}_{\mathbf{P}^3}(2)) \to H^0(\mathcal{O}_X(2))$. Here $\dim H^0(\mathcal{O}_{\mathbf{P}^3}(2)) = 10$ and $\deg \mathcal{O}_X(2) = 8 > 2g - 2 = 0$, so $\ell = 8$ by Riemann–Roch. Hence $\dim H^0(\mathcal{I}_X(2)) \geq 2$: there are two linearly independent quadratic forms Q_1, Q_2 vanishing on X.

Every nonzero quadratic form Q vanishing on X is irreducible: otherwise $Q = \lambda\lambda'$ and $X \subseteq \{\lambda = 0\} \cup \{\lambda' = 0\}$ lies in a plane (X irreducible). Let $Y = V(Q_1)$, an irreducible quadric surface of degree 2; $Q_2 \notin (Q_1)$ (non-proportional of equal degree), so $Y \not\subseteq V(Q_2)$. All components Z_j of $Y \cap V(Q_2)$ have dimension 1 (I, 7.2), and Bezout (I, 7.7) gives $\sum_j i(Y, V(Q_2); Z_j) \deg Z_j = 2 \cdot 2 = 4$. The irreducible curve X lies in $Y \cap V(Q_2)$, hence is one of the Z_j, say Z_1, with $\deg Z_1 = 4$. Thus $i_1 = 1$ and there is no other component: $Z := \operatorname{Proj} k[x_0, \ldots, x_3]/(Q_1, Q_2)$ has X as its only irreducible component and is reduced at the generic point of X ($i_1 = 1$ is the length of the local ring there).

Now Q_1, Q_2 is a regular sequence in the UFD $S = k[x_0, \ldots, x_3]$, so $S/(Q_1, Q_2)$ is Cohen–Macaulay of dimension 2; hence Z has no embedded points, and being generically reduced with the single generic point of X, Z is reduced, so $Z = X$ as schemes. Moreover depth $2 \geq 1$

means (Q_1, Q_2) is saturated, so $I(X) = (Q_1, Q_2)$. Thus X is the complete intersection of the two irreducible quadrics $V(Q_1), V(Q_2)$. □

Exercise 3.7

> **Problem.** In view of (3.10), one might ask conversely, is every plane curve with nodes a projection of a nonsingular curve in $\mathbf{P}^3$? Show that the curve $xy + x^4 + y^4 = 0$ (assume $\operatorname{char} k \neq 2$) gives a counterexample.

Solution. Reading: "projection" means a projection $\varphi : X \to \mathbf{P}^2$ from a point $O \notin X$ of a nonsingular curve $X \subseteq \mathbf{P}^3$ which is birational onto the plane curve in question (as in (3.10)). The affine curve $xy + x^4 + y^4 = 0$ has projective closure $\Gamma = V(F)$, $F = xyz^2 + x^4 + y^4$, in $\mathbf{P}^2$ with coordinates $(x : y : z)$.

Γ *has exactly one singular point, a node.* In the chart $z = 1$: $f = xy + x^4 + y^4$, $f_x = y + 4x^3$, $f_y = x + 4y^3$. At a singular point, $y = -4x^3$ and $x = -4y^3 = 256x^9$. If $x \neq 0$ then $256x^8 = 1$, and then $f = -4x^4 + x^4 + 256x^{12} = -4x^4 + x^4 + x^4 = -2x^4 \neq 0$ (char $\neq 2$), a contradiction. So $x = 0$, hence $y = 0$: the only singular point in the chart is the origin. There $f = xy +$(terms of degree 4), so the multiplicity is 2 with tangent cone $xy = 0$, two distinct lines: a node. At infinity, $z = 0$, the points satisfy $x^4 + y^4 = 0$, so $x, y \neq 0$, and $F_x = yz^2 + 4x^3 = 4x^3 \neq 0$: nonsingular.

Γ *is irreducible.* If $F = GH$ with $\deg G, \deg H \geq 1$, then G, H meet in $\deg G \deg H \geq 3$ points (counted with multiplicity), all singular points of Γ (and no component is repeated, since the singular set is finite). So G, H meet only at the origin, which has multiplicity 2 on Γ; so both pass through it with multiplicity 1 and the tangent cones multiply to xy, so their tangents are the two different lines $x = 0, y = 0$, and the local intersection multiplicity at the origin is 1. Then $\deg G \deg H = 1$, a contradiction.

Suppose now $\Gamma = \varphi(X)$ with $X \subseteq \mathbf{P}^3$ a nonsingular curve and φ birational. Then X is the normalization of Γ (finite birational from a nonsingular curve), so by (F2) $g(X) = p_a(\Gamma) - \delta_O = 3 - 1 = 2$, and by (F3) $\deg X = \deg \varphi^* \mathcal{O}(1) = \deg \Gamma = 4$ (the divisor $\varphi^* \mathcal{O}(1) = \mathcal{O}_X(1)$ has degree $\deg X$). But a curve of degree 4 in $\mathbf{P}^3$ has $g \in \{0, 1, 3\}$ by Ex. 3.6(a) (if X lies in a plane, $g = 3$). So no such X exists. □

Exercise 3.8

> **Problem.** We say a (singular) integral curve in $\mathbf{P}^n$ is *strange* if there is a point which lies on all the tangent lines at nonsingular points of the curve.
>
> (a) There are many singular strange curves, e.g., the curve given parametrically by $x = t$, $y = t^p$, $z = t^{2p}$ over a field of characteristic $p > 0$.
>
> (b) Show, however, that if $\operatorname{char} k = 0$, there aren't even any singular strange curves besides $\mathbf{P}^1$.

Solution. (a) (The statement asks us to check that this curve is a singular strange curve.) With homogeneous coordinates $(w : x : y : z)$ and affine coordinates $x/w, y/w, z/w$, let C be the image of $\mathbf{P}^1 \to \mathbf{P}^3$, $(s : t) \mapsto (s^{2p} : s^{2p-1}t : s^p t^p : t^{2p})$ (the entries s^{2p}, t^{2p} have no common zero, so

this is a morphism). C is an integral closed curve, and $x/w = t/s$ is a coordinate on $\mathbf{P}^1$, so $k(C) = k(t)$, $\mathbf{P}^1 \to C$ is birational. The only point of C with $w = 0$ is $Q = (0:0:0:1)$ ($s = 0$), and the affine part is $\{(t, t^p, t^{2p})\} = \{y = x^p, z = x^{2p}\} \cong \mathbf{A}^1$, nonsingular.

C is singular at Q. In the chart $z \neq 0$ with parameter $s = 1/t$, the coordinates are $(w, x, y)/z = (s^{2p}, s^{2p-1}, s^p)$, so $\mathcal{O}_{C,Q}$ is the localization of $k[s^{2p}, s^{2p-1}, s^p]$ at the maximal ideal (s^{2p}, s^{2p-1}, s^p). Every element of $\mathcal{O}_{C,Q}$ has a power series expansion in s with no term s^j for $0 < j < p$ (true for the polynomials, and for inverses of units $c + h$, $h \in s^p k[[s]]$). As $p \geq 2$, $s \notin \mathcal{O}_{C,Q}$. But the normalization of C is $\mathbf{P}^1$, whose local ring at $s = 0$ is $k[s]_{(s)} \ni s$. So $\mathcal{O}_{C,Q}$ is not integrally closed, hence not regular: Q is a singular point.

Strangeness. At a nonsingular point $P = (a, a^p, a^{2p})$ the curve is cut out in $\mathbf{A}^3$ by $y - x^p$, $z - x^{2p}$, whose differentials are $dy - px^{p-1}dx = dy$ and dz (char p). So the tangent space is the x-direction, and the tangent line is $\{y = a^p, z = a^{2p}\}$, which contains the point at infinity $A = (0:1:0:0)$ of the x-axis. Thus A lies on all tangent lines at nonsingular points: C is a singular strange curve.

(b) Let $\operatorname{char} k = 0$. I show more generally:

Lemma D. Let $\nu : Y \to \mathbf{P}^n$ be a morphism from a curve, birational onto its image C. Assume that there is a point A and a cofinite set of $P \in Y$ such that the tangent line L_P to ν at P (the line through $\nu(P)$ in the direction $d\nu_P \neq 0$) passes through A. Then C is a line.

Applying it to the normalization of a strange integral curve X (at a nonsingular point of X, ν is a local isomorphism, so L_P is the tangent line of X) shows that a strange curve in characteristic 0 is a line; lines are nonsingular, so no singular curve is strange, and $\mathbf{P}^1$ itself is strange (every point is on its only tangent line).

Proof of Lemma D. Choose coordinates with $A = (0:\cdots:0:1)$ and write $\nu = (f_0 : \cdots : f_n)$. If $f_0 = \cdots = f_{n-1} = 0$, $C = A$ is a point; so choose $j < n$ with $f_j \neq 0$ and put $u_i = f_i/f_j \in K(Y)$, $i < n$. For all but finitely many P the u_i are regular at P, $\nu(P) \neq A$ and $f_j(P) \neq 0$, and the chart $x_j \neq 0$ has affine coordinates $v_i = x_i/x_j$, with $v_i \circ \nu = u_i$ for $i \neq j, n$ and $v_n \circ \nu = f_n/f_j$. The tangent line at $\nu(P)$ is $\nu(P) + k \cdot (\frac{du_i}{dt}(P))_{i \neq j}$ (t a uniformizer at P), and its point at infinity in the hyperplane $x_j = 0$ is the direction vector $(\frac{du_i}{dt}(P))$. A is the point at infinity in the direction e_n, so $A \in L_P$ means $\frac{du_i}{dt}(P) = 0$ for all $i < n$, $i \neq j$, i.e. $du_i \in \mathfrak{m}_P \Omega_{Y,P}$. Suppose some u_i is nonconstant. Since $\operatorname{char} k = 0$, $K(Y)$ is separably generated over k and $du_i \neq 0$ in $\Omega_{K(Y)/k}$ (II, 8.6A); thus du_i is a nonzero rational section of the invertible sheaf Ω_Y, which vanishes at only finitely many points. This contradicts the vanishing at infinitely many P. Hence all u_i ($i < n$) are constants c_i, and $\nu(P) = (c_0 f_j : \cdots : f_j : \cdots : c_{n-1} f_j : f_n)$ lies on the line through $(c_0 : \cdots : c_{n-1} : 0)$ and A. So C is contained in a line, hence is a line. □

Exercise 3.9

> **Problem.** Prove the following lemma of Bertini: if X is a curve of degree d in $\mathbf{P}^3$, not contained in any plane, then for almost all planes $H \subseteq \mathbf{P}^3$ (meaning a Zariski open subset of the dual projective space $(\mathbf{P}^3)^*$), the intersection $X \cap H$ consists of exactly d distinct points, no three of which are collinear.

Solution. Recall that X is a nonsingular curve (Chapter IV convention). I first prove two lemmas, valid in all characteristics; they are also used in Ex. 3.10.

Setting. Let $n \geq 3$, Y a curve and $\nu : Y \to \mathbf{P}^n$ a morphism, birational onto its image C, with C not contained in a hyperplane. There is a cofinite open set $Y_0 \subseteq Y$ on which ν is an immersion ($d\nu_P \neq 0$; this holds generically because ν is separable) and $\nu^{-1}(\nu(P)) = \{P\}$. For $P \in Y_0$ let $L_P \subseteq \mathbf{P}^n$ be the tangent line. The map $P \mapsto L_P$ is a morphism $Y_0 \to G(1,n)$ and extends to a morphism $\tau : Y \to G(1,n)$ (Y nonsingular, G projective; I, 6.8); write $L_P = \tau(P)$ for every P. Let $K = \{(P,Q) \in Y \times Y : L_P \cap L_Q \neq \emptyset\}$, a closed subset (incidence of lines is closed). Let $W \subseteq Y^3$ be the locally closed set of triples (P,Q,R) of points with $\nu(P), \nu(Q), \nu(R)$ pairwise distinct and collinear. If $\nu(P) \neq \nu(Q)$, the line $\nu(P)\nu(Q)$ is not contained in C (C is not a line), so the fibres of $p_{12} : W \to Y \times Y$ are finite and $\dim W \leq 2$.

Lemma B. If $\dim W = 2$, then $L_P \cap L_Q \neq \emptyset$ for all $P, Q \in Y$, i.e. $K = Y \times Y$.

Proof. Let W_1 be an irreducible component of W of dimension 2. I claim $p_{12}(W_1) \subseteq K$. Granting this: $p_{12}(W_1)$ is constructible of dimension 2 (finite fibres) in the irreducible surface $Y \times Y$, hence dense, and K is closed, so $K = Y \times Y$.

For the claim, let $V = W_1 \cap p_{12}^{-1}(Y \times Y \setminus K)$, open in W_1; suppose $V \neq \emptyset$, so $\dim V = 2$. Look at $p_3 : V \to Y$. I show that every fibre of p_3 is finite, giving $\dim V \leq 1$, a contradiction. Fix $R \in Y$ and let $\psi : Y \to \mathbf{P}^{n-1}$ be the projection from $\nu(R)$ (a morphism, extended at R); it is nonconstant because C is not a line. The fibre V_R consists of pairs (P,Q) with $P \neq Q$, $\psi(P) = \psi(Q)$ and $(P,Q) \notin K$. I show that all but finitely many pairs (P,Q) with $\psi(P) = \psi(Q)$ are in K. Write $\psi = \pi_{\nu(R)} \circ \nu$ and let ψ factor as $Y \xrightarrow{f} \tilde{Y} \to C_R = \psi(Y)$, $\tilde{Y}$ the normalization of C_R.

Case 1: $K(Y)/K(\tilde{Y})$ is inseparable. If $u \in K(\tilde{Y})$ had $du \neq 0$ in $\Omega_{K(Y)/k}$, then $K(Y)$ would be separable over $k(u)$ and hence over $K(\tilde{Y})$. So $du = 0$ for all $u \in K(\tilde{Y})$, in particular for the affine coordinates of ψ; so $d\psi_P = 0$ for all P. As $d\psi_P = d\pi \circ d\nu_P$ and $\ker d\pi$ is the direction towards $\nu(R)$, each L_P ($P \in Y_0$, $\nu(P) \neq \nu(R)$) passes through $\nu(R)$, and two such lines meet. The remaining pairs (P,Q) have P or Q in the finite set $(Y \setminus Y_0) \cup \{R\}$, and each point has a finite fibre under ψ: finitely many pairs.

Case 2: f separable. Then f is ramified at only finitely many points (2.2), and C_R has finitely many singular points. Let $E \subseteq C_R$ be the finite set consisting of the singular points of C_R, the images of the ramification points of f, of $Y \setminus Y_0$ and of R. For (P,Q) with $\psi(P) = \psi(Q) = T \notin E$: T is a nonsingular point of C_R (so $\tilde{Y} \to C_R$ is an isomorphism near T), f is unramified at P and Q, so ψ^* of a local parameter at T is a uniformizer at P and $d\psi_P : T_P Y \to T_T C_R$ is an isomorphism. Since $d\psi_P = d\pi \circ d\nu_P$, $d\nu_P \neq 0$, $L_P \not\ni \nu(R)$, and $\pi(L_P)$ is the tangent line ℓ_T of C_R at T. Hence L_P, and likewise L_Q, lies in the plane $\Pi_T = \pi^{-1}(\ell_T) \cup \{\nu(R)\}$. Two lines in a plane meet: $(P,Q) \in K$. The pairs with $\psi(P) = \psi(Q) \in E$ are finite in number (finite fibres). $\square$

Lemma C. A set $\mathcal{L}$ of lines in $\mathbf{P}^n$, any two of which meet, is either concurrent (all through one point) or contained in one plane.

Proof. Suppose the lines are not concurrent. Then $\mathcal{L}$ contains two distinct lines L_1, L_2, meeting in a point a, and a line $L_3 \not\ni a$. L_3 meets L_1 and L_2 in points $b \neq c$ (if $b = c$ then $b \in L_1 \cap L_2 = \{a\}$), both in the plane $\Pi = \langle L_1, L_2 \rangle$, so $L_3 \subseteq \Pi$, and $L_1 \cap L_2 \cap L_3 = \emptyset$. If $M \in \mathcal{L}$ were not contained in Π, then $M \cap \Pi$ would be a single point, lying on each of L_1, L_2, L_3 (as M meets each and each lies in Π), which is impossible. So $\mathcal{L} \subseteq \Pi$. $\square$

Step 1: $\dim W \leq 1$ for our nonplanar nonsingular $X \subseteq \mathbf{P}^3$. Apply the setting with $Y = X$, ν

the inclusion. If $\dim W = 2$, then by Lemma B the tangent lines $T_P X$ pairwise meet, and by Lemma C they are either all in a plane Π — then $X \subseteq \Pi$ since $P \in T_P X$, contradiction — or all pass through a point A, i.e. X is strange. By Theorem 3.9 a strange curve is a line, or a conic in characteristic 2, hence planar. Contradiction. So $\dim W \leq 1$.

Step 2: the dimension count. Let $\mathcal{B} = \{(H, P, Q, R) : H \in (\mathbf{P}^3)^*, (P, Q, R) \in W, P, Q, R \in H\}$. A plane contains P, Q, R iff it contains the line PQ, so $\mathcal{B} \to W$ is a $\mathbf{P}^1$-bundle and $\dim \mathcal{B} \leq \dim W + 1 \leq 2$. The image Σ of $\mathcal{B}$ in $(\mathbf{P}^3)^*$ is constructible of dimension ≤ 2, hence contained in a proper closed subset: for H in a dense open set, $X \cap H$ contains no three distinct collinear points.

Step 3: distinctness. By Bertini (II, 8.18) there is a dense open set of $H \in (\mathbf{P}^3)^*$ with $H \not\supseteq X$ and $H \cap X$ regular at every point, i.e. the zero-dimensional scheme $H \cap X$ is reduced. It has length $\deg(X \cdot H) = d$, so $X \cap H$ consists of exactly d distinct points.

The intersection of the two dense open sets of Steps 2 and 3 is dense open, and for H in it $X \cap H$ consists of d distinct points, no three collinear. □

Exercise 3.10

> **Problem.** Generalize the statement that "not every secant is a multisecant" as follows. If X is a curve in $\mathbf{P}^n$, not contained in any $\mathbf{P}^{n-1}$, and if $\operatorname{char} k = 0$, show that for almost all choices of $n-1$ points $P_1, \ldots, P_{n-1}$ on X, the linear space L^{n-2} spanned by the P_i does not contain any further points of X.

Solution. For $n = 2$ there is one point and nothing to prove, so let $n \geq 3$. "Almost all" means: for $(P_1, \ldots, P_{n-1})$ in a dense open subset of X^{n-1}. I will prove this for the larger class of data (Y, ν): Y a curve and $\nu : Y \to \mathbf{P}^n$ a morphism, birational onto its image and with image in no hyperplane (call this class $\mathcal{C}_n$); $X \subseteq \mathbf{P}^n$ is the case $\nu =$ inclusion. For $(Y, \nu) \in \mathcal{C}_n$ and $1 \leq k \leq n-1$ let $B_k(Y, \nu) \subseteq Y^k$ be the set of $(P_1, \ldots, P_k)$ such that either the points $\nu(P_1), \ldots, \nu(P_k)$ do not span a linear space of dimension $k-1$, or there is $R \in Y \setminus \{P_1, \ldots, P_k\}$ with $\nu(R)$ in their span. It is constructible.

Claim $P(n, k)$**:** $\dim B_k(Y, \nu) \leq k - 1$ for every $(Y, \nu) \in \mathcal{C}_n$ and $1 \leq k \leq n-1$.

For $k = n-1$ and ν the inclusion this gives the exercise: the complement of B_{n-1} is constructible of dimension $n-1$ in the irreducible variety X^{n-1}, hence contains a dense open set, and for (P_i) in it, the P_i span a $\mathbf{P}^{n-2} = L^{n-2}$ with $L \cap X = \{P_1, \ldots, P_{n-1}\}$.

$k = 1$. $B_1 = \{P : \nu^{-1}\nu(P) \neq \{P\}\}$ is finite since ν is birational.

$k = 2$, $n \geq 3$. B_2 consists of pairs with $\nu(P_1) = \nu(P_2)$ (the diagonal plus finitely many pairs: dimension ≤ 1), and pairs (P_1, P_2) such that $\nu(R) \in$ line $\nu(P_1)\nu(P_2)$ for some $R \notin \{P_1, P_2\}$. If moreover $\nu(R) \in \{\nu(P_1), \nu(P_2)\}$, then P_1 or P_2 lies in the finite set of points over which ν is not injective, giving dimension ≤ 1. Otherwise $(P_1, P_2, R) \in W$ (notation of Ex. 3.9), and by finiteness of the fibres of p_{12} this part has dimension $\leq \dim W$. I claim $\dim W \leq 1$: if $\dim W = 2$, Lemmas B and C show that the tangent lines L_P are all in a plane (then C lies in a plane, contradicting non-degeneracy, $n \geq 3$) or pass through a common point A; in the latter case Lemma D (char 0, Ex. 3.8(b)) says C is a line, again contradicting non-degeneracy. So $\dim B_2 \leq 1$.

Induction step $P(n-1,k) \Rightarrow P(n,k+1)$*, for* $2 \le k$*,* $k+1 \le n-1$. Let $(Y,\nu) \in \mathcal{C}_n$ and suppose $\dim B_{k+1} = k+1$. Then B_{k+1} contains a nonempty open $V \subseteq Y^{k+1}$. For $O \in Y$ let $V_O = \{(P_1,\dots,P_k) : (P_1,\dots,P_k,O) \in V\}$, open in Y^k; it is nonempty for O in a dense open subset of Y. Since $\dim B_2 \le 1$, for all but finitely many O the set $\{P : (P,O) \in B_2\}$ is finite; also $\nu^{-1}\nu(O) = \{O\}$ for all but finitely many O. Fix O satisfying all three conditions. Let $\psi = \psi_O : Y \to \mathbf{P}^{n-1}$ be the projection from $\nu(O)$. For $P, R \ne O$ with $(P,O) \notin B_2$, $\psi(R) = \psi(P)$ forces $R = P$; so ψ is injective on a cofinite set, and as $\operatorname{char} k = 0$ it is birational onto its image. The image is in no hyperplane of $\mathbf{P}^{n-1}$ (else C would lie in a hyperplane through $\nu(O)$). So $(Y,\psi) \in \mathcal{C}_{n-1}$ and, as $k \le (n-1)-1$, $P(n-1,k)$ gives $\dim B_k(Y,\psi) \le k-1$: its complement contains a nonempty open set $G \subseteq Y^k$. Choose $(P_1,\dots,P_k) \in V_O \cap G$ with all $P_i \ne O$. I show $(P_1,\dots,P_k,O) \notin B_{k+1}$, contradicting $V \subseteq B_{k+1}$.

Independence. Note $\nu(P_i) \ne \nu(O)$. If $c_0\nu(O) + \sum c_i\nu(P_i) = 0$ for vector representatives, applying the linear projection with kernel $\langle \nu(O) \rangle$ gives $\sum c_i \psi(P_i) = 0$, so $c_i = 0$ by goodness of (P_i) for ψ, then $c_0 = 0$.

Span. Let $\Lambda = \operatorname{span}(\nu(P_1),\dots,\nu(P_k),\nu(O))$ and let $R \in Y \setminus \{P_1,\dots,P_k,O\}$ with $\nu(R) \in \Lambda$. Then $\nu(R) \ne \nu(O)$ and $\psi(R) \in \operatorname{span}(\psi(P_1),\dots,\psi(P_k))$, with $R \notin \{P_i\}$, contradicting $(P_i) \in G$.

This proves $P(n,k+1)$. Since $P(m,2)$ holds for all $m \ge 3$ and $P(m,1)$ for all m, induction on k gives $P(n,k)$ for all $1 \le k \le n-1$. □

Exercise 3.11

Problem.

(a) If X is a nonsingular variety of dimension r in $\mathbf{P}^n$, and if $n > 2r+1$, show that there is a point $O \notin X$, such that the projection from O induces a closed immersion of X into $\mathbf{P}^{n-1}$.

(b) If X is the Veronese surface in $\mathbf{P}^5$, which is the 2-uple embedding of $\mathbf{P}^2$ (I, Ex. 2.13), show that each point of every secant line of X lies on infinitely many secant lines. Therefore, the secant variety of X has dimension 4, and so in this case there is a projection which gives a closed immersion of X into $\mathbf{P}^4$ (II, Ex. 7.7). (A theorem of Severi [1] states that the Veronese surface is the only surface in $\mathbf{P}^5$ for which there is a projection giving a closed immersion into $\mathbf{P}^4$. Usually one obtains a finite number of double points with transversal tangent planes.)

Solution. (a) *Criterion.* For $O \notin X$ let $\pi : X \to \mathbf{P}^{n-1}$ be the projection; it is the morphism defined by the base-point-free linear system of hyperplanes through O. As in the proof of (3.4), by (II, 7.3) and (II, 7.8.2), π is a closed immersion iff this system separates points and tangent vectors. It separates points $P \ne Q$ iff $\pi(P) \ne \pi(Q)$ iff O is not on the line PQ. It separates tangent vectors at P iff $d\pi_P$ is injective on T_PX; since $\ker d\pi_P$ is the direction of the line OP, this holds iff the line OP is not contained in the embedded tangent space $\mathbf{T}_P(X) \subseteq \mathbf{P}^n$ (r-dimensional linear space), i.e. iff $O \notin \mathbf{T}_P(X)$. So π is a closed immersion iff O lies on no secant line PQ ($P \ne Q$) and on no $\mathbf{T}_P(X)$.

Dimension count. The set $I = \{(P,Q,x) : P \ne Q,\ x \in PQ\} \subseteq (X \times X \setminus \Delta) \times \mathbf{P}^n$ is a $\mathbf{P}^1$-bundle over the irreducible variety $X \times X \setminus \Delta$ of dimension $2r$, so $\dim I = 2r+1$, and its image $S_1 \subseteq \mathbf{P}^n$ (the points on secant lines) is constructible of dimension $\le 2r+1$. The set

$T = \{(P, x) : P \in X, x \in \mathbf{T}_P(X)\} \subseteq X \times \mathbf{P}^n$ is closed with all fibres over X of dimension r (X nonsingular), so $\dim T \leq 2r$ and its image S_2 (the union of the embedded tangent spaces) has dimension $\leq 2r$. Since $n > 2r + 1$, $S_1 \cup S_2 \neq \mathbf{P}^n$. Take $O \notin S_1 \cup S_2$; then $O \notin X$ (as $X \subseteq S_2$), and by the criterion the projection from O is a closed immersion into $\mathbf{P}^{n-1}$.

(b) $X = v(\mathbf{P}^2) \subseteq \mathbf{P}^5$, with $v(a) = (a_i a_j)_{0 \leq i \leq j \leq 2}$ for $a \in k^3 \setminus 0$ (coordinates z_{ij}, $z_{ji} := z_{ij}$). Valid in all characteristics.

A secant meets X only in its two defining points. Let $p, q \in k^3$ be independent and suppose $\alpha v(p) + \beta v(q) = \lambda v(r)$ with $\alpha, \beta \neq 0$, $\lambda \neq 0$ (vectors in k^6). For $\xi, \eta \in (k^3)^*$ the form $z(\xi, \eta) = \sum \xi_i \eta_j z_{ij}$ is linear in z, and $z(\xi, \eta)(v(a)) = \xi(a)\eta(a)$. Taking $\xi = \eta$ vanishing on $\langle p, q \rangle$ gives $\lambda \xi(r)^2 = 0$, so $r \in \langle p, q \rangle$, $r = xp + yq$. Choose ξ, η with $(\xi(p), \xi(q)) = (1, 0)$ and $(\eta(p), \eta(q)) = (0, 1)$: $\lambda xy = 0$. So $r \propto p$ or q, which forces $\beta = 0$ or $\alpha = 0$. Hence the line through $v(p), v(q)$ meets X in exactly these two points, and different secant lines through Z correspond to different pairs of points.

Points of X. Each $v(p) \in X$ lies on infinitely many secants (join it to any other point of X).

Other points. Let $Z = \alpha v(p) + \beta v(q)$ with $\alpha, \beta \neq 0$ be a point of a secant, not in X. As k is algebraically closed choose $a = \sqrt{\alpha} p$, $b = \sqrt{\beta} q$ (independent vectors), so $Z = v(a) + v(b)$ as vectors. For any $c, d \in k$ with $c^2 + d^2 = 1$ put $a' = ca + db$, $b' = -da + cb$. A direct computation of the (i, j) entry gives $a'_i a'_j + b'_i b'_j = (c^2 + d^2)(a_i a_j + b_i b_j) = a_i a_j + b_i b_j$ (the cross terms $\pm cd(a_i b_j + b_i a_j)$ cancel), so $v(a') + v(b') = Z$. Also $\det \begin{pmatrix} c & d \\ -d & c \end{pmatrix} = 1$, so a', b' are independent and Z lies on the secant through $v(a'), v(b')$. The conic $c^2 + d^2 = 1$ has infinitely many points (k algebraically closed), and the map $(c, d) \mapsto (c : d)$ is at most 2-to-1 on it, so the points $[a'] = [ca + db]$ of the line $\mathbf{P}(\langle a, b \rangle)$ take infinitely many values. By the first part these give infinitely many distinct secant lines through Z.

Dimension of the secant variety. Every point $Z = sv(a) + tv(b)$ of a secant line, written as the symmetric matrix $(sa_i a_j + tb_i b_j)$, has rank ≤ 2, so $\det(z_{ij}) = 0$ on S_1 (the union of the secant lines): the secant variety $\overline{S_1}$ lies in the cubic hypersurface $\det(z_{ij}) = 0$, so $\dim \overline{S_1} \leq 4$. This is also what the first part gives: $I' = \{(P, Q, x) \in I : x \notin X\}$ is irreducible of dimension 5, and every nonempty fibre of $I' \to \mathbf{P}^5$ is infinite, hence of dimension ≥ 1, so the image has dimension ≤ 4. In fact the fibres are exactly 1-dimensional, so $\dim \overline{S_1} = 4$: if $Z = v(p) + v(q) = v(a) + v(b)$ with independent p, q and a, b, then the bilinear forms $B(\xi, \eta) = \xi(p)\eta(p) + \xi(q)\eta(q) = \xi(a)\eta(a) + \xi(b)\eta(b)$ on $(k^3)^*$ have radical $\langle p, q \rangle^\perp = \langle a, b \rangle^\perp$, so $\langle p, q \rangle = \langle a, b \rangle$, and comparing coefficients, $(p, q) = (a, b)A$ with A invertible and $AA^t = 1$, a 1-dimensional family (a conic) of possibilities.

Projection into $\mathbf{P}^4$. By the criterion in (a), the projection from O is a closed immersion into $\mathbf{P}^4$ whenever $O \notin S_1 \cup S_2$. Here $\dim \overline{S_1} = 4$ and $\dim S_2 \leq 2r = 4$ (from (a)), so $S_1 \cup S_2$ lies in a proper closed subset of $\mathbf{P}^5$ and such O exists. (An explicit projection is given in II, Ex. 7.7(b).) □

Exercise 3.12

> **Problem.** For each value of $d = 2, 3, 4, 5$ and r satisfying $0 \leq r \leq \frac{1}{2}(d-1)(d-2)$, show that there exists an irreducible plane curve of degree d with r nodes and no other singularities.

Solution. A *node* is a double point with two distinct tangent directions. The cases ($\delta(d)$ is the

maximal r): $d=2$, $r=0$; $d=3$, $r=0,1$; $d=4$, $r=0,\dots,3$; $d=5$, $r=0,\dots,6$. Three constructions cover everything.

$r=0$. A nonsingular plane curve of degree d exists for every d (Bertini, II, 8.20.2); it is irreducible. For $d=2$ this finishes the proof.

Lemma 1 (projection of curves of genus g). Let X be a curve of genus g, and $d \geq 2g+1$ with $g < \delta(d)$. Then there is an irreducible plane curve of degree d with exactly $\delta(d)-g$ nodes and no other singularities.

Proof. By (3.2)(b) a divisor D of degree d is very ample, giving $X \hookrightarrow \mathbf{P}^N$; projecting from points off X repeatedly (3.5, as in Cor. 3.6) we get a closed immersion $X \subseteq \mathbf{P}^3$ with $\mathcal{O}_X(1)$ of degree d (each projection pulls $\mathcal{O}(1)$ back to $\mathcal{O}_X(1)$). X is in no plane: a nonsingular plane curve of degree d has genus $\delta(d) \neq g$. By (3.10) there is $O \notin X$ such that the projection $\varphi : X \to \Gamma \subseteq \mathbf{P}^2$ is birational onto Γ and Γ has at most nodes. Γ is irreducible, $\deg \Gamma = d$ by (F3), so $p_a(\Gamma) = \delta(d)$, and by (F2), $\delta(d) = g + \#\{\text{nodes}\}$ since each node has $\delta_P = 1$. □

Thus: $g=0$ (take $X=\mathbf{P}^1$, any d) gives $r=\delta(d)$, i.e. $(d,r)=(3,1),(4,3),(5,6)$; $g=1$ (a smooth plane cubic) gives $(4,2)$ and $(5,5)$; $g=2$ ($d=5$) gives $(5,4)$, once a curve of genus 2 is available, see below.

Lemma 2 ($r=1,2$ and $d \geq 2r+1$). For $r \in \{1,2\}$ and $d \geq 2r+1$ there is an irreducible plane curve of degree d with exactly r nodes and no other singularities. Moreover its normalization has genus $\delta(d)-r$.

Proof. Let $p_1=(0:0:1)$, $p_2=(1:0:0)$ and use $(x_0:x_1:x_2)$. Let $S \subseteq \mathbf{P}^2 \times (\mathbf{P}^1)^r$ be the closed subscheme defined by $x_0t_1 = x_1s_1$ (and for $r=2$ also $x_1t_2 = x_2s_2$), $(s_i:t_i)$ coordinates on the i-th $\mathbf{P}^1$ (the pencil of lines through p_i). Let $\pi : S \to \mathbf{P}^2$ and $f_i : S \to \mathbf{P}^1$ be the projections.

Structure of S. Over $U = \mathbf{P}^2 \setminus \{p_1,p_2\}$, $(s_1:t_1) = (x_0:x_1)$ and $(s_2:t_2) = (x_1:x_2)$ are determined, so $\pi^{-1}(U) \cong U$. Over p_1 the equations leave $(s_1:t_1)$ free and force $(s_2:t_2)=(0:1)$, so $E_1 = \pi^{-1}(p_1) \cong \mathbf{P}^1$, and $f_1|_{E_1}$ is an isomorphism; similarly $E_2 = \pi^{-1}(p_2) \cong \mathbf{P}^1$ with $f_2|_{E_2}$ an isomorphism. Near E_1 ($x_2=1$, $u=x_0$, $v=x_1$) the second equation reads $s_2 = vt_2$ and just expresses s_2 as a function of v (graph), while the first is $ut_1 = vs_1$: the standard blow-up of the origin. So S is a nonsingular surface (similarly near E_2), irreducible because every component has dimension ≥ 2 and so meets $\pi^{-1}(U)$. S is projective: $S = \mathrm{Bl}_{\{p_1,\dots,p_r\}} \mathbf{P}^2$. (For $r=1$ only p_1, E_1 are used; π is an isomorphism near p_2, and everything below about p_2, E_2 is void.)

Embedding. Put $a_0 = d-2r \geq 1$. The composite of the a_0-uple embedding of $\mathbf{P}^2$, the 2-uple embeddings of the $\mathbf{P}^1$'s and the Segre embedding is a closed immersion $\mathbf{P}^2 \times (\mathbf{P}^1)^r \hookrightarrow \mathbf{P}^N$ pulling $\mathcal{O}(1)$ back to $\mathcal{O}(a_0,2,\dots,2)$ (II, Ex. 5.13, 5.11). So $S \subseteq \mathbf{P}^N$ is a nonsingular projective surface, and each E_i is embedded by $\mathcal{O}_{\mathbf{P}^1}(2)$: a conic.

Bertini. By (II, 8.18) applied to S and to each conic E_i, there is a hyperplane $H' \subseteq \mathbf{P}^N$ such that $D = H' \cap S$ is regular at every point and $H' \cap E_i$ is regular, $i=1,\dots,r$ (finite intersection of dense open sets of hyperplanes). D is a nonsingular, connected (indeed irreducible) curve (III, 7.9.1). $H' \cap E_i$ is a reduced scheme of length $\deg E_i = 2$: two distinct points $a_i \neq b_i$, and $D \cap E_i = H' \cap E_i$, so D meets E_i transversally in a_i, b_i.

The plane curve. $D \neq E_i$, so $D \not\subseteq \pi^{-1}\{p_1,p_2\}$; $C = \pi(D)$ is an integral plane curve, $\pi|_D : D \to C$ is finite and birational (π is an isomorphism over U), so D is the normalization of C. *Degree:* take a line $\ell \subseteq \mathbf{P}^2$ missing p_1, p_2 and $\tilde{\ell} = \pi^{-1}(\ell) \cong \ell$. The section π^*(equation of ℓ) of $\pi^*\mathcal{O}(1)$

restricted to D vanishes on $D\cap\tilde\ell$ (and $D\not\supseteq\tilde\ell$ as D meets E_1 and $\tilde\ell$ does not), so $\deg(\pi^*\mathcal{O}(1)|_D) = \operatorname{length}(D\cap\tilde\ell) = \operatorname{length}(H'\cap\tilde\ell) = \deg(\mathcal{O}(1)|_{\tilde\ell})$ for the embedding. On $\tilde\ell\cong\ell$ the maps π and f_i are isomorphisms onto ℓ, $\mathbf{P}^1$ (projection of ℓ from $p_i\notin\ell$), so $\mathcal{O}(a_0,2,\dots,2)|_{\tilde\ell} = \mathcal{O}_{\mathbf{P}^1}(a_0+2r) = \mathcal{O}(d)$. By (F3), $\deg C = d$.

Singularities. $C\setminus\{p_1,p_2\}\cong D\setminus(E_1\cup E_2)$ is nonsingular. Over p_i the normalization $D\to C$ has the two preimages $a_i\neq b_i$; at these points $\pi|_D$ is an immersion: locally π is the blow-up map $(u,t)\mapsto(u,ut)$, whose differential on $\{u=0\}$ has kernel exactly the tangent line to E_i, and D is transversal to E_i. So C has at p_i two smooth branches. The tangent line of the branch through a_i is the line through p_i of slope given by the limit of the pencil parameter: $f_i(a_i)$; similarly $f_i(b_i)$; these are distinct because $f_i|_{E_i}$ is injective. So p_i is a node, and C has exactly r nodes. The genus of D is $p_a(C)-r = \delta(d)-r$ by (F2). □

In particular $(3,1),(4,1),(5,1),(5,2)$ are done, and for $(d,r)=(4,1)$ the curve D is a curve of genus 2.

Genus 2. Taking $X=D$ in Lemma 1 with $d=5=2g+1$: $(5,4)$ is done.

Lemma 3 ($(d,r)=(5,3)$). Let $\Gamma\subseteq\mathbf{P}^2$ be a nonsingular quartic (genus 3). Not being a line, or a conic in characteristic 2, Γ is not strange (Theorem 3.9); so for every point P of $\mathbf{P}^2$ the set $\{A\in\Gamma : P\in T_A\Gamma\}$ is finite. Choose successively $p_1,p_2,p_3\in\Gamma$ such that (i) they are not collinear; (ii) $T_{p_i}\Gamma\not\ni p_j$ for $j\neq i$; (iii) for each pair $i\neq j$, the line $L_{ij}=p_ip_j$ is not tangent to Γ at any point other than p_i,p_j. Each condition excludes only finitely many points at the stage where a point is chosen: for (ii), the points of $\Gamma\cap T_{p_i}\Gamma$ and the finite set $\{A : p_i\in T_A\Gamma\}$; for (iii), if $L_{ij}=T_A\Gamma$ for some $A\neq p_i,p_j$ then $A\in\{A : p_i\in T_A\Gamma\}$ (finite) and $p_j\in T_A\Gamma\cap\Gamma$ (finite). By (ii), L_{ij} is transversal to Γ at p_i and at p_j, so $L_{ij}\cdot\Gamma = p_i+p_j+a_{ij}+b_{ij}$ with $a_{ij}\neq b_{ij}$, both different from p_i,p_j (and from p_k, by (i)).

Choose coordinates $p_1=(1:0:0), p_2=(0:1:0), p_3=(0:0:1)$ and let $T(x)=(x_2x_3 : x_1x_3 : x_1x_2)$. On $U=\{x_1x_2x_3\neq0\}$, T is invertible with $T\circ T=\mathrm{id}$ ($T(T(x))=x_1x_2x_3\cdot x$). $\Gamma\not\subseteq$ the coordinate triangle, so $T|_{\Gamma\cap U}$ extends to a morphism $\varphi:\Gamma\to\mathbf{P}^2$, birational onto the integral curve $\Gamma'=\overline{T(\Gamma\cap U)}$ (the rational map T restricted to Γ' is the inverse).

Degree. φ is given by the sections $s_1=x_2x_3, s_2=x_1x_3, s_3=x_1x_2$ of $\mathcal{O}_\Gamma(2)$, whose only common zeros are the points p_1,p_2,p_3. At p_1, with a local parameter, $\operatorname{ord}(x_2/x_1)=\operatorname{ord}(x_3/x_1)=1$ because $T_{p_1}\Gamma$ is neither $\{x_2=0\}=L_{13}$ nor $\{x_3=0\}=L_{12}$; so s_1,s_2,s_3 have orders $2,1,1$ at p_1, and after removing the common factor the divisor is lowered by exactly p_1. The same holds at p_2,p_3. Hence $\varphi^*\mathcal{O}(1)\cong\mathcal{O}_\Gamma(2)(-p_1-p_2-p_3)$ has degree $8-3=5$ and $\deg\Gamma'=5$ by (F3). Since Γ is the normalization of Γ', (F2) gives $\sum\delta_P = p_a(\Gamma')-3 = 6-3 = 3$.

Three nodes. The points of Γ with $x_3=0$ are $p_1,p_2,a=a_{12},b=b_{12}$. At a, $s_3=x_1x_2\neq0$ and $s_1=s_2=0$, so $\varphi(a)=\varphi(b)=q_3:=(0:0:1)$; and $\varphi(p_1),\varphi(p_2)\neq q_3$ (in the chart $x_1=1$, with $x_2=\alpha t+\cdots$, $x_3=\beta t+\cdots$, $\alpha\beta\neq0$, the sections are $(\alpha\beta t^2 : \beta t : \alpha t)$, so $\varphi(p_1)=(0:\beta:\alpha)$; likewise $\varphi(p_2)$ has first and third coordinates nonzero and second coordinate 0, and $\varphi(p_3)$ has third coordinate 0, by the same transversality, so neither equals q_3). Since $s_1=s_2=0\neq s_3$ forces $x_3=0$, $x_1x_2\neq0$, no other point maps to q_3. So q_3 has exactly the two preimages a,b. Near q_3 the affine coordinates are $X=s_1/s_3=x_3/x_1$, $Y=s_2/s_3=x_3/x_2$. As $L_{12}=\{x_3=0\}$ meets Γ transversally at a and b, x_3 has a simple zero there, so φ is an immersion at a and b and the two branches at q_3 are smooth, with tangent slopes $Y/X=x_1/x_2$ evaluated at a and at b. These differ, because $a\neq b$ are points of the line $x_3=0$ and are determined by $(x_1:x_2)$.

So q_3 is a node. By symmetry q_1, q_2 are nodes as well. Each node contributes $\delta = 1$, so these three nodes account for $\sum \delta_P = 3$: there are no other singular points. Thus Γ' is an irreducible plane quintic with exactly 3 nodes. □

Summary.

- $d = 2$: $r = 0$ (smooth).
- $d = 3$: $r = 0$ (smooth), $r = 1$ (Lemma 2, or Lemma 1 with $g = 0$).
- $d = 4$: $r = 0$ (smooth), $r = 1$ (Lemma 2), $r = 2$ (Lemma 1, $g = 1$), $r = 3$ (Lemma 1, $g = 0$).
- $d = 5$: $r = 0$ (smooth), $r = 1, 2$ (Lemma 2), $r = 3$ (Lemma 3), $r = 4$ (Lemma 1, $g = 2$), $r = 5$ (Lemma 1, $g = 1$), $r = 6$ (Lemma 1, $g = 0$).

All hypotheses of the lemmas are met: Lemma 1 needs $d \geq 2g+1$ and $g < \delta(d)$, Lemma 2 needs $d \geq 2r + 1$ ($3 \geq 3$, $4 \geq 3$, $5 \geq 3$, $5 \geq 5$). The characteristic is arbitrary. □

4. Elliptic curves

Conventions. Throughout, k is algebraically closed and curves are complete nonsingular curves over k. As in the text, characteristic 2 is excluded wherever the cited results need it (Prop. 4.6, Cor. 4.7, Prop. 4.10 all assume $\operatorname{char} k \neq 2$); Ex. 4.6 assumes $\operatorname{char} k = 0$. For an elliptic curve (X, P_0) the group law is the one transported from $\operatorname{Pic}^0 X$ by $P \mapsto \mathcal{L}(P - P_0)$ (the discussion before Prop. 4.8); we write τ_a for translation by a, $\rho = -1_X$ for the inverse, and n_X for multiplication by n. A divisor $D = \sum n_i P_i$ of degree 0 corresponds to the point $\sum n_i P_i$ computed in the group law, and a divisor D of degree e to the point $\sum n_i P_i$ of the class $D - eP_0$. We write $j(\lambda) = 2^8(\lambda^2 - \lambda + 1)^3/(\lambda^2(\lambda - 1)^2)$ for the function of Thm. 4.1; it is constant on the Σ_3-orbits of Lemma 4.5.

Exercise 4.1

> **Problem.** Let X be an elliptic curve over k, with $\operatorname{char} k \neq 2$, let $P \in X$ be a point, and let R be the graded ring $R = \bigoplus_{n \geq 0} H^0(X, \mathcal{O}_X(nP))$. Show that for suitable choice of t, x, y,
>
> $$R \cong k[t, x, y]/(y^2 - x(x - t^2)(x - \lambda t^2))$$
>
> as a graded ring, where $k[t, x, y]$ is graded by setting $\deg t = 1$, $\deg x = 2$, $\deg y = 3$.

Solution. Write $R_n = H^0(X, \mathcal{O}_X(nP)) \subseteq K(X)$; multiplication $R_n \times R_m \to R_{n+m}$ is the product of functions. Let $t = 1 \in R_1$ (multiplication by t is the inclusion $R_n \subseteq R_{n+1}$). By the construction in the proof of Prop. 4.6 (with $P_0 = P$) there are $x \in R_2$ and $y \in R_3$, with $1, x$ a basis of R_2 and $1, x, y$ a basis of R_3, such that in $K(X)$

$$y^2 = x(x - 1)(x - \lambda), \qquad \lambda \neq 0, 1.$$

(In that proof x, y are changed only by $y \mapsto y + \frac{1}{2}(a_1x + a_3)$, $x \mapsto (x-a)/(b-a)$, which stay in R_3, R_2 and keep the pole orders.) Since $x \in R_2 \setminus R_1$ and $y \in R_3 \setminus R_2$, and x, y are regular off P, x has a pole of order exactly 2 at P and y of order exactly 3. Homogenizing, we get in R_6 the relation $y^2 = x(x-t^2)(x-\lambda t^2)$ (here t^2x is just the function x viewed in degree 4, etc.). So there is a graded homomorphism

$$\varphi : S := k[t,x,y]/(F) \to R, \qquad F = y^2 - x(x-t^2)(x-\lambda t^2).$$

φ_n is surjective. For $n \geq 1$, $\dim R_n = n$ by Riemann–Roch (as $\deg nP = n > 0 = 2g-2$). For $c \in \{0,1\}$ and $b \geq 0$ with $2b+3c \leq n$ the function x^by^c lies in R_n and has pole order exactly $2b+3c$ at P. Every integer $m \geq 2$ is uniquely of the form $2b+3c$ with $c \in \{0,1\}$ (take $c \equiv m \bmod 2$), while $1 = 2b+3c$ is impossible; so the pole orders occurring are exactly $0, 2, 3, \ldots, n$, all distinct. Functions with distinct pole orders at P are linearly independent, so these are n (for $n \geq 2$; for $n = 1$ just $\{1\}$) linearly independent elements of R_n, hence a basis. Thus R_n is spanned by the images of the monomials $t^{n-2b-3c}x^by^c$, $c \leq 1$, and φ_n is onto (for $n = 0$ it is trivial).

φ_n is injective. In S we may use F to replace y^2 by $x(x-t^2)(x-\lambda t^2)$, which is homogeneous of the same degree 6; so S_n is spanned by the monomials $t^ax^by^c$ with $c \leq 1$ and $a+2b+3c = n$. By the count just made there are n such monomials if $n \geq 1$ (and one if $n = 0$). Hence $\dim S_n \leq n = \dim R_n$, and since φ_n is onto it is an isomorphism. Thus φ is an isomorphism of graded rings, i.e. $\ker(k[t,x,y] \to R) = (F)$. □

Exercise 4.2

> **Problem.** If D is any divisor of degree ≥ 3 on the elliptic curve X, and if we embed X in $\mathbf{P}^n$ by the complete linear system $|D|$, show that the image of X in $\mathbf{P}^n$ is projectively normal. *Note.* It is true more generally that if D is a divisor of degree $\geq 2g+1$ on a curve of genus g, then the embedding of X by $|D|$ is projectively normal (Mumford [4, p. 55]).

Solution. Let $d = \deg D \geq 3$, so $n = d-1$ ($h^0(D) = d$ by Riemann–Roch) and D is very ample (Cor. 3.2); $\mathcal{O}_X(1) = \mathcal{L}(D)$. Let $S = k[x_0, \ldots, x_n]/I(X)$ be the homogeneous coordinate ring and S_m its degree m part. Since X is normal, by II, Ex. 5.14 ($\Gamma_*(\mathcal{O}_X) = \bigoplus_m H^0(X, \mathcal{O}_X(m))$ is the integral closure of S) X is projectively normal if and only if the maps $S_m \to H^0(X, \mathcal{L}(mD))$ are surjective for all $m \geq 0$. Since S_m is the image of $\operatorname{Sym}^m H^0(\mathcal{L}(D))$, it suffices to prove:

$$(*) \qquad H^0(\mathcal{L}(D)) \otimes H^0(\mathcal{L}(mD)) \to H^0(\mathcal{L}((m+1)D)) \text{ is surjective for all } m \geq 1;$$

then by induction every element of $H^0(\mathcal{L}(mD))$ is a polynomial in $H^0(\mathcal{L}(D))$ (the cases $m = 0, 1$ being trivial: $m = 1$ since $|D|$ is complete).

Recall that on X (genus 1, $K = 0$) a divisor E of positive degree has $H^1(\mathcal{L}(E)) \cong H^0(\mathcal{L}(-E))^\vee = 0$, $h^0(E) = \deg E$.

Case $m \geq 2$. Since $|D|$ is base point free and k is infinite we can choose $s_1, s_2 \in H^0(\mathcal{L}(D))$ without common zero (take $s_1 \neq 0$, then s_2 outside the finitely many hyperplanes $\{s : s(p) = 0\}$, $p \in \operatorname{Supp} \operatorname{div} s_1$). The Koszul sequence

$$0 \to \mathcal{L}(-D) \xrightarrow{(s_2, -s_1)} \mathcal{O}_X^2 \xrightarrow{(s_1, s_2)} \mathcal{L}(D) \to 0$$

is exact; tensoring with $\mathcal{L}(mD)$ and taking cohomology we get that $H^0(\mathcal{L}(mD))^2 \to H^0(\mathcal{L}((m+1)D))$ is onto as soon as $H^1(\mathcal{L}((m-1)D)) = 0$, true because $m \geq 2$. A fortiori $(*)$ holds.

Case $m = 1$. By Bertini (II, 8.18) there is a hyperplane $H \not\supseteq X$ meeting X in a reduced divisor $E = p_1 + \cdots + p_d$ (p_i distinct), $E \in |D|$, cut out by a section $s \in H^0(\mathcal{L}(D))$. Let $W \subseteq H^0(\mathcal{L}(2D))$ be the image of $H^0(\mathcal{L}(D)) \otimes H^0(\mathcal{L}(D))$. From $0 \to \mathcal{L}(D) \xrightarrow{s} \mathcal{L}(2D) \to \mathcal{L}(2D)|_E \to 0$ and $H^1(\mathcal{L}(D)) = 0$ we get that $H^0(\mathcal{L}(2D)) \to \mathcal{L}(2D)|_E$ is onto with kernel $sH^0(\mathcal{L}(D)) \subseteq W$. So it suffices to show that W maps onto $\mathcal{L}(2D)|_E$. Fix a generator ε_i of the fibre of $\mathcal{L}(D)$ at p_i; this identifies $\mathcal{L}(D)|_E \cong k^d$ and $\mathcal{L}(2D)|_E \cong k^d$ (via ε_i^2) so that the multiplication becomes the coordinatewise product. Let $U \subseteq k^d$ be the image of $H^0(\mathcal{L}(D)) \to \mathcal{L}(D)|_E$; the kernel is ks (sections of $\mathcal{L}(D-E) \cong \mathcal{O}_X$), so $\dim U = d-1$: $U = \{a : \sum c_i a_i = 0\}$ is a hyperplane. The image of W is the span of $U \cdot U$.

Claim: all $c_i \neq 0$, i.e. $e_i \notin U$. If $e_i \in U$ there is $u \in H^0(\mathcal{L}(D))$ vanishing at p_j for $j \neq i$ but not at p_i. Then $\operatorname{div} u = \sum_{j \neq i} p_j + q$ for a point q, and $\operatorname{div} u \sim E$ gives $q \sim p_i$, so $q = p_i$ (distinct points of a curve of genus 1 are not linearly equivalent, as $X \not\cong \mathbf{P}^1$), so $\operatorname{div} u = E$ and $u(p_i) = 0$, a contradiction.

Now, as $d \geq 3$, for each i pick $j \neq k$ both different from i. Then $c_j e_i - c_i e_j \in U$ and $c_k e_i - c_i e_k \in U$, and their product is $c_j c_k e_i$. Hence all $e_i \in \operatorname{span}(U \cdot U)$, so W maps onto k^d and $W = H^0(\mathcal{L}(2D))$.

This proves $(*)$, hence projective normality. □

Exercise 4.3

> **Problem.** Let the elliptic curve X be embedded in $\mathbf{P}^2$ so as to have the equation $y^2 = x(x-1)(x-\lambda)$. Show that any automorphism of X leaving $P_0 = (0,1,0)$ fixed is induced by an automorphism of $\mathbf{P}^2$ coming from the affine (x,y)-plane given by
>
> $$\begin{cases} x' = ax + b \\ y' = cy. \end{cases}$$
>
> In each of the four cases of (4.7), describe these automorphisms of $\mathbf{P}^2$ explicitly, and hence determine the structure of the group $G = \operatorname{Aut}(X, P_0)$.

Solution. Throughout $\operatorname{char} k \neq 2$. Let $x, y \in K(X)$ be the affine coordinates; $\{1, x\}$ is a basis of $H^0(\mathcal{O}(2P_0))$, $\{1, x, y\}$ of $H^0(\mathcal{O}(3P_0))$, $X \subset \mathbf{P}^2$ is embedded by $|3P_0|$, and x, y have poles of exact orders $2, 3$ at P_0 and nowhere else (proof of Prop. 4.6).

Reduction to affine maps. Let $\sigma \in \operatorname{Aut}(X)$ with $\sigma(P_0) = P_0$. Then $\sigma^* x = x \circ \sigma$ has a pole of order exactly 2 at $\sigma^{-1}(P_0) = P_0$ and no other pole, so $\sigma^* x \in H^0(\mathcal{O}(2P_0))$, $\sigma^* x = ax + b$ with $a \neq 0$. Similarly $\sigma^* y \in H^0(\mathcal{O}(3P_0))$ with exact pole order 3: $\sigma^* y = cy + dx + e$, $c \neq 0$. Since σ is an automorphism, $(\sigma^* y)^2 = (\sigma^* x)(\sigma^* x - 1)(\sigma^* x - \lambda)$, i.e.

$$c^2 y^2 + 2cy(dx + e) + (dx + e)^2 = (ax + b)(ax + b - 1)(ax + b - \lambda).$$

Now $y^2 = x(x-1)(x-\lambda) \in k[x]$ and $K(X)$ has basis $1, y$ over $k(x)$; comparing the coefficients of y gives $2c(dx + e) = 0$, so $d = e = 0$ (char $\neq 2$). Thus $\sigma^* x = ax + b$, $\sigma^* y = cy$, i.e. $\sigma(x_0, y_0) = (ax_0 + b, cy_0)$: σ is induced by the automorphism $[x : y : z] \mapsto [ax + bz : cy : z]$ of $\mathbf{P}^2$ (which fixes $(0,1,0)$).

Which (a,b,c) *occur.* The map $(x,y)\mapsto(ax+b,cy)$ gives an automorphism of X iff $c^2x(x-1)(x-\lambda)=\prod_{u\in\{0,1,\lambda\}}(ax+b-u)=a^3\prod_u\big(x-\frac{u-b}{a}\big)$ as polynomials, i.e. iff $c^2=a^3$ and $\ell(x)=ax+b$ permutes $\{0,1,\lambda\}$ (the roots $\frac{u-b}{a}=\ell^{-1}(u)$ must be $\{0,1,\lambda\}$). So $G=\{(a,b,c):c^2=a^3,\ \ell_{a,b}\{0,1,\lambda\}=\{0,1,\lambda\}\}$, with composition $(a,b,c)\circ(a',b',c')=(aa',ab'+b,cc')$. Sending (a,b,c) to $\ell_{a,b}$ is a homomorphism of G onto the group H of affine maps $x\mapsto ax+b$ permuting $\{0,1,\lambda\}$ (each a has the two square roots of a^3 available as c); its kernel is $\{(1,0,\pm1)\}=\{1,\rho\}$, where $\rho:y\mapsto -y$ is the inversion. So $|G|=2|H|$ and $G/\{1,\rho\}\cong H\subseteq\Sigma_3$. An affine map is determined by the images p,q of $0,1$ and then sends λ to $p+\lambda(q-p)$, which must be the remaining element. Going through the six permutations:

- identity: always.
- $0\leftrightarrow 1$ $(x\mapsto 1-x)$: need $1-\lambda=\lambda$, i.e. $\lambda=\frac12$.
- $1\leftrightarrow\lambda$ $(x\mapsto\lambda x)$: need $\lambda^2=1$, i.e. $\lambda=-1$.
- $0\leftrightarrow\lambda$ $(x\mapsto\lambda+(1-\lambda)x)$: need $2\lambda-\lambda^2=0$, i.e. $\lambda=2$.
- $0\to1\to\lambda\to0$ $(x\mapsto1+(\lambda-1)x)$ and its inverse $0\to\lambda\to1\to0$ $(x\mapsto\lambda-\lambda x)$: need $\lambda^2-\lambda+1=0$.

This gives exactly the four cases of Cor. 4.7:

(i) $\lambda\notin\{-1,2,\frac12\}$ *and* $\lambda^2-\lambda+1\neq0$ (generic j): $H=1$, $G=\{(x,y)\mapsto(x,\pm y)\}\cong\mathbf{Z}/2$.

(ii) $\lambda\in\{-1,2,\frac12\}$ (these are three distinct values if $\operatorname{char}k\neq3$ and give $j=2^63^3=1728$; then $\lambda^2-\lambda+1\neq0$): $H=\{1,\ell\}$ with $\ell=-x,\ 2-x,\ 1-x$ respectively, i.e. $a=-1$ and $c^2=a^3=-1$. The automorphisms are $(x,y)\mapsto(\ell(x),\pm iy)$, $i^2=-1$ (together with $(x,\pm y)$). The element $g=(-1,b,i)$ satisfies $g^2=(1,0,-1)=\rho\neq1$, so g has order 4 and $G=\langle g\rangle\cong\mathbf{Z}/4$. For $\lambda=-1$, $g:(x,y)\mapsto(-x,iy)$.

(iii) $\lambda^2-\lambda+1=0$, $\operatorname{char}k\neq3$ $(\lambda=-\omega,-\omega^2,\ j=0)$: $H=\langle\ell\rangle\cong\mathbf{Z}/3$, $\ell=1+(\lambda-1)x$. Here $a=\lambda-1=\lambda^2$, $a^3=\lambda^6=1$ (as $\lambda^3=\lambda\cdot\lambda^2=\lambda^2-\lambda=-1$), so $c=\pm1$ and the automorphisms are $(x,y)\mapsto(\ell^r(x),\pm y)$. For $g:(x,y)\mapsto(\ell(x),-y)$, $g^3=\rho$, so g has order 6 and $G=\langle g\rangle\cong\mathbf{Z}/6$.

(iv) $\operatorname{char}k=3$, $\lambda=-1$ $(j=0=1728)$: then $-1=2=\frac12$ and $\lambda^2-\lambda+1=(\lambda+1)^2=0$, so all six conditions hold and H is the full group of affine permutations $x\mapsto\pm x+b$, $b\in\mathbf{F}_3$, of $\{0,1,-1\}=\mathbf{F}_3$, $|G|=12$. Put $t=(1,1,1):(x,y)\mapsto(x+1,y)$ (order 3) and $s=(-1,0,i):(x,y)\mapsto(-x,iy)$, where $i\in\mathbf{F}_9$, $i^2=-1$. Then $s^2=\rho$, so s has order 4, and a direct computation gives $sts^{-1}=t^{-1}$. Since $\langle t\rangle\cap\langle s\rangle=1$ and $3\cdot4=12$, $G=\langle t\rangle\rtimes\langle s\rangle\cong\mathbf{Z}/3\rtimes\mathbf{Z}/4$ with s acting by inversion (the nonabelian "dicyclic" group of order 12; it is a central extension of Σ_3 by $\{1,\rho\}$).

Here $\{1,\rho\}$ is central in all cases, which is why in (i)–(iii) G is abelian ($G/\{1,\rho\}=H$ is cyclic). □

Exercise 4.4

Problem. Let X be an elliptic curve in $\mathbf{P}^2$ given by an equation of the form

$$y^2+a_1xy+a_3y=x^3+a_2x^2+a_4x+a_6.$$

Show that the j-invariant is a rational function of the a_i, with coefficients in $\mathbf{Q}$. In particular, if the a_i are all in some field $k_0 \subseteq k$, then $j \in k_0$ also. Furthermore, for every $\alpha \in k_0$, there exists an elliptic curve defined over k_0 with j-invariant equal to α.

Solution. As throughout this section of the text, $\operatorname{char} k \neq 2$ (the definition of j requires it). (The statement as printed omits the characteristic hypothesis; I use this standing assumption.)

Step 1: reduction to $y^2 =$ cubic. Put $\bar{y} = y + \frac{1}{2}(a_1x + a_3)$. Then $\bar{y}^2 = f(x)$ with

$$f(x) = x^3 + Ax^2 + Bx + C, \quad A = a_2 + \tfrac{1}{4}a_1^2,\ B = a_4 + \tfrac{1}{2}a_1a_3,\ C = a_6 + \tfrac{1}{4}a_3^2.$$

Let $P_0 = [0:1:0]$ be the point at infinity (the only point on $z = 0$). It is nonsingular, and in the chart $y = 1$ with coordinates $u = x/y$, $v = z/y$ the curve is $v + a_1uv + a_3v^2 = u^3 + \cdots$, whose tangent at the origin is $v = 0$; so u is a local parameter, $\operatorname{ord} v = 3$ and $x/z = u/v$ has a pole of order exactly 2 at P_0 and no other pole. Thus $x \in H^0(\mathcal{O}(2P_0))$ is nonconstant and defines the morphism of degree 2, $x : X \to \mathbf{P}^1$, i.e. the morphism of $|2P_0|$, with $x(P_0) = \infty$. Its branch points are ∞ and the points with $\bar{y} = 0$, i.e. the roots e_1, e_2, e_3 of f; since X is nonsingular they are distinct (a multiple root of f gives a singular point $(e, 0)$). Applying the affine map $x \mapsto (x - e_1)/(e_2 - e_1)$ we get the λ of the text, $\lambda = (e_3 - e_1)/(e_2 - e_1)$, and by definition (and Thm. 4.1(a)) $j(X) = j(\lambda)$.

Step 2: j as a symmetric function of the roots. Substituting $\lambda = \frac{e_3 - e_1}{e_2 - e_1}$, $\lambda - 1 = \frac{e_3 - e_2}{e_2 - e_1}$ gives

$$j(X) = 2^8 \frac{N^3}{\Delta_f}, \qquad N = (e_2 - e_1)^2 - (e_2 - e_1)(e_3 - e_1) + (e_3 - e_1)^2, \quad \Delta_f = \prod_{i<j} (e_i - e_j)^2.$$

Now $N = \sum e_i^2 - \sum_{i<j} e_ie_j = s_1^2 - 3s_2 = A^2 - 3B$ and Δ_f is the discriminant of f: $\Delta_f = A^2B^2 - 4B^3 - 4A^3C + 18ABC - 27C^2$ (both are symmetric polynomials with integer coefficients in the e_i, hence polynomials in $s_1 = -A, s_2 = B, s_3 = -C$). Therefore

$$j(X) = \frac{2^8 (A^2 - 3B)^3}{A^2B^2 - 4B^3 - 4A^3C + 18ABC - 27C^2},$$

a rational function of a_1, a_2, a_3, a_4, a_6 with rational coefficients (only $\frac{1}{2}, \frac{1}{4}$ occur), whose denominator $\Delta_f \neq 0$ because X is nonsingular. (In the usual notation $b_2 = a_1^2 + 4a_2$, $b_4 = 2a_4 + a_1a_3$, $b_6 = a_3^2 + 4a_6$, $c_4 = b_2^2 - 24b_4$ one has $A^2 - 3B = c_4/16$, $\Delta_f = \Delta/16$ and this is $j = c_4^3/\Delta$; I checked this with a computer algebra system.) If all $a_i \in k_0$ then $A, B, C \in k_0$ and $j \in k_0$.

Step 3: every $\alpha \in k_0$ occurs. We exhibit curves of the given form with coefficients in k_0 (these are then nonsingular cubics, i.e. elliptic curves, defined over k_0); the nonsingularity is the condition $\Delta \neq 0$, where $\Delta = -b_2^2b_8 - 8b_4^3 - 27b_6^2 + 9b_2b_4b_6$, $b_8 = a_1^2a_6 + 4a_2a_6 - a_1a_3a_4 + a_2a_3^2 - a_4^2$ (so $\Delta = 16\Delta_f$).

- $\alpha \neq 0, 1728$: $E_\alpha :\ y^2 + xy = x^3 - \dfrac{36}{\alpha - 1728}x - \dfrac{1}{\alpha - 1728}$. Here $a_1 = 1$, $a_2 = a_3 = 0$, and a direct computation gives $c_4 = \dfrac{\alpha}{\alpha - 1728}$, $\Delta = \dfrac{\alpha^2}{(\alpha - 1728)^3} \neq 0$, so $j = c_4^3/\Delta = \alpha$.
- $\alpha = 1728$: $y^2 = x^3 - x$, with $c_4 = 48$, $\Delta = 64 \neq 0$, $j = 48^3/64 = 1728$ (this is Example 4.6.1; $\lambda = -1$).

- $\alpha = 0 \neq 1728$ (i.e. $\operatorname{char} k \neq 3$): $y^2 + y = x^3$, with $c_4 = 0$, $\Delta = -27 \neq 0$, $j = 0$.

If $\operatorname{char} k = 3$ then $1728 = 0$, so the case $\alpha = 0$ is covered by the second line and the others by the first. □

Exercise 4.5

Problem. Let X, P_0 be an elliptic curve having an endomorphism $f : X \to X$ of degree 2.

(a) If we represent X as a 2-1 covering of $\mathbf{P}^1$ by a morphism $\pi : X \to \mathbf{P}^1$ ramified at P_0, then as in (4.4), show that there is another morphism $\pi' : X \to \mathbf{P}^1$ and a morphism $g : \mathbf{P}^1 \to \mathbf{P}^1$, also of degree 2, such that $\pi \circ f = g \circ \pi'$.

(b) For suitable choices of coordinates in the two copies of $\mathbf{P}^1$, show that g can be taken to be the morphism $x \mapsto x^2$.

(c) Now show that g is branched over two of the branch points of π, and that g^{-1} of the other two branch points of π consists of the four branch points of π'. Deduce a relation involving the invariant λ of X.

(d) Solving the above, show that there are just three values of j corresponding to elliptic curves with an endomorphism of degree 2, and find the corresponding values of λ and j. [*Answers*: $j = 2^6 \cdot 3^3$; $j = 2^6 \cdot 5^3$; $j = -3^3 \cdot 5^3$.]

Solution. Throughout $\operatorname{char} k \neq 2$; "endomorphism" means $f(P_0) = P_0$ (any morphism $X \to X$ of degree 2 becomes one after composing with a translation, which does not change degrees), so f is a homomorphism (Lemma 4.9). Here π is the morphism defined by $|2P_0|$, $\pi(P_0) = \infty$, with branch points $\pi(X[2])$ (the ramification points of π are the 4 points R with $2R \sim 2P_0$, i.e. $R = -R$).

Preliminaries. Let $\rho = -1_X$, so $\pi\rho = \pi$ (the fibres of π are the divisors $R + \rho R \in |2P_0|$, Lemma 4.2) and $K(X)^\rho = \pi^* K(\mathbf{P}^1)$ (both have index 2 in $K = K(X)$).

Cross ratios. For 4 distinct points $z_1, \ldots, z_4 \in \mathbf{P}^1$ let $c(z_1, z_2, z_3, z_4) = \frac{(z_4 - z_1)(z_2 - z_3)}{(z_4 - z_3)(z_2 - z_1)}$, the image of z_4 under the Moebius map sending z_1, z_2, z_3 to $0, 1, \infty$. When the labelling of the four points is permuted, c runs through a Σ_3-orbit as in Lemma 4.5 (classical), and $j(c)$ is constant on it. *Fact (R):* if $\varphi : X \to \mathbf{P}^1$ is any morphism of degree 2, then it has exactly 4 branch points (Hurwitz) and $j(X) = j(c(z_1, \ldots, z_4))$ for these points in any order. Indeed, if P_1 is a ramification point of φ then $\varphi^*\varphi(P_1) = 2P_1$, so φ is (up to an automorphism of $\mathbf{P}^1$) the morphism defined by $|2P_1|$ (II, 7.8.1); apply the construction of j with P_1 as base point and Thm. 4.1(a) (the proof of (4.4)). Sending $\varphi(P_1)$ to ∞ and then two other branch points to $0, 1$ by an affine map gives $c(a, b, \infty, c_3) = \frac{c_3 - a}{b - a} = \lambda$.

(a) Since $\deg f = 2$ and $\operatorname{char} k \neq 2$, f is separable, so $\ker f = \{0, T\}$ with T a point of order 2, $T = -T$. Let $K = K(X)$ and $h = \pi \circ f : X \to \mathbf{P}^1$, $\deg h = 4$, $L = h^* K(\mathbf{P}^1) \subseteq K$. Consider the automorphisms ρ, τ_T, $\sigma = \rho\tau_T : x \mapsto -x - T = T - x$ of X and $G' = \{1, \rho, \tau_T, \sigma\} \cong (\mathbf{Z}/2)^2$ (these are four distinct maps: ρ, σ have fixed points, τ_T none, and $\rho \neq \sigma$). We have $h\rho = \pi f \rho = \pi\rho f = \pi f = h$ (as f is a homomorphism, $f\rho = \rho f$) and $h\tau_T = \pi \circ (f(x) + f(T)) = h$. So $L \subseteq K^{G'}$, and $[K : K^{G'}] = 4 = [K : L]$, hence $L = K^{G'}$ and K/L is Galois with group G'; h is the quotient map $X \to X/G' \cong \mathbf{P}^1$.

Let $Y = X/\langle\sigma\rangle$ (the curve with function field K^σ) and $\pi' : X \to Y$ the quotient. σ has fixed points $\{y : 2y = T\}$, four of them (2_X is a surjective homomorphism with kernel $X[2]$, which consists of the 4 ramification points of π, so $\{y : 2y = T\}$ is a coset of $X[2]$), so Hurwitz for the tame degree 2 cover π' gives $0 = 2(2g(Y) - 2) + 4$, $g(Y) = 0$, $Y \cong \mathbf{P}^1$. Since $L \subseteq K^\sigma \subseteq K$, $h = g \circ \pi'$ for a morphism $g : Y \to X/G' = \mathbf{P}^1$ of degree $[K^\sigma : L] = 2$. So $\pi \circ f = g \circ \pi'$ with π' of degree 2 and g of degree 2. (Also $K^\rho \supseteq L$ gives $\pi f = g_1 \pi$ for the map g_1 induced by f; the new factorization through π' is the one wanted.)

(b) A degree-2 morphism $g : \mathbf{P}^1 \to \mathbf{P}^1$ has two ramification points r_1, r_2 (Hurwitz: $-2 = 2(-2) + \deg R$, tame), with distinct images $b_1 \neq b_2$ (else the fibre over b_1 would have degree ≥ 4). Choose coordinates s on the source with $r_1 = 0, r_2 = \infty$, and t on the target with $b_1 = 0, b_2 = \infty$. Then $g^*(0) = 2 \cdot 0$, $g^*(\infty) = 2 \cdot \infty$, so $t \circ g = \kappa s^2$ and replacing s by $\kappa^{-1/2} s$ we get $g(s) = s^2$.

(c) Now work with the fixed points of the three involutions: $F_\rho = X[2]$, $F_\sigma = \{2y = T\}$, $F_\tau = \varnothing$. The ramification points of h are the points with nontrivial stabilizer in G', i.e. $F_\rho \cup F_\sigma$ (disjoint: $2x = 0 \neq T$), 8 points. The G'-orbit of $x \in F_\rho$ is $\{x, x + T\}$, similarly for F_σ, and h identifies exactly the G'-orbits. So $B_\rho = h(F_\rho)$ and $B_\sigma = h(F_\sigma)$ are disjoint sets of 2 points each, and $B = B_\rho \cup B_\sigma$ is the branch locus of h (4 points). Since $f(F_\rho \cup F_\sigma) \subseteq X[2]$ ($2f(x) = f(2x) \in \{0, f(T)\} = 0$), we have $B = \pi f(F_\rho \cup F_\sigma) \subseteq \pi(X[2])$, and both sets have 4 points; so B is exactly the set of branch points of π.

For $x \in X$ with image $y = \pi'(x) \in Y$ we have $e_x(h) = |\mathrm{Stab}_{G'}(x)|$ and $e_x(\pi') = |\mathrm{Stab}(x) \cap \langle\sigma\rangle|$, hence $e_y(g) = e_x(h)/e_x(\pi')$. This is 2 if $x \in F_\rho$, and 1 otherwise. So g is branched exactly over B_ρ, two of the four branch points of π. The branch points of π' are the 4 points $\pi'(y)$, $y \in F_\sigma$ (distinct, since σ fixes them); they lie over $h(F_\sigma) = B_\sigma$, and $g^{-1}(B_\sigma)$ has at most $2 \cdot 2 = 4$ points. Hence $g^{-1}(B_\sigma)$ consists exactly of the four branch points of π': *g is branched over two branch points of π, and the preimage of the other two is the set of the four branch points of π'.*

Normalize as in (b): $g(s) = s^2$ with $B_\rho = \{0, \infty\}$; scaling t by a constant (and s by its square root) we may assume $B_\sigma = \{1, \nu\}$ with $\nu \neq 0, 1, \infty$. So the branch points of π are $0, 1, \nu, \infty$, those of π' are $\pm 1, \pm\mu$ with $\mu^2 = \nu$. Fact (R) applied to π and to π' gives $j(X) = j(\nu)$ (as $c(0, 1, \infty, \nu) = \nu$) and

$$j(X) = j(c(-1, 1, -\mu, \mu)), \qquad c(-1, 1, -\mu, \mu) = \frac{(\mu + 1)(1 + \mu)}{(2\mu)\, 2} = \frac{(1 + \mu)^2}{4\mu}.$$

Relation. The invariant λ of X (the class of ν under Σ_3) satisfies

$$j(\nu) = j\Big(\frac{(1 + \mu)^2}{4\mu}\Big), \qquad \mu^2 = \nu.$$

(d) We compute: $c = \frac{(1+\mu)^2}{4\mu}$ gives $c - 1 = \frac{(1-\mu)^2}{4\mu}$, $c^2 - c + 1 = \frac{\mu^4 + 14\mu^2 + 1}{16\mu^2}$, so with $\nu = \mu^2$

$$j(c) = 2^8 \frac{(\mu^4 + 14\mu^2 + 1)^3/(16\mu^2)^3}{(1 - \mu^2)^4/(256\mu^4)} = \frac{2^4(\nu^2 + 14\nu + 1)^3}{\nu(\nu - 1)^4}.$$

Thus the relation reads $\dfrac{2^8(\nu^2 - \nu + 1)^3}{\nu^2(\nu - 1)^2} = \dfrac{2^4(\nu^2 + 14\nu + 1)^3}{\nu(\nu - 1)^4}$, i.e. $P(\nu) = 0$ with

$$P(\nu) = 16(\nu^2 - \nu + 1)^3(\nu - 1)^2 - \nu(\nu^2 + 14\nu + 1)^3 = (\nu + 1)^2(\nu^2 - 6\nu + 1)(\nu^2 - \nu + 16)(16\nu^2 - \nu + 1)$$

(the factorization is checked by expanding; $P(0) \neq 0 \neq P(1)$). The values of $j(\nu)$ at the roots:

- $\nu = -1$: $j = 2^8 \cdot 27/4 = 1728 = 2^6 \cdot 3^3$ (the Σ_3-orbit is $\{-1, 2, \frac{1}{2}\}$).
- $\nu^2 - 6\nu + 1 = 0$, $\nu = 3 \pm 2\sqrt{2}$: here $\nu^2 - \nu + 1 = 5\nu$ and $(\nu-1)^2 = 4\nu$, so $j = 2^8 \cdot 125\nu^3/(4\nu^3) = 8000 = 2^6 \cdot 5^3$.
- $\nu^2 - \nu + 16 = 0$, $\nu = \frac{1 \pm 3\sqrt{-7}}{2}$: $\nu^2 - \nu + 1 = -15$ and $\nu^2(\nu-1)^2 = (\nu^2 - \nu)^2 = 256$, so $j = 2^8(-15)^3/256 = -3375 = -3^3 \cdot 5^3$; the roots of $16\nu^2 - \nu + 1$ are the reciprocals of these, in the same orbit, with the same j.

So necessarily $j(X) \in \{1728, 8000, -3375\}$ (in characteristic 0; in characteristic $p > 2$ the same computation gives the reductions of these values).

Converse. Conversely let ν be a root of P and $\mu^2 = \nu$. Put $X_\nu : w^2 = t(t-1)(t-\nu)$ (so $j(X_\nu) = j(\nu)$) and let X' be the curve of genus 1 with $K(X') = k(s, v)$, $v^2 = (s^2 - 1)(s^2 - \nu)$, a double cover of $\mathbf{P}^1_s$ branched over $\pm 1, \pm\mu$; by (R) and the computation above $j(X') = j((1 + \mu)^2/4\mu) = j(\nu) = j(X_\nu)$, so $X' \cong X_\nu$ (Thm. 4.1(b)). The inclusion $k(t, w) \subseteq k(s, v)$, $t \mapsto s^2$, $w \mapsto sv$ (note $s^2v^2 = s^2(s^2-1)(s^2-\nu)$) has degree 2, so it gives a morphism $X' \to X_\nu$ of degree 2, hence after composing with the isomorphism and a translation an endomorphism of degree 2 of X_ν. So exactly the three values $j = 2^6 \cdot 3^3,\ 2^6 \cdot 5^3,\ -3^3 \cdot 5^3$ occur (in characteristic 0), with $\lambda \in \{-1, 2, \frac{1}{2}\}$, $\{3 \pm 2\sqrt{2}, \dots\}$ and $\{\frac{1 \pm 3\sqrt{-7}}{2}, \dots\}$ respectively. □

Exercise 4.6

Problem. (a) Let X be a curve of genus g embedded birationally in $\mathbf{P}^2$ as a curve of degree d with r nodes. Generalize the method of (Ex. 2.3) to show that X has $6(g-1)+3d$ inflection points. A node does not count as an inflection point. Assume $\operatorname{char} k = 0$.

(b) Now let X be a curve of genus g embedded as a curve of degree d in $\mathbf{P}^n$, $n \geq 3$, not contained in any $\mathbf{P}^{n-1}$. For each point $P \in X$, there is a hyperplane H containing P, such that P counts at least n times in the intersection $H \cap X$. This is called an *osculating hyperplane* at P, a generalization of the tangent line for curves in $\mathbf{P}^2$. If P counts $n+1$ times in $H \cap X$, we say H is a *hyperosculating hyperplane* and P is a *hyperosculation point*. Use Hurwitz's theorem and induction on n to show that X has $n(n+1)(g-1) + (n+1)d$ hyperosculation points.

(c) If X is an elliptic curve, for any $d \geq 3$, embed X as a curve of degree d in $\mathbf{P}^{d-1}$, and conclude that X has exactly d^2 points of order d in its group law.

Solution. Throughout $\operatorname{char} k = 0$ (Hurwitz's formula then reads $2g-2 = m(2g'-2) + \sum(e_P - 1)$ for a finite morphism of degree m). The counts are made with multiplicities, as in Ex. 2.3(e); the precise multiplicities are given below.

(a) Let $\nu : X \to C \subset \mathbf{P}^2$ be the birational morphism onto the plane curve C of degree d with r nodes as only singularities; ν is an immersion (each branch at a node is smooth), and $\nu^*\mathcal{O}(1)$ has degree d. We assume $d \geq 2$ (for a line every tangent line is the line itself, the Gauss map is constant and the formula fails). For $P \in X$ let T_P be the tangent line to the branch of C at P and $r_P = \operatorname{ord}_P(\nu^* T_P) \geq 2$ its contact order with the branch. We say P is an *inflection point* if $r_P \geq 3$ and count it $r_P - 2$ times (nodes themselves do not count; only the branch through P

enters). There are only finitely many inflection points (as in Ex. 2.3(a); this also follows from the Wronskian argument in (b), or from the local formula below, which shows that they are among the ramification points of the nonconstant morphism ϕ defined below). The map $\gamma : P \mapsto T_P$ is a rational map $X \to (\mathbf{P}^2)^*$, hence a morphism (II, 6.8), and by the local computation below its value at every P is T_P.

Local computation. Fix P and a line $L \neq T_P$, and let $Q_0 = L \cap T_P$. Choose affine coordinates (X_1, Y_1) on $\mathbf{P}^2$ (line at infinity not through P or Q_0) with $P = (0,0)$, $T_P = \{Y_1 = 0\}$, so $Q_0 = (q_1, 0)$. The branch at P is parametrized by $t \mapsto (t, y(t))$ with t a local parameter at P and $y = ct^r + \cdots$, $r = r_P$, $c \neq 0$ (formal derivative $y' = dy/dt$). The tangent line at the point with parameter t is $Y_1 - y(t) = y'(t)(X_1 - t)$. Write points of L as $Q_0 + uv$ with $v = (v_X, v_Y)$, $v_Y \neq 0$ (as $L \neq T_P$). Then $T_t \cap L$ has parameter

$$u(t) = \frac{y + y'(q_1 - t)}{v_Y - y' v_X}.$$

The denominator is a unit. If $q_1 \neq 0$ ($P \notin L$) the numerator has order exactly $r - 1$ (because $y'q_1 = rcq_1 t^{r-1} + \cdots$ and y has order r; here $\operatorname{char} k = 0$ is used). If $q_1 = 0$ ($P \in L$) the numerator is $y - ty' = c(1 - r)t^r + \cdots$ of order exactly r.

The Gauss map to a line. Choose a line L which meets C transversally in d distinct smooth points, passes through no node and no inflection point, and is not tangent to any branch; define $\phi : X \to L$, $\phi(P) = T_P \cap L$ (a morphism, being the composite of γ with the projection of $(\mathbf{P}^2)^*$ from the point $[L]$). By the local computation $\phi(P) = Q_0$ corresponds to $u(0) = 0$ and the ramification index of ϕ at P is $e_P = r_P - 1$ if $P \notin L$ and $e_P = r_P = 2$ if $P \in L$ (an inflection point is not on L, and for $P \in L$ the contact is 2). In particular ϕ is ramified exactly at the d points of $\nu^{-1}(L)$ (with $e = 2$) and at the inflection points.

Degree of ϕ. Let $Q \in L$ be general and $\psi_Q : X \to \mathbf{P}^1$ the projection from Q (a morphism, since $Q \notin C$): $P \mapsto$ the line QP. Its fibre over a line $\ell \ni Q$ is $\nu^*(\ell)$, so $\deg \psi_Q = d$, and the ramification index at P is the contact order of $\ell = QP$ with the branch at P; so ψ_Q is ramified at P iff QP is tangent to the branch, i.e. $Q \in T_P$, and then $e_P = r_P$. For Q outside the finitely many tangent lines at inflection points, $e_P = 2$ at all these points, and Hurwitz gives

$$2g - 2 = -2d + \#\{P : Q \in T_P\}, \quad \text{so} \quad \#\{P : Q \in T_P\} = 2g - 2 + 2d.$$

On the other hand $\phi^{-1}(Q) = \{P : Q \in T_P\}$ (as $T_P \neq L$), and for Q outside the finite branch set of ϕ this fibre is reduced; so $m := \deg \phi = 2g - 2 + 2d$. (Check: for C smooth, $g = \frac{(d-1)(d-2)}{2}$ and $m = d(d-1)$, as in Ex. 2.3(e).)

Hurwitz for ϕ. With $\Phi = \sum_P (r_P - 2)$, the number of inflection points counted with multiplicity,

$$2g - 2 = -2m + d + \Phi, \quad \text{hence} \quad \Phi = 2g - 2 + 2(2g - 2 + 2d) - d = 6(g-1) + 3d.$$

So X has $6(g-1) + 3d$ inflection points. Since $g = \frac{(d-1)(d-2)}{2} - r$, this is $3d(d-2) - 6r$ (for a smooth curve $3d(d-2)$, as in Ex. 2.3(e); for a nodal cubic, 3). (The same number is obtained from (b) with $n = 2$.)

(b) Let $V \subseteq H^0(X, \mathcal{L})$, $\mathcal{L} = \mathcal{O}_X(1)$ of degree d, be the $(n+1)$-dimensional image of $H^0(\mathbf{P}^n, \mathcal{O}(1))$ (injective because X lies in no hyperplane); V is base point free. For $P \in X$ the orders $\operatorname{ord}_P(s)$, $s \in V \setminus 0$, take exactly $n + 1$ values $0 = a_0(P) < a_1(P) < \cdots < a_n(P)$ (as $\dim\{s : \operatorname{ord}_P s \geq j\}$

drops by 0 or 1 when $j \to j+1$); $a_i \geq i$. A hyperplane $H = \{s = 0\}$ meets X at P with multiplicity $\operatorname{ord}_P s$. The sections of order $\geq n$ at P form a space of dimension ≥ 1 (n linear conditions), so there is a hyperplane in which P counts at least n times, and the osculating hyperplane (max contact $a_n(P)$) is unique. P is a hyperosculation point iff $a_n(P) \geq n+1$ iff $w(P) := \sum_{i=0}^{n}(a_i(P) - i) > 0$ (if $a_n = n$ then $a_i = i$ for all i since the a_i are strictly increasing). We count P with multiplicity $w(P)$ (for $n = 2$ this is the $r_P - 2$ of (a); for a point with $(a_i) = (0, 1, \ldots, n-1, n+1)$ it is 1). We prove:

$$\sum_{P \in X} w(P) = (n+1)d + n(n+1)(g-1).$$

(This replaces the inductive argument by the Wronskian, which proves the whole formula at once; for $n = 1$ it is exactly Hurwitz's theorem $\sum(e_P - 1) = 2d + 2g - 2$.)

Let $s_0, \ldots, s_n$ be a basis of V. At a point P choose a local parameter z and a generator e of $\mathcal{L}$, write $s_i = f_i e$ with $f_i \in \mathcal{O}_{X,P}$, and let $D = d/dz$ be the derivation of $K(X)$ with $Dz = 1$ (exists since $\operatorname{char} k = 0$). Define the Wronskian $W = \det\big(D^j f_i\big)_{0 \leq i,j \leq n} \in \mathcal{O}_{X,P}$.

(1) *$W\, e^{\otimes(n+1)}(dz)^{\otimes n(n+1)/2}$ is independent of the choices.* If $e' = ue$ (u a unit) then $f_i' = f_i/u$, and by Leibniz $D^j(f_i/u) = \sum_{l \leq j} \binom{j}{l} D^{j-l}(u^{-1}) D^l f_i$, so $(D^j f_i') = (D^l f_i) \cdot M$ with M triangular with diagonal entries u^{-1}; thus $W' = u^{-(n+1)} W$ and $W' e'^{n+1} = W e^{n+1}$. If z' is another parameter, $d/dz' = (dz/dz')D$, and $(d/dz')^j f = (dz/dz')^j D^j f + \sum_{l<j}(*) D^l f$, so $W' = (dz/dz')^{n(n+1)/2} W$, which compensates $(dz')^{n(n+1)/2} = (dz'/dz)^{n(n+1)/2}(dz)^{n(n+1)/2}$. Hence the local expressions glue to a global section $\mathcal{W}$ of $\mathcal{L}^{\otimes(n+1)} \otimes \Omega_X^{\otimes n(n+1)/2}$ (regular, as $W \in \mathcal{O}_{X,P}$), of degree $(n+1)d + \frac{n(n+1)}{2}(2g-2) = (n+1)d + n(n+1)(g-1)$.

(2) $\mathcal{W} \neq 0$. The constants of D in $K(X)$ are k (if $Df = 0$ for some $f \notin k$, then z is algebraic over $k(f)$, hence separable over it, and D vanishing on $k(f)$ would force $Dz = 0$, contradicting $Dz = 1$; here $\operatorname{char} k = 0$ is used), and the classical Wronskian criterion says $W = 0$ iff $f_0, \ldots, f_n$ are linearly dependent over the constants. Since $s_0, \ldots, s_n$ are linearly independent over k, $W \neq 0$.

(3) $\operatorname{ord}_P(W) = w(P)$. W is multiplied by a nonzero constant if the basis of V is changed (multilinearity), so we may take a basis with $\operatorname{ord}_P f_i = a_i$, $f_i = c_i z^{a_i} + \cdots$, $c_i \neq 0$. Then $D^j f_i = z^{a_i - j}\big(c_i\, a_i(a_i-1)\cdots(a_i-j+1) + O(z)\big)$, hence

$$W = z^{\sum a_i - n(n+1)/2}\Big(\prod_i c_i \cdot \det\big(a_i^{\underline{j}}\big) + O(z)\Big), \quad a^{\underline{j}} = a(a-1)\cdots(a-j+1).$$

Since $a^{\underline{j}}$ is monic of degree j in a, $\det(a_i^{\underline{j}}) = \prod_{i<l}(a_l - a_i) \neq 0$ (Vandermonde, characteristic 0). So $\operatorname{ord}_P W = \sum a_i - \sum_{j=0}^{n} j = w(P)$.

Therefore $\sum_P w(P) = \deg \mathcal{W} = (n+1)d + n(n+1)(g-1)$, which is the asserted number of hyperosculation points, counted with multiplicity $w(P)$.

(c) Let X be elliptic, $d \geq 3$, and embed X by the complete linear system $|dP_0|$ into $\mathbf{P}^{d-1}$ (Cor. 3.2; $h^0(dP_0) = d$ so the image lies in no hyperplane). Take $n = d-1$, $g = 1$: $\sum w(P) = (n+1)d = d^2$.

For $P \in X$ and $0 \leq k \leq d-1$, $\deg(dP_0 - kP) = d - k \geq 1$ so $h^0(dP_0 - kP) = d - k$ (Riemann–Roch); for $k = d$, $h^0(dP_0 - dP) = 1$ if $dP \sim dP_0$ and 0 otherwise. If $dP \not\sim dP_0$ the dimension of

$\{s : \operatorname{ord}_P s \geq k\}$ drops by 1 at each step $k = 0, \ldots, d-1$, so the orders are $0, 1, \ldots, d-1$ and $w(P) = 0$. If $dP \sim dP_0$ the dimensions are $d, d-1, \ldots, 1$ for $k = 0, \ldots, d-1$ and 1 for $k = d$, so there is no drop at $k = d-1$ and the orders are $0, 1, \ldots, d-2, d$: $a_n = d$, $w(P) = 1$. Thus the hyperosculation points are exactly the P with $dP \sim dP_0$ (a hyperplane section is a divisor in $|dP_0|$ of degree d, so P counts d times in some $H \cap X$ iff $H \cdot X = dP$), each of weight exactly 1. Since $dP \sim dP_0$ means $d(P - P_0) \sim 0$, i.e. $dP = 0$ in the group law, we conclude that X has exactly d^2 points P with $dP = 0$ ("of order d", meaning order dividing d). □

Exercise 4.7

Problem. *The Dual of a Morphism.* Let X and X' be elliptic curves over k, with base points P_0, P_0'.

(a) If $f : X \to X'$ is any morphism, use (4.11) to show that $f^* : \operatorname{Pic} X' \to \operatorname{Pic} X$ induces a homomorphism $\hat{f} : (X', P_0') \to (X, P_0)$. We call this the *dual* of f.

(b) If $f : X \to X'$ and $g : X' \to X''$ are two morphisms, then $(g \circ f)^\wedge = \hat{f} \circ \hat{g}$.

(c) Assume $f(P_0) = P_0'$, and let $n = \deg f$. Show that if $Q \in X$ is any point and $f(Q) = Q'$, then $\hat{f}(Q') = n_X(Q)$. (Do the separable and purely inseparable cases separately, then combine.) Conclude that $\hat{f} \circ f = n_X$ and $f \circ \hat{f} = n_{X'}$.

(d) If $f, g : X \to X'$ are two morphisms preserving the base points P_0, P_0', then $(f+g)^\wedge = \hat{f} + \hat{g}$. [*Hints*: It is enough to show for any $\mathcal{L} \in \operatorname{Pic} X'$, that $(f+g)^*\mathcal{L} \cong f^*\mathcal{L} \otimes g^*\mathcal{L}$. For any f, let $\Gamma_f : X \to X \times X'$ be the graph morphism. Then it is enough to show (for $\mathcal{L}' = p_2^*\mathcal{L}$) that $\Gamma_{f+g}^*(\mathcal{L}') = \Gamma_f^*\mathcal{L}' \otimes \Gamma_g^*\mathcal{L}'$. Let $\sigma : X \to X \times X'$ be the section $x \mapsto (x, P_0')$. Define a subgroup of $\operatorname{Pic}(X \times X')$ as follows: $\operatorname{Pic}_\sigma = \{\mathcal{L} \in \operatorname{Pic}(X \times X') \mid \mathcal{L}$ has degree 0 along each fibre of p_1, and $\sigma^*\mathcal{L} = 0$ in $\operatorname{Pic} X\}$. Note that this subgroup is isomorphic to the group $\operatorname{Pic}^0(X'/X)$ used in the definition of the Jacobian variety. Hence there is a 1-1 correspondence between morphisms $f : X \to X'$ and elements $\mathcal{L}_f \in \operatorname{Pic}_\sigma$ (this defines $\mathcal{L}_f$). Now compute explicitly to show that $\Gamma_g^*(\mathcal{L}_f) = \Gamma_f^*(\mathcal{L}_g)$ for any f, g. Use the fact that $\mathcal{L}_{f+g} = \mathcal{L}_f \otimes \mathcal{L}_g$, and the fact that for any $\mathcal{L}$ on X', $p_2^*\mathcal{L} \in \operatorname{Pic}_\sigma$, to prove the result.] (Part (d) is starred in the book.)

(d) Using (d), show that for any $n \in \mathbf{Z}$, $\widehat{n_X} = n_X$. Conclude that $\deg n_X = n^2$.

(e) Show for any f that $\deg \hat{f} = \deg f$.

Solution. (As in the text, for results on End we use $\operatorname{char} k \neq 2$ via Prop. 4.10; for (d), whose printed hint is somewhat cryptic, I give a direct proof along the same lines, working with the universal property of Thm. 4.11 for the family $\mathcal{L}_f$ rather than with the graph morphisms.) Recall the group law: $P \mapsto \mathcal{L}(P - P_0)$ is an isomorphism $(X, P_0) \to \operatorname{Pic}^0 X$; and by Thm. 4.11, $J = X$ with $\mathcal{L} = \mathcal{L}(\Delta) \otimes p_1^*\mathcal{L}(-P_0)$ on $X \times X$ is a Jacobian: its restriction to $X \times \{t\}$ is $\mathcal{L}(t - P_0)$, and for every scheme T of finite type over k and every $\mathcal{M} \in \operatorname{Pic}^0(X/T) = \{\mathcal{M} \in \operatorname{Pic}(X \times T)$ of degree 0 on all fibres$\}/p_2^* \operatorname{Pic} T$ there is a unique morphism $\varphi : T \to X$ with $\varphi^*\mathcal{L} \cong \mathcal{M}$ in $\operatorname{Pic}^0(X/T)$; moreover (Remark 4.10.6 and Thm. 4.11) the group law μ of X is the one for which $\mu^*\mathcal{L} \cong p_1^*\mathcal{L} \otimes p_2^*\mathcal{L}$ in $\operatorname{Pic}^0(X/X \times X)$.

(a) Let $\mathcal{L}'$ be the universal sheaf on $X' \times X'$ and let $\mathcal{M} = (f \times 1_{X'})^*\mathcal{L}'$ on $X \times X'$. Its restriction to $X \times \{t'\}$ is $f^*\mathcal{L}(t' - P_0')$, of degree 0 (as $\deg f^*D = \deg f \cdot \deg D$). So $\mathcal{M} \in \operatorname{Pic}^0(X/X')$ and by Thm. 4.11 there is a unique morphism $\hat{f} : X' \to X$ with $\hat{f}^*\mathcal{L} \cong \mathcal{M}$ in $\operatorname{Pic}^0(X/X')$. Restricting to $X \times \{t'\}$: $\mathcal{L}(\hat{f}(t') - P_0) \cong f^*\mathcal{L}(t' - P_0')$, that is

$$\hat{f}(t') - P_0 \sim f^*(t' - P_0') \qquad (t' \in X').$$

So $\hat f$ is the morphism corresponding to $f^*:\operatorname{Pic}^0 X'\to\operatorname{Pic}^0 X$ under the isomorphisms $\operatorname{Pic}^0 X'\cong X'$, $\operatorname{Pic}^0 X\cong X$; as f^* is a homomorphism, $\hat f$ is one (it sends $P_0'\mapsto P_0$; also Lemma 4.9). It depends only on f and the base points, and $\hat f(t')$ is the group-law sum $\sum n_iR_i-\sum n_i'R_i'$ of $f^*(t')=\sum n_iR_i$ and $f^*(P_0')=\sum n_i'R_i'$. (If f is constant, f^* of a degree-0 divisor is 0 and $\hat f=0$.)

(b) For $Q''\in X''$: $\hat g(Q'')-P_0'\sim g^*(Q''-P_0'')$ and $\hat f(\hat gQ'')-P_0\sim f^*(\hat gQ''-P_0')\sim f^*g^*(Q''-P_0'')=(gf)^*(Q''-P_0'')$. By uniqueness in (a), $(g\circ f)^\wedge(Q'')=\hat f(\hat g(Q''))$. (If f or g is constant both sides are 0, since $\hat f(P_0')=P_0$ and constants have dual 0 by (a).)

(c) Assume f nonconstant (if $f=0$ everything is 0). Let $Q\in X$, $Q'=f(Q)$. Since $f^*(Q')-f^*(P_0')$ has degree 0, by (a) and the group law

$$\hat f(Q')=\sum f^*(Q')-\sum f^*(P_0')\quad\text{(sums in the group law, with multiplicities).}\tag{1}$$

f separable. f is a homomorphism, so $f\tau_a=\tau_{f(a)}f$ and the ramification index e_R is the same at all points; as f is separable, $e=1$, so $\ker f$ has exactly n points and $f^*(Q')=\sum_{\kappa\in\ker f}(Q+\kappa)$, $f^*(P_0')=\sum_\kappa\kappa$. Then (1) gives $\hat f(Q')=nQ+\sum\kappa-\sum\kappa=nQ$.

f purely inseparable ($n=p^e$). Then f is a bijection on points, $f^{-1}(Q')=\{Q\}$, so $f^*(Q')=nQ$ and $f^*(P_0')=nP_0$, and (1) gives $\hat f(Q')=nQ-nP_0=nQ$ (here nQ is the multiple in the group law, $P_0=0$).

General case. Let K_s be the separable closure of $f^*K(X')$ in $K(X)$; then $K(X)/K_s$ is purely inseparable of degree p^e, so $K(X)^{p^e}\subseteq K_s$ and, comparing degrees ($[K:K^{p^e}]=p^e$), $K_s=K(X)^{p^e}$. Thus $f=h\circ F$ where $F:X\to X^{(p^e)}$ is the p^e-th power Frobenius ($X^{(p^e)}$ is again an elliptic curve with base point $F(P_0)$) and $h:X^{(p^e)}\to X'$ is separable, $h(F(P_0))=P_0'$, $n=\deg h\cdot p^e$. By (b), $\hat f=\hat F\circ\hat h$, and

$$\hat f(f(Q))=\hat F\big(\hat h(h(FQ))\big)=\hat F\big((\deg h)\cdot FQ\big)=\hat F\big(F((\deg h)Q)\big)=p^e\deg h\cdot Q=nQ,$$

using the separable case for h, that F is a homomorphism, and the inseparable case for F. So $\hat f\circ f=n_X$.

Then $f\circ\hat f\circ f=f\circ n_X=n_{X'}\circ f$ (f is a homomorphism), i.e. $(f\hat f-n_{X'})\circ f=0$; as f is surjective, $f\circ\hat f=n_{X'}$.

(d) Let $f,g:X\to X'$ preserve base points; $f+g=\mu'\circ(f,g)$ with μ' the group law of X'. For a morphism $h:X\to X'$ put $\mathcal L_h:=(1_{X'}\times h)^*\mathcal L'\in\operatorname{Pic}(X'\times X)$, where $\mathcal L'=\mathcal L(\Delta')\otimes p_1^*\mathcal L(-P_0')$. Then $\mathcal L_h\cong\mathcal O(\Gamma_h)\otimes q^*\mathcal O(-P_0')$, where $q:X'\times X\to X'$ is the projection and $\Gamma_h=\{(h(x),x)\}$ is the (transposed) graph; its restriction to $X'\times\{x\}$ is $\mathcal L(h(x)-P_0')$, of degree 0, and its restriction to $\{Q'\}\times X$ is $\mathcal O_X(h^*Q')$ (as $\Gamma_h\cap(\{Q'\}\times X)=h^{-1}(Q')$ scheme-theoretically). By Thm. 4.11 applied to the curve X' with parameter space $T=X$ we have $\mathcal L_h=(h)^*\mathcal L'$ in $\operatorname{Pic}^0(X'/X)$, and for morphisms $f,g:T\to X'=J(X')$ the group law gives

$$(f+g)^*\mathcal L'=(f,g)^*\mu'^*\mathcal L'=(f,g)^*(p_1^*\mathcal L'\otimes p_2^*\mathcal L')=f^*\mathcal L'\otimes g^*\mathcal L'\quad\text{in }\operatorname{Pic}^0(X'/T).$$

Taking $T=X$: $\mathcal L_{f+g}\cong\mathcal L_f\otimes\mathcal L_g\otimes r^*N$ for some $N\in\operatorname{Pic}X$ ($r:X'\times X\to X$ the projection). Restrict to $\{Q'\}\times X$ (note r^*N restricts to N) and write $E_h(Q')=\mathcal L_h|_{\{Q'\}\times X}$; then

$$E_{f+g}(Q')\cong E_f(Q')\otimes E_g(Q')\otimes N\qquad(Q'\in X').$$

Taking $Q' = P_0'$ and dividing the two relations eliminates N:

$$E_{f+g}(Q') \otimes E_{f+g}(P_0')^{-1} \cong \big(E_f(Q') \otimes E_f(P_0')^{-1}\big) \otimes \big(E_g(Q') \otimes E_g(P_0')^{-1}\big).$$

For h nonconstant, $E_h(Q') = \mathcal{O}_X(h^*Q')$, so $E_h(Q') \otimes E_h(P_0')^{-1} = \mathcal{O}_X(h^*(Q' - P_0'))$; for h constant ($= P_0'$) we have $\mathcal{L}_h \cong \mathcal{O}$ (the pull-back of Δ' is $\{P_0'\} \times X$, cancelled by $q^*\mathcal{O}(-P_0')$), and the class is trivial, which we write $h^*(Q' - P_0') = 0$. Hence in all cases

$$(f+g)^*(Q' - P_0') \sim f^*(Q' - P_0') + g^*(Q' - P_0') \qquad \text{for all } Q'.$$

By the characterization of the dual in (a) (with the above convention for constants, whose dual is 0) this says $(f+g)^\wedge(Q') = \hat{f}(Q') + \hat{g}(Q')$, i.e. $(f+g)^\wedge = \hat{f} + \hat{g}$ (equal on closed points, hence equal as morphisms). (This is the content of the hint: $\mathcal{L}_{f+g} = \mathcal{L}_f \otimes \mathcal{L}_g$ in Pic_σ, and the term N is eliminated by normalizing at P_0'.)

(e) For the identity, $\hat{1}(Q) - P_0 \sim Q - P_0$, so $\hat{1}_X = 1_X$; for the zero map $\hat{0} = 0$. By (d), $(n \cdot 1)^\wedge = n\hat{1} = n_X$ for $n \geq 0$ by induction, and $0 = (1 + (-1))^\wedge = \hat{1} + \widehat{(-1)}$ gives $\widehat{(-1)} = -1$; so $\widehat{n_X} = n_X$ for all $n \in \mathbf{Z}$. For $n \neq 0$, apply (c) to $f = n_X$ (a nonconstant endomorphism, Prop. 4.10): $\widehat{n_X} \circ n_X = (\deg n_X)_X$, i.e. $(n^2)_X = n_X \circ n_X = (\deg n_X)_X$. Since $m \mapsto m_X$ is injective $\mathbf{Z} \to \mathrm{End}(X, P_0)$ (Prop. 4.10), $\deg n_X = n^2$.

(f) If $f(P_0) = P_0'$ and f is nonconstant, (c) gives $\hat{f} \circ f = n_X$ with $n = \deg f$; taking degrees, $\deg \hat{f} \cdot \deg f = \deg(n_X) = n^2 = (\deg f)^2$ by (e), so $\deg \hat{f} = \deg f$. For a general nonconstant f, write $f = \tau_a \circ f_0$ with $f_0(P_0) = P_0'$; then $\deg f = \deg f_0$ and $f^*(Q' - P_0') = f_0^*\big((Q' - a) - (P_0' - a)\big) \sim f_0^*(Q' - P_0')$ because the divisor $[Q' - a] - [P_0' - a]$ and $[Q'] - [P_0']$ both correspond to the point Q' of $\mathrm{Pic}^0 X' \cong X'$; hence $\hat{f} = \hat{f}_0$ and $\deg \hat{f} = \deg f$. □

Exercise 4.8

Problem. For any curve X, the *algebraic fundamental group* $\pi_1(X)$ is defined as $\varprojlim \mathrm{Gal}(K'/K)$, where K is the function field of X, and K' runs over all Galois extensions of K such that the corresponding curve X' is etale over X (III, Ex. 10.3). Thus, for example, $\pi_1(\mathbf{P}^1) = 1$ (2.5.3). Show that for an elliptic curve X,

$$\begin{aligned}
&\pi_1(X) = \textstyle\prod_{l \text{ prime}} \mathbf{Z}_l \times \mathbf{Z}_l && \text{if } \operatorname{char} k = 0;\\
&\pi_1(X) = \textstyle\prod_{l \neq p} \mathbf{Z}_l \times \mathbf{Z}_l && \text{if } \operatorname{char} k = p \text{ and Hasse } X = 0;\\
&\pi_1(X) = \mathbf{Z}_p \times \textstyle\prod_{l \neq p} \mathbf{Z}_l \times \mathbf{Z}_l && \text{if } \operatorname{char} k = p \text{ and Hasse } X \neq 0,
\end{aligned}$$

where $\mathbf{Z}_l = \varprojlim \mathbf{Z}/l^n$ is the l-adic integers. [*Hints*: Any Galois etale cover X' of an elliptic curve is again an elliptic curve. If the degree of X' over X is relatively prime to p, then X' can be dominated by the cover $n_X : X \to X$ for some n with $(n, p) = 1$. The Galois group of the covering n_X is $\mathbf{Z}/n \times \mathbf{Z}/n$. Etale covers of degree divisible by p can occur only if the Hasse invariant of X is not zero.]

Solution. Let $p = \operatorname{char} k$ ($p \neq 2$ in this section; the arguments below work for $p = 0$ or p odd), $K = K(X)$.

Step 0: etale covers are isogenies. Let $X' \to X$ be a connected etale cover (with K'/K Galois). By Hurwitz ($\deg R = 0$) $2g(X') - 2 = 0$, so X' is an elliptic curve; choose P_0' over P_0. Then $f : X' \to X$ is a separable homomorphism of degree N (the group law is the one with these base

points), and $\ker f$ has N points; the translations τ_κ, $\kappa \in \ker f$, are N distinct automorphisms of X' over X, so K'/K is Galois with group $\ker f$ (a finite abelian group). Conversely, a separable isogeny f is etale: ramification indices are constant by translation ($f\tau_a = \tau_{f(a)}f$) and equal 1 at a generic point. Moreover, by Ex. 4.7(c), for $f : X' \to X$ of degree N there is $\hat{f} : X \to X'$ with $f \circ \hat{f} = N_X$ and $\hat{f} \circ f = N_{X'}$ (and $\deg \hat{f} = N$).

Since all groups $\ker f$ are abelian, $\pi_1(X) = \varprojlim \ker f$; we compute this limit over a cofinal family of covers, each cover in the family dominating a given one; a cover Y dominating Y' induces a *surjection* of Galois groups.

Step 1: torsion of prime-to-p order. **Lemma A.** If $p \nmid n$ (any $n \geq 1$ in char 0), then n_X is separable of degree n^2 and $X[n] = \ker n_X \cong \mathbf{Z}/n \times \mathbf{Z}/n$. (This is the statement in Example 4.8.1.) *Proof.* $\deg n_X = n^2$ by Ex. 4.7(e). Let $\omega \neq 0$ be a regular differential on X ($g = 1$). Since $\Omega_X \cong \mathcal{O}_X$ (via ω), the pull-back of ω by the group law $\mu : X \times X \to X$, as a section of the relative differentials $\Omega_{X\times X/X} = p_1^*\Omega_X \cong \mathcal{O}_{X\times X}$ (relative to p_2), is a global section of $\mathcal{O}_{X\times X}$, hence a constant c times $p_1^*\omega$; restricting to $X \times \{P_0\}$ gives $c = 1$, and restricting to $X \times \{a\}$ gives $\tau_a^*\omega = \omega$: ω is translation invariant. For $h_1, h_2 : X \to X$, the differential of $h_1 + h_2 = \mu \circ (h_1, h_2)$ at x is $d\tau_{h_2(x)} \circ dh_{1,x} + d\tau_{h_1(x)} \circ dh_{2,x}$ (as $\mu(\cdot, b) = \tau_b$, $\mu(a, \cdot) = \tau_a$), so by invariance $(h_1+h_2)^*\omega = h_1^*\omega+h_2^*\omega$. By induction $n_X^*\omega = n\omega \neq 0$. If n_X were inseparable it would factor through Frobenius and $n_X^*\omega = 0$; so n_X is separable, and a separable homomorphism has $|\ker| = \deg = n^2$. Applying this to every divisor e of n, $|X[e]| = e^2$. A finite abelian group A of order n^2 with $|A[e]| = e^2$ for all $e \mid n$ is $\cong (\mathbf{Z}/n)^2$: for a prime $l \mid n$, $l^v \| n$, the l-part is $\bigoplus \mathbf{Z}/l^{b_i}$ with $\sum b_i = 2v$, $|A[l]| = l^2$ forces exactly two summands $b_1 + b_2 = 2v$, and $|A[l^v]| = l^{\min(b_1,v)+\min(b_2,v)} = l^{2v}$ forces $b_1, b_2 \geq v$, so $b_1 = b_2 = v$. □

Step 2: characteristic 0. Every etale connected cover $f : X' \to X$ of degree N is dominated by $N_X : X \to X$, since $N_X = f\circ\hat{f}$. Thus $\{N_X\}$ is cofinal and $\pi_1(X) = \varprojlim_N X[N]$, $X[N] \cong (\mathbf{Z}/N)^2$, the transition maps being surjections. Decomposing $N = \prod l^{v_l}$ and using the Chinese remainder theorem, $\pi_1(X) = \prod_l \varprojlim_v A_v^{(l)}$ with $A_v^{(l)} = X[l^v] \cong (\mathbf{Z}/l^v)^2$ and surjective transition maps. **Lemma B.** Such an inverse limit is $\cong \mathbf{Z}_l^2$. *Proof.* A surjection $\theta : (\mathbf{Z}/l^{v+1})^2 \to (\mathbf{Z}/l^v)^2$ induces a surjection, hence an isomorphism, on the quotients by l (both $\cong (\mathbf{Z}/l)^2$); so any lifts $\tilde{e}, \tilde{e}'$ of a basis e, e' of the target generate the source modulo l and therefore (Nakayama) form a basis of $(\mathbf{Z}/l^{v+1})^2$. Thus compatible bases (e_v, e'_v) of all A_v can be chosen inductively, giving an isomorphism of the inverse system with $(\mathbf{Z}/l^v)^2$ and the natural reductions, whose limit is $\mathbf{Z}_l^2$. □ Hence $\pi_1(X) = \prod_l \mathbf{Z}_l \times \mathbf{Z}_l$ in characteristic 0.

Step 3: characteristic p; Hasse invariant and p-torsion. We use Ex. 4.15, applied to $Y = X^{(p)}$ (so $Y_p = X$ and $F : X \to Y$ is the k-linear Frobenius of the exercise): the dual $\hat{F} : X^{(p)} \to X$ is separable iff the Hasse invariant of $X^{(p)}$ is 1. Now $X^{(p)}$ is X with the k-structure twisted by the Frobenius of k, hence has the same underlying ringed space and the same absolute Frobenius, so $X^{(p)}$ and X have the same Hasse invariant (as in the proof of 4.15). Thus $\hat{F}$ is separable iff the Hasse invariant of X is 1. Write $V = \hat{F}$; then $V \circ F = p_X$ (Ex. 4.7(c)), F is bijective on points, and separable degree and kernel satisfy $|\ker g| = \deg_s g$ for a homomorphism g (translation invariance). Hence $|X[p]| = \deg_s V = p$ if the Hasse invariant is 1 and $= 1$ if it is 0.

Lemma C. If $f : X' \to X$ is any isogeny, then $|X'[p]| = |X[p]|$. *Proof.* From $f \circ p_{X'} = p_X \circ f$ and multiplicativity of separable degrees, $\deg_s f \cdot \deg_s p_{X'} = \deg_s p_X \cdot \deg_s f$, and $\deg_s p_Y = |Y[p]|$ (as $p_Y = VF$). □ In particular $X^{(p^a)}$ (isogenous to X via F^a) has $|X^{(p^a)}[p]| = |X[p]|$: it is ordinary iff X is. Also an etale cover of degree divisible by p gives a point of order p on X'

(Cauchy in $\ker f$), so $|X'[p]| \neq 1$, hence by Lemma C $|X[p]| \neq 1$:

If $\operatorname{Hasse} X = 0$, *every etale cover has degree prime to* p. Then $f : X' \to X$ of degree N, $p \nmid N$, satisfies $f \circ \hat{f} = N_X$, which is separable (Lemma A), so $\hat{f}$ is separable; N_X factors through f, i.e. $\{N_X : p \nmid N\}$ is cofinal. By Lemma A, $\ker N_X \cong (\mathbf{Z}/N)^2$, and as in Step 2 (with $l \neq p$) $\pi_1(X) = \prod_{l \neq p} \mathbf{Z}_l^2$.

If $\operatorname{Hasse} X \neq 0$. Put $X_a = X^{(p^a)}$, an ordinary curve with $V_a := (F^a)^\wedge = \hat{F}_X \hat{F}_{X_1} \cdots : X_a \to X$ (Ex. 4.7(b)); each factor is separable (each X_j is ordinary), so V_a is separable of degree p^a, etale, and $V_a \circ F^a = (p^a)_X$, $F^a \circ V_a = (p^a)_{X_a}$. Since $\deg_s p_{X_a} = p$ we get $|X_a[p^a]| = p^a$ and $|X_a[p]| = p$, so $X_a[p^a]$ is cyclic; $\ker V_a \subseteq X_a[p^a]$ (as $(p^a)_{X_a} = F^a V_a$) has order p^a, so $\ker V_a = X_a[p^a] \cong \mathbf{Z}/p^a$. For $a \geq 0$ and m prime to p let

$$c_{a,m} = V_a \circ m_{X_a} : X_a \to X,$$

etale (both factors are separable), of degree $p^a m^2$, with Galois group $G_{a,m} = \ker c_{a,m}$ of order $p^a m^2$. The p-Sylow of $G_{a,m}$ is $\ker V_a \cong \mathbf{Z}/p^a$ (it is contained in $G_{a,m}$ since $V_a(mx) = mV_a x$) and the prime-to-p part is $\ker m_{X_a} \cong (\mathbf{Z}/m)^2$ (Lemma A for X_a): $G_{a,m} \cong \mathbf{Z}/p^a \times (\mathbf{Z}/m)^2$.

Cofinality. Let $f : X' \to X$ be etale of degree $N = p^a m$ with $p \nmid m$. Then $f \circ \hat{f} = N_X = m_X \circ V_a \circ F^a = V_a \circ m_{X_a} \circ F^a = c_{a,m} \circ F^a$. The inseparable degree of N_X is p^a (as m_X, V_a are separable), f is separable, so $\hat{f}$ has inseparable degree exactly p^a; therefore (as in (c) of Ex. 4.7) $\hat{f} = h \circ F^a$ with $h : X_a \to X'$. Then $f \circ h \circ F^a = c_{a,m} \circ F^a$, so $f \circ h = c_{a,m}$ (F^a is surjective): $c_{a,m}$ dominates f. Thus the $c_{a,m}$ ($a \geq 0$, $p \nmid m$; ordered by $a' \leq a$, $m' \mid m$) form a cofinal system (directed: $V_a = V_{a'} \circ W$ with $W = \hat{F}_{X_{a'}} \cdots \hat{F}_{X_{a-1}} : X_a \to X_{a'}$, so for $a' \leq a$, $m' \mid m$ one has $c_{a,m} = c_{a',m'} \circ W \circ (m/m')_{X_a}$, a morphism over X).

The limit. The chain $(a_k, m_k) = (k, k!_{p'})$, $k \geq 1$ ($k!_{p'}$ the prime-to-p part of $k!$) is cofinal in the directed set above, so $\pi_1(X) = \varprojlim_k G_k$, $G_k = G_{k,k!_{p'}} = P_k \times Q_k$ with $P_k = \ker V_k \cong \mathbf{Z}/p^k$ and $Q_k = \ker(k!_{p'})_{X_k} \cong (\mathbf{Z}/k!_{p'})^2$. The transition maps are the surjective homomorphisms induced by the dominations (restrictions of $W \circ (\cdot)_{X_k}$ above, which are homomorphisms); they respect the decomposition into p-primary and prime-to-p parts. Hence $\pi_1(X) = \varprojlim P_k \times \varprojlim Q_k$. The P_k are cyclic p-groups with surjective transitions, so compatible generators can be chosen inductively and $\varprojlim P_k = \mathbf{Z}_p$. Decomposing $Q_k = \bigoplus_{l \neq p} Q_k^{(l)}$, $Q_k^{(l)} \cong (\mathbf{Z}/l^{v_l(k)})^2$, with surjective transitions, we get $\varprojlim Q_k = \prod_{l \neq p} \mathbf{Z}_l^2$ by Lemma B (and since $v_l(k) \to \infty$). Hence $\pi_1(X) = \mathbf{Z}_p \times \prod_{l \neq p} \mathbf{Z}_l \times \mathbf{Z}_l$. □

Exercise 4.9

Problem. We say two elliptic curves X, X' are *isogenous* if there is a finite morphism $f : X \to X'$.

(a) Show that isogeny is an equivalence relation.

(b) For any elliptic curve X, show that the set of elliptic curves X' isogenous to X, up to isomorphism, is countable. [*Hint*: X' is uniquely determined by X and $\ker f$.]

Solution. A finite morphism of curves is a nonconstant morphism. Composing with a translation of X' we may always assume $f(P_0) = P_0'$, so that f is a homomorphism (Lemma 4.9), a *finite homomorphism* with finite kernel. As above $\operatorname{char} k \neq 2$ where Prop. 4.10 is used.

(a) Reflexive: 1_X. Transitive: the composition of nonconstant morphisms is nonconstant. Symmetric: if $f : X \to X'$ is finite with $f(P_0) = P_0'$, $n = \deg f \geq 1$, then by Ex. 4.7(c) $\hat{f} \circ f = n_X$, which is nonconstant (Prop. 4.10); hence $\hat{f} : X' \to X$ is nonconstant, so finite, and X' is isogenous to X implies X is isogenous to X'.

(b) Let $p = \operatorname{char} k$ ($p = 0$ allowed). Every nonconstant $f : X \to X'$ factors as $X \xrightarrow{F^a} X^{(p^a)} \xrightarrow{h} X'$ with F^a the p^a-th power Frobenius and h separable (the separable closure of $f^*K(X')$ in $K(X)$ equals $K(X)^{p^a}$, as shown in Ex. 4.7(c); $a = 0$ if $p = 0$); $X^{(p^a)}$ is an elliptic curve depending only on X and a, and h is a separable homomorphism (base point $F^a(P_0) \mapsto P_0'$) with kernel $\Lambda = \ker h$, a finite subgroup of $X^{(p^a)}(k)$, $|\Lambda| = \deg h$.

Claim: X' is determined up to isomorphism by (a, Λ). Let $Y = X^{(p^a)}$ and let $h_i : Y \to X_i'$ ($i = 1, 2$) be separable homomorphisms with the same kernel Λ. The translations τ_κ, $\kappa \in \Lambda$, satisfy $h_i\tau_\kappa = h_i$, so $h_i^*K(X_i') \subseteq K(Y)^\Lambda$; as $[K(Y) : h_i^*K(X_i')] = \deg h_i = |\Lambda| = [K(Y) : K(Y)^\Lambda]$, we get $h_i^*K(X_i') = K(Y)^\Lambda$. Hence $K(X_1') \cong K(X_2')$ over k and $X_1' \cong X_2'$ (equivalence of curves and function fields).

Countability. The set of finite subgroups of $Y(k)$ is countable: every element of a finite subgroup is a torsion point, and $Y[m] = \ker m_Y$ is finite (of order $\leq \deg m_Y = m^2$, Ex. 4.7(e)) for each $m \geq 1$, so the torsion subgroup $\bigcup_m Y[m]$ is countable, and a countable set has countably many finite subsets. So the set of pairs (a, Λ) ($a \geq 0$ an integer, $\Lambda \subseteq X^{(p^a)}(k)$ finite subgroup) is countable. By (a), every X' isogenous to X receives a finite $f : X \to X'$, giving a pair (a, Λ) from which X' is determined up to isomorphism. Hence there are at most countably many isomorphism classes of elliptic curves isogenous to X. □

Exercise 4.10

Problem. If X is an elliptic curve, show that there is an exact sequence

$$0 \to p_1^* \operatorname{Pic} X \oplus p_2^* \operatorname{Pic} X \to \operatorname{Pic}(X \times X) \to R \to 0,$$

where $R = \operatorname{End}(X, P_0)$. In particular, we see that $\operatorname{Pic}(X \times X)$ is bigger than the sum of the Picard groups of the factors. Cf. (III, 12.6), (V, Ex. 1.6).

Solution. Let $p_1, p_2 : X \times X \to X$ be the projections; we regard p_2 as the structure map of a family of curves $X \times \{t\} \cong X$ (first factor), parametrized by $t \in X$ (second factor). For $\mathcal{L} \in \operatorname{Pic}(X \times X)$ write $\mathcal{L}_t = \mathcal{L}|_{X \times \{t\}} \in \operatorname{Pic} X$.

Definition of the map $\Psi : \operatorname{Pic}(X \times X) \to R$. Put $\mathcal{M} = \mathcal{L} \otimes p_1^*(\mathcal{L}_{P_0})^{-1}$. Then $\mathcal{M}_t = \mathcal{L}_t \otimes \mathcal{L}_{P_0}^{-1}$ has degree 0 for all t ($\deg \mathcal{L}_t$ is constant in t: $\mathcal{L}$ is flat over X and the degree is locally constant, III, 9.9). So $\mathcal{M} \in \operatorname{Pic}^0(X/X)$, and by Thm. 4.11 (universal property of $J = X$, with $T = X$) there is a unique morphism $\Psi(\mathcal{L}) : X \to X$ with $\mathcal{L}(\Psi(\mathcal{L})(t) - P_0) \cong \mathcal{M}_t = \mathcal{L}_t \otimes \mathcal{L}_{P_0}^{-1}$ for all t. Since $\mathcal{M}_{P_0} = \mathcal{O}$, $\Psi(\mathcal{L})(P_0) = P_0$, so $\Psi(\mathcal{L}) \in \operatorname{End}(X, P_0) = R$. In words: $\Psi(\mathcal{L})(t)$ is the point of $X \cong \operatorname{Pic}^0 X$ corresponding to $\mathcal{L}_t \otimes \mathcal{L}_{P_0}^{-1}$.

Ψ is a homomorphism. $(\mathcal{L} \otimes \mathcal{L}')_t \otimes (\mathcal{L} \otimes \mathcal{L}')_{P_0}^{-1} = \left(\mathcal{L}_t\mathcal{L}_{P_0}^{-1}\right) \otimes \left(\mathcal{L}_t'\mathcal{L}_{P_0}'^{-1}\right)$, and the identification $X \cong \operatorname{Pic}^0 X$ is a group isomorphism, so $\Psi(\mathcal{L} \otimes \mathcal{L}')(t) = \Psi(\mathcal{L})(t) + \Psi(\mathcal{L}')(t)$, i.e. $\Psi(\mathcal{L} \otimes \mathcal{L}') = \Psi(\mathcal{L}) + \Psi(\mathcal{L}')$ in R (the sum in R being pointwise).

Ψ *is surjective.* Let $\alpha \in R$ and $\mathcal{L} = (1_X \times \alpha)^*\big(\mathcal{L}(\Delta) \otimes p_1^*\mathcal{L}(-P_0)\big)$. Then $\mathcal{L}_t = \mathcal{L}(\alpha(t) - P_0)$, and $\mathcal{L}_{P_0} = \mathcal{L}(\alpha(P_0) - P_0) = \mathcal{O}$, so $\Psi(\mathcal{L})(t)$ is the point corresponding to $\mathcal{L}(\alpha(t) - P_0)$, namely $\alpha(t)$: $\Psi(\mathcal{L}) = \alpha$.

The image of $\iota = p_1^* \oplus p_2^* : \operatorname{Pic} X \oplus \operatorname{Pic} X \to \operatorname{Pic}(X \times X)$ *is* $\ker \Psi$. If $\mathcal{L} = p_1^*A \otimes p_2^*B$, then $\mathcal{L}_t \cong A$ for every t, so $\mathcal{M}_t \cong \mathcal{O}$ and $\Psi(\mathcal{L}) = 0$ (as $\Psi(\mathcal{L})(t) = P_0$). Conversely if $\Psi(\mathcal{L}) = 0$, then $\mathcal{M}_t \cong \mathcal{O}_X$ for all $t \in X$, so by the criterion recalled in the text (III, Ex. 12.4: for T integral of finite type, $\mathcal{M} \in \operatorname{Pic}(X \times T)$ is trivial on every fibre $X \times \{t\}$ iff $\mathcal{M} \in p_2^* \operatorname{Pic} T$) $\mathcal{M} \cong p_2^*B$ for some $B \in \operatorname{Pic} X$. Then $\mathcal{L} \cong p_1^*(\mathcal{L}_{P_0}) \otimes p_2^*B \in \operatorname{im} \iota$.

ι *is injective.* If $p_1^*A \otimes p_2^*B \cong \mathcal{O}$, restricting to $X \times \{t\}$ gives $A \cong \mathcal{O}$ (as p_2^*B is trivial on it), and restricting to $\{s\} \times X$ gives $B \cong \mathcal{O}$.

So the sequence is exact. Since $R \supseteq \mathbf{Z} \cdot 1_X \neq 0$ (indeed $\mathbf{Z} \hookrightarrow R$ by Prop. 4.10), $\operatorname{Pic}(X \times X)$ is strictly larger than $p_1^* \operatorname{Pic} X \oplus p_2^* \operatorname{Pic} X$; e.g. $\mathcal{O}(\Delta)$ (with $\Psi(\mathcal{O}(\Delta)) = 1_X \neq 0$; likewise $\mathcal{O}(\Gamma_f)$ for a nonconstant $f \in R$, where $\Psi = \hat{f} \neq 0$) is not of the form $p_1^*A \otimes p_2^*B$. □

Exercise 4.11

Problem. Let X be an elliptic curve over $\mathbf{C}$, defined by the elliptic functions with periods $1, \tau$. Let R be the ring of endomorphisms of X.

(a) If $f \in R$ is a nonzero endomorphism corresponding to complex multiplication by α, as in (4.18), show that $\deg f = |\alpha|^2$.

(b) If $f \in R$ corresponds to $\alpha \in \mathbf{C}$ again, show that the dual $\widehat{f}$ of (Ex. 4.7) corresponds to the complex conjugate $\bar{\alpha}$.

(c) If $\tau \in \mathbf{Q}(\sqrt{-d})$ happens to be integral over $\mathbf{Z}$, show that $R = \mathbf{Z}[\tau]$.

Solution. Let $\Lambda = \mathbf{Z} + \mathbf{Z}\tau$ (with $\tau \notin \mathbf{R}$), so $X \cong \mathbf{C}/\Lambda$ as a group and, by Prop. 4.18 and its proof, $R \to \mathbf{C}$, $f \mapsto \alpha_f$, is an injective ring homomorphism whose image is $\{\alpha \in \mathbf{C} : \alpha\Lambda \subseteq \Lambda\}$; under it $n_X \mapsto n$ (Remark 4.18.1).

(a) Let $f \neq 0$, so $\alpha = \alpha_f \neq 0$. In characteristic 0 the morphism f is finite and separable, and it is a group homomorphism, so $\deg f = \#\ker f$ (as in the proof of Cor. 4.17). Now

$$\ker f = \{z \in \mathbf{C}/\Lambda : \alpha z \in \Lambda\} = \alpha^{-1}\Lambda/\Lambda,$$

and multiplication by α is an isomorphism $\alpha^{-1}\Lambda/\Lambda \to \Lambda/\alpha\Lambda$, so $\deg f = [\Lambda : \alpha\Lambda]$. Write $\Lambda = \mathbf{Z}\omega_1 \oplus \mathbf{Z}\omega_2$ and $\alpha\omega_i = \sum_j m_{ij}\omega_j$ with $M = (m_{ij}) \in M_2(\mathbf{Z})$; the sublattice $\alpha\Lambda$ has index $|\det M|$. Since ω_1, ω_2 are also an $\mathbf{R}$-basis of $\mathbf{C}$ ($\tau \notin \mathbf{R}$), M is the matrix of the $\mathbf{R}$-linear map "multiplication by α" of $\mathbf{C} = \mathbf{R}^2$, whose determinant is $|\alpha|^2$ (in the basis $1, i$ it is $\begin{pmatrix} a & b \\ -b & a \end{pmatrix}$ for $\alpha = a + bi$, and determinants do not depend on the basis). Hence $\deg f = |\alpha|^2$.

(b) The dual $\widehat{f}$ is again in R (Ex. 4.7(a), with $X' = X$, $P_0' = P_0$). If $f = 0$ then f^* sends every degree 0 class to the trivial class, so $\widehat{f} = 0$ and the claim is trivial. Let $f \neq 0$, $n = \deg f = |\alpha|^2 = \alpha\bar{\alpha}$ by (a). By Ex. 4.7(c), $\widehat{f} \circ f = n_X$. Applying the injective ring homomorphism $R \to \mathbf{C}$ and writing $\beta = \alpha_{\widehat{f}}$, we get $\beta\alpha = n = \alpha\bar{\alpha}$. Since $\alpha \neq 0$, $\beta = \bar{\alpha}$.

(c) Write $\tau = r + s\sqrt{-d}$ with $r, s \in \mathbf{Q}$ ($s \neq 0$ since $\tau \notin \mathbf{R}$). Since τ is integral over $\mathbf{Z}$, $\tau \cdot 1 = \tau \in \Lambda$ and $\tau \cdot \tau = \tau^2$; the minimal polynomial of τ over $\mathbf{Q}$ is $x^2 - 2rx + (r^2 + ds^2)$, which lies in $\mathbf{Z}[x]$

because $\mathbf{Z}$ is integrally closed. So $\tau^2 = 2r\tau - (r^2 + ds^2) \in \mathbf{Z} + \mathbf{Z}\tau = \Lambda$, i.e. $\tau\Lambda \subseteq \Lambda$, i.e. $\tau \in R$ (identifying R with its image in $\mathbf{C}$). Hence $\mathbf{Z}[\tau] = \mathbf{Z} + \mathbf{Z}\tau = \Lambda \subseteq R$. Conversely every $\alpha \in R$ satisfies $\alpha = \alpha \cdot 1 \in \alpha\Lambda \subseteq \Lambda$. Therefore $R = \Lambda = \mathbf{Z}[\tau]$. (This is consistent with Thm. 4.19: here $2r \in \mathbf{Z}$ and $r^2 + ds^2 \in \mathbf{Z}$, so the conditions $2br$, $b(r^2 + ds^2) \in \mathbf{Z}$ hold for every $b \in \mathbf{Z}$.) $\square$

Exercise 4.12

Problem. Again let X be an elliptic curve over $\mathbf{C}$ determined by the elliptic functions with periods $1, \tau$, and assume that τ lies in the region G of (4.15B).

(a) If X has any automorphisms leaving P_0 fixed other than ± 1, show that either $\tau = i$ or $\tau = \omega$, as in (4.20.1), (4.20.2). This gives another proof of the fact (4.7) that there are only two curves, up to isomorphism, having automorphisms other than ± 1.

(b) Now show that there are exactly three values of τ for which X admits an endomorphism of degree 2. Can you match them with the three values of j determined in (Ex. 4.5)? [Answers: $\tau = i$; $\tau = \sqrt{-2}$; $\tau = \frac{1}{2}(-1 + \sqrt{-7})$.]

Solution. Throughout $\Lambda = \mathbf{Z} + \mathbf{Z}\tau$, $\omega = \frac{1}{2}(-1 + \sqrt{-3})$ (so $\omega^3 = 1$), and $\operatorname{Im}\tau > 0$ for $\tau \in G$ (Fig. 16). Both i and ω lie in G ($\operatorname{Re} \le 0$ and absolute value 1). We identify $R = \operatorname{End}(X, P_0)$ with a subring of $\mathbf{C}$ as in Prop. 4.18 (its image is $\{\alpha : \alpha\Lambda \subseteq \Lambda\}$), and use Thm. 4.15B as follows: if $\Lambda = \gamma(\mathbf{Z} + \mathbf{Z}\theta)$ for some $\gamma \in \mathbf{C}^*$ and θ with $\operatorname{Im}\theta > 0$, then expressing the basis $(1, \tau)$ and $(\gamma, \gamma\theta)$ of Λ through one another we find $\theta = (a\tau + b)/(c\tau + d)$ with $\begin{pmatrix} a & b \\ c & d \end{pmatrix} \in GL_2(\mathbf{Z})$, and $\operatorname{Im}\theta = (ad - bc)\operatorname{Im}\tau/|c\tau + d|^2 > 0$ forces $ad - bc = 1$, so $J(\tau) = J(\theta)$ by (4.15B), and, if $\theta \in G$ too, the uniqueness statement of (4.15B) gives $\tau = \theta$. $(*)$

A lemma on lattices. Let O be one of the rings $\mathbf{Z}[i]$, $\mathbf{Z}[\omega]$, $\mathbf{Z}[\sqrt{-2}]$, $\mathbf{Z}[\frac{1+\sqrt{-7}}{2}]$, and let $\Lambda \subset \mathbf{C}$ be a lattice with $O\Lambda \subseteq \Lambda$. Then $\Lambda = \gamma O$ for a shortest nonzero vector γ of Λ. Indeed, every $z \in \mathbf{C}$ is at distance < 1 from O: the lattice O is tiled by a square of side 1, an equilateral triangle of side 1, a rectangle $1 \times \sqrt{2}$, resp. acute triangles with sides $1, \sqrt{2}, \sqrt{2}$, and the largest distance from a vertex is at most $\frac{\sqrt{2}}{2}, \frac{1}{\sqrt{3}}, \frac{\sqrt{3}}{2}, \frac{2}{\sqrt{7}}$ respectively (half diagonal, resp. circumradius of an acute triangle), all < 1. Now let $v \in \Lambda$. Choose $\beta \in O$ with $|v/\gamma - \beta| < 1$. Then $v - \beta\gamma \in \Lambda$ (as $\beta\gamma \in O\Lambda \subseteq \Lambda$) and $|v - \beta\gamma| < |\gamma|$, so $v = \beta\gamma$ by minimality of $|\gamma|$. Thus $\Lambda = O\gamma$.

(a) Let $\sigma \neq \pm 1$ be an automorphism of X with $\sigma(P_0) = P_0$. By Lemma 4.9 it is a homomorphism, so $\sigma \in R$, and $\sigma^{-1} \in R$, i.e. σ is a unit of R. Let $\alpha \in \mathbf{C}$ correspond to it; then $\alpha \neq \pm 1$, $\alpha^{-1} \in R$, and by Ex. 4.11(a), $|\alpha|^2 = \deg\sigma = 1$. As $|\alpha| = 1$ and $\alpha \neq \pm 1$, $\alpha \notin \mathbf{R}$. Since $R \cap \mathbf{R} = \mathbf{Z}$ (proof of (4.19)), $R \neq \mathbf{Z}$, so X has complex multiplication and by Thm. 4.19, R is a subring of the ring of integers $\mathcal{O}_K$ of an imaginary quadratic field K. So α is an algebraic integer in K, $t = \alpha + \bar{\alpha} \in \mathbf{Z}$ (its trace), $\alpha\bar{\alpha} = 1$, and $|t| = 2|\operatorname{Re}\alpha| \le 2$. Thus α is a root of $x^2 - tx + 1$ with $t \in \{0, \pm 1\}$ (for $t = \pm 2$ we would get $\alpha = \pm 1$), i.e. $\alpha \in \{\pm i,\ \omega, \omega^2,\ -\omega, -\omega^2\}$. If $\alpha = \pm i$ then $i \in R$, and if $\alpha \in \{\omega, \omega^2, -\omega, -\omega^2\}$ then $\omega \in \{\alpha, \alpha^2, \alpha^4\} \subseteq R$ (indeed $\alpha = \omega$; $\alpha = \omega^2 \Rightarrow \alpha^2 = \omega$; $\alpha = -\omega \Rightarrow \alpha^4 = \omega^4 = \omega$; $\alpha = -\omega^2 \Rightarrow \alpha^2 = \omega$). So Λ is stable under $\mathbf{Z}[i]$ or under $\mathbf{Z}[\omega]$, and by the Lemma $\Lambda = \gamma\mathbf{Z}[i] = \gamma(\mathbf{Z} + \mathbf{Z}i)$, resp. $\Lambda = \gamma(\mathbf{Z} + \mathbf{Z}\omega)$. By $(*)$ and $i, \omega \in G$ we obtain $\tau = i$, resp. $\tau = \omega$.

Conversely, for $\tau = i$, $i\Lambda = \Lambda$ and for $\tau = \omega$, $\omega\Lambda = \Lambda$, so these two curves do have the automorphisms $\pm i$, resp. $\pm\omega, \pm\omega^2$ (Examples 4.20.1, 4.20.2). Every elliptic curve over $\mathbf{C}$ is

isomorphic to some X_τ with $\tau \in G$ (4.14B, 4.15B), and $X_\tau \cong X_{\tau'}$ with $\tau, \tau' \in G$ forces $\tau = \tau'$ (by J and 4.15B, 4.1(b)). Hence, up to isomorphism, there are exactly two elliptic curves over $\mathbf{C}$ with an automorphism fixing P_0 other than ± 1: X_i ($j = 1728$) and X_ω ($j = 0$). This reproves Cor. 4.7.

(b) *Which τ.* If $f \in R$ has degree 2, with $\alpha \in \mathbf{C}$ corresponding to it, then $|\alpha|^2 = 2$ by Ex. 4.11(a). Since $\pm\sqrt{2} \notin \mathbf{Z} = R \cap \mathbf{R}$, $\alpha \notin \mathbf{R}$, so X has complex multiplication and α is an algebraic integer in an imaginary quadratic field (Thm. 4.19). Then $t = \alpha + \bar{\alpha} \in \mathbf{Z}$, $\alpha\bar{\alpha} = 2$, $t^2 < 8$, so $t \in \{0, \pm 1, \pm 2\}$ and α is a root of $x^2 - tx + 2$:

$$\alpha = \pm\sqrt{-2} \quad (t = 0), \qquad \alpha = \tfrac{1}{2}(\pm 1 \pm \sqrt{-7}) \quad (t = \pm 1), \qquad \alpha = \pm 1 \pm i \quad (t = \pm 2).$$

The subring $\mathbf{Z}[\alpha]$ is, respectively, $O = \mathbf{Z}[\sqrt{-2}]$, $O = \mathbf{Z}[\frac{1+\sqrt{-7}}{2}]$ (note $\frac{-1+\sqrt{-7}}{2} = \frac{1+\sqrt{-7}}{2} - 1$), $O = \mathbf{Z}[i]$. As $\alpha\Lambda \subseteq \Lambda$ we have $O\Lambda \subseteq \Lambda$, so by the Lemma $\Lambda = \gamma O = \gamma(\mathbf{Z} + \mathbf{Z}\theta)$ with $\theta = \sqrt{-2}$, $\frac{1+\sqrt{-7}}{2}$, i respectively. By $(*)$ (note $\mathbf{Z}+\mathbf{Z}\frac{1+\sqrt{-7}}{2} = \mathbf{Z}+\mathbf{Z}\frac{-1+\sqrt{-7}}{2}$) we get $\tau = \theta'$, where $\theta' = \sqrt{-2}$, $\frac{-1+\sqrt{-7}}{2}$, i, which all lie in G ($|\theta'|^2 = 2, 2, 1$ and $\operatorname{Re}\theta' = 0, -\frac{1}{2}, 0$).

Conversely, for these three τ there is an endomorphism of degree 2: for $\tau = i$, $\alpha = 1 + i$ satisfies $\alpha\Lambda \subseteq \Lambda$; for $\tau = \sqrt{-2}$, $\alpha = \tau$ ($\tau \cdot \tau = -2 \cdot 1$); for $\tau = \frac{-1+\sqrt{-7}}{2}$, $\alpha = \tau$ since $\tau^2 + \tau + 2 = 0$ gives $\tau \cdot \tau = -\tau - 2 \in \Lambda$. In each case $|\alpha|^2 = 2$, so by Prop. 4.18 and Ex. 4.11(a) there is an endomorphism of degree 2. So there are exactly three values of $\tau \in G$: i, $\sqrt{-2}$, $\frac{1}{2}(-1 + \sqrt{-7})$.

Matching with j. By (Ex. 4.5(d)) the curves with an endomorphism of degree 2 have $j \in \{2^6 \cdot 3^3,\ 2^6 \cdot 5^3,\ -3^3 \cdot 5^3\} = \{1728, 8000, -3375\}$. The three τ's give three pairwise non-isomorphic curves (4.15B, 4.1(b)), so j is injective on them, and since each of the three curves has j in the above set, the matching is a bijection. For $\tau = i$ we have $j = 1728$ (Example 4.20.1). For the other two we use the real structure. Let g_2, g_3 be those of Λ and e_1, e_2, e_3 the values of $\wp$ at the three nonzero half periods $\frac{1}{2}, \frac{\tau}{2}, \frac{1+\tau}{2}$ (distinct, the roots of $4x^3 - g_2 x - g_3$), so $\Delta = g_2^3 - 27g_3^2 = 16\prod_{i<j}(e_i - e_j)^2$ and $j = 1728 g_2^3/\Delta$, $j - 1728 = 1728 \cdot 27\, g_3^2/\Delta$. Since $\bar{\Lambda} = \Lambda$ (for $\tau = \sqrt{-2}$, $\bar{\tau} = -\tau$; for $\tau = \frac{-1+\sqrt{-7}}{2}$, $\bar{\tau} = -1 - \tau$), we have $\wp(\bar{z}) = \overline{\wp(z)}$, so g_2, g_3 are real and complex conjugation permutes the e_i according to its action on the half periods mod Λ.

- $\tau = \sqrt{-2}$: conjugation fixes each of $\frac{1}{2}, \frac{\tau}{2}, \frac{1+\tau}{2}$ mod Λ (e.g. $\overline{(1+\tau)/2} = (1-\tau)/2 \equiv (1+\tau)/2$), so all e_i are real and distinct, $\Delta > 0$, hence $j - 1728 \geq 0$.
- $\tau = \frac{-1+\sqrt{-7}}{2}$: $\overline{\tau/2} = \frac{-1-\tau}{2} \equiv \frac{1+\tau}{2}$, so $e_2 = \wp(\tau/2)$ and $e_3 = \wp(\frac{1+\tau}{2})$ are complex conjugate, and not real since $e_2 \neq e_3$, while e_1 is real. Then $(e_1 - e_2)(e_1 - e_3) = |e_1 - e_2|^2 > 0$ and $(e_2 - e_3)^2 < 0$ ($e_2 - e_3$ is purely imaginary and nonzero), so $\Delta < 0$, hence $j - 1728 \leq 0$.

As $j \in \{1728, 8000, -3375\}$ and $j \neq 1728$ for these two curves (they are not isomorphic to X_i), the first gives $j = 8000$ and the second $j = -3375$. (Numerical confirmation: from $j = q^{-1} + 744 + 196884q + \cdots$, $q = e^{2\pi i\tau}$, one gets $j(\sqrt{-2}) \approx 8000$ and $j(\frac{-1+\sqrt{-7}}{2}) \approx -3375$; I also checked this with the Eisenstein series to high precision.) Hence

$$\tau = i \leftrightarrow j = 2^6 \cdot 3^3 = 1728, \qquad \tau = \sqrt{-2} \leftrightarrow j = 2^6 \cdot 5^3 = 8000, \qquad \tau = \tfrac{1}{2}(-1+\sqrt{-7}) \leftrightarrow j = -3^3 \cdot 5^3 = -3375.$$

□

Exercise 4.13

Problem. If $p = 13$, there is just one value of j for which the Hasse invariant of the corresponding curve is 0. Find it. [Answer: $j = 5 \pmod{13}$.]

Solution. Let $k = \overline{\mathbf{F}}_{13}$. Every elliptic curve over k is isomorphic to some $X_\lambda : y^2 = x(x-1)(x-\lambda)$ with $\lambda \neq 0, 1$ (Prop. 4.6), with $j = j(\lambda) = 2^8(\lambda^2-\lambda+1)^3/(\lambda^2(\lambda-1)^2)$, and by Cor. 4.22 X_λ has Hasse invariant 0 iff $h(\lambda) = 0$, where for $p = 13$, $k' = 6$,

$$h(\lambda) = \sum_{i=0}^{6} \binom{6}{i}^2 \lambda^i = 1+36\lambda+225\lambda^2+400\lambda^3+225\lambda^4+36\lambda^5+\lambda^6 \equiv \lambda^6+10\lambda^5+4\lambda^4+10\lambda^3+4\lambda^2+10\lambda+1 \pmod{13}.$$

Consider the polynomial $g(\lambda) = 2^8(\lambda^2-\lambda+1)^3 - 5\lambda^2(\lambda-1)^2$, so that for $\lambda \neq 0, 1$, $j(\lambda) = 5 \iff g(\lambda) = 0$. Expanding, $(\lambda^2-\lambda+1)^3 = \lambda^6 - 3\lambda^5 + 6\lambda^4 - 7\lambda^3 + 6\lambda^2 - 3\lambda + 1$ and $5\lambda^2(\lambda-1)^2 = 5\lambda^4 - 10\lambda^3 + 5\lambda^2$, so

$$g = 256\lambda^6-768\lambda^5+1531\lambda^4-1782\lambda^3+1531\lambda^2-768\lambda+256 \equiv 9\lambda^6-\lambda^5+10\lambda^4-\lambda^3+10\lambda^2-\lambda+9 \pmod{13},$$

using $256 \equiv 9$, $768 \equiv 1$, $1531 \equiv 10$, $1782 \equiv 1$. On the other hand $-4h \equiv -4\lambda^6 - 40\lambda^5 - 16\lambda^4 - 40\lambda^3 - 16\lambda^2 - 40\lambda - 4 \equiv 9\lambda^6 - \lambda^5 + 10\lambda^4 - \lambda^3 + 10\lambda^2 - \lambda + 9$. Hence $g \equiv -4h \pmod{13}$, i.e. $h \equiv 3g$. Neither polynomial vanishes at $\lambda = 0$ ($h(0) = 1$, $g(0) = 9$) or at $\lambda = 1$ ($h(1) = \binom{12}{6} = 924 \equiv 1$, $g(1) = 256 \equiv 9$). Therefore for $\lambda \in k \setminus \{0, 1\}$:

$$h(\lambda) = 0 \iff g(\lambda) = 0 \iff j(\lambda) = 5.$$

Since h has degree 6 it has roots in k, all different from $0, 1$, so there exist curves with Hasse invariant 0, and by the equivalence they are exactly the curves with $j = 5$. So the unique value is $j = 5 \pmod{13}$. (Mod 13 one finds $h = (\lambda^2+4\lambda-4)(\lambda^2-6\lambda+1)(\lambda^2-\lambda+3)$, whose six roots lie in $\mathbf{F}_{169}$ and form a single Σ_3-orbit, consistent with j having degree 6 in λ and with Cor. 4.23: $[13/12] + 2 = 3 \geq 1$.) □

Exercise 4.14

Problem. The Fermat curve $X : x^3 + y^3 = z^3$ gives a nonsingular curve in characteristic p for every $p \neq 3$. Determine the set $\mathfrak{P} = \{p \neq 3 \mid X_{(p)} \text{ has Hasse invariant } 0\}$, and observe (modulo Dirichlet's theorem) that it is a set of primes of density $\frac{1}{2}$.

Solution. Let $f = x^3 + y^3 - z^3$. In characteristic $p \neq 3$ the partial derivatives $3x^2, 3y^2, -3z^2$ vanish simultaneously only at $x = y = z = 0$, so $X_{(p)}$ is nonsingular (also for $p = 2$). Proposition 4.21 applies to any nonsingular plane cubic (its proof does not use $p \neq 2$): $X_{(p)}$ has Hasse invariant 0 iff the coefficient of $(xyz)^{p-1}$ in f^{p-1} is 0 in k. By the multinomial theorem

$$f^{p-1} = \sum_{a+b+c=p-1} \frac{(p-1)!}{a!\,b!\,c!}(-1)^c\, x^{3a} y^{3b} z^{3c}.$$

The monomial $(xyz)^{p-1}$ occurs iff $3a = 3b = 3c = p-1$, which is possible iff $p \equiv 1 \pmod{3}$, with $a = b = c = \frac{p-1}{3}$. In that case the coefficient is $(-1)^{(p-1)/3}(p-1)!/((\frac{p-1}{3})!)^3$, a quotient of products of integers all smaller than p, hence a unit mod p: the Hasse invariant is 1. If $p \not\equiv 1$

(mod 3) (and $p \neq 3$), i.e. $p \equiv 2 \pmod 3$, the monomial does not occur, the coefficient is 0 and the Hasse invariant is 0. (For $p = 2$: f^1 has no xyz term, so $X_{(2)}$ is supersingular.) Therefore

$$\mathfrak{P} = \{p \text{ prime} : p \equiv 2 \pmod 3\}.$$

By Dirichlet's theorem on primes in arithmetic progressions the primes $\equiv 1$ and $\equiv 2 \pmod 3$ each have density $\frac{1}{\varphi(3)} = \frac{1}{2}$, so $\mathfrak{P}$ has density $\frac{1}{2}$. (This matches Remark 4.23.4 and Example 4.20.2: X has $j = 0$ and complex multiplication by $\mathbf{Z}[\omega]$, and $p \equiv 2 \pmod 3$ iff p stays prime in $\mathbf{Z}[\omega]$; it also matches the Note after Cor. 4.23: $j = 0$ has Hasse invariant 0 iff $p \equiv 2 \pmod 3$.) □

Exercise 4.15

> **Problem.** Let X be an elliptic curve over a field k of characteristic p. Let $F' : X_p \to X$ be the k-linear Frobenius morphism (2.4.1). Use (4.10) to show that the dual morphism $\widehat{F'} : X \to X_p$ is separable if and only if the Hasse invariant of X is 1. Now use (Ex. 4.7) to show that if the Hasse invariant is 1, then the subgroup of points of order p on X is isomorphic to $\mathbf{Z}/p$; if it is 0, it is 0.

Solution. (k is algebraically closed.) X_p is the same scheme as X with the structure morphism composed with Frobenius of k; since k is perfect it is again an elliptic curve, with base point P_0 (the same point), and F' is the identity on underlying topological spaces and $F'^{\#}(s) = s^p$; it has degree p (Example 2.4.3), so by Ex. 4.7(f) $\deg \widehat{F'} = p$.

Preliminaries on tangent maps (used also in 4.16 and 4.20). For a homomorphism $g : Y \to Z$ of elliptic curves let $dg : T_0Y \to T_0Z$ be the map on Zariski tangent spaces at the origins (both 1-dimensional).

(T1) $d(g + h) = dg + dh$, $d(g \circ h) = dg \circ dh$, $d(n_Y) = n \cdot \mathrm{id}$. Indeed $g + h = \mu_Z \circ (g, h)$ and $d\mu_Z(u, v) = u + v$ on $T_{(0,0)}(Z \times Z) = T_0Z \oplus T_0Z$ because $\mu_Z(z, 0) = z = \mu_Z(0, z)$. Also $d\rho = -\mathrm{id}$ from $\mu \circ (1, \rho) = \mathrm{const}$; induction gives $d(n_Y) = n$.

(T2) A nonconstant homomorphism $g : Y \to Z$ is separable iff $dg \neq 0$. ($\Rightarrow$) If g is separable, it is ramified at only finitely many points (Prop. 2.2(a)). Since $g \circ \tau_y = \tau_{g(y)} \circ g$ for translations τ_y (automorphisms, Prop. 4.8), the ramification index of g is the same at all points; so g is unramified at 0, i.e. g^* maps a uniformizer at $0 \in Z$ to a uniformizer at $0 \in Y$, i.e. $dg \neq 0$. ($\Leftarrow$) If g is inseparable, write $g = s \circ h$ with s separable and $h : Y \to Y'$ purely inseparable of degree p^r, $r \geq 1$. By Prop. 2.5, h is a composition of k-linear Frobenius morphisms and isomorphisms; a k-linear Frobenius sends a local parameter u to u^p, so $h^*(\mathfrak{m}_{h(0)}) \subseteq \mathfrak{m}_0^p \subseteq \mathfrak{m}_0^2$, i.e. $dh = 0$ and $dg = ds \circ dh = 0$.

(T3) Let $f : Y \to Z$ be a homomorphism and $\widehat{f} : Z \to Y$ its dual (Ex. 4.7(a)), induced by $f^* : \mathrm{Pic}^0 Z \to \mathrm{Pic}^0 Y$ (under $X \cong J$, Thm. 4.11). By Remark 4.10.7 the tangent space of J at 0 is $H^1(\mathcal{O})$, tangent vectors being the classes $1 + \varepsilon a$, $a \in H^1(\mathcal{O})$, in $\mathrm{Pic}(\cdot \times \mathrm{Spec}\, k[\varepsilon]/\varepsilon^2)$; f^* sends $1 + \varepsilon a$ to $1 + \varepsilon f^* a$. Hence $d\widehat{f} = f^* : H^1(Z, \mathcal{O}_Z) \to H^1(Y, \mathcal{O}_Y)$.

First assertion. Apply (T3) to $f = F' : X_p \to X$: the tangent map of $\widehat{F'} : X \to X_p$ is $F'^* : H^1(X, \mathcal{O}_X) \to H^1(X_p, \mathcal{O}_{X_p})$. Since F' and the Frobenius F of X are the same morphism of ringed spaces, this map is, as a map of abelian groups, $F^* : H^1(X, \mathcal{O}_X) \to H^1(X, \mathcal{O}_X)$. By

definition the Hasse invariant is 1 iff $F^* \neq 0$. So $d\widehat{F'} \neq 0$ iff Hasse $= 1$, and by (T2) $\widehat{F'}$ is separable iff Hasse $= 1$.

Points of order p. Ex. 4.7(c) with $f = F'$ gives $F' \circ \widehat{F'} = p_X$. Since F' is the identity on points, $F'(y) = P_0$ iff $y = P_0$. Hence for a closed point x: $px = 0 \iff F'(\widehat{F'}x) = P_0 \iff \widehat{F'}(x) = P_0$, so $X[p] = \ker \widehat{F'}$ as groups of closed points. The degree of $\widehat{F'}$ is p. If it is separable ($\iff$ Hasse $= 1$) then $\#\ker \widehat{F'} = \deg \widehat{F'} = p$ (fibres of a homomorphism are cosets of the kernel), so $X[p]$ is a group of order p, i.e. $\mathbf{Z}/p$. If it is inseparable ($\iff$ Hasse $= 0$) then, p being prime, it is purely inseparable, hence injective on points, so $X[p] = 0$. □

Exercise 4.16

Problem. Again let X be an elliptic curve over k of characteristic p, and suppose X is defined over the field $\mathbf{F}_q$ of $q = p^r$ elements, i.e., $X \subseteq \mathbf{P}^2$ can be defined by an equation with coefficients in $\mathbf{F}_q$. Assume also that X has a rational point over $\mathbf{F}_q$. Let $F' : X_q \to X$ be the k-linear Frobenius with respect to q.

(a) Show that $X_q \cong X$ as schemes over k, and that under this identification, $F' : X \to X$ is the map obtained by the qth-power map on the coordinates of points of X, embedded in $\mathbf{P}^2$.

(b) Show that $1_X - F'$ is a separable morphism and its kernel is just the set $X(\mathbf{F}_q)$ of points of X with coordinates in $\mathbf{F}_q$.

(c) Use (Ex. 4.7) to show that $\widehat{F'} + F' = a_X$ for some integer a, and that $N = q - a + 1$, where $N = \#X(\mathbf{F}_q)$.

(d) Use the fact that $\deg(m + nF') > 0$ for all $m, n \in \mathbf{Z}$ to show that $|a| \leq 2\sqrt{q}$. This is Hasse's proof of the analogue of the Riemann hypothesis for elliptic curves (App. C, Ex. 5.6).

(e) Now assume $q = p$, and show that the Hasse invariant of X is 0 if and only if $a \equiv 0 \pmod{p}$. Conclude for $p \geq 5$ that X has Hasse invariant 0 if and only if $N = p + 1$.

Solution. Let $f \in \mathbf{F}_q[x, y, z]$ be an equation of X and $S = k[x, y, z]/(f)$, so $X = \operatorname{Proj} S$ with base point $P_0 \in X(\mathbf{F}_q)$. X_q is the same scheme $\operatorname{Proj} S$ with k acting through $a \mapsto a^q$; write S_q for S with this k-algebra structure ($a * s = a^q s$), so $X_q = \operatorname{Proj} S_q$ and F' corresponds to the ring map $S \to S_q$, $s \mapsto s^q$ (it is k-linear: $(as)^q = a * s^q$).

(a) Define $\Psi : S \to S$ by $\Psi(\sum c_I x^I) = \sum c_I^q x^I$ ($c_I \in k$). It is well defined on S because f has coefficients in $\mathbf{F}_q$ (so $\Psi(f) = f$), and it is a graded ring automorphism (k is perfect). Since $\Psi(as) = a^q \Psi(s) = a * \Psi(s)$, $\Psi : S \to S_q$ is an isomorphism of graded k-algebras. Hence $\Theta : X \to X_q$, the morphism of schemes over k corresponding to Ψ^{-1}, is an isomorphism. The composite $F' \circ \Theta : X \to X$ corresponds to the k-algebra map $S \to S_q \to S$, $s \mapsto s^q \mapsto \Psi^{-1}(s^q)$, which sends $x_i \mapsto \Psi^{-1}(x_i^q) = x_i^q$; so $F' \circ \Theta$ is the morphism $\varphi : (x : y : z) \mapsto (x^q : y^q : z^q)$. Write $\varphi = F' \circ \Theta$. As F' is the identity on points, Θ is the map of points $P \mapsto \varphi(P)$; in particular $\Theta(P_0) = \varphi(P_0) = P_0$ since P_0 has coordinates in $\mathbf{F}_q$.

Below we write $\varphi \in R = \operatorname{End}(X, P_0)$ for this endomorphism (it is a homomorphism by Lemma 4.9 as $\varphi(P_0) = P_0$); "F'" in the statements (b)–(e) means φ under the identification Θ. Its degree is q (that of F'; Θ is an isomorphism).

(b) A closed point $P = (x : y : z)$ satisfies $\varphi(P) = P$ iff $(x^q : y^q : z^q) = (x : y : z)$, i.e. $x_i^q = \lambda x_i$ for all i and some $\lambda \neq 0$; normalizing a nonzero coordinate to 1 gives $\lambda = 1$, so all $x_i^q = x_i$, i.e. $x_i \in \mathbf{F}_q$. Hence $\ker(1_X - \varphi) = X(\mathbf{F}_q)$ (as a set of closed points). Now $d\varphi = 0$ since φ^* sends a local parameter u to $u^q \in \mathfrak{m}^2$ (or by (T2), φ being purely inseparable). By (T1) $d(1_X - \varphi) = \mathrm{id} - d\varphi = \mathrm{id} \neq 0$. In particular $1_X - \varphi$ is nonconstant, and by (T2) it is separable. As a separable homomorphism, $\deg(1_X - \varphi) = \#\ker(1_X - \varphi) = \#X(\mathbf{F}_q) = N$.

(c) Recall Ex. 4.7: $\widehat{\ }$ is additive (d), contravariantly multiplicative (b), $g \circ \widehat{g} = \widehat{g} \circ g = [\deg g]$ (c) for $g \in R$ (trivially also for $g = 0$, whose dual is 0), $\widehat{n_X} = n_X$ (e), $\deg \widehat{g} = \deg g$ (f), and $\widehat{1_X} = 1_X$. Moreover integers are central in R, since $n_X \circ g = g \circ n_X$ for a homomorphism g. Dualizing the additive element $1 - \varphi$: $\widehat{1 - \varphi} = 1 - \widehat{\varphi}$ (from $\varphi + (-\varphi) = 0$ one gets $\widehat{-\varphi} = -\widehat{\varphi}$). By (b) and Ex. 4.7(c),

$$N = \deg(1 - \varphi) \quad \text{and} \quad (1 - \varphi)(1 - \widehat{\varphi}) = [N].$$

Expanding, $(1 - \varphi)(1 - \widehat{\varphi}) = 1 - \varphi - \widehat{\varphi} + \varphi\widehat{\varphi} = 1 - (\varphi + \widehat{\varphi}) + q$, so

$$\varphi + \widehat{\varphi} = [q + 1 - N] = a_X \quad \text{with } a = q + 1 - N \in \mathbf{Z}, \qquad \text{i.e. } N = q - a + 1,$$

using that $\mathbf{Z} \to R$, $n \mapsto n_X$, is injective ($\deg n_X = n^2 > 0$ for $n \neq 0$ by Ex. 4.7(e)).

(d) For $m, n \in \mathbf{Z}$ put $g = m + n\varphi \in R$. Then $\widehat{g} = m + n\widehat{\varphi}$ and

$$[\deg g] = g\widehat{g} = m^2 + mn(\varphi + \widehat{\varphi}) + n^2\varphi\widehat{\varphi} = [\, m^2 + a\, mn + q\, n^2\,].$$

So $\deg(m + n\varphi) = m^2 + amn + qn^2$ (injectivity of $\mathbf{Z} \to R$ again). A degree is ≥ 0 (it is > 0 unless $m + n\varphi$ is constant). Taking $(m, n) = (-a, 2)$ gives $a^2 - 2a^2 + 4q = 4q - a^2 \geq 0$, i.e. $|a| \leq 2\sqrt{q}$. (Remark: the printed "> 0 for all m, n" must exclude $(0, 0)$, and also fails when $m + n\varphi = 0$ for some $(m, n) \neq (0, 0)$, which can happen for supersingular curves over $\mathbf{F}_{p^{2s}}$ with $\varphi = \pm p^s$; in that case $n\varphi = -m$, $n\widehat{\varphi} = -m$ give $na = -2m$, $n^2 q = m^2$, so $a^2 = 4q$ and the bound still holds, with equality. The argument with ≥ 0 covers both cases.)

(e) Let $q = p$, so $X_q = X_p$ and $F'\Theta = \varphi$. Then $\widehat{\varphi} = \widehat{\Theta} \circ \widehat{F'}$ by Ex. 4.7(b), and $\widehat{\Theta} : X_p \to X$ is an isomorphism ($\widehat{\Theta}\Theta = 1$). Hence $\widehat{\varphi}$ is separable iff $\widehat{F'}$ is, iff (Ex. 4.15) the Hasse invariant is 1. By (T1) and (c), $d\widehat{\varphi} = d(a_X) - d\varphi = a \cdot \mathrm{id} - 0 = a \cdot \mathrm{id}$, which is the multiplication by the image of a in $\mathbf{F}_p \subseteq k$. By (T2), $\widehat{\varphi}$ is separable iff $a \not\equiv 0 \pmod p$. Hence: Hasse invariant $0 \iff a \equiv 0 \pmod p$.

For $p \geq 5$, (d) gives $|a| \leq 2\sqrt{p} < p$ (as $p > 4$), so $a \equiv 0 \pmod p$ iff $a = 0$, iff $N = p - a + 1 = p + 1$. Thus for $p \geq 5$: Hasse invariant $0 \iff N = p + 1$. (For $p = 2, 3$ this fails to be forced: $|a| = p$ is possible.) □

Exercise 4.17

Problem. Let X be the curve $y^2 + y = x^3 - x$ of (4.23.8).

(a) If $Q = (a, b)$ is a point on the curve, compute the coordinates of the point $P + Q$, where $P = (0, 0)$, as a function of a, b. Use this formula to find the coordinates of nP, $n = 1, 2, \ldots, 10$. [Check: $6P = (6, 14)$.]

(b) This equation defines a nonsingular curve over $\mathbf{F}_p$ for all $p \neq 37$.

Solution. The curve in $\mathbf{P}^2$ is $y^2z + yz^2 = x^3 - xz^2$; the line $z = 0$ meets it in $x^3 = 0$, i.e. in $3P_0$ with $P_0 = (0, 1, 0)$, which is the origin of the group law (4.23.8). So (embedding by $|3P_0|$) three points are collinear iff they sum to 0. For a point (x, y) the vertical line $x = \text{const}$ meets X again at $(x, -1 - y)$ (the sum of the two roots of $y^2 + y - c$ is -1), so $-(x, y) = (x, -1 - y)$.

(a) Let $Q = (a, b)$ with $a \neq 0$ (the points with $x = 0$ are only $P = (0, 0)$ and $-P = (0, -1)$). The line PQ is $y = mx$, $m = b/a$. Substituting: $m^2x^2 + mx = x^3 - x$, i.e. $x\,(x^2 - m^2x - (m+1)) = 0$, whose roots are 0, a and a third value x_3 with $a + x_3 = m^2$. So the third point of intersection is $R = (x_3, mx_3)$ with $x_3 = \frac{b^2}{a^2} - a = \frac{b^2 - a^3}{a^2}$. From $b^2 + b = a^3 - a$ we get $b^2 - a^3 = -a - b$, so $x_3 = -\frac{a+b}{a^2}$. Then $P + Q = -R = (x_3, -1 - mx_3)$, with $-1 - mx_3 = -1 + \frac{b(a+b)}{a^3} = \frac{b^2 + ab - a^3}{a^3} = \frac{ab - a - b}{a^3}$. Hence

$$\boxed{P + Q = \Big(-\frac{a+b}{a^2},\ \frac{ab - a - b}{a^3}\Big) \qquad (a \neq 0).}$$

For $Q = P$ we compute the tangent at P: $(2y + 1)\,dy = (3x^2 - 1)\,dx$ gives slope -1, the line $y = -x$ gives $x^2 - x = x^3 - x$, i.e. $x^2(x - 1) = 0$, so the third point is $(1, -1)$ and $2P = (1, 0)$. Also $P + (-P) = 0$.

Now $nP = P + (n - 1)P$; for $n \geq 3$ the formula applies as long as $x((n-1)P) \neq 0$, which we see from the values themselves:

$$\begin{array}{lll} 1P = (0, 0), & 2P = (1, 0), & 3P = (-1, -1), \\ 4P = (2, -3), & 5P = (\tfrac{1}{4}, -\tfrac{5}{8}), & 6P = (6, 14), \\ 7P = (-\tfrac{5}{9}, \tfrac{8}{27}), & 8P = (\tfrac{21}{25}, -\tfrac{69}{125}), & 9P = (-\tfrac{20}{49}, -\tfrac{435}{343}), \\ 10P = (\tfrac{161}{16}, -\tfrac{2065}{64}). & & \end{array}$$

Samples: $3P = P + 2P$ with $(a, b) = (1, 0)$: $(-1,\ (0 - 1 - 0)/1) = (-1, -1)$. $4P$ from $(a, b) = (-1, -1)$: $(-(-2)/1,\ (1 + 1 + 1)/(-1)) = (2, -3)$. $5P$ from $(2, -3)$: $(1/4,\ (-6 - 2 + 3)/8) = (\frac{1}{4}, -\frac{5}{8})$. $6P$ from $(\frac{1}{4}, -\frac{5}{8})$: $a + b = -\frac{3}{8}$, $a^2 = \frac{1}{16}$, so $x = 6$; $ab - a - b = -\frac{5}{32} + \frac{3}{8} = \frac{7}{32}$, $a^3 = \frac{1}{64}$, so $y = 14$ ($6P = (6, 14)$ as in the check; indeed $196 + 14 = 210 = 216 - 6$). $7P$ from $(6, 14)$: $(-\frac{20}{36}, \frac{84-20}{216}) = (-\frac{5}{9}, \frac{8}{27})$; and the remaining ones are obtained in the same way (each result lies on the curve, which is a further check; e.g. for $10P$, $\frac{2065^2}{64^2} - \frac{2065}{64} = \frac{4264225 - 132160}{4096} = \frac{4132065}{4096} = \big(\frac{161}{16}\big)^3 - \frac{161}{16}$). (All of this was also verified with exact rational arithmetic.) None of $x(nP)$, $n = 2, \ldots, 9$, is 0, which justifies using the formula at each step.

(b) Put $f = y^2 + y - x^3 + x$. Affine points: a singular point satisfies $f_y = 2y + 1 = 0$, $f_x = -3x^2 + 1 = 0$, $f = 0$. If $p = 2$, $f_y = 1 \neq 0$ always; if $p = 3$, $f_x = 1 \neq 0$ always; so there is no singular affine point for $p = 2, 3$. For $p \neq 2, 3$: $y = -\frac{1}{2}$, $x^2 = \frac{1}{3}$, and $f = \frac{1}{4} - \frac{1}{2} - x(x^2 - 1) = -\frac{1}{4} + \frac{2}{3}x = 0$ gives $x = \frac{3}{8}$; then $x^2 = \frac{9}{64} = \frac{1}{3}$ iff $27 = 64$ iff $37 = 0$ in $\mathbf{F}_p$. So for $p \neq 37$ there is no singular affine point. At infinity, $z = 0$ forces $x^3 = 0$, so the only point is $(0, 1, 0)$, where $\partial F/\partial z = y^2 + 2yz + 2xz = 1 \neq 0$ for $F = y^2z + yz^2 - x^3 + xz^2$: nonsingular. Hence $X_{(p)}$ is nonsingular for every $p \neq 37$. (For $p = 37$ the point $(5, 18)$ is singular: $f(5, 18) = 222 = 6 \cdot 37$, $f_x = -74$, $f_y = 37$.) $\square$

Exercise 4.18

Problem. Let X be the curve $y^2 = x^3 - 7x + 10$. This curve has at least 26 points with integer coordinates. Find them (use a calculator), and verify that they are all contained in the subgroup (maybe equal to all of $X(\mathbf{Q})$?) generated by $P = (1, 2)$ and $Q = (2, 2)$.

Solution. Take the point at infinity $P_0 = (0,1,0)$ as origin (it is a flex, $z = 0$ meeting X in $3P_0$), so collinear points sum to 0 and $-(x,y) = (x,-y)$. For $A = (x_1,y_1) \neq \pm B = (x_2,y_2)$, the line through them has slope $m = \frac{y_2-y_1}{x_2-x_1}$, the third intersection point has abscissa $x_3 = m^2 - x_1 - x_2$ and ordinate $y_1 + m(x_3 - x_1)$, and $A + B = (x_3, -y_1 - m(x_3 - x_1))$; for doubling $m = \frac{3x_1^2-7}{2y_1}$.

The integer points. The cubic is negative for $x < -3.2$, so one searches $x \geq -3$; a computer search over $-3 \leq x \leq 10^5$ for $x^3 - 7x + 10$ a perfect square gives (and there are no others in this range) $x \in \{-3,-2,-1,1,2,3,5,9,13,31,41,67,302\}$, each with $y = \pm$ the following: $y = 2,4,4,2,2,4,10,26,46,172,262,548,5248$. So there are 26 integer points:

$(-3,\pm2)$, $(-2,\pm4)$, $(-1,\pm4)$, $(1,\pm2)$, $(2,\pm2)$, $(3,\pm4)$, $(5,\pm10)$, $(9,\pm26)$, $(13,\pm46)$, $(31,\pm172)$, $(41,\pm262)$, $(67$

(Check, e.g., $302^3 - 7 \cdot 302 + 10 = 27543608 - 2114 + 10 = 27541504 = 5248^2$.)

They lie in $\langle P,Q\rangle$. Since $-(x,y) = (x,-y)$ it suffices to express one point of each $\pm$ pair. With $P = (1,2)$, $Q = (2,2)$:

$$P+Q:\ m=0,\ x_3=-3,\ \text{third point } (-3,2),\ P+Q=(-3,-2).$$
$$2P:\ m=-1,\ x_3=-1,\ \text{third } (-1,4),\ 2P=(-1,-4).$$
$$P-Q=P+(2,-2):\ m=-4,\ x_3=13,\ \text{third } (13,-46),\ P-Q=(13,46).$$
$$3P=2P+P:\ m=3,\ x_3=9,\ \text{third } (9,26),\ 3P=(9,-26).$$
$$2P+Q=(-1,-4)+(2,2):\ m=2,\ x_3=3,\ \text{third } (3,4),\ 2P+Q=(3,-4).$$
$$3P+Q=(9,-26)+(2,2):\ m=-4,\ x_3=5,\ \text{third } (5,-10),\ 3P+Q=(5,10).$$
$$4P+Q=P+(3P+Q)=(1,2)+(5,10):\ m=2,\ x_3=-2,\ \text{third } (-2,-4),\ 4P+Q=(-2,4).$$
$$2P+2Q=(2P+Q)+Q=(3,-4)+(2,2):\ m=-6,\ x_3=31,\ \text{third } (31,-172),\ 2P+2Q=(31,172).$$
$$5P+2Q=(4P+Q)+(P+Q)=(-2,4)+(-3,-2):\ m=6,\ x_3=41,\ \text{third } (41,262),\ 5P+2Q=(41,-262).$$
$$6P+Q=(3P+Q)+3P=(5,10)+(9,-26):\ m=-9,\ x_3=67,\ \text{third } (67,-548),\ 6P+Q=(67,548).$$
$$4P-Q=3P+(P-Q)=(9,-26)+(13,46):\ m=18,\ x_3=302,\ \text{third } (302,5248),\ 4P-Q=(302,-5248).$$

Together with $P = (1,2)$, $Q = (2,2)$ these are the 13 points P, Q, $P+Q$, $2P$, $P-Q$, $3P$, $2P+Q$, $3P+Q$, $4P+Q$, $2P+2Q$, $5P+2Q$, $6P+Q$, $4P-Q$ of the 13 pairs, so all 26 integer points lie in the subgroup generated by P and Q. (I verified these sums by exact arithmetic as well.) Whether $\langle P,Q\rangle$ is all of $X(\mathbf{Q})$ would require a descent (determination of the Mordell–Weil group), which I have not carried out; the exercise only asks for the verification above. By Ex. 4.19 below the group $X(\mathbf{Q})$ is torsion-free. □

Exercise 4.19

Problem. Let X, P_0 be an elliptic curve defined over $\mathbf{Q}$, represented as a curve in $\mathbf{P}^2$ by an equation with integer coefficients. Then X can be considered as the fibre over the generic point of a scheme $\overline{X}$ over $\operatorname{Spec}\mathbf{Z}$. Let $T \subseteq \operatorname{Spec}\mathbf{Z}$ be the open subset consisting of all primes $p \neq 2$ such that the fibre $X_{(p)}$ of $\overline{X}$ over p is nonsingular. For any n, show that $n_X : X \to X$ is defined over T, and is a flat morphism. Show that the kernel of n_X is also flat over T. Conclude that for any $p \in T$, the natural map $X(\mathbf{Q}) \to X_{(p)}(\mathbf{F}_p)$ induced on the groups of rational points, maps the n-torsion points of $X(\mathbf{Q})$ injectively into the torsion subgroup of $X_{(p)}(\mathbf{F}_p)$, for $(n,p) = 1$.

By this method one can show easily that the groups $X(\mathbf{Q})$ in (Ex. 4.17) and (Ex. 4.18) are torsion-free.

Solution. This exercise needs several results from scheme theory beyond the text; I name them where used, and indicate where the proof of a theorem of the text is used over a base other than a field.

Setup. Let F be the integral cubic form, $\overline{X} = \operatorname{Proj} \mathbf{Z}[x,y,z]/(F) \subseteq \mathbf{P}^2_{\mathbf{Z}}$, $X = \overline{X}_{\mathbf{Q}}$, and let $P_0 \in X(\mathbf{Q})$ be given by coprime integers; it defines a section $e : \operatorname{Spec} \mathbf{Z} \to \overline{X}$ ($F(P_0) = 0$ in $\mathbf{Z}$). T is open and nonempty (the discriminant is a nonzero integer), so $T = \operatorname{Spec} \mathbf{Z}[S^{-1}]$ is connected, regular of dimension 1. Put $\overline{X}_T = \overline{X} \times_{\mathbf{Z}} T$, $\pi : \overline{X}_T \to T$. Its fibres over $p \in T$ are the nonsingular plane cubics $X_{(p)}$ (nonzero mod p), over the generic point X; the Hilbert polynomial of all fibres is $3m$, so π is flat (III, Thm. 9.9) and projective. A flat morphism of finite type with nonsingular geometric fibres is smooth (EGA IV 17.5.1; III, 10.2 is the case over a field), here of relative dimension 1. In particular $\overline{X}_T$ is a regular scheme (smooth over the regular scheme T; alternatively Matsumura 23.7), of dimension 2 at closed points, and all fibres are elliptic curves (over $\mathbf{Q}$ resp. $\mathbf{F}_p$, geometrically).

(1) n_X is defined over T. We redo the proof of Thm. 4.11 with base T in place of $\operatorname{Spec} k$, to show that $(\overline{X}_T, e)$ is a group scheme over T. Let T' be a T-scheme of finite type, $X' = \overline{X}_T \times_T T'$ with projections $p : X' \to T'$, $q : X' \to \overline{X}_T$, and let $\mathcal{M} \in \operatorname{Pic}^0(X'/T')$ (invertible, of degree 0 on all fibres). Put $\mathcal{M}' = \mathcal{M} \otimes q^*\mathcal{L}(e(T))$, where $e(T) \subseteq \overline{X}_T$ is an effective Cartier divisor (a section of the smooth morphism π of relative dimension 1 is a regular immersion of codimension 1; likewise for the diagonal $\Delta \subseteq \overline{X}_T \times_T \overline{X}_T$, which is used at the end).

- p is projective and flat with geometrically integral fibres of genus 1, and $\mathcal{M}'$ has degree 1 on each fibre. By Riemann–Roch on the geometric fibre $X'_t \otimes \overline{k(t)}$ (which does not change h^0, h^1), $h^0(X'_t, \mathcal{M}'_t) = 1$ and $h^1 = 0$ for every point $t \in T'$ (closed or not).
- By cohomology and base change (III, 12.11) we get $R^1p_*\mathcal{M}' = 0$ and $p_*\mathcal{M}'$ locally free of rank 1, compatible with base change. (Only closed points need the explicit computation: T' is of finite type over $\mathbf{Z}$, so every nonempty closed subset of T' contains a closed point, and $\{t : h^1(\mathcal{M}'_t) \geq 1\}$ is closed by semicontinuity, III, 12.8.)
- Replacing $\mathcal{M}$ by $\mathcal{M} \otimes p^*(p_*\mathcal{M}')^{-1}$ (this does not change its class in $\operatorname{Pic}^0(X'/T')$) we may assume $p_*\mathcal{M}' \cong \mathcal{O}_{T'}$. The section 1 gives $s \in \Gamma(X', \mathcal{M}')$, nonzero on every fibre (base change), hence a nonzerodivisor on each integral fibre. As X' is flat over T', its zero scheme Z is an effective Cartier divisor flat over T' (local criterion of flatness), whose fibres are single reduced points ($\deg \mathcal{M}'_t = 1$). Thus $Z \to T'$ is finite, flat of degree 1, i.e. an isomorphism, and gives a section $T' \to X'$, hence $f : T' \to \overline{X}_T$ (compose with q).
- Since Z is the graph of f, $Z = (f \times 1)^*\Delta$ and $\mathcal{M}' \cong \mathcal{O}(Z) \cong (f \times 1)^*\mathcal{L}(\Delta)$ (up to a pullback from T'); twisting by $-e(T)$ gives $f^*\mathcal{L} \cong \mathcal{M}$ with $\mathcal{L} = \mathcal{L}(\Delta) \otimes p_2^*\mathcal{L}(-e(T))$ in $\operatorname{Pic}(X'/T')$. Uniqueness of f is clear for the same reasons (the zero scheme of s determines f).

Hence $(\overline{X}_T, \mathcal{L})$ represents $T' \mapsto \operatorname{Pic}^0(X'/T')$ on T-schemes of finite type, and, as in Remark 4.10.6, it is a group scheme over T with identity e, inverse ρ and multiplication μ (Yoneda: the group structure of $\operatorname{Pic}^0(\cdot/T')$ is functorial in T'). Its fibres are the elliptic curves $X_{(p)}$ (after base change to $\overline{\mathbf{F}}_p$) with the group law of Prop. 4.8 relative to $e(p)$ (last statement of Thm. 4.11 over the algebraically closed field), and the generic fibre is (X, P_0). Then n_X, the n-fold iterate of μ (composed with ρ if $n < 0$), is a T-morphism $\overline{X}_T \to \overline{X}_T$ restricting to n_X on the generic fibre (and to $n_{X_{(p)}}$ on the fibre over p). Because $p \neq 2$ for $p \in T$, Prop. 4.10 (which assumes

char $\neq 2$), applied after base change to $\overline{\mathbf{F}}_p$, resp. $\overline{\mathbf{Q}}$, shows that for $n \neq 0$ each restriction $n_{X_{(p)}}$ and n_X are finite surjective morphisms of curves.

(2) n_X is flat. Let $n \neq 0$ and write $g = n_X : \overline{X}_T \to \overline{X}_T$. (For $n = 0$, $0_X = e \circ \pi$ is not flat, as its image $e(T)$ is not open; so "for any n" must be read as $n \neq 0$.) The morphism g is projective (a T-morphism between projective T-schemes) and quasi-finite (on each fibre over T it is a finite morphism of curves), hence finite (III, Ex. 11.2). Let $x \in \overline{X}_T$, $y = g(x)$, $A = \mathcal{O}_y \to B = \mathcal{O}_x$. Both are regular local rings, and $B/\mathfrak{m}_A B$ has dimension 0 since the fibre of g over y is finite. Furthermore $\dim A = \dim B$: since π is flat, $\dim \mathcal{O}_x = \dim \mathcal{O}_{T,\pi(x)} + \dim \mathcal{O}_{X_{\pi(x)},x}$, and the same for y (note $\pi(x) = \pi(y)$); the finite surjective morphism g of the fibre curve maps closed points to closed points and the generic point to the generic point, so $\dim \mathcal{O}_{X_t,x} = \dim \mathcal{O}_{X_t,y}$. By the "miracle flatness" criterion (Matsumura, *Commutative Ring Theory*, Thm. 23.1: A regular, B Cohen–Macaulay, $\dim B = \dim A + \dim B/\mathfrak{m}_A B$ imply B flat over A) g is flat at every point. So $n_X : \overline{X}_T \to \overline{X}_T$ is finite and flat; being flat and finite it is locally free of finite rank.

(3) The kernel. Let $K = \overline{X}_T \times_{n_X, \overline{X}_T, e} T = \ker(n_X)$ with structure map $K \to T$. This is the base change of n_X along $e : T \to \overline{X}_T$, hence $K \to T$ is finite and flat, i.e. $K = \operatorname{Spec} \mathcal{A}$ with $\mathcal{A}$ a locally free $\mathcal{O}_T$-algebra of finite rank; the rank is constant (T connected) and equals $\deg n_X = n^2$ (generic fibre = $X[n]$ has n^2 points over $\overline{\mathbf{Q}}$, since $\deg n_X = n^2$ by Ex. 4.7(e) and n_X is separable in characteristic 0).

(4) Injectivity on torsion. Let $p \in T$ with $p \nmid n$ and $R_p = \mathbf{Z}_{(p)}$. Each $x \in X(\mathbf{Q})$ extends uniquely to an R_p-point $\tilde{x}$ of $\overline{X}$ (valuative criterion, $\overline{X}$ proper), and the natural map is $x \mapsto \bar{x} = \tilde{x} \bmod p \in X_{(p)}(\mathbf{F}_p)$; it is a homomorphism because μ is defined over T and $\tilde{x} + \tilde{y}$ is the extension of $x + y$. Let $x \in X(\mathbf{Q})$ with $nx = 0$. Then x is a $\mathbf{Q}$-point of K, and since $K \times R_p = \operatorname{Spec} B$ with B finite over R_p, the ring homomorphism $B \to \mathbf{Q}$ has image integral over R_p, hence contained in the integrally closed ring R_p: $\tilde{x}$ is an R_p-point of K. Next, $K \times \mathbf{F}_p = \ker(n_{X_{(p)}})$ is *reduced*: $d(n_{X_{(p)}}) = n \cdot \mathrm{id} \neq 0$ in $\mathbf{F}_p$ (T1 in Ex. 4.15), so $n_{X_{(p)}}$ is unramified at 0, and for $Q \in X_{(p)}[n]$, translation by Q identifies neighbourhoods of 0 and of Q compatibly with $n_{X_{(p)}}$. Hence $K \times \overline{\mathbf{F}}_p$ is a finite reduced scheme over an algebraically closed field, i.e. a disjoint union of n^2 reduced points, so $K \times \mathbf{F}_p \to \operatorname{Spec} \mathbf{F}_p$ is etale. Since $K \times R_p \to \operatorname{Spec} R_p$ is finite flat with etale closed fibre, it is etale (the etale locus is open and contains the closed fibre, and every point of the finite scheme $K \times R_p$ specializes into the closed fibre). Now suppose $\bar{x} = 0$. Then the two sections $\tilde{x}$ and e of $K \times R_p \to \operatorname{Spec} R_p$ agree at the closed point. The locus where two sections of a separated etale morphism agree is open and closed, and $\operatorname{Spec} R_p$ is connected, so $\tilde{x} = e$, and so $x = 0$. Thus the homomorphism $X(\mathbf{Q})[n] \to X_{(p)}(\mathbf{F}_p)$ is injective, with image in the n-torsion subgroup $X_{(p)}(\mathbf{F}_p)[n]$ (it is a homomorphism). In particular, if $x \in X(\mathbf{Q})$ has order m with $p \nmid m$, then $\bar{x}$ has order m, so m divides $\#X_{(p)}(\mathbf{F}_p)$.

Application to (4.17), (4.18). Suppose $X(\mathbf{Q})$ had a point of prime order ℓ. For every $p \in T$ with $p \neq \ell$ we would get $\ell \mid N_p := \#X_{(p)}(\mathbf{F}_p)$.

(4.17) $y^2 + y = x^3 - x$: $p = 3, 5 \in T$ (nonsingular for $p \neq 37$, Ex. 4.17(b)). Counting points (including the point at infinity): over $\mathbf{F}_3$: $x = 0, 1, 2$ each give 2 points, so $N_3 = 6 + 1 = 7$; over $\mathbf{F}_5$: $x^3 - x = 0, 0, 1, 4, 0$ for $x = 0, 1, 2, 3, 4$ and $y^2 + y = c$ has $1 + \left(\frac{1+4c}{5}\right)$ solutions, giving $2, 2, 1, 0, 2$ points, $N_5 = 7 + 1 = 8$. If $\ell \neq 3$ then $\ell \mid 7$, so $\ell = 7$, but $7 \neq 5$ and $7 \nmid 8$; if $\ell = 3$ then $3 \nmid N_5 = 8$. Contradiction: $X(\mathbf{Q})$ is torsion-free.

(4.18) $y^2 = x^3 - 7x + 10$: the discriminant is $-16(4(-7)^3 + 27 \cdot 10^2) = -2^8 \cdot 83$, so $3, 5, 7 \in T$,

and counting points mod $3, 5, 7$ gives $N_3 = 7$, $N_5 = 10$, $N_7 = 13$. If $\ell \neq 3$, $\ell \mid 7$, so $\ell = 7$; but $7 \neq 5$ and $7 \nmid 10$. If $\ell = 3$: $3 \nmid N_5 = 10$. So $X(\mathbf{Q})$ is torsion-free. (For $\ell = 2$ one can also see directly that $x^3 - 7x + 10$ has no rational root.) $\square$

Exercise 4.20

Problem. Let X be an elliptic curve over a field k of characteristic $p > 0$, and let $R = \mathrm{End}(X, P_0)$ be its ring of endomorphisms.

(a) Let X_p be the curve over k defined by changing the k-structure of X (2.4.1). Show that $j(X_p) = j(X)^{1/p}$. Thus $X \cong X_p$ over k if and only if $j \in \mathbf{F}_p$.

(b) Show that p_X in R factors into a product $\pi\widehat{\pi}$ of two elements of degree p if and only if $X \cong X_p$. In this case, the Hasse invariant of X is 0 if and only if π and $\widehat{\pi}$ are associates in R (i.e., differ by a unit). (Use (2.5).)

(c) If $\mathrm{Hasse}(X) = 0$ show in any case $j \in \mathbf{F}_{p^2}$.

(d) For any $f \in R$, there is an induced map $f^* : H^1(\mathcal{O}_X) \to H^1(\mathcal{O}_X)$. This must be multiplication by an element $\lambda_f \in k$. So we obtain a ring homomorphism $\varphi : R \to k$ by sending f to λ_f. Show that any $f \in R$ commutes with the (nonlinear) Frobenius morphism $F : X \to X$, and conclude that if $\mathrm{Hasse}(X) \neq 0$, then the image of φ is $\mathbf{F}_p$. Therefore, R contains a prime ideal $\mathfrak{p}$ with $R/\mathfrak{p} \cong \mathbf{F}_p$.

Solution. (k is algebraically closed.) In (a) the j-invariant is the one of the text, defined for $p \neq 2$ via Prop. 4.6; for $p = 2$ the same proof works with the usual j-invariant of a Weierstrass equation, whose formula has coefficients in $\mathbf{F}_2$. The printed exponent is read as $1/p$ (the scan shows "$j(X)^{1\,p}$"), which is what the proof gives; it implies the stated equivalence.

(a) Let $\sigma : k \to k$ be the automorphism $a \mapsto a^{1/p}$. Let $X = \operatorname{Proj} S$, $S = k[x, y, z]/(f)$, $f = \sum c_I x^I$, and let X^σ be the curve defined by $f^\sigma = \sum \sigma(c_I) x^I$. As in Ex. 4.16(a), $X_p = \operatorname{Proj} S_p$ with S_p the k-algebra S with $a * s = a^p s$. The ring map $\Psi : k[x, y, z]/(f^\sigma) \to S_p$, $\sum d_I x^I \mapsto \sum d_I^p x^I$, is well defined ($\Psi(f^\sigma) = \sum \sigma(c_I)^p x^I = f$), graded, bijective, and k-linear for the structure $*$ ($\Psi(as) = a^p \Psi(s) = a * \Psi(s)$). Hence $X_p \cong X^\sigma$ over k. For $p \neq 2$ take $X : y^2 = x(x-1)(x-\lambda)$ (Prop. 4.6); then $X_p \cong y^2 = x(x-1)(x-\sigma(\lambda))$, so $\lambda(X_p) = \sigma(\lambda)$ up to Σ_3, and since $j(\lambda) = 2^8(\lambda^2 - \lambda + 1)^3/(\lambda^2(\lambda - 1)^2)$ has coefficients in $\mathbf{F}_p$,

$$j(X_p) = j(\sigma(\lambda)) = \sigma(j(\lambda)) = j(X)^{1/p}.$$

By Thm. 4.1(b), $X \cong X_p$ iff $j^{1/p} = j$ iff $j^p = j$ iff $j \in \mathbf{F}_p$.

(b) Recall (Ex. 4.7(c),(f)): every $\pi \in R$ of degree m satisfies $\pi\widehat{\pi} = \widehat{\pi}\pi = m_X$ and $\deg \widehat{\pi} = m$; so $p_X = \pi\widehat{\pi}$ with both of degree p holds for *every* $\pi \in R$ of degree p, and the first claim says: R contains an element of degree p iff $X \cong X_p$. Also $p_X = F'\widehat{F'}$ (proof of Ex. 4.15).

($\Leftarrow$) Let $\theta : X_p \to X$ be an isomorphism; composing with a translation we may assume $\theta(P_0) = P_0$. Then $\pi = F' \circ \theta^{-1} \in R$ has degree $\deg F' = p$.

($\Rightarrow$) Let $\pi \in R$ have degree p. By (T1), $0 = p \cdot \mathrm{id} = d(p_X) = d\pi \circ d\widehat{\pi}$ in k, so $d\pi = 0$ or $d\widehat{\pi} = 0$, i.e. one of $\pi, \widehat{\pi}$ is inseparable by (T2), hence (degree p prime) purely inseparable of degree p. By Prop. 2.5 (with $r = 1$, $Y = X$) such a morphism is, up to an isomorphism, the k-linear Frobenius $X_p \to X$, and in particular $X \cong X_p$ over k. (Concretely: if $L = \rho^* K(X) \subseteq K(X)$ for ρ purely inseparable of degree p, then $K(X)^p \subseteq L$ and $[K(X) : K(X)^p] = p$, so $L = K(X)^p$.)

Hasse invariant. Assume $X \cong X_p$ and let $\pi \in R$ be any element of degree p. The statement is symmetric in $\pi, \widehat{\pi}$ (as $\widehat{\widehat{\pi}} = \pi$), and by the argument of ($\Rightarrow$) at least one of $\pi, \widehat{\pi}$ is purely inseparable; replacing π by $\widehat{\pi}$ if necessary we may assume π is purely inseparable of degree p. By Prop. 2.5 it is then $\pi = F'\theta^{-1}$ for an isomorphism $\theta : X_p \to X$, and $\theta(P_0) = P_0$ because $\pi(P_0) = P_0$ and F' is the identity on points. Since θ^{-1} is an isomorphism, $\widehat{\theta^{-1}} = \theta$ ($\widehat{g}g = 1$), so $\widehat{\pi} = \theta\widehat{F'}$ (Ex. 4.7(b)). Now π is purely inseparable of degree p, and $\widehat{\pi}$ is separable iff $\widehat{F'}$ is, iff Hasse $= 1$ (Ex. 4.15). If Hasse$(X) = 1$, $\widehat{\pi}$ is separable and π is not, so they are not associates (composing with a unit preserves (in)separability). If Hasse$(X) = 0$ both π and $\widehat{\pi}$ are purely inseparable of degree p, so $\pi^*K(X) = K(X)^p = \widehat{\pi}^*K(X)$, and then $v = (\pi^*)^{-1} \circ \widehat{\pi}^*$ is an automorphism of the field $K(X)$ with $\widehat{\pi}^* = \pi^* \circ v$. By the equivalence of categories (I, 6.12) $v = u^*$ for an automorphism u of X, and $\widehat{\pi}^* = \pi^* u^* = (u \circ \pi)^*$, i.e. $\widehat{\pi} = u \circ \pi$. Moreover $u(P_0) = u(\pi(P_0)) = \widehat{\pi}(P_0) = P_0$. So u is a unit of R and $\widehat{\pi} = u\pi$: $\pi, \widehat{\pi}$ are associates. This proves: Hasse $= 0$ iff $\pi, \widehat{\pi}$ are associates.

(c) If Hasse $= 0$, $\widehat{F'} : X \to X_p$ is purely inseparable of degree p (not separable, Ex. 4.15). By Prop. 2.5 applied to $\widehat{F'}$, $X \cong (X_p)_p = X_{p^2}$ over k. By (a) applied twice, $j(X_{p^2}) = j(X)^{1/p^2}$, so $j = j^{1/p^2}$, i.e. $j^{p^2} = j$, i.e. $j \in \mathbf{F}_{p^2}$.

(d) For $f \in R$, f is a k-morphism so $f^* : H^1(X, \mathcal{O}_X) \to H^1(X, \mathcal{O}_X)$ is k-linear, and H^1 is 1-dimensional, so $f^* = \lambda_f \cdot \mathrm{id}$. $\varphi(1) = 1$; $(fg)^* = g^*f^*$ gives $\varphi(fg) = \varphi(f)\varphi(g)$; additivity: by (T3) (Ex. 4.15) the tangent map of $\widehat{f}$ is f^*, and $(f+g)^\wedge = \widehat{f} + \widehat{g}$ (Ex. 4.7(d)), so by (T1) $(f+g)^* = d(\widehat{f} + \widehat{g}) = f^* + g^*$. So $\varphi : R \to k$ is a ring homomorphism.

The absolute Frobenius commutes with every morphism of schemes: $f \circ F = F \circ f$ (both send $s \mapsto f^\#(s)^p = f^\#(s^p)$). Hence $F^*f^* = f^*F^*$ on $H^1(X, \mathcal{O}_X)$. Take $a \in H^1$ and recall F^* is p-linear: $F^*(\lambda_f a) = \lambda_f^p F^* a$, while $f^*(F^*a) = \lambda_f F^* a$. So $(\lambda_f^p - \lambda_f)F^*(a) = 0$ for all a. If Hasse$(X) \neq 0$, F^* is bijective and we can choose a with $F^*a \neq 0$, so $\lambda_f^p = \lambda_f$, i.e. $\lambda_f \in \mathbf{F}_p$. Thus $\varphi : R \to \mathbf{F}_p$, and it is onto since $\varphi(1) = 1$. Its kernel $\mathfrak{p}$ is a two-sided ideal with $R/\mathfrak{p} \cong \mathbf{F}_p$ a field, hence a (maximal and) prime ideal. (It contains p_X.) □

Exercise 4.21

> **Problem.** Let O be the ring of integers in a quadratic number field $\mathbf{Q}(\sqrt{-d})$. Show that any subring $R \subseteq O$, $R \neq \mathbf{Z}$, is of the form $R = \mathbf{Z} + f \cdot O$, for a uniquely determined integer $f \geq 1$. This integer f is called the conductor of the ring R.

Solution. Here $d > 0$ is squarefree (the field is imaginary quadratic; this is the case of interest, cf. Thm. 4.19) and subrings contain 1. Then $O = \mathbf{Z} \oplus \mathbf{Z}\eta$ with $\eta = \sqrt{-d}$ if $d \equiv 1, 2 \pmod 4$ and $\eta = \frac{1+\sqrt{-d}}{2}$ if $d \equiv 3 \pmod 4$. Let $\pi : O \to \mathbf{Z}$, $a + b\eta \mapsto b$, a surjective homomorphism of groups with kernel $\mathbf{Z} \cdot 1$.

Let $R \subseteq O$ be a subring, $R \neq \mathbf{Z}$. Then $\mathbf{Z} \subseteq R$ and $\pi(R) = f\mathbf{Z}$ for a unique integer $f \geq 0$. If $f = 0$ then $R \subseteq \ker \pi = \mathbf{Z}$, so $R = \mathbf{Z}$, excluded; hence $f \geq 1$. Since $\mathbf{Z} \subseteq R$ and $\pi(R) = f\mathbf{Z}$, we have $R = \pi^{-1}(f\mathbf{Z}) = \mathbf{Z} \oplus \mathbf{Z}f\eta$ (every element $a + b\eta$ of R has $f \mid b$; conversely there is $r_0 \in R$ with $\pi(r_0) = f$, $r_0 = a_0 + f\eta$, and $a_0 \in \mathbf{Z} \subseteq R$, so $f\eta \in R$ and then $a + bf\eta \in R$ for all $a, b \in \mathbf{Z}$). On the other hand

$$\mathbf{Z} + fO = \mathbf{Z} + \mathbf{Z}f + \mathbf{Z}f\eta = \mathbf{Z} + \mathbf{Z}f\eta.$$

So $R = \mathbf{Z} + fO$. (Conversely $\mathbf{Z} + fO$ is a subring for every $f \geq 1$: $(fu)(fv) = f(fuv) \in fO$.)

Uniqueness. $[O : \mathbf{Z} + fO] = [\mathbf{Z}\eta : \mathbf{Z}f\eta] = f$, because $O/R \cong \mathbf{Z}/\pi(R) = \mathbf{Z}/f$ (via π, as $R = \pi^{-1}(f\mathbf{Z})$). So $f = [O : R]$ is determined by R. □

Exercise 4.22

> **Problem.** If $X \to \mathbf{A}^1_{\mathbf{C}}$ is a family of elliptic curves having a section, show that the family is trivial. [Hints: Use the section to fix the group structure on the fibres. Show that the points of order 2 on the fibres form an etale cover of $\mathbf{A}^1_{\mathbf{C}}$, which must be trivial, since $\mathbf{A}^1_{\mathbf{C}}$ is simply connected. This implies that λ can be defined on the family, so it gives a map $\mathbf{A}^1_{\mathbf{C}} \to \mathbf{A}^1_{\mathbf{C}} - \{0, 1\}$. Any such map is constant, so λ is constant, so the family is trivial.] (Starred in the book.)

Solution. *Interpretation.* The scan prints the base as $\mathbf{A}^1_{\mathbf{C}}$ (affine line over $\mathbf{C}$, as the hint about simple connectedness requires). A family of elliptic curves is a smooth projective morphism $\pi : X \to \mathbf{A}^1 = \operatorname{Spec} A$, $A = \mathbf{C}[t]$, with all fibres of genus 1, and a section e; "trivial" means $X \cong E \times \mathbf{A}^1$ over $\mathbf{A}^1$ for an elliptic curve E (compatibly with the section). I give a direct proof in which the three hinted steps are replaced by a Weierstrass equation over A: the role of "the 2-torsion is a trivial cover" is played by the fact that A is a PID, and "λ is constant" by the fact that a family with a Weierstrass equation over A has constant discriminant, forcing a map from $\mathbf{A}^1$ to a curve of genus 1 to be constant.

Step 1: a global Weierstrass equation. Since π is smooth and e is a section, $D = e(\mathbf{A}^1)$ is an effective Cartier divisor on X, flat over A. Put $\mathcal{L}_n = \mathcal{O}_X(nD)$ and $M_n = H^0(X, \mathcal{L}_n)$. On each fibre X_t (a genus 1 curve with origin $e(t)$), $\mathcal{L}_n|_{X_t} = \mathcal{O}(n\,e(t))$ has degree $n \geq 1$, so by Riemann–Roch $h^0 = n$ and $h^1 = 0$ for all t (also for non-closed points, by semicontinuity). By cohomology and base change (III, 12.11, or Grauert 12.9), $\pi_*\mathcal{L}_n$ is locally free of rank n and $M_n \otimes k(t) = H^0(X_t, \mathcal{L}_n|_{X_t})$ for every t. A is a PID, so M_n is free of rank n. The inclusions $M_n \subseteq M_{n+1}$ stay injective on all fibres $H^0(X_t, \mathcal{O}(ne)) \subseteq H^0(X_t, \mathcal{O}((n+1)e))$, so M_{n+1}/M_n is locally free of rank 1, hence free. We have $M_1 = A \cdot 1$. Choose $x \in M_2$ with $M_2 = A \oplus Ax$ and $y \in M_3$ with $M_3 = M_2 \oplus Ay$. On every fibre x_t has a pole of order exactly 2 at $e(t)$ and y_t of order exactly 3 (they are not in the previous spaces), so x^2, xy, x^3, y^2 have exact pole orders $4, 5, 6, 6$ on every fibre, and by Nakayama x^2, xy generate M_4/M_3, M_5/M_4, and x^3 generates M_6/M_5. Therefore M_6 has A-basis $1, x, y, x^2, xy, x^3$ and $y^2 = u\,x^3 +$ (combination of $1, x, y, x^2, xy$) with $u \in A$ nowhere vanishing (the pole order of y_t^2 is exactly 6), i.e. $u \in A^* = \mathbf{C}^*$. Replacing (x, y) by (ux, uy) gives

$$y^2 + a_1xy + a_3y = x^3 + a_2x^2 + a_4x + a_6, \qquad a_i \in A.$$

As char $= 0$, replacing y by $y + \frac{1}{2}(a_1x + a_3)$ and x by $x - \frac{1}{3}a_2$ (changes inside M_3, M_2) we get $y^2 = x^3 + ax + b$ with $a, b \in A = \mathbf{C}[t]$.

The sections $1, x, y$ of $\mathcal{L}_3$ generate it on every fibre and give on each fibre X_t the closed embedding $X_t \hookrightarrow \mathbf{P}^2$ by $|3e(t)|$ ((3.3.3), as in the proof of Prop. 4.6). Hence $\iota : X \to \mathbf{P}^2_A$ is a proper morphism which is a closed immersion on every fibre, hence a closed immersion (proper + injective on points + unramified; II, 7.3 / EGA IV 18.12.6), with image inside $W = \{y^2z = x^3 + axz^2 + bz^3\} \subseteq \mathbf{P}^2_A$. Both X and W are flat over A (the cubic form is nonzero on every fibre) and ι is an isomorphism of X_t onto W_t for every closed t (X_t is an integral curve contained in the integral plane cubic W_t — $y^2z -$ (cubic in x, z) is irreducible — so it has the same underlying

space, and a closed subscheme of the reduced scheme W_t with the same underlying space is W_t). If $\mathcal{I}$ is the ideal of X in W, then $\mathcal{I} \otimes k(t) = 0$ for all closed t (flatness of $\mathcal{O}_X$ gives injectivity of $\mathcal{I} \otimes k(t) \to \mathcal{O}_{W_t}$ with image the ideal of X_t in W_t, which is 0), so $\mathcal{I} = 0$ by Nakayama and since the support of $\mathcal{I}$ is closed and every nonempty closed subset of W contains a closed point lying over a closed point of $\mathbf{A}^1$. Thus $X \cong W$ over A.

Step 2: the discriminant is constant. Each fibre X_t is nonsingular, so the cubic $x^3+a(t)x+b(t)$ has distinct roots: $\delta(t) = 4a(t)^3 + 27b(t)^2 \neq 0$ for all $t \in \mathbf{C}$. So $\delta \in A = \mathbf{C}[t]$ has no zeros, hence $\delta \equiv \delta_0 \in \mathbf{C}^*$ is a nonzero constant.

Step 3: a, b are constant. The affine curve $C : 4a^3 + 27b^2 = \delta_0$ in $\mathbf{A}^2$ is nonsingular (the partials $12a^2, 54b$ vanish only at $(0,0) \notin C$), and its closure $\overline{C} : 4a^3 + 27b^2c = \delta_0 c^3$ in $\mathbf{P}^2$ (coordinates a, b, c) is a nonsingular plane cubic (the only point at infinity, $c = 0$, is $(0:1:0)$ since then $a^3 = 0$, and there $\partial/\partial c = 27b^2 - 3\delta_0 c^2 = 27 \neq 0$), so $\overline{C}$ is irreducible (two components of a plane curve would meet in a singular point) of genus 1 ((I, Ex. 7.2) / (II, 8.20.3)). The polynomial map $t \mapsto (a(t), b(t))$ is a morphism $\mathbf{A}^1 \to C \subseteq \overline{C}$, which extends to a morphism $\mathbf{P}^1 \to \overline{C}$ ($\mathbf{P}^1$ is a nonsingular curve and $\overline{C}$ is projective). If it were nonconstant, it would be a finite morphism and Example 2.5.4 would give $g(\mathbf{P}^1) = 0 \geq g(\overline{C}) = 1$, which is absurd. Hence $a(t) \equiv a_0$, $b(t) \equiv b_0$ are constants.

Conclusion. $X \cong \{y^2z = x^3 + a_0xz^2 + b_0z^3\} \times \mathbf{A}^1 = E \times \mathbf{A}^1$ over $\mathbf{A}^1$, where E is the elliptic curve $y^2 = x^3 + a_0x + b_0$ (nonsingular since $\delta_0 \neq 0$), and the section e corresponds to the point at infinity $(0:1:0) \times \mathbf{A}^1$ (the pole of x, y). So the family is trivial. (The λ-invariant of the family is then constant, as in the hint: in the notation of Step 1 the roots of $x^3 + ax + b$ are the abscissas of the points of order 2, and they are constants.) $\square$

5. The canonical embedding

Conventions. k is algebraically closed; curves are nonsingular projective curves; K is a canonical divisor, $h^0(D) = \dim H^0(X, \mathcal{O}_X(D))$ and $\dim |D| = h^0(D) - 1$. Results of the text are quoted by number (Prop. 5.2, Thm. 5.4, Ex. 4.6, ...). Several exercises (5.2, 5.3, 5.5, 5.7) use that $\operatorname{Aut} X$ is finite, Bertini's theorem, Hurwitz' theorem and $|\operatorname{Aut} X| \leq 84(g-1)$ (Ex. 2.5); there I assume $\operatorname{char} k = 0$, as the book's Ex. 2.5 does. Two standard facts beyond the text are used and named where needed: the Plücker formula for the total weight of the Weierstrass points (this is the weighted form of Ex. 4.6(b)), and the elementary Castelnuovo–Severi inequality, which is proved below as Lemma A.

Lemma A (Castelnuovo–Severi, simple form). Let f, f' be base-point-free pencils on a curve X of genus g, of degrees a, b, i.e. morphisms $f, f' : X \to \mathbf{P}^1$. Then either $g \leq (a-1)(b-1)$, or there is a morphism $h : X \to Z$ of degree $m > 1$ with $m \mid a$, $m \mid b$, through which both f and f' factor. In particular: (i) if $\gcd(a,b) = 1$ then $g \leq (a-1)(b-1)$; (ii) if $a = b = 3$ and $g \geq 5$ then $f = \alpha \circ h$, $f' = \beta \circ h$ with $h : X \to \mathbf{P}^1$ of degree 3 and $\alpha, \beta \in \operatorname{Aut} \mathbf{P}^1$, so f and f' define the same linear system.

Proof. Let $\varphi = (f, f') : X \to \mathbf{P}^1 \times \mathbf{P}^1$, $Y = \varphi(X)$, $m = \deg(X \to Y)$, $\nu : \tilde{Y} \to Y$ the normalization, so $\varphi = \nu \circ h$ with $h : X \to \tilde{Y}$ of degree m. The projections $\mathrm{pr}_1 \circ \nu$, $\mathrm{pr}_2 \circ \nu : \tilde{Y} \to \mathbf{P}^1$ are finite of degrees a/m, b/m, so $m \mid a, b$. If $m = 1$, $X = \tilde{Y}$ is the normalization of the integral curve $Y \subset \mathbf{P}^1 \times \mathbf{P}^1$. Y is an effective divisor of some type (a', b'); intersecting with the two

rulings shows $\{a', b'\} = \{a, b\}$, and by (III, Ex. 5.6) $p_a(Y) = (a-1)(b-1)$. By (Ex. 1.8) $g \le p_a(Y)$. If $m > 1$ we are in the second alternative. For (ii): $m = 3$, so $\tilde{Y} \to \mathbf{P}^1$ has degree 1 and is finite onto $\mathbf{P}^1$ with $\tilde{Y}$ normal, hence an isomorphism; so $f = \alpha \circ h$ with $\alpha : \tilde{Y} \cong \mathbf{P}^1$, and likewise $f' = \beta \circ h$ with $\beta : \tilde{Y} \cong \mathbf{P}^1$ (here $h : X \to \tilde{Y}$). □

Exercise 5.1

> **Problem.** Show that a hyperelliptic curve can never be a complete intersection in any projective space. Cf. (Ex. 3.3).

Solution. Let X be hyperelliptic (so $g \ge 2$) and suppose $X \subseteq \mathbf{P}^n$ is a (nonsingular) complete intersection of hypersurfaces of degrees $d_1, \dots, d_{n-1}$. By (II, Ex. 8.4(c)), $\omega_X \cong \mathcal{O}_X(a)$ with $a = \sum d_i - n - 1$. Since $\deg \omega_X = 2g - 2 > 0$ and $\deg \mathcal{O}_X(1) > 0$ we get $a \ge 1$.

We show that $|K| = |aH|$ (H a hyperplane section) is very ample, contradicting Prop. 5.2. By (3.1) we must show that for all $P, Q \in X$ (possibly equal) $h^0(K - P - Q) = h^0(K) - 2$, i.e. that the evaluation map $H^0(X, \mathcal{O}_X(a)) \to H^0(\mathcal{O}_{P+Q} \otimes \mathcal{O}_X(a))$ onto the 2-dimensional space of $\mathcal{O}_X(a)$ restricted to the subscheme $P + Q$ is surjective. Choose a linear form ℓ_0 not vanishing at P or Q and let $V \subseteq H^0(\mathcal{O}_X(a))$ be the image of $\{\ell_0^{a-1}\ell : \ell \text{ a linear form}\}$. Since ℓ_0^{a-1} is a unit near P, Q, the image of V in $H^0(\mathcal{O}_{P+Q} \otimes \mathcal{O}_X(a))$ is the same as that of the linear forms ℓ in $H^0(\mathcal{O}_{P+Q} \otimes \mathcal{O}_X(1))$ up to a unit. But linear forms on $X \subseteq \mathbf{P}^n$ separate points and tangent vectors ($\mathcal{O}_X(1)$ is very ample), i.e. they surject onto $H^0(\mathcal{O}_{P+Q} \otimes \mathcal{O}_X(1))$. Hence $H^0(\mathcal{O}_X(a)) \to H^0(\mathcal{O}_{P+Q} \otimes \mathcal{O}_X(a))$ is surjective, so K is very ample, contradicting Prop. 5.2 since X is hyperelliptic. □

Exercise 5.2

> **Problem.** If X is a curve of genus ≥ 2 over a field of characteristic 0, show that the group $\operatorname{Aut} X$ of automorphisms of X is finite. [*Hint*: If X is hyperelliptic, use the unique g^1_2 to show that $\operatorname{Aut} X$ permutes the ramification points of the 2-fold covering $X \to \mathbf{P}^1$. If X is not hyperelliptic, show that $\operatorname{Aut} X$ permutes the hyperosculation points (Ex. 4.6) of the canonical embedding.] Cf. (Ex. 2.5).

Solution. A fixed point bound. *If $\sigma \ne \mathrm{id}$ is an automorphism of X (genus $g \ge 2$), then σ has at most $2g+2$ fixed points.* The fixed set is closed and proper (an equalizer of two morphisms to a separated curve), hence finite, so there is Q with $\sigma Q \ne Q$. By Riemann–Roch $h^0((g+1)Q) \ge 2$, so there is a nonconstant f with poles only at Q, of order $m \le g+1$ (a nonconstant function has a pole). Then $f \circ \sigma$ has its only pole at $\sigma^{-1}Q \ne Q$, so $h = f - f \circ \sigma \ne 0$ has poles of total degree $2m \le 2g + 2$, hence at most $2g + 2$ zeros. If $P \ne Q$ is fixed by σ then $P \ne \sigma^{-1}Q$ (otherwise $Q = \sigma P = P$), f is regular at P and $h(P) = f(P) - f(\sigma P) = 0$. So σ has at most $2g + 2$ fixed points besides possibly Q (which is not fixed). (∗)

Hyperelliptic case. Let $f : X \to \mathbf{P}^1$ be of degree 2 and $L = f^*\mathcal{O}(1)$. Any g^1_2 is complete (Clifford: a g^r_2 has $r \le 1$, so $|L|$ is the complete system), and by Prop. 5.3 the g^1_2 is unique. If $\sigma \in \operatorname{Aut} X$ then $\sigma^*|L|$ is a g^1_2, so $\sigma^* L \cong L$ and σ^* acts on the 2-dimensional space $H^0(L)$ (after choosing an isomorphism) by some $A \in GL_2$ up to scalar; thus $f \circ \sigma = \tau_\sigma \circ f$ with $\tau_\sigma \in PGL_2$ uniquely determined (because f is surjective), and $\sigma \mapsto \tau_\sigma$ is a homomorphism $\operatorname{Aut} X \to \operatorname{Aut} \mathbf{P}^1$.

Let $R \subseteq X$ be the ramification points of f and $B = f(R)$. By Hurwitz (II, 2.4 / the formula $2g-2 = 2(-2) + \deg R$, valid in characteristic 0) $|B| = 2g+2 \geq 6$. f is ramified at P iff $f \circ \sigma = \tau_\sigma f$ is ramified at $\sigma^{-1}P$, so σ permutes R and τ_σ permutes B. An element of $\operatorname{Aut} \mathbf{P}^1$ fixing three points is the identity, hence the image of $\sigma \mapsto \tau_\sigma$ embeds into the finite group $\operatorname{Sym}(B)$. The kernel consists of σ with $f \circ \sigma = f$, i.e. of elements of the Galois group of the degree-2 extension $K(X)/K(\mathbf{P}^1)$: at most 2 elements. So $|\operatorname{Aut} X| \leq 2\,(2g+2)!$.

Nonhyperelliptic case ($g \geq 3$; for $g = 2$ the curve is hyperelliptic). Let $W = \{P \in X : h^0(K - gP) > 0\}$, i.e. the points through which some canonical divisor has multiplicity $\geq g$. In the canonical embedding $X \subset \mathbf{P}^{g-1}$ (Prop. 5.2) this says that some hyperplane meets X at P with multiplicity $\geq g = n+1$ ($n = g-1$): the hyperosculation points of Ex. 4.6(b). If $\sigma \in \operatorname{Aut} X$ then $\sigma^* K \sim K$, so σ maps canonical divisors to canonical divisors, and therefore $\sigma(W) = W$. Hence there is a homomorphism $\operatorname{Aut} X \to \operatorname{Sym}(W)$ once we know W is finite, and its kernel consists of automorphisms with $|W|$ fixed points. By $(*)$ the kernel is trivial as soon as $|W| \geq 2g+3$. So it suffices to show: W is finite and $|W| \geq 2g+3$.

Let $P \in X$ and $0 = a_0 < a_1 < \cdots < a_{g-1}$ the vanishing orders at P of the g sections of $H^0(K)$; $h^0(K - aP) = \#\{i : a_i \geq a\}$. Define the weight $w(P) = \sum_i (a_i - i) \geq 0$; $P \in W$ iff $a_{g-1} \geq g$ iff $w(P) > 0$. By Ex. 4.6(b) (here $n = g-1$, $d = 2g-2$, so $n(n+1)(g-1) + (n+1)d = g(g-1)(g+1)$ hyperosculation points counted with weights), $\sum_P w(P) = g(g-1)(g+1)$. For completeness, this is the following Wronskian argument: if $s_1, \ldots, s_g$ is a basis of $H^0(\omega_X)$ and $s_j = f_j\,dz$ in a local parameter z, the Wronskian $\det(d^i f_j / dz^i)_{0 \leq i < g}$ is a global section of $\omega_X^{\otimes g(g+1)/2}$ (it is multiplied by the right power of the Jacobian under a change of parameter), of degree $\frac{1}{2}g(g+1)(2g-2) = g(g-1)(g+1)$; in characteristic 0 it is not identically 0 because the f_j are linearly independent over k, and its order of vanishing at P is $\sum_i (a_i - i) = w(P)$. In particular W is finite and nonempty.

Weight bound. For each $j \geq 0$ with $a := a_{g-1-j}$, the divisor $D = K - aP$ is effective and special (as $K - D = aP$ is effective) with $\dim |D| = j$ and $\deg D = 2g-2-a$. By Clifford (Thm. 5.4) $j \leq \frac{1}{2}(2g-2-a)$, i.e. $a_{g-1-j} \leq 2g-2-2j$, i.e. $a_i \leq 2i$ for all i. So $w(P) \leq \sum i = g(g-1)/2$. If $a_i = 2i$ for some $1 \leq i \leq g-2$, then for $D = K - a_i P$ we have equality in Clifford, with $D \neq 0$ ($\deg D = 2g-2-2i > 0$) and $D \neq K$; by Thm. 5.4, X would be hyperelliptic, which it is not. So $a_i \leq 2i-1$ for every $1 \leq i \leq g-2$ (and $g \geq 3$ so this range is nonempty), whence $w(P) \leq g(g-1)/2 - 1$ for every P (in particular for $P \in W$). (For $g = 3$ this is 2.) Therefore

$$|W| \;\geq\; \frac{g(g-1)(g+1)}{g(g-1)/2 - 1} \;>\; \frac{g(g-1)(g+1)}{g(g-1)/2} = 2g+2,$$

and $|W| \geq 2g+3$. (For $g = 3$ this reads $|W| \geq 12$; indeed the flexes of a smooth quartic are 24 counted with weight.) This proves finiteness of $\operatorname{Aut} X$. □

Exercise 5.3

> **Problem.** *Moduli of Curves of Genus 4.* The hyperelliptic curves of genus 4 form an irreducible family of dimension 7. The nonhyperelliptic ones form an irreducible family of dimension 9. The subset of those having only one g^1_3 is an irreducible family of dimension 8. [*Hint*: Use (5.2.2) to count how many complete intersections $Q \cap F_3$ there are.]

Solution. Throughout $\operatorname{char} k = 0$ (Ex. 5.2: finiteness of Aut, Bertini), and "dimension of a

family" is the dimension of its image in the (coarse) moduli space $\mathfrak{M}_4$, as in Examples 5.5.5, 5.5.6.

Hyperelliptic. By Example 5.5.5 a hyperelliptic curve of genus g is the double cover of $\mathbf{P}^1$ branched in $2g+2$ distinct points, and by Prop. 5.3 the g^1_2 and hence the branch set B is determined by X up to $\operatorname{Aut}\mathbf{P}^1$ (and conversely B determines X: $y^2 = \prod(x-b_i)$). The space of branch sets is an open subset of $\operatorname{Sym}^{10}\mathbf{P}^1 = \mathbf{P}^{10}$ (irreducible, dim 10) for $g=4$, and PGL_2 (dim 3) acts with finite stabilizers (a set of ≥ 3 points). So the family has dimension $10-3=7$ and is irreducible (image of an irreducible variety).

Nonhyperelliptic. By Example 5.2.2, the canonical models are exactly the nonsingular complete intersections $X = Q \cap F$ of a quadric Q and a cubic F in $\mathbf{P}^3$ (conversely such X has $\omega_X = \mathcal{O}_X(2+3-4) = \mathcal{O}_X(1)$, degree $6 = 2g-2$, $g=4$, and $H^0(\mathcal{O}_{\mathbf{P}^3}(1)) \to H^0(\omega_X)$ is injective of dimension $4=g$, so it is the canonical embedding). X determines Q (unique quadric through X, 5.2.2). By the Koszul resolution $0 \to \mathcal{O}(-5) \to \mathcal{O}(-2)\oplus\mathcal{O}(-3) \to \mathcal{I}_X \to 0$ and $H^1(\mathcal{O}_{\mathbf{P}^3}(-2)) = 0$ we get $H^0(\mathcal{I}_X(3)) = Q \cdot H^0(\mathcal{O}(1)) \oplus kF$ of dimension 5; so F is determined by X modulo $Q\cdot$(linear forms) and scalars. Hence the set of such curves is parametrised by an open subset $\mathcal{U}$ (nonsingularity is open, nonempty by Bertini) of the $\mathbf{P}^{15}$-bundle over $\mathbf{P}^9$ whose fibre over Q is $\mathbf{P}(S_3/Q\cdot S_1)$, $\dim S_3 = 20$. So $\mathcal{U}$ is irreducible of dimension $9+15=24$. Two such curves are isomorphic iff they differ by an element of PGL_4 (canonical embedding is unique up to PGL_4), and stabilizers are the finite groups $\operatorname{Aut}X$ (Ex. 5.2). So $\mathcal{U} \to \mathfrak{M}_4$ has fibres of dimension $\dim PGL_4 = 15$, and the image is irreducible of dimension $24-15=9$.

Exactly one g^1_3. By Example 5.5.2, $X = Q \cap F$ has exactly two g^1_3 if Q is nonsingular and exactly one if Q is a quadric cone. Quadric cones form the (irreducible) determinant hypersurface of rank ≤ 3 quadrics in $\mathbf{P}^9$, of dimension 8. Such a Q with vertex V: for F general, nonsingular $X = Q\cap F$ exist: $|\mathcal{O}_Q(3)|$ is base point free so a general F misses V, and by Bertini (II 8.18) a general $F\cap Q$ is nonsingular on $Q\setminus V$. (Note: if $V \in X$ then X is singular at V, since Q is singular there; so for nonsingular X the vertex is not on X.) The same count gives an irreducible family of dimension $8+15=23$ in the parameter space, and $23-15=8$ in $\mathfrak{M}_4$. □

Exercise 5.4

> **Problem.** Another way of distinguishing curves of genus g is to ask, what is the least degree of a birational plane model with only nodes as singularities (3.11)? Let X be nonhyperelliptic of genus 4. Then:
>
> (a) if X has two g^1_3's, it can be represented as a plane quintic with two nodes, and conversely;
>
> (b) if X has only one g^1_3, then it can be represented as a plane quintic with a tacnode (I, Ex. 5.14d), but the least degree of a plane representation with only nodes is 6.

Solution. Here "represented as a plane curve X'" means a birational morphism $\nu : X \to X' \subset \mathbf{P}^2$ (so X is the normalization). We use: for an integral plane curve X' of degree e with normalization of genus g, $g = \frac{1}{2}(e-1)(e-2) - \sum \delta_x$, $\delta_x = \operatorname{length}(\tilde{\mathcal{O}}_x/\mathcal{O}_x)$ (Ex. 1.8); a singular point consisting of two smooth branches has δ equal to their intersection multiplicity, so $\delta = 1$ iff it is a node. Since X is nonhyperelliptic of genus 4 every g^1_3 is complete and base-point free (a base point would give a g^1_2; Clifford gives dim ≤ 1).

Write $X \subset Q \cap F \subset \mathbf{P}^3$ for the canonical model (5.2.2), $K = H$ the hyperplane class.

Preliminary: projection from a point. For $P \in X$, $|K - P|$ has dimension 2 and is base point free: $h^0(K - P - Q') = 2$ for all Q', since $h^0(P + Q') = 1$ (X is nonhyperelliptic) and Riemann–Roch give $h^0(K - P - Q') = h^0(P + Q') - (2 + 1 - 4) = 2$. It is the projection ψ_P from P of the canonical curve to $\mathbf{P}^2$; it is birational onto a quintic X_P (the image has degree m with $m \mid 5$ and cannot be a line, so $m = 5$). $p_a(X_P) = 6$, so $\sum \delta = 2$. Two points $R, R' \neq P$ have the same image iff P, R, R' are collinear, and $\psi_P(P) = \psi_P(R)$ iff R lies on the tangent line T_P; ψ_P fails to be an immersion at $R \neq P$ iff $T_R \ni P$, at P iff T_P meets X with multiplicity ≥ 3 at P. A line meeting $X = Q \cap F$ in ≥ 3 points (counted with multiplicity) meets Q in ≥ 3 points, hence lies in Q.

(a) Suppose X has two g^1_3. By Example 5.5.2, Q is nonsingular, $Q \cong \mathbf{P}^1 \times \mathbf{P}^1$, and X has type $(3, 3)$ on Q (II 6.6.1); the two g^1_3 are cut out by the two rulings. Choose $P \in X$ such that P is not a ramification point of either projection $p_i : X \to \mathbf{P}^1$ (finitely many exceptions, char 0) and such that the two ruling lines L_1, L_2 through P meet X in three distinct points each: $L_i \cap X = \{P, Q_i, R_i\}$ (the fibre of p_i through P is reduced for P outside the finitely many branch fibres). Since $L_1 \cap L_2 = \{P\}$, the points Q_1, R_1, Q_2, R_2 are distinct and different from P. Also $T_P \neq L_i$ (transversality), $T_P \subset T_P Q$, and $T_P \cap Q = \{P\}$ with multiplicity 2, so T_P meets X only at P (a double point there). Lines through P meeting X in two more points or tangent elsewhere lie in Q, so are L_1, L_2, which meet X transversally. Consequently ψ_P is an immersion everywhere (at P: if T_P met X with multiplicity 3 at P, then, as $Q(\gamma(t)) = 0$ along a local parametrization with the line T_P contained in $T_P Q$, $Q|_{T_P}$ vanishes to order ≥ 3 and being quadratic it vanishes identically, so $T_P \subset Q$, contradiction), injective except $\psi_P(Q_i) = \psi_P(R_i)$, $i = 1, 2$. So X_P has exactly two singular points, each with two smooth branches, hence $\delta \geq 1$ at each; as $\sum \delta = 2$, both are nodes. So X is a plane quintic with two nodes N_1, N_2.

Conversely, let $\nu : X \to X'$ be a quintic with two nodes N_1, N_2 and H the pullback of the line class. If $\nu^{-1}(N_i) = \{N_i', N_i''\}$, the linear forms vanishing at N_i give a 2-dimensional subspace of $H^0(H - N_i' - N_i'')$ (degree 3), i.e. a g^1_3 D_i. If $D_1 \sim D_2$ then $N_1' + N_1'' \sim N_2' + N_2''$ is a divisor of degree 2 with $h^0 \geq 2$ unless the two divisors coincide ($h^0 = 1$ since X is nonhyperelliptic), which is impossible as $N_1 \neq N_2$. So X has two distinct g^1_3.

(b) Suppose X has only one g^1_3. By Example 5.5.2, Q is a quadric cone with vertex V, $V \notin X$ (see 5.3), and the lines of the cone cut out the g^1_3 (X meets each line of the cone in three points). Take $P \in X$ general so that the line L of the cone through P and V meets X in three distinct points P, Q', R' transversally and $T_P \neq L$. The only lines in Q through P is L. As in (a), ψ_P is an immersion everywhere, injective except $\psi_P(Q') = \psi_P(R')$. So X_P has a unique singular point c, with two smooth branches (images of neighbourhoods of Q', R'), and $\delta_c = 2$, i.e. the two smooth branches have intersection multiplicity 2, hence are tangent, and the singularity is analytically $y(y - x^2) = 0$: this is a tacnode (I, Ex. 5.14d). So X is a plane quintic with a tacnode.

Least degree with only nodes. If $\nu : X \to X'$ is a birational plane model of degree e with only nodes, then $4 = \frac{1}{2}(e - 1)(e - 2) - \#\text{nodes}$, so $\frac{1}{2}(e - 1)(e - 2) \geq 4$ forces $e \geq 5$, and $e = 5$ forces exactly two nodes, so by the converse in (a) X would have two g^1_3, contrary to hypothesis. Hence $e \geq 6$. On the other hand projecting the canonical sextic in $\mathbf{P}^3$ from a general point of $\mathbf{P}^3$ (Thm. 3.10 and the proof of 3.6) gives a birational morphism onto a plane sextic with only nodes. So the least degree is 6. □

Exercise 5.5

Problem. *Curves of Genus 5.* Assume X is not hyperelliptic.

(a) The curves of genus 5 whose canonical model in $\mathbf{P}^4$ is a complete intersection $F_2 \cdot F_2 \cdot F_2$ form a family of dimension 12.

(b) X has a g^1_3 if and only if it can be represented as a plane quintic with one node. These form an irreducible family of dimension 11. [*Hint*: If $D \in g^1_3$, use $K - D$ to map $X \to \mathbf{P}^2$.]

*(c) In that case, the conics through the node cut out the canonical system (not counting the fixed points at the node). Mapping $\mathbf{P}^2 \to \mathbf{P}^4$ by this linear system of conics, show that the canonical curve X is contained in a cubic surface $V \subseteq \mathbf{P}^4$, with V isomorphic to $\mathbf{P}^2$ with one point blown up (II, Ex. 7.7). Furthermore, V is the union of all the trisecants of X corresponding to the g^1_3 (5.5.3), so V is contained in the intersection of all the quadric hypersurfaces containing X. Thus V and the g^1_3 are unique.

Note. Conversely, if X does not have a g^1_3, then its canonical embedding is a complete intersection, as in (a). More generally, a classical theorem of Enriques and Petri shows that for any nonhyperelliptic curve of genus $g \geq 3$ the canonical model is projectively normal and is an intersection of quadric hypersurfaces unless X has a g^1_3 or $g = 6$ and X has a g^2_5 (see Saint-Donat [1]). (Starred in the book: part (c).)

Solution. Again $\operatorname{char} k = 0$. (Only the dimension counts need this.)

(a) A complete intersection $X = Q_1 \cap Q_2 \cap Q_3 \subset \mathbf{P}^4$ of three quadrics has $\omega_X = \mathcal{O}_X(2+2+2-5) = \mathcal{O}_X(1)$ (II, Ex. 8.4), so $\deg \omega_X = 8 = 2g-2$, $g = 5$. The map $H^0(\mathcal{O}_{\mathbf{P}^4}(1)) \to H^0(\omega_X)$ is injective (X nondegenerate, as $I_X(1) = 0$ by the Koszul complex) between spaces of dimension $5 = g$, so $X \subset \mathbf{P}^4$ is the canonical embedding (and X is nonhyperelliptic since K is very ample). By the Koszul complex $H^0(\mathcal{I}_X(2)) = \langle Q_1, Q_2, Q_3 \rangle$ is 3-dimensional, so X determines a point of the Grassmannian $G(3, 15)$ of 3-dimensional subspaces of the 15-dimensional space of quadratic forms, $\dim G(3, 15) = 3 \cdot 12 = 36$. Nets with nonsingular intersection form a nonempty open subset $\mathcal{U}$ (Bertini applied to the Veronese embedding $v_2(\mathbf{P}^4) \subset \mathbf{P}^{14}$ three times), irreducible of dimension 36. Two such curves are isomorphic iff their nets are PGL_5-conjugate (uniqueness of the canonical embedding up to projective equivalence), and stabilizers are finite (Ex. 5.2). So the image in $\mathfrak{M}_5$ is irreducible of dimension $36 - 24 = 12$ (which is $\dim \mathfrak{M}_5 = 3g - 3$, i.e. these curves are the general curves of genus 5).

(b) *($\Leftarrow$)* Suppose $\nu : X \to X' \subset \mathbf{P}^2$ is a plane quintic with a node N, $\nu^{-1}(N) = \{N', N''\}$, H the line class. The pencil of lines through N gives a 2-dimensional subspace of $H^0(H - N' - N'')$, of degree 3: so X has a g^1_3.

($\Rightarrow$) Let D be a g^1_3; it is complete with $h^0(D) = 2$ (Clifford, $\dim |D| \leq 3/2$). $\deg(K - D) = 5$ and Riemann–Roch gives $h^0(D) - h^0(K - D) = 3 + 1 - 5$, so $h^0(K - D) = 3$: $|K - D|$ is a g^2_5. It has no base points: if P were one, $h^0(K - D - P) = 3$ and Riemann–Roch gives $h^0(D + P) = h^0(K - D - P) + 4 + 1 - 5 = 3$, so $D + P$ is a g^2_4, contradicting Clifford (equality $\dim |D + P| = \frac{1}{2} \deg$ forces $D + P = 0$, K or hyperelliptic). So $\psi = \varphi_{|K-D|} : X \to \mathbf{P}^2$ is a morphism, nondegenerate (the system is complete with $h^0 = 3$), of image X' of degree m with $m \cdot \deg \psi = 5$ and $m \geq 2$: so $m = 5$, ψ is birational onto a plane quintic X'. $p_a(X') = 6$, so $\sum \delta = 1$: X' has exactly one singular point, a double point with $\delta = 1$, i.e. a node or an ordinary cusp.

Remark (honest correction). The cusp can occur, so the literal "only if" needs "node or ordinary cusp". Let X' be a plane quintic with an ordinary cusp c and no other singularity (such quintics

exist: a general member of the linear system of quintics with an ordinary cusp at a given point, by Bertini), and let X be its normalization, of genus 5; c has one preimage P_0. The lines through c cut a base-point-free g^1_3 $D = H - 2P_0$, so by Lemma A(i) with $(a,b) = (2,3)$ X is not hyperelliptic. For a plane quintic whose only singularity is a double point of $\delta = 1$ one has $K = 2H - C$ with C the conductor divisor ($N' + N''$ for a node, $2P_0$ for a cusp; adjunction on the plane), so in both cases $D = H - C$ and $K - D = H$: the model is $\varphi_{|K-D|}$ for the pencil D given by the lines through the double point. Now if this X also had a nodal quintic model, its pencil of lines through the node would be a second base-point-free g^1_3 D', equal to D by Lemma A(ii) ($g = 5$); then both models are $\varphi_{|K-D|}$, hence projectively equivalent, which is absurd (one has a node, the other a cusp). So such X has a g^1_3 but no nodal quintic model: the nodal case is the general one (the cuspidal ones form a family of dimension $16 + 2 - 8 = 10$: quintics with an ordinary cusp at a given point form a $\mathbf{P}^{16}$, the cusp point varies in $\mathbf{P}^2$, and $\dim PGL_3 = 8$). In (c) below the arguments are identical for the cusp (conics through the cusp), and I assume the singular point is a node as in the statement.

Dimension 11. The plane quintics with a node at a point N form a linear system of dimension $21 - 3 = 18$, i.e. $\mathbf{P}^{17}$; as N varies we get an irreducible variety (a $\mathbf{P}^{17}$-bundle over $\mathbf{P}^2$) of dimension 19, in which the curves with exactly one node, an ordinary one, and nonsingular elsewhere form a nonempty open subset (nonempty by Bertini on the blow-up of N with the very ample system $|5L - 2E|$). Isomorphism classes of the normalizations correspond to PGL_3-orbits: if X'_1, X'_2 have isomorphic normalizations X, the g^1_3 is unique (Lemma A(ii), $g = 5$), so the model is $\varphi_{|K-D|}$ for the same D, hence unique up to PGL_3; stabilizers are finite (Ex. 5.2). So the family is irreducible of dimension $19 - 8 = 11$.

(c) Let $\nu : X \to X'$ be a nodal quintic as in (b), $H = K - D$, $\nu^{-1}(N) = \{N', N''\}$. The pencil of lines through N gives a g^1_3 $D' \subseteq |H - N' - N''|$; by Lemma A(ii) $D' = D$ (unique g^1_3, $g = 5$), so $K = H + D = 2H - N' - N''$. Conics through N form a 5-dimensional space which injects into $H^0(\mathcal{O}_X(2H - N' - N'')) = H^0(K)$ (X' is not on a conic as it is an irreducible quintic), which has dimension 5; hence the conics through N cut out exactly $|K|$ outside N', N''.

Put $\mathbf{P}^2$ with coordinates x, y, z and $N = (0:0:1)$. The conics through N are spanned by x^2, xy, y^2, xz, yz, and $\theta = (x^2 : xy : y^2 : xz : yz) : \mathbf{P}^2 \dashrightarrow \mathbf{P}^4$ is the canonical map composed with ν. Let $\pi : \tilde{P} \to \mathbf{P}^2$ be the blow-up at N, with exceptional curve E, line class L; then θ extends to $\tilde{\theta} : \tilde{P} \to \mathbf{P}^4$ given by $|2L - E|$ (the blow-up resolves the base point). As ν separates the two branches at the node, ν lifts to $\tilde{\nu} : X \to \tilde{P}$ onto the proper transform $\tilde{X}'$, of class $5L - 2E$, and $\tilde{\theta} \circ \tilde{\nu}$ is the canonical map. Put $V = \tilde{\theta}(\tilde{P}) \supseteq X$.

V is the cubic scroll, $\tilde{\theta}$ is an isomorphism onto V. Write $(a_0, \dots, a_4) = (x^2, xy, y^2, xz, yz)$. The three quadrics $a_0a_2 - a_1^2$, $a_0a_4 - a_1a_3$, $a_1a_4 - a_2a_3$ (the 2×2 minors of $\begin{pmatrix} a_0 & a_1 & a_3 \\ a_1 & a_2 & a_4 \end{pmatrix}$) vanish on θ, hence on V. Conversely let a be a point where all minors vanish. If $(a_3, a_4) = (0,0)$ then $a_0a_2 = a_1^2$, $(a_0 : a_1 : a_2) = (x^2 : xy : y^2)$ and $a = \theta(x : y : 0)$. If $(a_3, a_4) = (u, v) \neq 0$, the minors give $va_0 = ua_1$, $va_1 = ua_2$, so $(a_0, a_1, a_2) = t(u^2, uv, v^2)$; for $t \neq 0$, $a = \theta(u : v : 1/t)$, and for $t = 0$, $a = (0:0:0:u:v)$ is the limit point of θ along the exceptional direction. So $V = V(\text{minors})$ as a set, and V is the union of $\theta(\mathbf{P}^2 \setminus N)$ and the line $\{a_0 = a_1 = a_2 = 0\} = \tilde{\theta}(E)$. Now $\tilde{\theta}$ is injective on points by this description (a point a with $(a_3, a_4) = (u, v) \neq 0$ determines $(u : v)$ and t, hence the point $(u : v : 1/t)$ of $\mathbf{P}^2 \setminus N$, or, if $t = 0$, the direction $(u : v)$ at N; a point with $(a_3, a_4) = 0$ determines $(x : y)$). It is an immersion: on $\mathbf{P}^2 \setminus N$ use the charts $x = 1$ and $y = 1$ (which do not contain N): in the chart $x = 1$ the map is $(y, z) \mapsto (1 : y : y^2 : z : yz)$,

and $a_1/a_0 = y$, $a_3/a_0 = z$ are local coordinates on the target, so it is an immersion (likewise for $y = 1$); the two charts $x = 1$, $y = 1$ cover $\mathbf{P}^2 \setminus N$. Along the exceptional curve use the blow-up chart $y = xs$ (coordinates (x, s), $E = \{x = 0\}$), where the map reads $(x, s) \mapsto (x : xs : xs^2 : 1 : s)$, i.e. (x, xs, xs^2, s) in the affine chart $a_3 = 1$, with Jacobian columns $(1, s, s^2, 0)$ and $(0, x, 2xs, 1)$ always independent; the chart $x = ys'$ is symmetric. A proper morphism that is injective and immersive is a closed immersion, so $\tilde{\theta} : \tilde{P} \to V$ is an isomorphism and $V \cong \mathbf{P}^2$ blown up at a point; $\deg V = (2L - E)^2 = 3$.

V is the union of the trisecants. For a line $\ell \ni N$ in $\mathbf{P}^2$ let $\tilde{\ell}$ be its proper transform, of class $L-E$. Then $(2L-E)\cdot(L-E) = 1$, so $\tilde{\theta}(\tilde{\ell})$ is a line Λ_ℓ in $\mathbf{P}^4$, and $\tilde{X}'\cdot\tilde{\ell} = (5L-2E)(L-E) = 3$, so Λ_ℓ meets X in a divisor of degree 3: it is the trisecant spanned by the members of the divisor of the g^1_3 cut by the pencil of lines through N (the divisor D' above; compare 5.5.3). The lines Λ_ℓ, together with the line $\tilde{\theta}(E)$ (the image of the directrix, limit of Λ_ℓ), sweep out V: $V = \bigcup_\ell \Lambda_\ell$. If Q' is any quadric containing X, then $\tilde{\theta}^*Q'$ is a section of $\mathcal{O}_{\tilde{P}}(2(2L - E))$ vanishing on $\tilde{X}'$; its restriction to $\tilde{\ell} \cong \mathbf{P}^1$ is a section of $\mathcal{O}(2)$ vanishing on the degree-3 divisor $\tilde{X}' \cap \tilde{\ell}$, hence is zero. So $\Lambda_\ell \subseteq Q'$ for every ℓ, hence $V \subseteq Q'$. Thus $V \subseteq \bigcap_{Q' \supseteq X} Q'$. Since the three minors above are quadrics containing $V \supseteq X$ and cut out V set-theoretically, the intersection of all quadrics containing X *equals* V (as a set). So V depends only on $X \subset \mathbf{P}^4$, and the g^1_3 is unique by Lemma A(ii). □

Exercise 5.6

> **Problem.** Show that a nonsingular plane curve of degree 5 has no g^1_3. Show that there are nonhyperelliptic curves of genus 6 which cannot be represented as a nonsingular plane quintic curve.

Solution. Let $X \subset \mathbf{P}^2$ be a nonsingular quintic, H the line class. Then $g = 6$ and $\omega_X = \mathcal{O}_X(2)$ (II, 8.20.3 / Ex. 8.4). The restriction $H^0(\mathbf{P}^2, \mathcal{O}(2)) \to H^0(\mathcal{O}_X(2)) = H^0(K)$ is injective (X not in a conic) between spaces of dimension 6, hence bijective: *sections of K are the conics.*

Suppose D is a g^1_3; by Clifford D is complete with $h^0(D) = 2$ (and we may replace D by an effective member). Riemann–Roch: $h^0(K - D) = h^0(D) - (3 + 1 - 6) = 4$. By the above, $h^0(K - D)$ is the dimension of the space of conics containing the scheme $D \subset X \subset \mathbf{P}^2$ (i.e. whose restriction to X vanishes on D). We show that conics through a degree-3 divisor D on X form a space of dimension exactly 3, a contradiction.

Claim. If D is an effective divisor of degree 3 on a nonsingular plane curve X, the conics containing D form a 3-dimensional space. Let ℓ_D be a line with $D \le \ell_D \cdot X$ if one exists. (i) If $D \le \ell \cdot X$ for a line ℓ: a conic containing D restricts to ℓ as a binary quadratic form vanishing on the degree-3 subscheme $D \subset \ell$, so it vanishes on ℓ and equals $\ell \cdot \ell'$; these form a 3-dimensional space. (ii) Otherwise pick $P \in \operatorname{Supp} D$ and $D' = D - P$ (degree 2); D' lies on a unique line ℓ' and imposes 2 independent conditions on conics (conics surject onto binary quadratics on ℓ', and then onto the length-2 subscheme), so conics containing D' form a 4-dimensional space. Since $D \not\le \ell' \cdot X$ (else we are in (i)), choose a line ℓ'' missing $\operatorname{Supp} D$: then $\ell'\ell''$ contains D' but not D (its divisor on X is $\ell' X + \ell'' X$, which is $\ge D$ iff $\ell' X \ge D$). So conics containing D form a hyperplane of the 4-dimensional space: dimension 3. This proves the claim and hence that X has no g^1_3.

Second part. A nonsingular curve of type $(3,4)$ on $Q = \mathbf{P}^1 \times \mathbf{P}^1$ exists (the system $|\mathcal{O}_Q(3,4)|$ is very ample, so a general member is nonsingular by Bertini, and it is connected, hence irreducible, being an ample divisor on a nonsingular surface, III 7.9), and has genus $(3-1)(4-1) = 6$ (III, Ex. 5.6). Its projections to the two factors are base-point-free pencils of degrees 3 and 4 (intersection with the rulings), in particular X has a g^1_3. X is not hyperelliptic: a base-point-free g^1_2 together with the base-point-free g^1_3 would give by Lemma A(i) $g \le (2-1)(3-1) = 2$, but $g = 6$. So X is nonhyperelliptic of genus 6 with a g^1_3, hence by the first part it cannot be represented as a nonsingular plane quintic. □

Exercise 5.7

Problem.

(a) Any automorphism of a curve of genus 3 is induced by an automorphism of $\mathbf{P}^2$ via the canonical embedding.

*(b) Assume $\operatorname{char} k \neq 3$. If X is the curve given by

$$x^3y + y^3z + z^3x = 0,$$

the group $\operatorname{Aut} X$ is the simple group of order 168, whose order is the maximum $84(g-1)$ allowed by (Ex. 2.5). See Burnside [1, Section 232] or Klein [1].

*(c) Most curves of genus 3 have no automorphisms except the identity. [*Hint*: For each n, count the dimension of the family of curves with an automorphism T of order n. For example, if $n = 2$, then for suitable choice of coordinates, T can be written $x \to -x$, $y \to y$, $z \to z$. Then there is an 8-dimensional family of curves fixed by T; changing coordinates there is a 4-dimensional family of such T, so the curves having an automorphism of degree 2 form a family of dimension 12 inside the 14-dimensional family of all plane curves of degree 4.]

Note. More generally it is true (at least over $\mathbf{C}$) that for any $g \ge 3$, a "sufficiently general" curve of genus g has no automorphisms except the identity (Baily [1]). (Starred in the book: parts (b), (c).)

Solution. (a) Let X have genus $g \ge 2$ (in particular $g = 3$), $V = H^0(X, \omega_X)$, $\varphi : X \to \mathbf{P}(V^\vee)$ the canonical morphism: $\varphi(P)$ is the hyperplane $V_P = \{s : s(P) = 0\} \subset V$ (a base-point-free system, Lemma 5.1). An automorphism σ acts on differentials by pull-back $\rho = \sigma^* : V \to V$, a linear isomorphism, and for $s \in V$: $(\sigma^* s)(P) = 0 \iff s(\sigma P) = 0$. Hence $\rho(V_{\sigma P}) = V_P$. Let A be the projective linear map of $\mathbf{P}(V^\vee)$ (the space of hyperplanes of V) induced by ρ^{-1} acting on subspaces of V: it sends the hyperplane V_P to $\rho^{-1}(V_P) = V_{\sigma P}$. That is, $\varphi \circ \sigma = A \circ \varphi$. For $g = 3$ nonhyperelliptic, φ is an embedding onto a plane quartic (Example 5.2.1), and σ is the restriction of $A \in PGL_3 = \operatorname{Aut} \mathbf{P}^2$. (For hyperelliptic X the same equation $\varphi\sigma = A\varphi$ holds with φ the canonical $2:1$ map onto a conic.) □

(b) *Hypotheses.* The hypothesis printed in the book is $\operatorname{char} k \neq 3$. I prove the statement in characteristic 0 (hence over $\mathbf{C}$), which is what Ex. 2.5 and Ex. 5.2 (finiteness of $\operatorname{Aut} X$, the bound $84(g-1)$, Hurwitz) require. Two remarks on the printed hypothesis: the curve is singular in characteristic 7 (the point $(1:2:4)$ is a singular point), so $\operatorname{char} k \neq 7$ at least is needed for X to be a genus-3 curve; and I have not tried to prove the statement in positive characteristic, where $|\operatorname{Aut} X| \le 84(g-1)$ is not available in general.

Let $F = x^3y + y^3z + z^3x$. *X is nonsingular:* $F_x = 3x^2y + z^3$, $F_y = x^3 + 3y^2z$, $F_z = y^3 + 3z^2x$. At a singular point $3x^2y = -z^3$, $3y^2z = -x^3$, $3z^2x = -y^3$; multiplying, $27x^3y^3z^3 = -x^3y^3z^3$, so $28x^3y^3z^3 = 0$, $xyz = 0$, say $x = 0$, then $z = 0$ (first equation), then $y = 0$ (third); similarly in the other cases. So X is a nonsingular quartic, $g = 3$, nonhyperelliptic.

By (a) $G = \operatorname{Aut} X = \{A \in PGL_3 : F \circ A = cF \text{ for a scalar } c\}$ (as F is irreducible), a finite group with $|G| \le 168$ (Ex. 5.2, 2.5).

Elements. Let ζ be a primitive 7-th root of unity. $S : (x, y, z) \mapsto (y, z, x)$ and $T = \operatorname{diag}(\zeta, \zeta^4, \zeta^2)$ lie in G (check on monomials: $x^3y \mapsto \zeta^{3+4} = 1$, $y^3z \mapsto \zeta^{12+2} = 1$, $z^3x \mapsto \zeta^{6+1} = 1$). $STS^{-1} = T^4$, so $H = \langle S, T\rangle$ has order 21. Put $s_a = \zeta^a - \zeta^{-a}$ and $p = s_1$, $q = s_2$, $r = s_4$, and $M = \begin{pmatrix} p & q & r \\ q & r & p \\ r & p & q \end{pmatrix}$ (so $M_{ij} = s_{a_ia_j}$ with $(a_1, a_2, a_3) = (1, 2, 4)$ and indices mod 7, $s_{-a} = -s_a$). We show $F(Mv) = 49F(v)$. Let $\delta = p + q + r = \eta - \bar\eta$, $\eta = \zeta + \zeta^2 + \zeta^4$; then $\delta^2 = -7$ ($\eta + \bar\eta = -1$, $\eta\bar\eta = 2$). Direct computation in $\mathbf{Z}[\zeta]$ using $\zeta^7 = 1$, $\sum \zeta^i = 0$: $p^2 + q^2 + r^2 = (\zeta^2 + \zeta^5) + (\zeta^4 + \zeta^3) + (\zeta + \zeta^6) - 6 = -7$, hence $pq + qr + rp = 0$; $pq = \zeta^3 + \zeta^4 - \zeta - \zeta^6$, $qr = \zeta^6 + \zeta - \zeta^5 - \zeta^2$, $rp = \zeta^5 + \zeta^2 - \zeta^3 - \zeta^4$, whence $pqr = \delta$. So $e_1 = \delta$, $e_2 = 0$, $e_3 = \delta$ and $t^3 = \delta t^2 + \delta$ for $t = p, q, r$. Moreover $u := p^2q + q^2r + r^2p = -\delta$ (expand with $p^2 = \zeta^2 + \zeta^5 - 2$, $q^2 = \zeta^3 + \zeta^4 - 2$, $r^2 = \zeta + \zeta^6 - 2$: $u = -\zeta - \zeta^2 - \zeta^4 + \zeta^3 + \zeta^5 + \zeta^6 = -\delta$) and $v := pq^2 + qr^2 + rp^2 = -3\delta - u = -2\delta$. The linear forms $X' = px + qy + rz$, $Y' = qx + ry + pz$, $Z' = rx + py + qz$ are permuted cyclically by $(x, y, z) \mapsto (y, z, x)$, so $F(X', Y', Z')$ is invariant under this cyclic permutation. Expanding, the coefficients are (cyclic classes): x^4: $p^3q + q^3r + r^3p$; x^3y: $p^3r + q^3p + r^3q + 3(p^2q^2 + q^2r^2 + r^2p^2)$; x^3z: $p^4 + q^4 + r^4 + 3pqr(p + q + r)$; x^2y^2: $3(p^3r + q^3p + r^3q + pqr(p + q + r))$; x^2yz: $3(p + q + r)(u + pqr)$. Using $t^3 = \delta(t^2 + 1)$: $p^3q + q^3r + r^3p = \delta(u + \delta) = 0$; $p^3r + q^3p + r^3q = \delta(v + \delta) = -\delta^2 = 7$; $p^2q^2 + q^2r^2 + r^2p^2 = e_2^2 - 2e_1e_3 = -2\delta^2 = 14$; $p^4 + q^4 + r^4 = -3\delta^2 = 21$ (Newton: $P_1 = \delta$, $P_2 = -7$, $P_3 = -4\delta$, $P_4 = 21$). So the coefficients are 0, $7 + 42 = 49$, $21 + 3\delta^2 = 0$, $3(7 + \delta^2) = 0$, $3\delta(-\delta + \delta) = 0$. Hence $F(Mv) = 49(x^3y + y^3z + z^3x)$. Also $M^2 = -7I$: $(M^2)_{ac} = \sum_{b\in\{1,2,4\}} s_{ab}s_{bc} = \sum_{b\in(\mathbf{Z}/7)^*} \zeta^{b(a+c)} - \sum_{b\in(\mathbf{Z}/7)^*} \zeta^{b(a-c)} = 7[a \equiv -c] - 7[a \equiv c] = -7[a = c]$ (squares mod 7 contain no a, c with $a \equiv -c$). So $R = M/\delta \in G$ has $F \circ R = F$, $R^2 = I$, and R is not monomial (all entries nonzero).

$|G| = 168$. Let $P = \langle T\rangle$. The eigenvalues ζ, ζ^4, ζ^2 of T are distinct, so any A with $ATA^{-1} \in \langle T\rangle$ (up to scalar) maps coordinate axes to coordinate axes: $N_G(P)$ consists of monomial matrices. A monomial $A = DP_\pi$ (D diagonal, P_π permutation) satisfies $F \circ A = cF$ iff π preserves the set of monomials $\{x^3y, y^3z, z^3x\}$, i.e. $\pi \in \langle S\rangle$, and $D = \operatorname{diag}(a, b, c')$ satisfies $a^3b = b^3c' = c'^3a$; normalizing $a = 1$: $b = \lambda$, $c' = \lambda^{-2}$, $\lambda^{-6} = \lambda$, i.e. $\lambda^7 = 1$: 7 solutions. So the monomial subgroup of G is H, of order 21, and $N_G(P) = H$ (it contains H and is monomial). Since $R \in G$ is not monomial, $R \notin N_G(P)$, so P is not normal and $n_7 = |G : N_G(P)| = |G|/21 \ge 2$. By Sylow $n_7 \equiv 1 \pmod 7$ and n_7 divides $|G|/7 \le 24$: so $n_7 = 8$ and $|G| = 168$.

Simplicity. Facts: $N_G(P) = H$ of order 21; $C_G(P) = P$ (in H, S does not commute with T); $n_7 = 8$, so G has 48 elements of order 7; every element of G commuting with an element of order 7 lies in the corresponding Sylow 7-subgroup, so an element whose order is divisible by 7 has order exactly 7. Also PGL_3 has no subgroup $(\mathbf{Z}/2)^3$ (lift generators A_i with $A_i^2 = 1$; commutators are scalars ω with $\omega^3 = 1$ (determinants) and $\omega^2 = 1$ (conjugating A_j^2), so the A_i commute, are simultaneously diagonal with entries ± 1, and the image of $\{\pm 1\}^3$ in PGL_3 has order 4). Let $1 \ne N \lhd G$. *Case* $7 \nmid |N|$. Then $P \cap N = 1$, and P acts on N by conjugation; if

P centralized N then $N \subseteq C_G(P) = P$, so $N = 1$. So $P \to \operatorname{Aut} N$ is injective and $7 \mid |\operatorname{Aut} N|$. The group NP has $n_7(NP) = |N : N \cap H| \in \{|N|, |N|/3\}$ dividing 24 and $\equiv 1 \pmod 7$ so equal to 1 or 8. If 1, P is normal in NP and $[N, P] \subseteq N \cap P = 1$, so P centralizes N, contradiction. If 8: $|N| = 24$, or $|N| = 8$. If $|N| = 24$ then $G/N \cong \mathbf{Z}/7$, so all 144 elements outside N have order divisible by 7, hence order 7; but G has only 48 elements of order 7: contradiction. If $|N| = 8$, $7 \mid |\operatorname{Aut} N|$ forces $N \cong (\mathbf{Z}/2)^3$ (the automorphism groups of the other groups of order 8 have orders $4, 8, 8, 24$), impossible by the above. *Case* $7 \mid |N|$. N contains all 8 Sylow 7-subgroups, so $|N| \geq 49$ and $|N| \in \{56, 84, 168\}$. $|N| = 84$: $n_7(N) = 8$ must divide 12, no. $|N| = 56$: the 48 elements of order 7 leave 8 elements, forming the unique Sylow 2-subgroup of N, characteristic in N hence normal in G of order 8, contradicting the first case. So $N = G$: G is simple of order 168 (the unique such group, $PSL_2(\mathbf{F}_7) \cong GL_3(\mathbf{F}_2)$), and $168 = 84(3 - 1)$. □

(c) Let X be nonhyperelliptic of genus 3 (the hyperelliptic curves form a 5-dimensional family in the 6-dimensional $\mathfrak{M}_3$, Example 5.5.6, so they are not general). By (a) and Example 5.5.6, X is a nonsingular quartic and an automorphism is a linear map $A \in PGL_3$ with $F \circ A = cF$. We show that the quartics with a nontrivial automorphism lie in a finite union of constructible sets of dimension ≤ 13 in $\mathbf{P}^{14}$, so the general quartic has trivial automorphism group. (Then in $\mathfrak{M}_3$ the locus has dimension $\leq 13 - 8 = 5 < 6$.)

By Ex. 5.2 and 2.5 ($|\operatorname{Aut} X| \leq 168$) a nontrivial automorphism group contains an element T of prime order $p \leq 168$, so finitely many p. In characteristic 0 such $T \in PGL_3$ is diagonalizable: $T = \operatorname{diag}(1, \varepsilon^a, \varepsilon^b)$, ε a primitive p-th root of unity; this gives finitely many conjugacy classes of T (types). For a given type let $C \subset PGL_3$ be its conjugacy class and Cent the centralizer; $\dim C = 8 - \dim \operatorname{Cent} = 9 - \sum m_i^2$ where m_i are the multiplicities of the distinct eigenvalues ($\dim \operatorname{Cent} = \sum m_i^2 - 1$): $\dim C = 6$ if the eigenvalues are distinct and 4 if two coincide. The quartics with $F \circ T = cF$ are those whose form lies in an eigenspace E_χ of T on the 15-dimensional space of quartic forms: the monomial $x^i y^j z^k$ has eigenvalue ε^{aj+bk}. The union over conjugates gTg^{-1} of the curves invariant under T is the image of $PGL_3 \times \mathbf{P}(E_\chi) \to \mathbf{P}^{14}$, of dimension $\leq \dim C + \dim E_\chi - 1$. We check this is ≤ 13, i.e. $\dim E_\chi \leq 14 - \dim C$:

- Two equal eigenvalues, $T \sim \operatorname{diag}(1, 1, \varepsilon^c)$, need $\dim E_\chi \leq 10$: the eigenvalue is ε^{ck}, k the z-degree. For $p = 2$: k even gives $5 + 3 + 1 = 9$ monomials, k odd gives $4 + 2 = 6$ (so $\dim E_\chi \leq 9$: the 8-dimensional family of the hint, $8 + 4 = 12$). For $p = 3$: $k \equiv 0$: $5 + 2 = 7$; $k \equiv 1$: $4 + 1 = 5$; $k \equiv 2$: 3. For $p \geq 5$ the $k = 0, \ldots, 4$ are distinct mod p so $\dim E_\chi \leq 5$.
- Distinct eigenvalues ($p \geq 3$), need $\dim E_\chi \leq 8$: for $p = 3$, $T = \operatorname{diag}(1, \omega, \omega^2)$ and the eigenvalue of $x^i y^j z^k$ is ω^{j-k}; the 15 pairs (j, k), $j + k \leq 4$, give $5, 5, 5$ for the three classes of $j - k \bmod 3$. For $p \geq 5$ the eigenvalue is ε^{aj+bk} with $b \neq 0$, and j determines $k \in \{0, \ldots, 4\}$ uniquely mod p for a given value, so $\dim E_\chi \leq 5$.

In every case the dimension is $\leq 13 < 14$. Taking the finite union over types, over p, and over the characters χ, the quartics with a nontrivial automorphism lie in a constructible subset of dimension ≤ 13 of $\mathbf{P}^{14}$. Hence the general plane quartic, i.e. the general curve of genus 3, has no automorphism except the identity. □

6. Classification of curves in projective 3-space

Conventions. k is algebraically closed; "curve in $\mathbf{P}^3$" means an irreducible nonsingular projective curve embedded in $\mathbf{P}^3$, d its degree, g its genus, D (or H) its hyperplane section, $\mathcal{O}_X(n) = \mathcal{O}_X(nD)$. A curve is *nondegenerate* if it lies in no plane. A *complete intersection* of surfaces of degrees a, b has $\omega_X = \mathcal{O}_X(a+b-4)$, degree ab and genus $\frac{1}{2}ab(a+b-4)+1$ (II, Ex. 8.4); a nonsingular curve on $\mathbf{P}^1 \times \mathbf{P}^1 \cong$ a nonsingular quadric has type (a,b), degree $a+b$ and genus $(a-1)(b-1)$ (III, Ex. 5.6). Results of the text are quoted by number. The following lemma on quadric cones is stated in the text as Remark 6.4.1(d) (proved in V, Ex. 2.9); I include a short proof.

Lemma C (curves on a quadric cone). Let X be a nonsingular irreducible curve of degree d and genus g on a quadric cone $Q \subset \mathbf{P}^3$ with vertex V. If $d = 2a$ then $V \notin X$ and $g = (a-1)^2$; if $d = 2a+1$ then $V \in X$ and $g = a^2 - a$.

Proof. The blow-up of Q at V is the ruled surface $\tilde{Q} = \mathbf{F}_2$ (V, Section 2), with Pic $= \mathbf{Z}C_0 \oplus \mathbf{Z}f$, $C_0^2 = -2$, $C_0 f = 1$, $f^2 = 0$, $K = -2C_0 - 4f$, hyperplane class $H = C_0 + 2f$, exceptional curve C_0. Since X is nonsingular, its proper transform $\tilde{X} \cong X$, and $\tilde{X} \sim \alpha C_0 + \beta f$ for integers α, β. Then $d = \tilde{X} \cdot H = \beta$, and $\tilde{X} \cdot C_0 = \beta - 2\alpha = \text{mult}_V X \in \{0, 1\}$ (X nonsingular), so β is even iff $V \notin X$. By adjunction $2g-2 = \tilde{X}(\tilde{X}+K) = -2\alpha^2 + 2\alpha\beta - 2\beta$, i.e. $g = 1 - \alpha^2 + \alpha\beta - \beta$. If $\beta = 2a$, $\alpha = a$, $g = 1 - a^2 + 2a^2 - 2a = (a-1)^2$; if $\beta = 2a+1$, $\alpha = a$, $g = 1 - a^2 + a(2a+1) - 2a - 1 = a^2 - a$. □

Exercise 6.1

> **Problem.** A rational curve of degree 4 in $\mathbf{P}^3$ is contained in a unique quadric surface Q, and Q is necessarily nonsingular.

Solution. Let $X \cong \mathbf{P}^1$ be nonsingular of degree 4. X is nondegenerate (a plane curve of degree 4 has genus 3). $\mathcal{O}_X(2) = \mathcal{O}_{\mathbf{P}^1}(8)$ has $h^0 = 9 < 10 = \dim H^0(\mathcal{O}_{\mathbf{P}^3}(2))$, so some quadric Q contains X. Every quadric containing X is irreducible: otherwise it is a union of two planes (or a double plane) and the irreducible X would lie in a plane. If $Q \neq Q'$ both contain X, they have no common component, so $Q \cap Q'$ is a complete intersection curve of degree $4 = \deg X$ containing X; hence $Q \cap Q'$ has the same support as X with multiplicity 1, and, having no embedded points, it equals X scheme-theoretically (a complete intersection is Cohen–Macaulay and generically reduced, so reduced). Then X is a complete intersection of two quadrics and has genus $1 \neq 0$ (see conventions), a contradiction. So Q is unique.

Q is irreducible, so it is nonsingular or a cone. If Q were a cone, Lemma C with $d = 4 = 2a$, $a = 2$ gives $g = (2-1)^2 = 1 \neq 0$, a contradiction. So Q is nonsingular. (Then X has type (a,b) with $a+b=4$, $(a-1)(b-1) = 0$: type $(1,3)$.) □

Exercise 6.2

> **Problem.** A rational curve of degree 5 in $\mathbf{P}^3$ is always contained in a cubic surface, but there are such curves which are not contained in any quadric surface.

Solution. Let $X \cong \mathbf{P}^1$ have degree 5 in $\mathbf{P}^3$. Then $\mathcal{O}_X(3) = \mathcal{O}_{\mathbf{P}^1}(15)$ has $h^0 = 16 < 20 = \dim H^0(\mathcal{O}_{\mathbf{P}^3}(3))$, so the restriction map has a kernel of dimension ≥ 4: X lies on a cubic surface (indeed on a ≥ 4-dimensional space of cubic forms).

An example. Let $\varphi : \mathbf{P}^1 \to \mathbf{P}^3$, $(s : t) \mapsto (s^5 : s^4t : s^3t^2 + st^4 : t^5)$. It is a closed immersion: it has no base point ($s = 0$ gives $t^5 \neq 0$, $t = 0$ gives $s^5 \neq 0$); on the chart $s = 1$ it is $t \mapsto (1 : t : t^2 + t^4 : t^5)$, which is injective with $x_1/x_0 = t$ an immersion, and the point $t = \infty$ maps to $(0 : 0 : 0 : 1)$, not in the image of the chart $s = 1$ (where $x_0 = 1$); near $t = \infty$ (chart $t = 1$, coordinate s): $(s^5 : s^4 : s^3+s : 1)$ and $x_2/x_3 = s^3+s$ has derivative $1 \neq 0$ at $s = 0$. So $X = \varphi(\mathbf{P}^1)$ is a nonsingular rational quintic. A quadric $\sum c_{ij}x_ix_j$ contains X iff $\sum c_{ij}\varphi_i\varphi_j = 0$ identically; setting $s = 1$ the ten products are $1,\ t,\ t^2+t^4,\ t^5,\ t^2,\ t^3+t^5,\ t^6,\ t^4+2t^6+t^8,\ t^7+t^9,\ t^{10}$ (for $x_0^2, x_0x_1, x_0x_2, x_0x_3, x_1^2, x_1x_2, x_1x_3, x_2^2, x_2x_3, x_3^2$). These are linearly independent: the coefficients of $1, t, t^{10}$ force $c_{00} = c_{01} = c_{33} = 0$; t^9 and t^8 force $c_{23} = c_{22} = 0$; t^6 forces $c_{13} = 0$; t^3 forces $c_{12} = 0$; t^5 forces $c_{03} = 0$; and $t^2 + t^4$, t^2 are independent, so $c_{02} = c_{11} = 0$. Hence X lies on no quadric. □

Exercise 6.3

> **Problem.** A curve of degree 5 and genus 2 in $\mathbf{P}^3$ is contained in a unique irreducible quadric surface Q. Show that for any abstract curve X of genus 2, there exist embeddings of degree 5 in $\mathbf{P}^3$ for which Q is nonsingular, and there exist other embeddings of degree 5 for which Q is singular.

Solution. Uniqueness. Let $X \subset \mathbf{P}^3$ be of degree 5, genus 2. It is nondegenerate (a plane quintic has genus 6). $\mathcal{O}_X(2)$ has degree $10 > 2g - 2$, so $h^0 = 10 + 1 - 2 = 9 < 10$ and X lies on some quadric; any such quadric is irreducible (else X is in a plane). Two distinct quadrics containing X would meet in a curve of degree $4 < 5 = \deg X$ containing X (no common component, both irreducible), impossible. So Q is unique and irreducible.

Embeddings with Q nonsingular. Let X have genus 2 and let K be canonical; $|K|$ is the (unique, Prop. 5.3) g^1_2, $h^0(K) = 2$. Choose distinct points P_1, P_2, P_3 no two of which are conjugate under the hyperelliptic involution, and let $D_3 = P_1 + P_2 + P_3$. By Riemann–Roch $h^0(D_3) = 2$. The pencil $|D_3|$ has no base point: a base point must lie in D_3, say P_1, and then $h^0(D_3 - P_1) = 2$, so $P_2 + P_3 \sim K$ ($h^0 = 2$ for a degree 2 divisor forces $\sim K$), i.e. P_2, P_3 are conjugate (or equal Weierstrass points $2P \sim K$, excluded by $P_2 \neq P_3$); contradiction. Also $D_3 \not\sim K + P$ for all P (that would give $h^0(D_3 - P) = h^0(K) = 2$, so P is a base point). Now consider multiplication $\mu : H^0(K) \otimes H^0(D_3) \to H^0(K + D_3)$. It is injective: if $sa + tb = 0$ with s, t a basis of $H^0(K)$ (without common zeros) and $a, b \in H^0(D_3)$, then $sa = -tb$ and $\operatorname{div}(t) \leq \operatorname{div}(a) + \operatorname{div}(s)$, with $\operatorname{div}(t), \operatorname{div}(s)$ disjoint, so a vanishes on $\operatorname{div}(t)$, i.e. $a \in H^0(D_3 - K)$ which is 0 since $D_3 - K$ has degree 1 and $D_3 \not\sim K + P$; so $a = 0$, then $b = 0$. Both sides have dimension 4 ($h^0(K + D_3) = 5 + 1 - 2 = 4$), so μ is an isomorphism. The divisor $D = K + D_3$ has degree $5 = 2g + 1$, hence is very ample (3.2(b)) and $|D|$ embeds X in $\mathbf{P}^3$ with coordinates $x_{ij} = s_ia_j$ (s_0, s_1 a basis of $H^0(K)$, a_0, a_1 a basis of $H^0(D_3)$). Then $x_{00}x_{11} = x_{01}x_{10}$ on X: X lies on the nonsingular quadric $Q : x_{00}x_{11} = x_{01}x_{10}$, which by the uniqueness above is *the* quadric of X.

Embeddings with Q singular. Let $P \in X$ and $D = 2K + P$, degree 5, very ample (3.2(b)), $h^0(D) = 4$. Let $\sigma_P \in H^0(\mathcal{O}(P))$ be the section vanishing at P and let s, t be a basis of $H^0(K)$;

then $\sigma_P s^2, \sigma_P st, \sigma_P t^2$ are independent sections of D (they are $\sigma_P \cdot$ independent elements of $H^0(2K)$, as s^2, st, t^2 are independent: t/s is a nonconstant function), vanishing at P. Since $|D|$ is base point free, there is a section u with $u(P) \neq 0$; so $\{\sigma_P s^2, \sigma_P st, \sigma_P t^2, u\}$ is a basis of $H^0(D)$. In the embedding $(x_0 : x_1 : x_2 : x_3) = (\sigma_P s^2 : \sigma_P st : \sigma_P t^2 : u)$ the curve lies on the quadric cone $x_0 x_2 = x_1^2$ with vertex $(0 : 0 : 0 : 1) =$ the image of P. By uniqueness this is the quadric Q of X, and it is singular. □

Exercise 6.4

Problem. There is no curve of degree 9 and genus 11 in $\mathbf{P}^3$. [*Hint*: Show that it would have to lie on a quadric surface, then use (6.4.1).]

Solution. Let $X \subset \mathbf{P}^3$ be such a curve. It is nondegenerate (a plane curve of degree 9 has genus 28). $\mathcal{O}_X(3)$ has degree $27 > 2g - 2 = 20$, so $h^0(\mathcal{O}_X(3)) = 27 + 1 - 11 = 17 < 20$ and there are at least 3 linearly independent cubic forms F_1, F_2, F_3 vanishing on X.

X lies on a quadric. Suppose not. X is irreducible and lies in no plane, so for each i it lies in an irreducible component of $F_i = 0$ that is a surface of degree ≥ 2; and not of degree 2 (it would be an irreducible quadric containing X). So that component has degree 3, i.e. F_i is irreducible. Then F_1, F_2 are two non-proportional irreducible cubics without common component, so $F_1 \cap F_2$ is a complete intersection curve of degree $9 = \deg X$ containing X; as before it equals X, so X is a complete intersection of type $(3, 3)$, of genus $\frac{1}{2} \cdot 9 \cdot 2 + 1 = 10 \neq 11$. Contradiction. Hence X lies on an irreducible quadric Q.

Using 6.4.1. If Q is nonsingular, X has type (a, b), $a + b = 9$, $(a - 1)(b - 1) = 11$, i.e. $ab = 19$ with $a+b = 9$: $t^2 - 9t + 19$ has discriminant 5, not a square: impossible. If Q is a cone, Lemma C (Remark 6.4.1(d)) with $d = 9 = 2a + 1$, $a = 4$ gives $g = a^2 - a = 12 \neq 11$. So no such curve exists. □

Exercise 6.5

Problem. If X is a complete intersection of surfaces of degrees a, b in $\mathbf{P}^3$, then X does not lie on any surface of degree $< \min(a, b)$.

Solution. Let $X = F_a \cap F_b$ scheme-theoretically (the ideal sheaf $\mathcal{I}_X$ is generated locally by F_a, F_b, which form a regular sequence since X has codimension 2). The Koszul complex gives an exact sequence $0 \to \mathcal{O}(-a - b) \to \mathcal{O}(-a) \oplus \mathcal{O}(-b) \to \mathcal{I}_X \to 0$. Twisting by c and taking cohomology, using $H^1(\mathcal{O}_{\mathbf{P}^3}(c - a - b)) = 0$ (III, 5.1), we get that $H^0(\mathcal{I}_X(c))$ is the image of $H^0(\mathcal{O}(c - a)) \oplus H^0(\mathcal{O}(c - b))$, which is 0 for $c < \min(a, b)$. So no nonzero form of degree $c < \min(a, b)$ vanishes on X. □

Exercise 6.6

Problem. Let X be a projectively normal curve in $\mathbf{P}^3$, not contained in any plane. If $d = 6$, then $g = 3$ or 4. If $d = 7$, then $g = 5$ or 6. Cf. (II, Ex. 8.4) and (III, Ex. 5.6).

Solution. By (II, Ex. 5.14(d)) (X nonsingular, hence normal) projective normality means that $H^0(\mathbf{P}^3, \mathcal{O}(n)) \to H^0(X, \mathcal{O}_X(n))$ is surjective for all $n \geq 0$. For $n = 1$ it is injective (X nondegenerate), so $h^0(\mathcal{O}_X(1)) = 4$. Riemann–Roch: $4 = d + 1 - g + h^1(D)$, so $h^1(D) = g + 3 - d \geq 0$, i.e. $g \geq d - 3$. By Castelnuovo (6.4): $g \leq \frac{1}{4}d^2 - d + 1$ for d even and $g \leq \frac{1}{4}(d^2 - 1) - d + 1$ for d odd. For $d = 6$: $3 \leq g \leq 4$. For $d = 7$: $4 \leq g \leq 6$. It remains to exclude $g = 4$ for $d = 7$. For $n = 2$: $\deg \mathcal{O}_X(2) = 14 > 2g - 2 = 6$ so $h^0(\mathcal{O}_X(2)) = 14 + 1 - 4 = 11 > 10 = h^0(\mathcal{O}_{\mathbf{P}^3}(2))$, so the restriction map is not surjective: X is not projectively normal. Hence $g = 5$ or 6 when $d = 7$. □

Exercise 6.7

> **Problem.** The line, the conic, the twisted cubic curve, and the elliptic quartic curve in $\mathbf{P}^3$ have no multisecants. Every other curve in $\mathbf{P}^3$ has infinitely many multisecants. [*Hint*: Consider a projection from a point of the curve to $\mathbf{P}^2$.]

Solution. A *multisecant* is a line meeting X in at least three distinct points (text before 3.7); for X a line one excludes the line X itself.

No multisecants. *Line:* any other line meets it in at most one point. *Conic:* a line meeting the conic in three points lies in its plane and meets it in at most 2 points (Bezout). *Twisted cubic and elliptic quartic:* let X be nondegenerate of degree d and L a line meeting X in distinct points P_1, P_2, P_3. The linear forms vanishing on L form a 2-dimensional space, which injects into $H^0(\mathcal{O}_X(1))$ (X nondegenerate) and lands in $H^0(\mathcal{O}_X(D - P_1 - P_2 - P_3))$, so $h^0(D - P_1 - P_2 - P_3) \geq 2$, where $\deg(D - \sum P_i) = d - 3$. For $d = 3$ this is a divisor of degree 0, whose $h^0 \leq 1$; for $d = 4$, $g = 1$ it is a divisor of degree 1 on an elliptic curve, with $h^0 = 1$ by Riemann–Roch. Contradiction in both cases.

Every other curve has infinitely many. Let X be a curve in $\mathbf{P}^3$ other than these four. Throughout assume $\operatorname{char} k = 0$ for this part (separability of the pencils used).

Plane curves. If X is a plane curve of degree $d \geq 3$, a general line of its plane meets X transversally in $d \geq 3$ distinct points (Bertini, II 8.18), and there are infinitely many such lines.

Curves on a quadric. Let X be nondegenerate and lie on an irreducible quadric Q. If Q is nonsingular of type (a, b), $a \leq b$, then $d = a + b$, $g = (a - 1)(b - 1)$; the pair (a, b) is not $(1,1), (1,2), (2,2)$ (conic is planar, twisted cubic, elliptic quartic), so $b \geq 3$ (note $(2,2)$ has $g = 1$, $d = 4$, and $(1,2)$ has $g = 0$, $d = 3$; the pairs with $a = 0$ are lines). A general line of the ruling meeting X in b points meets it in $b \geq 3$ distinct points (general fibre of the degree-b morphism $X \to \mathbf{P}^1$, separable), giving infinitely many multisecants. If Q is a cone, by Lemma C: for $d = 2a$, the lines of the cone meet X in a points each, and $a \geq 3$ unless $d = 4$ ($a = 2$, $g = 1$: the elliptic quartic, excluded) or $d = 2$; for $d = 2a + 1$, $V \in X$ and each line of the cone through V meets X in V and a further points, $a \geq 2$ for $d \geq 5$ (and $d = 3$ is the twisted cubic). Again infinitely many multisecants.

General case: reduction via the hint. Let X be nondegenerate, not one of the four types, and suppose, for a contradiction, that X has only finitely many multisecants. They contain only finitely many points of X, so we may take $R \in X$ on none of them ("general" below means: outside a finite set). Project from R: $\psi_R : X \to \mathbf{P}^2$, given by the system of planes through R with the base point R removed (a g^2_{d-1} inside $|D - R|$; it has no base point, since a plane through R transversal to T_R meets X simply at R). If $\psi_R(Q') = \psi_R(Q'')$ with $Q', Q'' \neq R$ distinct, then

R, Q', Q'' are collinear, a multisecant through R: excluded. So ψ_R is injective on $X \setminus R$; in characteristic 0 it is separable, so a map of degree > 1 would have fibres with ≥ 2 points off R, and ψ_R is birational onto its image Y_R, a plane curve of degree $d-1$. Now ψ_R is defined by $|D - R|$, so by (3.1(b)) it is a closed immersion iff for all P, Q (possibly equal) planes through R separate them: this means

(1) no line through R meets X in two further distinct points (true by the choice of R);

(2) T_R meets X only at R, and with multiplicity exactly 2 (so R is not a flex);

(3) no tangent line T_Q, $Q \neq R$, passes through R.

If ψ_R were a closed immersion, $X \cong Y_R$ would be a nonsingular plane curve of degree $d-1$ and $g = \frac{1}{2}(d-2)(d-3)$. By Castelnuovo (6.4), $g \leq \lfloor d^2/4 \rfloor - d + 1$ and $\frac{1}{2}(d-2)(d-3) - (\frac{1}{4}d^2 - d + 1) = \frac{1}{4}(d-2)(d-4) > 0$ for $d \geq 5$, so $d \geq 5$ is impossible; $d = 4$ would give $g = 1$ (elliptic quartic, excluded), $d = 3$ would give $g = 0$ (twisted cubic, excluded), and $d \leq 2$ is a line or planar. Hence, for X not among the four types, *(2) or (3) fails for every R off a finite set.*

The remaining step (not proved). Flexes. If every point of X were a flex, i.e. $T_R \cdot X \geq 3$ at R for all R, then in a local parameter t one has $\gamma'' \parallel \gamma'$ identically; writing $\gamma'' = f\gamma'$ gives $\gamma' = e^{\int f} v$ with v constant (characteristic 0), so X is a line. So for general R the failure of (2),(3) is: *T_R meets X at a second point, or some T_Q ($Q \neq R$) passes through R.* Both say that the correspondence $\mathcal{C} = \{(Q, R) : Q \neq R,\ R \in T_Q \cap X\} \subset X \times X$ has a component of dimension 1 whose two projections are dominant (if the projection to the first factor were finite-to-one onto a finite set of Q's, the R's would lie in finitely many lines and could not fill X). That is: *the tangent line at a general point of X meets X again.* So, in characteristic 0, what remains to prove the second assertion in general is

Lemma T (not proved here). *If $X \subset \mathbf{P}^3$ is a nondegenerate curve over a field of characteristic 0, then $T_P \cap X = \{P\}$ for P in a dense open subset of X.*

Lemma T says that the tangent developable has multiplicity exactly 2 along X at a general point of X. I believe it is true (it is the classical fact that the tangent line at a general point of a non-planar space curve is not a "tangential secant"), but *I could not give a complete proof of it.* What I verified: it fails exactly for plane curves (so nondegeneracy is essential, and Lemma T is where it enters); it holds for nondegenerate curves on a quadric Q (a tangent line $T_P \not\subset Q$ satisfies $T_P \cdot Q = 2P$, so it meets $X \subset Q$ only at P, and $T_P \subset Q$ for only finitely many P); and it holds when X is the image of $t \mapsto (1 : t^a : t^b : t^c)$ for all small exponents (such a tangent line meets X again iff there is $\tau \neq 1$ with $\tau^a = 1 + a(\tau - 1)$, and likewise for b, c; a computer search over $|a|, |b|, |c| \leq 9$ found no solution, but this is only evidence). The argument above is therefore complete for: (i) the four exceptional curves having no multisecants; (ii) plane curves and curves on a quadric (nonsingular or a cone) having infinitely many multisecants; and for (iii) all other nondegenerate curves it is complete *only conditionally on Lemma T*, which is used exactly once, to pass from "(2) or (3) fails for general R" to a contradiction. I flag this as the one gap in Section 6. □

Exercise 6.8

Problem. A curve X of genus g has a nonspecial divisor D of degree d such that $|D|$ has no base points if and only if $d \geq g+1$.

Solution. *Convention.* $|D|$ is understood to be nonempty (and we assume $g \geq 1$, or $d \geq 1$ when $g = 0$: for $g = 0$ and $d = 0$ the trivial divisor $D = 0$ is nonspecial with no base points, an evident degenerate exception).

($\Rightarrow$) Let D be nonspecial of degree d with $|D|$ base point free. If $d \leq g$ then $h^0(D) = d+1-g \leq 1$, so $|D|$ consists of at most one divisor $E \geq 0$ of degree d and has base points $\operatorname{Supp} E$, unless $E = 0$, i.e. $d = 0$, $D \sim 0$; but $D \sim 0$ nonspecial means $h^0(K) = 0$, $g = 0$, the excluded case. So $d \geq g+1$.

($\Leftarrow$) Let $d \geq g+1$. *Reformulation.* If D is nonspecial then $|D|$ is base point free iff $h^0(D-P) = h^0(D) - 1$ for all P (3.1); Riemann–Roch gives $h^0(D-P) = d - g + h^0(K - D + P)$ and $h^0(D) = d + 1 - g$, so $|D|$ is base point free iff $h^0(K - D + P) = 0$ for all P (i.e. all $D - P$ are nonspecial).

Case $g = 0$: take $D = dP_0$, $d \geq 1$; $h^0 = d + 1$, base point free.

Case $g \geq 1$, $d = g + 1$. Choose points $P_1, \ldots, P_g$ inductively: if $h^0(K - P_1 - \cdots - P_{i-1}) \geq 1$ choose P_i outside the (finite) base locus of this complete system, so the h^0 drops by one. Then $h^0(K - D_g) = 0$ for $D_g = \sum_{i \leq g} P_i$ ($h^0(K) = g$). By Riemann–Roch $h^0(D_g) = 1$, so $|D_g| = \{D_g\}$. For a point P: $h^0(K - D_g + P) = h^0(D_g - P) - (g - 1 + 1 - g) = h^0(D_g - P)$ (degree $g-1$), which is ≤ 1 and equals 1 iff $D_g - P$ is linearly equivalent to an effective divisor, i.e. iff $P \in \operatorname{Supp} D_g$ (since $|D_g|$ is a point). For $P = P_i$ the system $|K - D_g + P_i|$ has a unique member F_i, effective of degree $g-1$. Choose $P_{g+1} \notin \bigcup_i F_i$, and put $D = D_g + P_{g+1}$ (degree $g+1$). Then for all P: if $P \notin \operatorname{Supp} D_g$, $h^0(K - D + P) \leq h^0(K - D_g + P) = 0$; if $P = P_i$, $h^0(K - D + P_i) = h^0(F_i - P_{g+1}) = 0$ since F_i is the unique member of its system and $P_{g+1} \notin F_i$. Also $h^0(K - D) \leq h^0(K - D_g) = 0$, so D is nonspecial; so D is nonspecial and base point free of degree $g+1$.

Case $d > g + 1$. Let D_0 be as above of degree $g+1$ and $E \geq 0$ any effective divisor of degree $d - g - 1$, $D = D_0 + E$. Then $h^0(K - D) = 0$ and $h^0(K - D + P) \leq h^0(K - D_0 + P) = 0$ for all P, so D is nonspecial and base point free. $\square$

Exercise 6.9

Problem. Let X be an irreducible nonsingular curve in $\mathbf{P}^3$. Then for each $m \gg 0$, there is a nonsingular surface F of degree m containing X. [*Hint*: Let $\pi : \tilde{P} \to \mathbf{P}^3$ be the blowing-up of X and let $Y = \pi^{-1}(X)$. Apply Bertini's theorem to the projective embedding of $\tilde{P}$ corresponding to $\mathcal{I}_Y \otimes \pi^* \mathcal{O}_{\mathbf{P}^3}(m)$.] (Starred in the book.)

Solution. (The ambient space is $\mathbf{P}^3$, as for a "surface F"; the superscript is blurred in the scan. The same proof gives a nonsingular hypersurface in $\mathbf{P}^r$, $r \geq 3$.)

Let $\pi : \tilde{P} \to \mathbf{P}^3$ be the blow-up along X. It is nonsingular (II, 8.24, the center X being nonsingular), $Y = \pi^{-1}(X)$ is an effective Cartier divisor, $\mathcal{I}_Y = \mathcal{O}_{\tilde{P}}(-Y)$ is invertible and π is an isomorphism off Y. By (II, 7.10(b)) for $m \gg 0$ the invertible sheaf $\mathcal{L}_m = \mathcal{I}_Y \otimes \pi^* \mathcal{O}(m)$ is very ample on $\tilde{P}$ (I used $\mathcal{I}_Y = \mathcal{I}_X \mathcal{O}_{\tilde{P}}$). Embed $\tilde{P} \subset \mathbf{P}^N$ by the complete system $|\mathcal{L}_m|$. By Bertini (II, 8.18) a general hyperplane section G of $\tilde{P}$ is nonsingular (and irreducible, $\dim \tilde{P} = 3$), and, Y being irreducible and $\mathcal{L}_m$ very ample, $G \not\supseteq Y$ for general G.

Identification of sections. A section of $\mathcal{L}_m$ is a section of $\pi^*\mathcal{O}(m)$ vanishing on Y; it restricts on $\tilde{P}\setminus Y \cong \mathbf{P}^3\setminus X$ to a section of $\mathcal{O}(m)$ and extends over X (codimension 2, $\mathbf{P}^3$ normal), so it comes from a form f of degree m, which vanishes on X. So $H^0(\tilde{P},\mathcal{L}_m) = H^0(\mathbf{P}^3,\mathcal{I}_X(m)) =: W_m$, and the hyperplane section G corresponding to f satisfies $\pi^*(F) = G+Y$, where $F = V(f) \supseteq X$ is a surface of degree m; G is the proper transform of F (as $G \not\supseteq Y$). For f general, G is nonsingular, so $F \setminus X \cong G \setminus Y$ is nonsingular.

Nonsingularity of F along X. Let m be large enough that $\mathcal{I}_X(m)$ is globally generated (II, 5.17), so $W_m \to \mathcal{I}_{X,P}/\mathfrak{m}_P\mathcal{I}_{X,P}$ is surjective for each $P \in X$ (a 2-dimensional space, as X is a nonsingular curve in a nonsingular threefold: $\mathcal{I}_{X,P} = (u,v)$ with u,v part of a regular system of parameters). Now $f \in \mathfrak{m}_P^2$ iff the class of f in $\mathcal{I}_P/\mathfrak{m}_P\mathcal{I}_P$ is 0 (because u,v are linearly independent in $\mathfrak{m}_P/\mathfrak{m}_P^2$): two independent linear conditions. So $B_P = \{f \in W_m : f \in \mathfrak{m}_P^2\}$ has codimension 2, and $B = \{(f,P) : f \in B_P\} \subseteq W_m \times X$ is closed of dimension $\dim W_m - 1$. Its image in W_m (a closed set, X projective) is a proper closed subset: for f outside it, $F = V(f)$ is nonsingular at every point of X (F is singular at $P \in X$ iff $f \in \mathfrak{m}_P^2$).

The set of $f \in W_m$ for which G_f is nonsingular and $\not\supseteq Y$ contains a nonempty open subset of $\mathbf{P}(W_m)$ (Bertini), and the set of f with F nonsingular along X is a nonempty open subset. $\mathbf{P}(W_m)$ being irreducible, the two meet: for such f, F is nonsingular off X and along X, so F is a nonsingular surface of degree m containing X. □